Proved outer bounds for the sector lengths of pure symmetric six-qubit states

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Abstract

We give an explicit defining system for a proved outer region $\mathcal{R}_4^{\mathrm{ref}}$ of pure symmetric six-qubit sector lengths and distinguish its constructively attained boundary patches from excluded or unresolved parts. The four-copy inequality $8q_2 + q_4^2 \geq 2q_0q_6$ has exactly one equality orbit: coherent product states. A total-spin-eight projection and Haar variance give the stronger condition $8q_2 + \frac{64}{81}q_4^2 \geq 2q_0q_6$, again with coherent products as the only equality states. The current figures use this refinement, whose curved lower patch and open crease arcs are therefore rigorously excluded, not merely uncertified. Its exact certificate, analytical refinement and sector-coordinate geometry are given here. The remaining bounds combine spin-multipole compatibility, positive angular-momentum projections, Gram matrices and the spectrum of an auxiliary spin-two reduction. The sharp plane $65S_2 + 63S_3 - 217S_1 \leq 1011$ has a fully attained equality segment; we give its state families and complete equality-orbit classification, including $|\omega\rangle$ and the exceptional C_2 state $|E_*\rangle$.

The 22 October strengthening retains two complementary paired-spectral remainders and lifts nine exact null-state supports, the algebraic C_4 threshold and zero-dipole bounds to finite neighborhoods. The zero-dipole flow lift now uses exact scalar comparison integrals, strictly improving its former exponential estimate. Ten gray junctions are excluded by equality-line, transition-quadrupole and flow arguments; the tiny junction is recomputed on the stronger cap. Explicit families determine almost the entire zero-dipole side, with a rigorous sector-length uncertainty below 3.53×10^{-8} on the two remaining small tails. The complete complex-C_2 null-roof image is an explicitly attained inner band for $0 \leq |F| \leq b_c$; its scalar inverse and all restricted equality orbits are determined. All six physical internal modes have the strict local extremum sign along the respective open arcs. Four finite strips have unrestricted complete fiber descriptions, including the enlarged interval $0.170799 \leq |F| \leq 0.175797$. Local stability is not substituted for global exclusion outside these strips. A critical cusp and an additional sharp off-side C_2 surface are proved; their neighborhood widths are not quantified and are not used as finite plotting widths.

The document also retains the previously completed paired-purity and quadrupole-free faces, the zero-overlap roof for $S_1 \geq 2b_c^2/3$ with the algebraic threshold b_c, and the complete attainable part of the lower spin-two plane. Cyclic families, coherent superpositions and collective filters provide explicit states and coverage proofs, not convex-hull interpolations. Three perspectives and magnifications show only exposed, nonredundant boundaries.

Definitions, analytical derivations and state checks are provided here. Certain unrestricted sharpness and continuation lemmas are computer-assisted exact proofs: their finite rational or directed- interval inputs, verifier structure and portable data supplement are identified explicitly. No conjecture or failed numerical fit is used as a proof. The resulting body is a proved outer bound, not a complete characterization of the achievable sector-length image.

1 Definitions and the defining-system convention

Let $\rho = |\psi\rangle\langle\psi|$ be normalized and supported on $\mathrm{Sym}^6(\mathbb{C}^2) \simeq V_3$, where V_j denotes the spin-j space. With $\sigma_0 = I$ and $\sigma_{1,2,3} = X, Y, Z$, define

$$S_k = \sum_{\mathrm{wt}(\mu)=k} \left[\mathrm{Tr}(\rho\sigma_{\mu_1} \otimes \cdots \otimes \sigma_{\mu_6})\right]^2, \qquad S_0 = 1. \tag{1}$$

Here $\mathrm{wt}(\mu)$ counts nonidentity factors. Write $x = S_1$, $y = S_2$, $z = S_3$, and $s(x, y, z) = 45{+}10x{-}2y{-}3z$. Purity and permutation symmetry give the elimination identities

$$\boxed{S_4 = s(x, y, z), \qquad S_5 = 3z - 9x, \qquad S_6 = 18 - 2x + y - z.} \tag{2}$$

In particular $\sum_{k=0}^{6} S_k = 64$.

Use the following constants and functions throughout:

$$s_* := 13 - \frac{8}{5}\sqrt{15}, \qquad b := \sqrt{s_* - 5}, \qquad d := \sqrt{10} - b, \qquad \gamma := \frac{14}{5}, \qquad [v]_+ := \max(v, 0), \tag{3}$$

$$\begin{aligned} D(x, y) &:= 5x + y - 5, \\ A(x, y) &:= \frac{\left[1 - \sqrt{10D(x, y)}\right]_+}{9}, \\ q(x, z) &:= 1 + \frac{x}{4} - \frac{z}{8}, \qquad r(x, z) := \sqrt{q(x, z)}, \\ \Delta(x, y, z) &:= 15 + 5x - y - z, \\ \mathcal{F}(x, r) &:= b + dr + \gamma\sqrt{x}\sqrt{\frac{1 - r}{1 + r}}. \end{aligned} \tag{4}$$

The functions D, A are retained only to state the historical cap and comparison boundaries; that cap is no longer a separate defining condition. Here $x \geq 0$ and $y \geq 5$ ensure $D \geq 0$; the overlap range $0 \leq r \leq 1$, derived below, makes the remaining radicals well defined. Define the orange envelope for every $y \geq 5$ by

$$\Psi(y) := \sup_{7/10 \leq t \leq 1} \frac{14y - 5 - 80t - 8(y - 5)/t}{3}. \tag{5}$$

Proposition 1 (Base defining system)**.** *Every pure symmetric six-qubit sector tuple belongs to the following base region* $\mathcal{R} \subset \mathbb{R}^3$*:*

$$x \geq 0, \qquad \boxed{z \geq 2x + \frac{12x}{4 + \sqrt{5(y-5)/2}}}, \tag{6a}$$

$$y \geq 5, \tag{6b}$$

$$q(x,z) \geq 0, \tag{6c}$$

$$5x + 5 \leq y + z, \tag{6d}$$

$$\boxed{\Delta(x,y,z) \geq 0}, \tag{6e}$$

$$\boxed{5y + 6z - 15x - 85 \leq \sqrt{40\Delta(x,y,z)}}, \tag{6f}$$

$$\boxed{s(x,y,z) \geq \Psi(y)}, \tag{6g}$$

$$\boxed{65y + 63z - 217x \leq 1011}, \tag{6h}$$

$$\boxed{10(y-5) + \frac{24x}{3+4t} + \frac{81q(x,z)}{t} \geq \frac{1}{1+t}} \quad (\forall t > 0), \tag{6i}$$

$$\boxed{\sqrt{y-5} \leq \mathcal{F}\big(x, r(x,z)\big)}. \tag{6j}$$

The support envelope has the explicit form

$$\Psi(y) = \begin{cases} \dfrac{6y-9}{7}, & 5 \leq y \leq \frac{99}{10}, \\ \dfrac{14y - 5 - 160\sqrt{(y-5)/10}}{3}, & \frac{99}{10} \leq y \leq 15, \\ 2y - 15, & y \geq 15. \end{cases} \tag{7}$$

The first two branches are denoted $\Phi(y)$ *when restricted to* $5 \leq y \leq 15$*. The extension beyond* 15 *avoids assuming an upper domain restriction that is actually implied by the system itself.*

There is also the strict conditional compatibility conclusion

$$q(x,z) = 0, \quad 0 < x < \frac{3}{32} \quad \Longrightarrow \quad y > \frac{51}{10} - \frac{4}{5}x, \tag{8}$$

proved in Theorem 6. The closed body $\mathcal{R}$ is retained as the continuous necessary-bound geometry; the excluded interior of this particular roof rim is marked separately. Compactness gives a positive excluded neighborhood at each of its interior points, but does not supply a sharp replacement curve or a plotting width. In contrast, the octupole-strengthened coefficient $\gamma = 14/5$ in (6j) is a global inequality. It defines the historical base region; the current region replaces it by the stronger unit-dipole coefficient $\gamma_{22} = \sqrt{3/2}$ in (377).

A second conditional compatibility conclusion determines the null paired rim exactly:

$$q = 0, \quad y = 7 + 3x \quad \Longrightarrow \quad \boxed{\frac{2}{27} \leq x \leq \frac{1}{6}}, \tag{9}$$

with every point in this interval attained. The lower endpoint has the unique state orbit η_6 of (205), as proved in Theorem 9. This is a statement *on that rim only*, not a universal lower bound on S_1. Its excluded part is distinguished from the attained segment without introducing an unsupported finite-width cut or altering the closed body $\mathcal{R}$.

The full teal family (6i) replaces the older single root cap. The domain condition $q \geq 0$ is explicit, including cases where the supporting family imposes no positive lower bound on q. The new lower bound gives $z \geq 2x$ and therefore $q \leq 1$; hence $0 \leq r \leq 1$ and the implicit red condition (6j) is well defined. It is imposed as written, without replacing the overlap-dependent factor by one. Likewise,

the linear low branch of (6g) is the manuscript's $a_4 \geq 0$ inequality. It is now dominated by (6h); only the surviving exposed part of the orange envelope is drawn, not an additional redundant flat facet. Redundancy certificates are collected in Section 22. In particular the old flat spectral inequality is implied by (6f) and is not another defining condition. The domain $\Delta \geq 0$ is explicit: the left side of (6f) is not assumed nonnegative and must not simply be squared.

Condition	Origin	Proof here
$x \geq 0$, $y \geq 5$	Nonnegative spin-multipole lengths	Sec. 4
New lower boundary	Three-vector Gram matrix and quadrupole norm	Sec. 15
$q \geq 0$	Squared time-reversal overlap, $q = \lvert\Theta\rvert^2$	Sec. 4
$y + z \geq 5x + 5$	Nonnegative two-copy spin-two weight p_2	Sec. 4
Paired-spectral curved cap	Auxiliary spectral gap and trace rearrangement	Sec. 17.3
Paired-purity plane	Auxiliary paired spectrum	Sec. 17.2
Orange envelope	Even four-qubit correlations and purity	Sec. 16
Sharp singlet-dependent plane	Positive irreducible spin-projection blocks, $a_4 \geq 9\lvert\Theta\rvert^2/286$	Sec. 23
Full teal supporting family	Singular and rotation Gram matrices; Cauchy–Schwarz	Sec. 18
Red implicit cap	Sharp null slice, octupole compatibility, balancing and time reversal	Sec. 14
Strict null-roof supplement	Common-kernel compatibility of the slice and rotation Gram matrices	Thm. 6
Sharp null paired rim	Flat auxiliary kernel, transition quadrupole and Cartesian compatibility	Sec. 14.5
Refined four-copy lower boundary	Positive four-copy compression, total-spin-eight projection and Haar variance	Sec. 2

The current full system. The latest figures show

$$\boxed{\begin{aligned} \mathcal{R}_4 &:= \mathcal{R}_{22} \cap \{\mathscr{C}_4 \geq 0\}, \\ \mathcal{R}_4^{\mathrm{ref}} &:= \mathcal{R}_{22} \cap \{\widehat{\mathscr{C}_4} \geq 0\} \ \subseteq \mathcal{R}_4. \end{aligned}} \tag{10}$$

where $\mathcal{R}_{22}$ is defined in Section 26 and $\mathscr{C}_4, \widehat{\mathscr{C}_4}$ in (13) and (20). Only the refined four-copy boundary is plotted; the weaker original bound remains a proved corollary, not a second surface. The refinement is added to the full previous system, not used as a substitute for any of its other bounds. The superscript "ref" denotes a refinement, not a claim of an optimal coefficient or a sharp attainable surface. To recover that previous system from (6), replace the historical implicit cap (6j) by (377), with $\gamma_{22} = \sqrt{3/2}$; replace the old paired-spectral row (6f) by the retained maximum (383); add the nine null-seed lifts (389)–(390), the critical C_4 input (391), and the guarded flow lift (395). The zero-dipole necessary envelope is specified in Section 27.2. Those sections give all domains, exact state selectors, origins and proofs. A dominated condition remains a useful corollary but does not produce a separate plotted surface. Existential, unquantified sharp-neighborhood theorems are not additional finite-width inequalities or color masks. Theorem 22 adds the full constructive C_2 null-roof band. The sharp *conditional* inequalities (455) describe unrestricted complete null fibers on the enlarged certified strip; they do not change this three-dimensional defining system or assert a finite positive-q width.

2 A four-copy inequality and the new lower-boundary patch

This strengthening uses compatibility of four identical pure spin-three vectors. It is not a marginal-purity positivity condition and is not obtained by taking a convex hull of attainable sector tuples.

2.1 Pair-spin probabilities and the sector inequality

Let Π_J project $V_3 \otimes V_3$ onto total spin J and define

$$q_J := \langle\psi|^{\otimes 2} \Pi_J |\psi\rangle^{\otimes 2}.$$

Only $J = 0, 2, 4, 6$ occur because the two copies are identical, and their probabilities sum to one. In the present normalization they are

$$\begin{aligned} q_0 &= \frac{8 + 2x - z}{56}, & q_2 &= \frac{5(y + z - 5x - 5)}{168}, \\ q_4 &= \frac{3(162 + 8x - 10y - 3z)}{616}, & q_6 &= \frac{401 + 137x + 35y + 5z}{1848}. \end{aligned} \tag{11}$$

In particular $q_0 = q(x, z)/7$: the subscripted singlet probability is *not* the unsubscripted overlap $q = |\Theta|^2$ used elsewhere.

Proposition 2 (Four-copy lower bound)**.** *Every normalized pure symmetric six-qubit state satisfies*

$$\boxed{8q_2 + q_4^2 \geq 2q_0 q_6.} \tag{12}$$

Equivalently its sector lengths obey

$$\boxed{\begin{aligned} \mathscr{C}_4(x, y, z) := \frac{5}{21}(y + z - 5x - 5) + \left[\frac{3(162 + 8x - 10y - 3z)}{616}\right]^2 \\ - \frac{(8 + 2x - z)(401 + 137x + 35y + 5z)}{51744} \geq 0. \end{aligned}} \tag{13}$$

Four-copy proof and exact finite certificate. Let P_4 be the orthogonal projector onto $\mathrm{Sym}^4(V_3)$. On this space consider the self-adjoint, rotation-invariant operator

$$T = \mathsf{P}_4 \big[8\Pi_2^{(12)} \otimes I^{(34)} + \Pi_4^{(12)} \otimes \Pi_4^{(34)} - 2\Pi_0^{(12)} \otimes \Pi_6^{(34)}\big] \mathsf{P}_4 \Big|_{\mathrm{Sym}^4(V_3)}. \tag{14}$$

Since $\psi^{\otimes 4} \in \mathrm{Sym}^4(V_3)$, its expectation is exactly $8q_2 + q_4^2 - 2q_0 q_6$. Thus it suffices to prove $T \succeq 0$.

Here is a completely specified finite spectral certificate. The pair projectors can be formed without Clebsch–Gordan phase conventions: with $K = (\mathbf{J}_1 + \mathbf{J}_2)^2$,

$$\Pi_J = \prod_{\substack{0 \leq \ell \leq 6 \\ \ell \neq J}} \frac{K - \ell(\ell+1) I}{J(J+1) - \ell(\ell+1)}.$$

Use unnormalized Dicke vectors $E_k = \sqrt{\binom{6}{k}}\, D_k$. An orthogonal basis of the total-magnetization-zero subspace is

$$v_{\boldsymbol{k}} = \sum_{\text{distinct permutations } \pi} E_{k_{\pi(1)}} \otimes \cdots \otimes E_{k_{\pi(4)}}, \qquad 0 \leq k_1 \leq \cdots \leq k_4 \leq 6, \quad \sum_i k_i = 12.$$

There are 18 such vectors. Their squared norms are $|\,\mathrm{Orb}(\boldsymbol{k})| \prod_i \binom{6}{k_i}$, so the restricted matrix T_0 is self-adjoint in this positive diagonal Gram matrix. Exact rational arithmetic gives

$$\det(tI + T_0) = \frac{t^3 \prod_{i=1}^{8} h_i(t)}{10241846214245335268089155296 19}, \tag{15}$$

where

$$\begin{aligned} h_1 &= 9t + 26, & h_2 &= 21t + 68, \quad h_3 = 63t + 152, \\ h_4 &= 99t^2 + 458t + 208, \\ h_5 &= 7623t^2 + 19351t + 7980, \\ h_6 &= 3361743t^2 + 23415966t + 37896248, \\ h_7 &= 3361743t^3 + 18065565t^2 + 23810696t + 268736, \\ h_8 &= 10085229t^3 + 76315176t^2 + 147641124t + 24816736. \end{aligned} \tag{16}$$

Each factor is strictly positive for $t > 0$. Consequently T_0 has no negative eigenvalue: a negative eigenvalue λ would make $\det(tI + T_0) = 0$ at $t = -\lambda > 0$. All irreducible spins in $\mathrm{Sym}^4(V_3)$ are integers and contain a zero-magnetization vector. Rotation invariance therefore ensures that every eigenvalue of T appears in T_0. This proves $T \succeq 0$ and (12). Substitution of (11) gives (13). The exact determinant identity is reproducible from the portable verifier `assets/verify_six_qubit_four_copy_inequality.py`; its input basis and projectors are precisely those specified above. The proof of positivity is finite and exact, not a numerical eigenvalue test or an optimization over state samples. □

2.2 Complete equality classification of the original bound

Theorem 1 (The coherent equality orbit)**.** *For a normalized pure symmetric six-qubit state, equality in* (12) *holds if and only if*

$$|\psi\rangle = e^{i\chi} u^{\otimes 6}, \qquad \langle u|u\rangle = 1. \tag{17}$$

Thus there is one equality orbit under collective rotations and global phase. Its full sector tuple is $(6, 15, 20, 15, 6, 1)$.

Proof. Write $F = 8q_2 + q_4^2 - 2q_0q_6 = \mathscr{C}_4(x, y, z)$. The positivity of T proved above implies that $F = 0$ forces $T\psi^{\otimes 4} = 0$. Let W_8 project onto total four-copy magnetization $M = 8$. It commutes with T and with copy symmetrization, so $v = W_8\psi^{\otimes 4}$ is still symmetric and lies in $\ker T$. The negative term $\Pi_0^{(12)} \otimes \Pi_6^{(34)}$ has total spin six and vanishes on this subspace. The two remaining terms are positive. In particular, on $M = 8$,

$$\Pi_4^{(12)} \otimes \Pi_4^{(34)} = |4, 4\rangle|4, 4\rangle\langle 4, 4|\langle 4, 4|.$$

Consequently $0 = \langle v, Tv\rangle$ implies $|\langle 4, 4|\psi^{\otimes 2}\rangle|^4 = 0$. The same argument applies after every collective rotation. The orbit of the highest-weight vector spans the irreducible space V_4, hence

$$\Pi_4\psi^{\otimes 2} = 0, \qquad q_4 = 0. \tag{18}$$

A short compatibility calculation now finishes the proof. Use the independently established sector bounds $x \le 6$, $y \le 15$ and put $\varepsilon = 6 - x \ge 0$, $\eta = 15 - y \ge 0$. From (11) and $q_4 = 0$,

$$z = \frac{162 + 8x - 10y}{3}, \qquad q_0 = \frac{\varepsilon - 5\eta}{84}, \qquad q_2 = \frac{5(\varepsilon + \eta)}{72}.$$

Equality $F = 0$ gives $4q_2 = q_0q_6 \le q_0$, because $0 \le q_6 \le 1$. Multiplication by 252 yields $67\varepsilon + 85\eta \le 0$. Both variables are nonnegative, so $x = 6$, $y = 15$ and $z = 20$. Since $x = 2|\langle \mathbf{J}\rangle|^2/3$, $x = 6$ means that the dipole has its maximal length three. In its direction $\boldsymbol{n}$, $J_{\boldsymbol{n}} \le 3I$ and $\langle J_{\boldsymbol{n}}\rangle = 3$ force the highest-weight state, which is exactly (17). Conversely a coherent product has $q_6 = 1$ and $q_0 = q_2 = q_4 = 0$, and saturates the bound. The argument uses only positive spin projections and compatibility; no equality ansatz or polynomial elimination is needed. An independent exact check of the magnetization-eight block and a spin-ladder version of the last step is supplied by `assets/spin_projection_rigidity.py`; it uses the same pair-projector construction as the original positivity verifier. □

2.3 A quantitative spin-eight refinement

Theorem 2 (Refined bound and its equality states)**.** *Every normalized pure symmetric six-qubit state satisfies*

$$F \ge \frac{17}{81} q_4^2, \qquad \textit{equivalently} \qquad \boxed{8q_2 + \frac{64}{81} q_4^2 \ge 2q_0q_6.} \tag{19}$$

Equality in the refined inequality holds exactly for the coherent products (17). *In sector coordinates, the refined gap is*

$$\boxed{\begin{aligned}\widehat{\mathscr{C}}_4(x,y,z) := \frac{5}{21}(y+z-5x-5) + \left[\frac{162+8x-10y-3z}{231}\right]^2 \\ - \frac{(8+2x-z)(401+137x+35y+5z)}{51744} \geq 0.\end{aligned}} \tag{20}$$

Proof. Let P_8 project onto total spin eight in the four-copy space and put $v_8 = \mathsf{P}_8\psi^{\otimes 4}$. Because T is positive and rotation invariant,

$$F \geq \langle v_8, T v_8\rangle.$$

The total-spin projector commutes with copy symmetrization; hence $v_8 \in \mathrm{Sym}^4(V_3)$ and the surrounding symmetrizers in (14) disappear from this expectation. The negative pair term has total spin six and vanishes. Discard the other positive term $8\Pi_2^{(12)} \otimes I^{(34)}$ and set $w = \Pi_4\psi^{\otimes 2} \in V_4$, so $\|w\|^2 = q_4$. Pair-spin projectors commute with total-spin projectors, giving

$$F \geq \|\mathsf{P}_8(w\otimes w)\|^2. \tag{21}$$

Here P_8 on $V_4 \otimes V_4$ is the maximum-spin, or Cartan-product, projector. This step does not assume that a pair projector commutes with the copy symmetrizer.

For normalized Haar measure dU on $SU(2)$, Schur averaging and the spin-eight coherent resolution of the identity give

$$\begin{aligned}\int |\langle 4,4|D^{(4)}(U)w\rangle|^2\, dU &= \frac{q_4}{9}, \\ \int |\langle 4,4|D^{(4)}(U)w\rangle|^4\, dU &= \frac{\|\mathsf{P}_8(w\otimes w)\|^2}{17}.\end{aligned} \tag{22}$$

The denominators are the dimensions of V_4 and V_8. Applying Jensen's inequality to the nonnegative function $h(U) = |\langle 4,4|D^{(4)}(U)w\rangle|^2$ proves

$$\|\mathsf{P}_8(w\otimes w)\|^2 = 17\int h^2\, dU \geq 17\left(\int h\, dU\right)^2 = \frac{17}{81}q_4^2.$$

Together with (21) this is (19); substituting (11) gives (20).

If $q_4 > 0$, Jensen's inequality here is *strict*. Indeed a nonzero $w \in V_4 \simeq \mathrm{Sym}^8(\mathbb{C}^2)$ has a coherent overlap given by a nonzero homogeneous polynomial of degree eight in a spinor. Such a polynomial has a zero on the complex projective line, possibly at infinity. Thus the continuous function h has a zero, but has positive average $q_4/9$; it cannot be constant. Consequently equality in the refined bound forces $q_4 = 0$. It then reduces to $F = 0$, and Theorem 1 proves coherence. Conversely coherent products saturate both four-copy bounds. This does not claim that 64/81 is an optimal coefficient. □

In particular *both* the original and the refined equality surfaces contain no pure-state sector point other than $(6, 15, 20)$. Every other point on the new curved patch and either open crease arc is rigorously excluded. For any nonempty pure-state fiber at fixed $(x, y) \neq (6, 15)$, compactness gives its minimum strictly above the refined root below; no explicit sharp replacement or uniform positive plotting width is inferred from this strictness alone.

2.4 A lower graph, not an additional upper cap

For comparison, the original gap has the polynomial form

$$\begin{aligned}1138368\,\mathscr{C}_4 &= 353z^2 + B_4(x,y)z + C_4(x,y), \\ B_4(x,y) &= 1498x + 2390y + 252738, \\ C_4(x,y) &= -4300x^2 - 5860xy - 1326972x + 2700y^2 + 177400y - 717188.\end{aligned} \tag{23}$$

The coefficient polynomial $C_4(x,y)$ is unrelated to a C_4-symmetric state family. The current refined gap is instead

$$\begin{aligned} 1707552\,\widehat{\mathscr{C}}_4 &= 453z^2 + \widehat{B}_4(x,y)z + \widehat{C}_4(x,y),\\ \widehat{B}_4(x,y) &= 2655x + 3075y + 387369,\\ \widehat{C}_4(x,y) &= -6994x^2 - 7430xy - 2012490x + 3200y^2 + 293640y - 1298856. \end{aligned} \tag{24}$$

On $x \ge 0, y \ge 5, z \ge 0$, both gaps increase strictly with z:

$$\begin{aligned} \frac{\partial \mathscr{C}_4}{\partial z} &= \frac{749x + 1195y + 353z + 126369}{569184} > 0,\\ \frac{\partial \widehat{\mathscr{C}}_4}{\partial z} &= \frac{885x + 1025y + 302z + 129123}{569184} > 0. \end{aligned} \tag{25}$$

Thus the condition cuts the *lower* boundary. On the plotting domain $0 \le x \le 6$, $5 \le y \le 15$, let

$$Z_4(x,y) := \frac{-2C_4(x,y)}{B_4(x,y) + \sqrt{B_4(x,y)^2 - 1412C_4(x,y)}}. \tag{26}$$

This historical larger root is rationalized to avoid cancellation. The refined root used in every current figure is

$$\widehat{Z}_4(x,y) := \frac{-2\widehat{C}_4(x,y)}{\widehat{B}_4(x,y) + \sqrt{\widehat{B}_4(x,y)^2 - 1812\widehat{C}_4(x,y)}}. \tag{27}$$

Both discriminants are positive on the domain: for the original, $B_4 \ge 252738$ and $C_4 \le 2551312$ suffice; for the refinement, $\widehat{B}_4 \ge 387369$ and $\widehat{C}_4 \le 3825744$ suffice. A negative root gives no additional restriction on $z \ge 0$. The current lower graph is exactly

$$\boxed{\widehat{Z}_-^{(4)}(x,y) = \max\left\{2x + \frac{12x}{4 + \sqrt{5(y-5)/2}},\quad 5x + 5 - y,\quad \widehat{Z}_4(x,y)\right\}.} \tag{28}$$

The upper graph is still the full $\mathcal{R}_{22}$ upper graph; no old upper condition is replaced or weakened.

The new inequality also retains the explicit projection witness used in the base geometry discussion. To check this without sampling, put $u = 3x/(y+3)$, $v = (y-5)/10$ and $z_{\mathrm{w}} = x[2 + 24/(y+3)]$. On $0 \le u, v \le 1$, direct substitution in the refined polynomial $\widehat{P}_4 = 1707552\widehat{\mathscr{C}}_4$ gives

$$\begin{aligned} \widehat{P}_4(x,y,z_{\mathrm{w}}) &= (1-u)^2 A(v) + 2u(1-u)B(v) + \frac{2508800}{9}u^2(1-v)^2,\\ A(v) &= 249344(1-v)^2 + 3755088v(1-v) + 3825744v^2,\\ B(v) &= \frac{604352}{3}(1-v)^2 + 1691648v(1-v) + 1788864v^2. \end{aligned} \tag{29}$$

Every term is nonnegative; the only zero on this square is the product point $u = v = 1$. This is an exact nonnegativity check for the added condition, not an assumption of convexity of the new lower graph.

The existing magnifications all have $0 \le x \le 1$, where the new cut is inactive on the entire old vertical interval. Indeed, put $w = \sqrt{5(y-5)/2}$ and let L_{G} denote the first entry of (28). Expanding the polynomial and using $0 \le x \le 1$, $w \ge 0$ term by term gives

$$\frac{(w+4)^2}{4}\widehat{P}_4(x,y,L_{\mathrm{G}}) \ge 285034w^2 - 781384w + 853696 \ge \frac{45346073400}{142517} > 0.$$

Monotonicity in z then proves the same statement above the old floor. Thus those cached magnified views remain valid without drawing an obsolete comparison boundary.

A strict, finite-region improvement. The rational tuple

$$(x, y, z) = \left(\frac{24}{5}, \frac{25}{2}, \frac{1653}{100}\right) \tag{30}$$

lies strictly inside the previous vertical interval but satisfies

$$\mathscr{C}_4 = -\frac{36821201}{3794560000} < 0, \qquad \widehat{\mathscr{C}}_4 = -\frac{222318673}{17075520000} < 0.$$

At its first two coordinates the pre-four-copy floor is approximately 16.5146604691, the original root is $Z_4 \simeq 16.5666396499$, and the refined root is $\widehat{Z}_4 \simeq 16.5790175768$, whereas the witness has $z = 16.53$. For an analytical check of the less elementary previous conditions, here $q = 107/800$, $\Delta = 997/100$ and $f^2 = 36/5$. The null-seed and balancing-flow guards are inactive. In (383), $G > 15$, whereas its two right-hand terms are bounded above by $8\Delta q < 11$ and $32q < 5$, respectively: use $K \geq 1/2$ and $H \geq \Delta/8$ in (381). The unit-dipole cap is strict since $r < 3/8$ and its right side is larger than $4/3 + \sqrt{24/11} > 11/4 > \sqrt{15/2}$. The full supporting family has the already positive term $10(y - 5) = 75$; the remaining affine and orange tests are direct substitutions. Thus all previous upper cuts are strict there. Continuity therefore excludes a finite region, not merely an isolated tuple. In contrast, the often-used midpoint $(3, 10, 10)$ is already excluded by the three-vector Gram bound; it is not used here to claim nonredundancy.

The refinement is also strict relative to the original four-copy region $\mathcal{R}_4$. At $(x, y, z) = (24/5, 25/2, 1657/100)$ the preceding upper tests remain strict, whereas

$$\mathscr{C}_4 = \frac{3377159}{3794560000} > 0, \qquad \widehat{\mathscr{C}}_4 = -\frac{40900753}{17075520000} < 0.$$

Thus the stronger coefficient removes an additional finite region, not just a previously excluded equality surface.

2.5 Exposed creases and the additional gray junction

The two old lower sheets meet on

$$c(v) = \left(\frac{2v(v+4)}{15},\ 5 + \frac{2v^2}{5},\ \frac{4v(v+10)}{15}\right), \qquad 0 \leq v \leq 5. \tag{31}$$

On this curve the original gap has the exact factorization

$$1138368\,\mathscr{C}_4(c(v)) = \frac{192}{25}(v - 5)(49v^3 + 497v^2 - 780v - 6180). \tag{32}$$

The refined gap, which governs the current figures, instead gives

$$\widehat{P}_4(c(v)) = \frac{64}{225}(v - 5)(1568v^3 + 17213v^2 - 21594v - 175320). \tag{33}$$

Let $\widehat{v}_4$ be the unique positive zero of this last cubic. It lies in $(3, 4)$ and is approximately 3.31933258350206. The old Gram/spin-two crease survives only for $0 \leq v \leq \widehat{v}_4$; its remaining open segment lies below the refined floor. The two genuine current creases are the exposed intersections of $\widehat{Z}_4$ with the old Gram and spin-two sheets. They meet at the product point $(6, 15, 20)$ and at

$$J_4 := c(\widehat{v}_4) \simeq (3.2393732179,\ 9.4071875200,\ 11.7896785693). \tag{34}$$

The former unrefined cut had its cubic zero at $v_4 \simeq 3.6547778508454$ and junction $c(v_4) \simeq (3.7302016723, 10.3429604556, 13.3080479060)$. That contact is superseded, not kept as a second plotted node or crease.

The new curved patch and both open crease arcs are gray because *every one of their points is rigorously unattainable*, by Theorem 2; only the coherent product endpoint is violet. In particular J_4

is excluded. Independently, it saturates the lower Gram bound with $x > 0$ and $5 < y < 15$, whereas the equality classification in Section 15 permits only $y = 5$ or $y = 15$ when $x > 0$. Thus neither the new gray node nor the gray patch relies on failed numerical fitting. This conclusion does not classify older gray surfaces on which $\widehat{\mathscr{C}_4} > 0$. All previously labeled attained states, the upper roof and the two vertical side faces are retained. Rear dashes and front solid strokes are recomputed against the new body, and no obsolete crease is drawn through the interior of the new patch.

3 The figure and its state labels

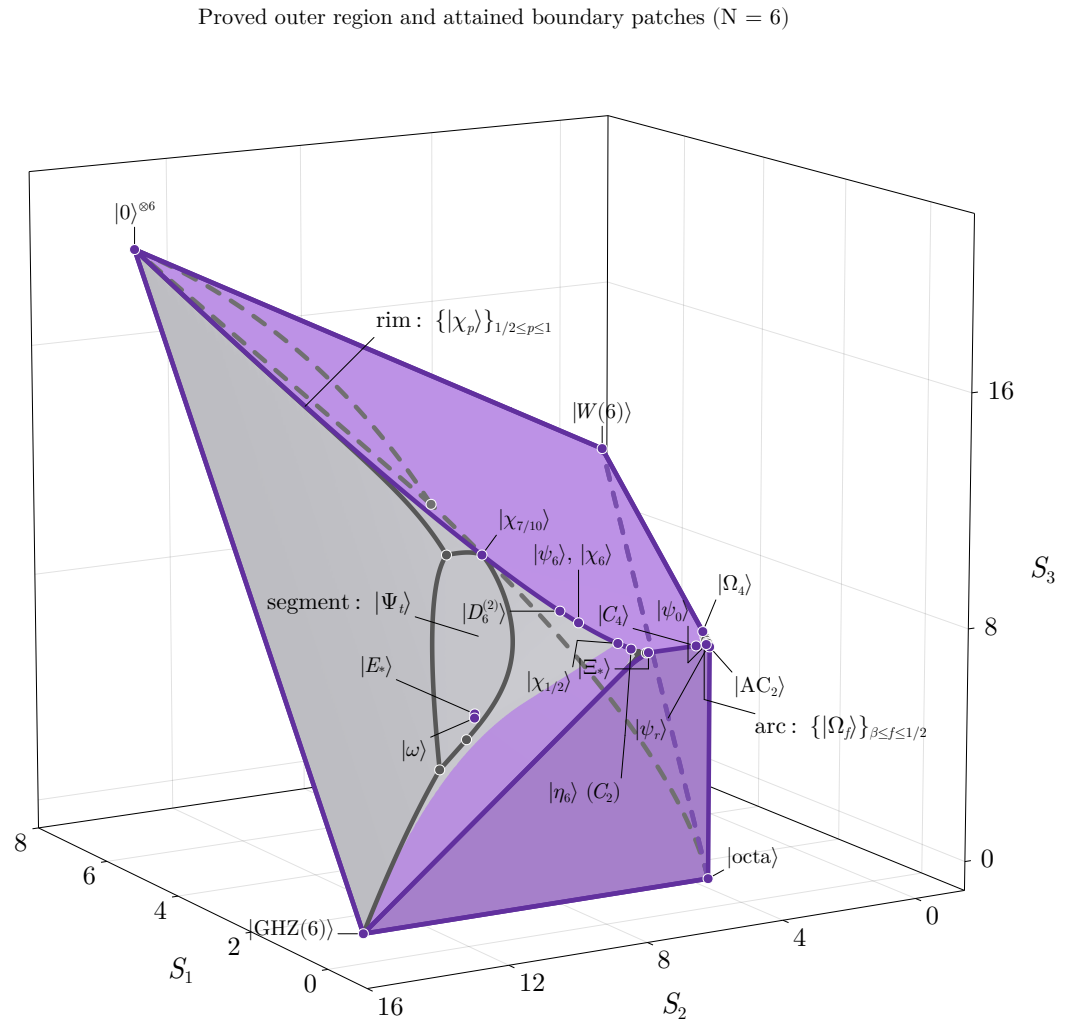


Figure 1: Current proved outer region $\mathcal{R}_4^{\mathrm{ref}}$: the complete 22 October system of Section 26 intersected with refined four-copy bound (20). Its new lower patch, open crease arcs and shifted gray junction J_4 are all rigorously excluded except at the product endpoint, by Theorem 2. The upper roof is unchanged. Section 32 treats the earlier ten gray junctions and the replacement microscopic flow contact. Violet denotes constructive attainment; gray denotes exclusion or unresolved attainment. The entire segment $\chi_{7/10}$–ω, including $|E_*\rangle$, is attained, not the surrounding supporting face (Theorem 14). Sections 27 and 29 add the finite unrestricted roof strips, full constructive C_2 inner band, exact C_4 threshold and almost complete zero-dipole side. The second strip has the widened width of Theorem 24. The inner band alone is not claimed to exhaust the unrestricted roof. Unquantified off-roof neighborhoods receive no invented width. Front edges are solid; rear edges are lighter and dashed, and rear nodes are lighter. Uniform flat nodes and border-connected leaders are retained. Strokes show only true body edges: interior family traces are omitted without changing attained fills, state nodes, labels or leaders. Named states need not be geometric junctions. This is an outer region, not the exact achievable set. The fold-outs (pp. 152, 153) retain all magnifications, including the E_*–ω detail and the resolved small-dipole roof.

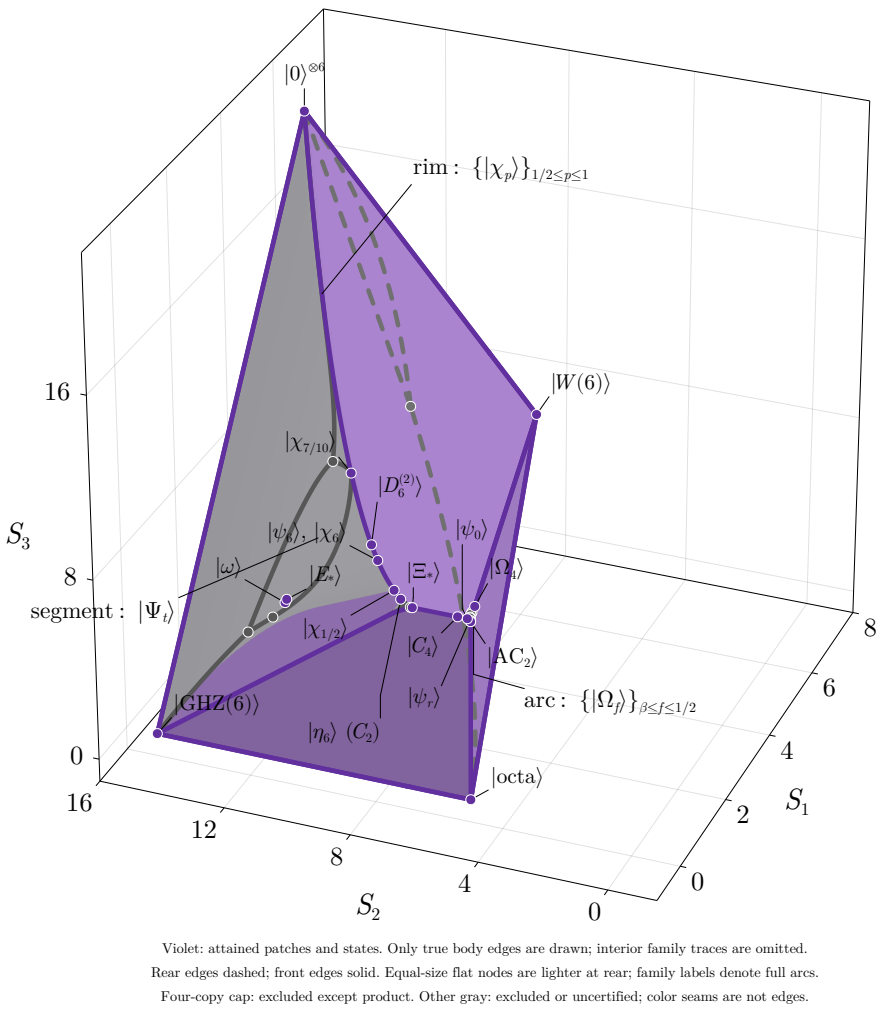


Figure 2: Alternative perspective of exactly the same outer region and attained sets as Fig. 1. The paired-spectral cap is evaluated from its exact nonlinear roof, without convexification. The state labels, uniform flat nodes and true-edge strokes, and camera-dependent rear dashes are retained. The corrected leaders are routed independently for this perspective, including the sharp C_2 paired-rim endpoint $|\eta_6\rangle$ at $S_1 = 2/27$, the attained sharp-plane family callout Ψ_t, and the distinct nearby $|E_*\rangle$ and $|\omega\rangle$ landmarks.

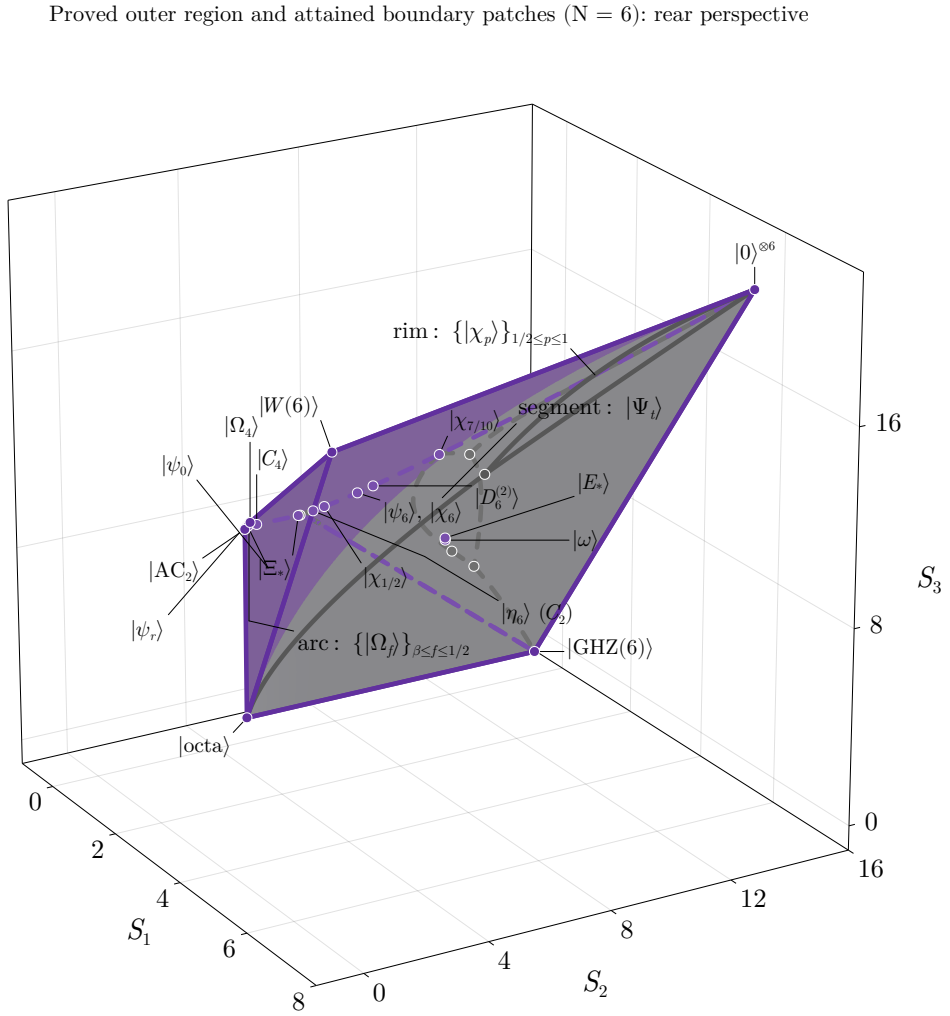


Figure 3: Rear-facing third perspective of the same outer region as Figs. 1 and 2. The camera is rotated by a half-turn in azimuth relative to the main view and raised slightly, revealing the previously rear-facing side. Hidden-edge visibility is recomputed against the exact analytical body, not inferred from the original perspective. All state labels, attained C_3, C_4 and C_5 rims, uniform flat nodes and boundary strokes are retained. The segment joining $\chi_{7/10}$ to ω is attained by Ψ_t but is not a body edge; its interior stroke is omitted. The additional $|\eta_6\rangle$ node marks the unique C_2 orbit at the sharp lower end of the null paired rim. The exceptional $|E_*\rangle$ landmark lies on the attained sharp-plane segment, close to but distinct from $|\omega\rangle$; rear nodes are lighter. Leaders remain visible and avoid neighboring nodes in this perspective. Violet nodes and curves are proved attained. The ten audited gray junctions are excluded in Section 32, including the replacement microscopic contact on the stronger exact-comparison flow cap. The older three points of Section 21 remain superseded. The refined lower patch, its open crease arcs and J_4 are excluded by Theorem 2. Other gray curves need not be unattainable merely because a neighboring node is excluded.

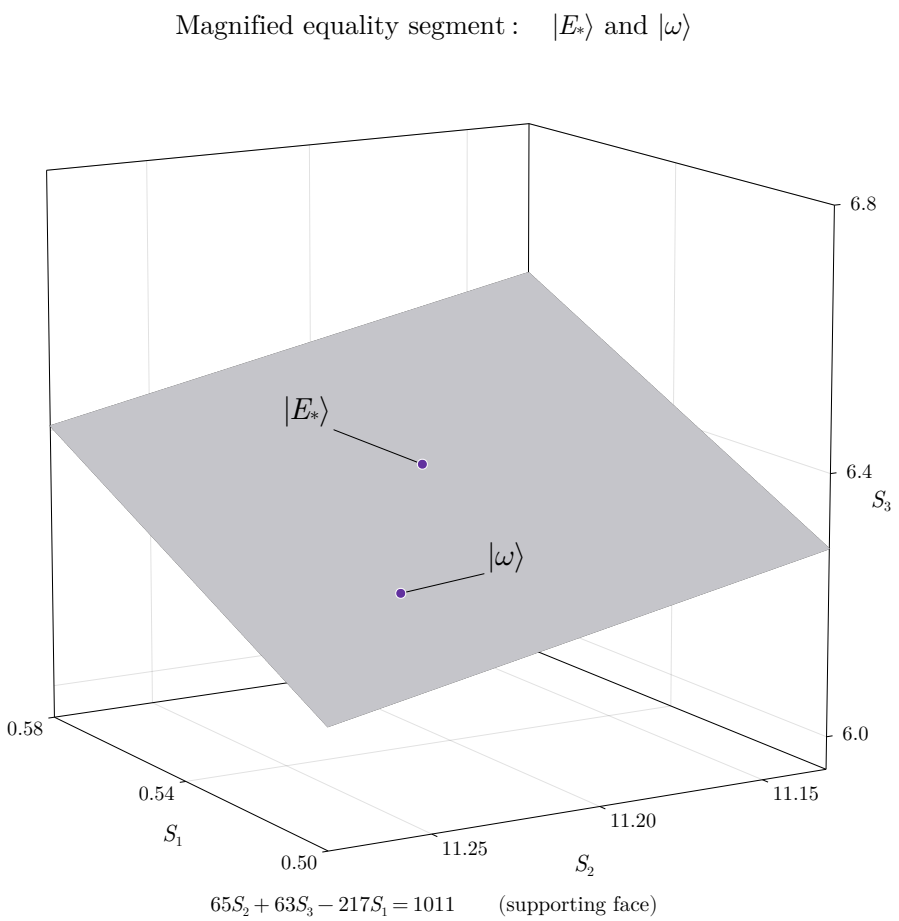


Figure 4: Magnified three-dimensional neighborhood of the exceptional C_2 state $|E_*\rangle$ and $|\omega\rangle$, with the same viewing angles as Fig. 1. Only the local coordinate window is rescaled: the exact sector positions and uniform marker size are unchanged. The two nodes and their separate black, border-connected leaders are therefore clearly resolved. The attained sharp-plane equality segment is not stroked in this edge-only view: $|E_*\rangle$ is an interior landmark, whereas $|\omega\rangle$ is its endpoint. Both lie inside the flat gray supporting face, not at geometric corners or creases. The gray face is sampled from the same exact outer roof; its rectangular window margins are not body edges. The exact points and explicit states are given by (35), (291) and (349). No further attained surface is implied.

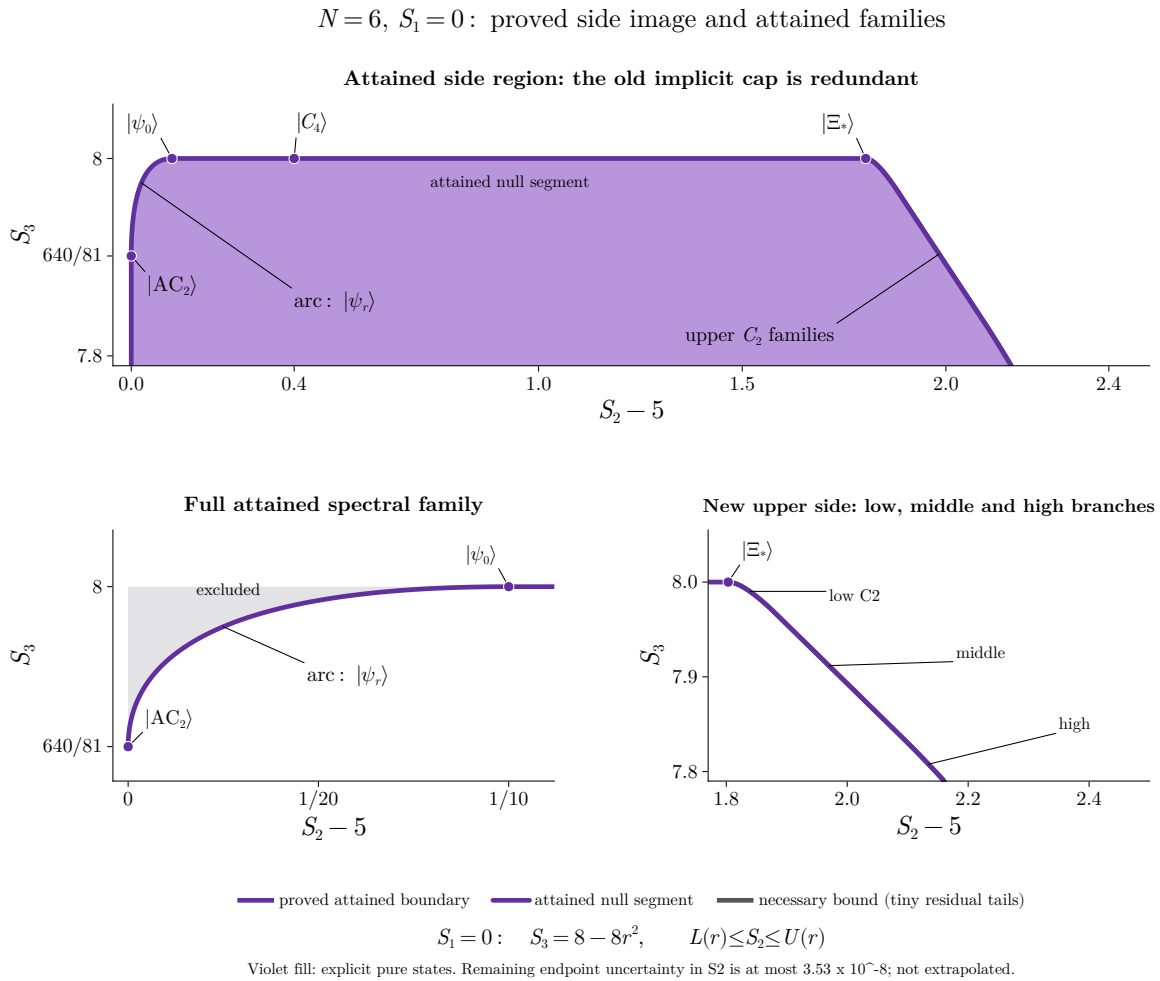


Figure 5: Magnified current $S_1 = 0$ boundary, using the sharp side branches and rigorous tiny-tail bracket of Theorem 20. The full C_3 rim and the attained segment through ψ_0, C_4 and Ξ_* are solid violet with the same stroke width. The old implicit-cap branch and its excluded gray join B are no longer exposed after the new side bounds; they are not drawn as current edges. The middle and high side families are given in (414) and (408). The detailed panels resolve the current curved endpoints. Each named-state label has a leader terminating at its node border; the callout arc : $|\psi_r\rangle$ denotes the full attained C_3 rim.

3.1 Resolving the improvement at small nonzero S_1

Fix $x_0 = 10^{-4}$. Figure 6 shows the actual upper coordinate $z = Z_+(x_0, y)$, rather than a difference from a historical roof. All the current conditions in Sections 26 and 2 are imposed before selecting the exposed curve; the four-copy cut is inactive in this small-S_1 window. The base implicit cap R of Lemma 11 and paired purity are retained only where the newer lifts do not dominate them. No obsolete comparison boundary is plotted.

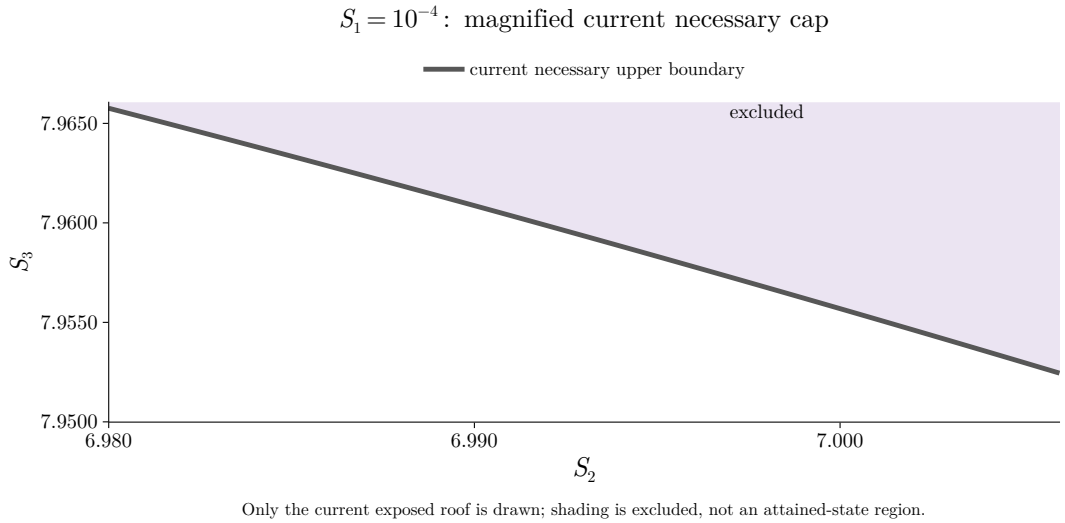


Figure 6: Current exposed upper boundary at $S_1 = 10^{-4}$, with $6.98 \le S_2 \le 7.006$. Both axes are actual sector lengths. The pale region above the gray curve is excluded. No historical or redundant boundary is shown, and no attainment is asserted.

The active branch may change after the new cuts. Each curve in this panel is recomputed from the same current necessary system as the three-dimensional body, rather than continuing a historical red sheet through a newly excluded region.

3.2 Nonzero-dipole slices resolving the new compatibility cuts

The new full teal family and the raised lower boundary have different geometric effects. The former lowers an upper roof near $y = 5$, while the latter raises a lower floor away from $x = 0$. To display them without perspective ambiguity, Fig. 7 fixes x. Its upper panel uses $x = 10^{-2}$ and the exact support envelope Z_{teal} in (257); its lower panel uses $x = 1$ and shows

$$Z_-(1, y) = \max\left\{2 + \frac{12}{4 + \sqrt{5(y-5)/2}}, 10 - y\right\}.$$

Each displayed boundary is clipped against all the other defining conditions. Shading lies on its excluded side. Previous weaker curves are not drawn.

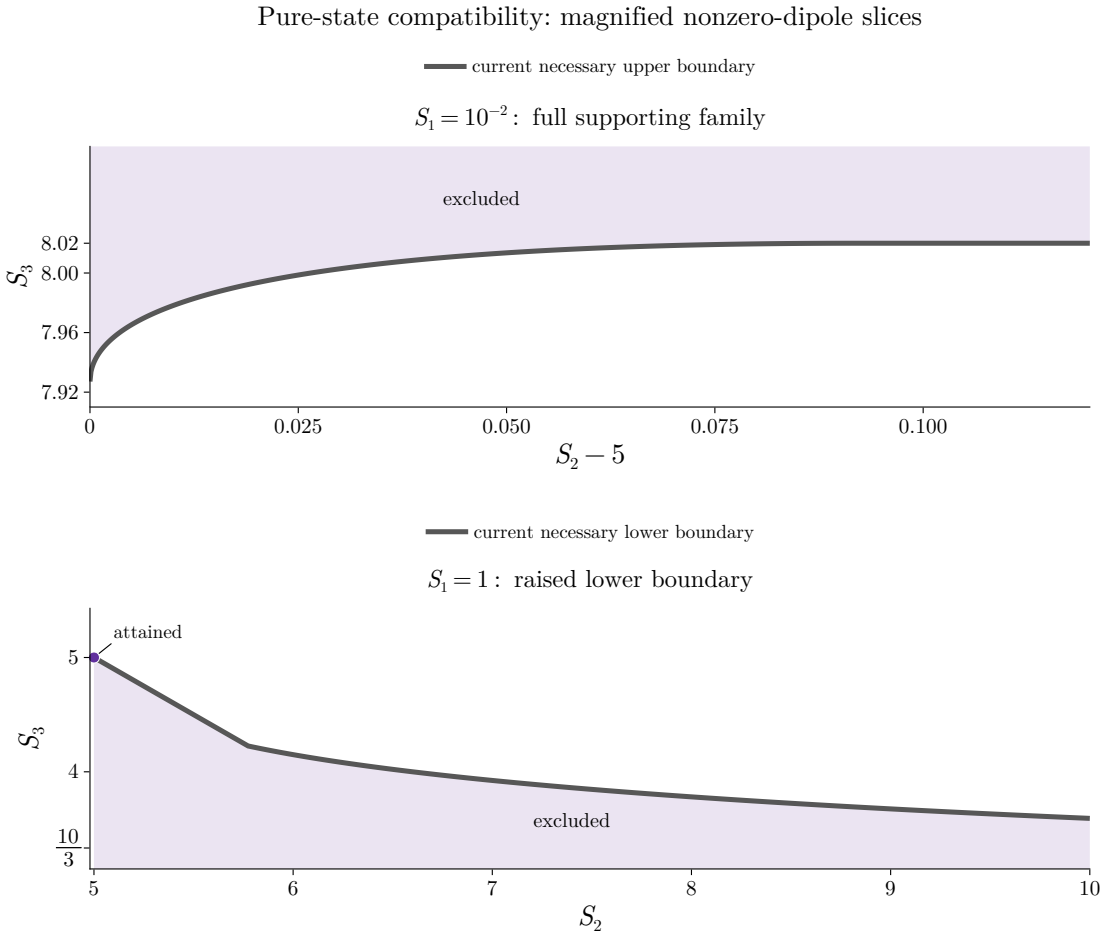


Figure 7: Magnified nonzero-S_1 slices showing the genuine new cuts. The upper boundary at $S_1 = 10^{-2}$ resolves the full supporting family (6i), rather than replacing it by its weaker root corollary. The lower boundary at $S_1 = 1$ resolves (6a); the exact witness $(1, 33/5, 7/2)$ is excluded. Only current exposed boundaries are drawn. Shading is excluded; the filled violet point alone asserts attainment.

3.3 Magnifying the paired-spectral cap and the sharp plane

Figure 8 fixes $x = 2/7$ and $x = 6/7$ near the two former corners. Both violate (6f), as proved in Section 17.3. They are outside the updated body and are therefore absent from the three-dimensional boundary-node list. The additional section at $x = 63/121$ resolves the new sharp plane (6h). The full attained equality segment intersects this fixed-x section only at its endpoint ω.

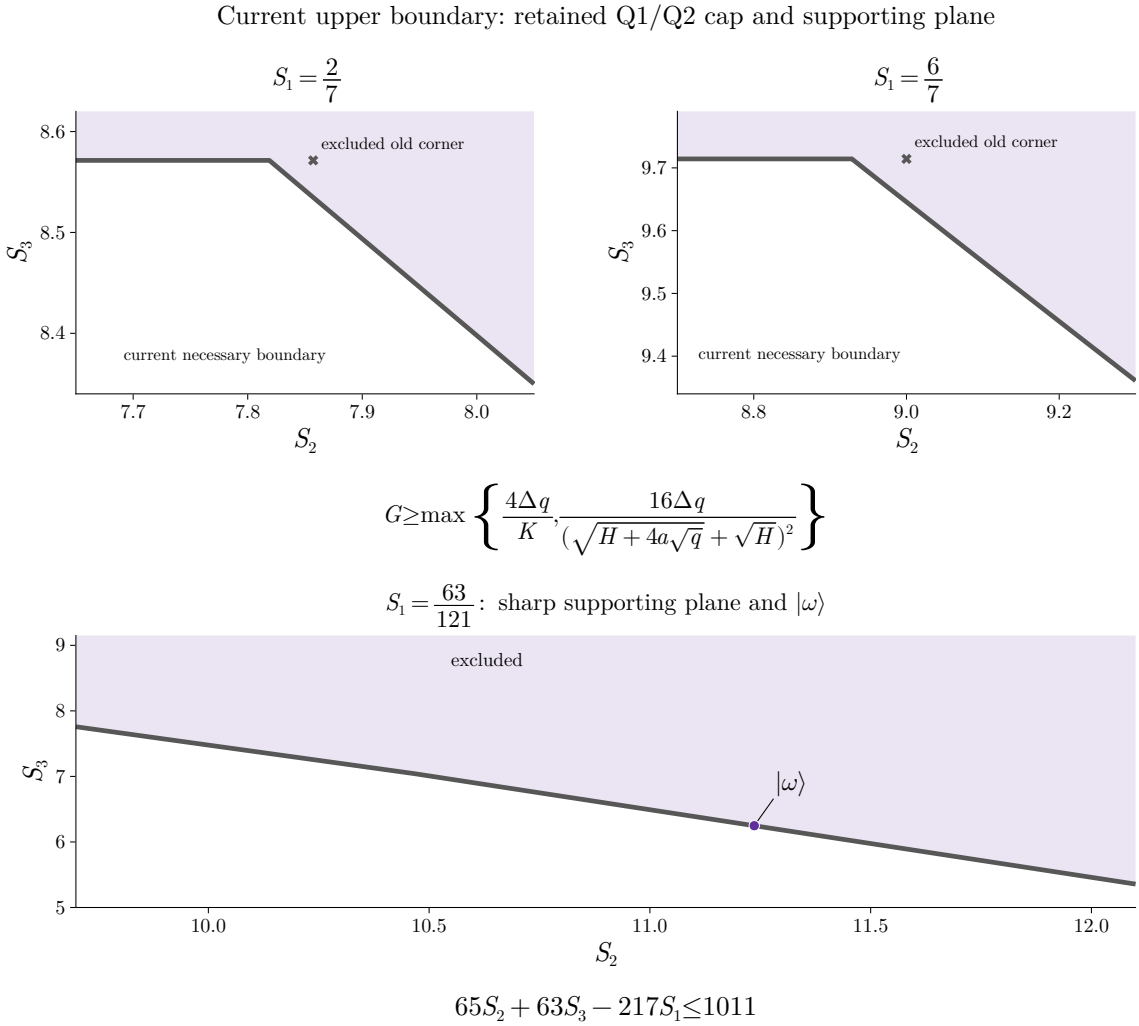


Figure 8: Exact vertical sections resolving the paired-spectral cap and the new sharp supporting plane (6h). Gray curves are the current exposed roof after all constraints are imposed. Crosses locate excluded former corners, not new boundary nodes. The old flat spectral facet and paired comparison lines are not drawn. The pale region above the current roof is excluded. In the lower panel, the violet ω point is the intersection with the full attained segment of Theorem 14. The adjacent gray plane boundary lies outside that segment and is not attained.

3.4 Known state labels

Write $D_k = |D_6^{(k)}\rangle = |3, 3-k\rangle$ for the normalized Dicke state with k excitations. Representatives of the labels are

Label	Representative	(S_1, S_2, S_3)
$\|\text{octa}\rangle$	$(D_1 + D_5)/\sqrt{2}$	$(0, 5, 0)$
$\|\text{GHZ}(6)\rangle$	$(D_0 + D_6)/\sqrt{2}$	$(0, 15, 0)$
$\|W(6)\rangle$	D_1	$(8/3, 5, 40/3)$
$\|0\rangle^{\otimes 6}$	D_0	$(6, 15, 20)$
$\|C_4\rangle$	$(D_0 + \sqrt{3}D_4)/2$	$(0, 27/5, 8)$
$\|\Omega_4\rangle$	$(\sqrt{3}D_0 + \sqrt{5}D_4)/\sqrt{8}$	$(1/6, 5, 25/3)$
$\|\text{AC}_2\rangle$	$(\sqrt{2}D_0 + \sqrt{5}D_3 + \sqrt{2}D_6)/3$	$(0, 5, 640/81)$
$\|\psi_6\rangle$	$(\sqrt{20}D_2 + D_5)/\sqrt{21}$	$\boldsymbol{p}_*$
$\|\chi_6\rangle$	$(2D_0 + \sqrt{3}D_5)/\sqrt{7}$	$\boldsymbol{p}_*$
$\|\chi_{1/2}\rangle$	$(D_0 + D_5)/\sqrt{2}$	$(1/6, 15/2, 25/3)$
$\|\eta_6\rangle$	Equation (205)	$(2/27, 65/9, 220/27)$
$\|\chi_{7/10}\rangle$	$(\sqrt{7}D_0 + \sqrt{3}D_5)/\sqrt{10}$	$(3/2, 99/10, 11)$
$\|\omega\rangle$	Equation (291)	$(63/121, 618/55, 756/121)$
$\|E_*\rangle$	Equation (349)	$\boldsymbol{p}_E$
$\|\psi_0\rangle$	$(D_0 + \sqrt{2}D_3 + D_6)/2$	$(0, 51/10, 8)$
$\|\Xi_*\rangle$	Equation (37)	$(0, s_*, 8)$

Here $\boldsymbol{p}_* = (24, 405, 440)/49$. The states ψ_6, χ_6 are not related by collective rotations, but have the same sector tuple and thus share one marker. For the exceptional C_2 state, with q_* defined in (344),

$$\boldsymbol{p}_E = \left(\frac{3}{2} - \frac{474}{169}q_*,\ \frac{99}{10} + \frac{3234}{845}q_*,\ 11 - \frac{2300}{169}q_*\right) \simeq (0.545436, 11.202557, 6.368150). \tag{35}$$

Its marker is at this exact point on the attained equality segment, not at ω and not at an additional geometric junction. Its explicit state and analytical verification are given in Proposition 8. The additional label $|\eta_6\rangle$ marks the sharp C_2 endpoint of the null paired rim: its complete equality classification is Theorem 9. The callout $\{|\Psi_t\rangle\}$ labels the entire sharp-plane equality segment, not a new isolated node. Its explicit Dicke coefficients and direct proof are in Section 23.4; at its upper parameter endpoint it is $-|\omega\rangle$. This real family covers the full sector segment but only one achiral branch of its state-orbit space. The complete classification, including the chiral states and an additional exceptional C_2 orbit, is Theorem 15. These states have the same already plotted sector image and do not require a new filled facet. The added $|E_*\rangle$ label identifies its exceptional orbit within that image; its leader is distinct from the nearby $|\omega\rangle$ leader. Figure 4 resolves these two nearby nodes by magnifying the coordinate window while retaining the original marker size. The full C_3 family of spectral-lemma equality states is

$$|\psi_r\rangle = \frac{\sqrt{1-r}}{2}(D_0 + D_6) + \sqrt{\frac{1+r}{2}}D_3, \qquad 0 \le r \le \frac{1}{9}, \tag{36}$$

$$(x, y, z) = \left(0, 5 + \frac{(1-9r)^2}{10}, 8(1-r^2)\right).$$

It joins $|\text{AC}_2\rangle$ to $|\psi_0\rangle$. The additional C_4 orbit also saturates the spectral lemma, but is an interior point of the longer achieved horizontal segment ending at Ξ_*. The historical red equality rim is superseded on this side. The newly attained side upper curves and their precise global-sharpness ranges are given in Theorem 20.

An explicit sharp-endpoint representative is

$$\begin{aligned} |\Xi_*\rangle = \frac{1}{2}\big[&\sqrt{(1+u_*)(1-2u_*)}\,D_0 + \sqrt{(1+u_*)(1+2u_*)}\,D_2 \\ &+ \sqrt{(1-u_*)(1-2u_*)}\,D_4 - \sqrt{(1-u_*)(1+2u_*)}\,D_6\big], \qquad u_* := \frac{\sqrt{5}-\sqrt{3}}{4}. \end{aligned} \tag{37}$$

3.5 An analytically attained nonzero-dipole C_4 arc

The new upper corner is not merely an outer point. It is attained by the fourfold-symmetric state

$$\boxed{|\Omega_4\rangle = \frac{\sqrt{3}\,D_0 + \sqrt{5}\,D_4}{\sqrt{8}}, \qquad (x,y,z) = \left(\frac{1}{6}, 5, \frac{25}{3}\right).} \tag{38}$$

In fact a whole adjacent arc is attained. Explicit normalized representatives are

$$\boxed{|\Omega_f\rangle = \frac{\sqrt{1+f}\,D_0 + \sqrt{3-f}\,D_4}{2}, \qquad b_c \le f \le \frac{1}{2}.} \tag{39}$$

Here $b_c = 0.366379159399607379\ldots$ is the exact algebraic threshold (425); the formerly established $3/8 \le f \le 1/2$ range is retained, and Section 27.4 proves its extension down to b_c. Here $f = |\langle \mathbf{J} \rangle|$, and the label Ω_4 denotes the endpoint $\Omega_{1/2}$. These states have C_4 rotational symmetry because the two Dicke levels differ by four. They are distinct from the original zero-dipole C_4 pyramid in the table: that state is the continuation of the same formula to $f = 0$, outside the interval in (39). The C_3 rim, which instead has $S_1 = 0$, is another attained family.

Direct analytical verification. The only occupied magnetic levels are $m = 3$ and $m = -1$, with probabilities $(1+f)/4$ and $(3-f)/4$. A spin dipole changes m by at most one, and a quadrupole by at most two. Their off-diagonal matrix elements therefore cannot connect these two occupied levels. It follows immediately that

$$\langle J_x \rangle = \langle J_y \rangle = 0, \qquad \langle J_z \rangle = f, \qquad \langle J_z^2 \rangle = 3 + 2f.$$

Using $J_x^2 + J_y^2 + J_z^2 = 12I$ and the absence of rank-two off-diagonal coherences gives the quadrupole in the manuscript's Cartesian notation:

$$Q = \operatorname{diag}\left(\frac{1-2f}{2}, \frac{1-2f}{2}, 2f-1\right), \qquad \tau = \frac{3}{2}(1-2f)^2. \tag{40}$$

Time reversal maps the support $\{D_0, D_4\}$ to the disjoint support $\{D_6, D_2\}$, so $\Theta = 0$ exactly. Substitution into (55) and (58) now gives

$$\boxed{x = \frac{2}{3}f^2, \qquad y = 5 + \frac{2}{5}(1-2f)^2, \qquad z = 8 + \frac{4}{3}f^2, \qquad q = 0.} \tag{41}$$

No numerical optimization or equality-state ansatz is needed for this verification.

Saturation of the full supporting family. For each $t > 0$, set

$$f(t) = \frac{3+4t}{8(1+t)}, \qquad t = \frac{8f-3}{4-8f}. \tag{42}$$

This maps $0 < t < \infty$ onto $3/8 < f < 1/2$. More explicitly, for these states and *any* $t > 0$, completing the square gives the exact residual identity

$$10(y-5) + \frac{24x}{3+4t} - \frac{1}{1+t} = \frac{[8(1+t)f - (3+4t)]^2}{(1+t)(3+4t)} \ge 0. \tag{43}$$

It vanishes precisely when $f = f(t)$; equivalently, $1 - 2f(t) = 1/[4(1+t)]$ makes the supporting-plane value exactly $1/(1+t)$. Thus every supporting plane is attained also at nonzero S_1 on this C_4 arc. The limiting endpoints are

$$|\Omega_{3/8}\rangle = \frac{\sqrt{11}\,D_0 + \sqrt{21}\,D_4}{\sqrt{32}}, \qquad (x,y,z) = \left(\frac{3}{32}, \frac{201}{40}, \frac{131}{16}\right), \tag{44}$$

$$|\Omega_{1/2}\rangle = |\Omega_4\rangle, \qquad (x,y,z) = \left(\frac{1}{6}, 5, \frac{25}{3}\right). \tag{45}$$

They correspond continuously to $t \downarrow 0$ and $t \to \infty$, respectively. These one-dimensional attained rims do not establish sharpness of the rest of the two-dimensional teal envelope.

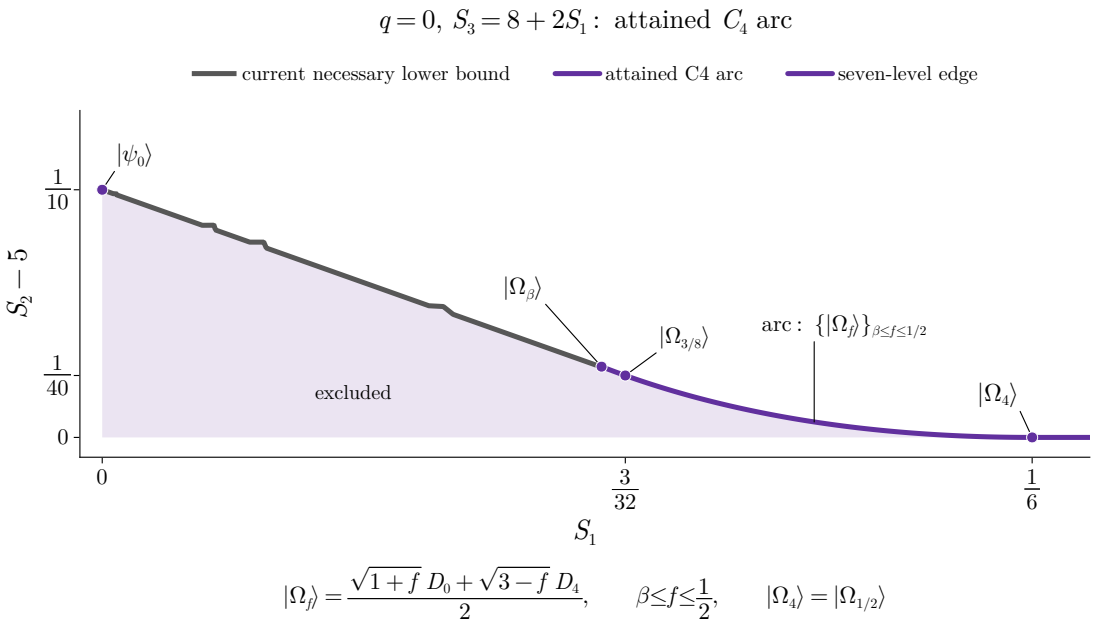


Figure 9: Magnified attained C_4 arc (39) on $q = 0$, shown in violet together with its two explicit endpoints. The callout arc : $\{|\Omega_f\rangle\}_{3/8\le f\le 1/2}$ labels the whole family, rather than duplicating the Ω_4 endpoint. Point labels are connected to node borders by clear leaders. In particular Ω_4 is a realized state, not a gray outer corner. The violet horizontal continuation to its right is the seven-level edge proved in Section 13; its full extent is visible in the three-dimensional views. The original $S_1 = 0$ pyramid C_4 and the separate attained C_3 rim must not be identified with this nonzero-S_1 arc. The dashed straight reference has no equality state in its interior, by Theorem 6; its endpoints are attained. The pale shaded region below the rim is excluded by the full compatibility family. No unproved finite-width replacement is drawn.

3.6 The full attained C_5 rim and its junction

The formerly gray junction $(3/2, 99/10, 11)$ is achievable, and so is the entire adjacent $q = 0$ rim from $(1/6, 15/2, 25/3)$ to the product vertex $(6, 15, 20)$. Explicit normalized pure representatives are

$$\boxed{|\chi_p\rangle = \sqrt{p}\, D_0 + \sqrt{1-p}\, D_5, \qquad \frac{1}{2} \le p \le 1.} \tag{46}$$

The parameter p is the probability on D_0, not a two-copy probability p_J. The two magnetic levels $m = 3, -2$ differ by five, so a collective rotation through $2\pi/5$ about z changes both amplitudes by the same global phase. A relative phase can likewise be removed by a collective z rotation. This explains the label C_5.

Proof sketch. The separation of the occupied Dicke levels removes all dipole and quadrupole coherences. Their moments are therefore weighted sums of two diagonal contributions. Time reversal sends the support to two disjoint levels, giving $\Theta = 0$. The sector conversions then give the rim explicitly. Substitution proves saturation of the new curved bound, and a single diagonal rank-four tensor verifies the special junction where a_4 vanishes.

Dipole, quadrupole and sector lengths. Since operators of rank one or two cannot connect the occupied levels,

$$F = (0, 0, 5p - 2), \qquad \langle J_z^2 \rangle = 4 + 5p.$$

Using $J_x^2 + J_y^2 + J_z^2 = 12I$ and the quadrupole convention in (54) gives

$$Q = \operatorname{diag}\left(-\frac{5p}{2}, -\frac{5p}{2}, 5p\right), \qquad f = 5p - 2, \qquad \tau = \frac{75}{2}p^2. \tag{47}$$

Time reversal sends $\{D_0, D_5\}$ to $\{D_6, D_1\}$, hence $\Theta = 0$ exactly. Equations (55) and (58) now give

$$\boxed{(x, y, z) = \left(\frac{2}{3}(5p-2)^2,\ 5 + 10p^2,\ 8 + \frac{4}{3}(5p-2)^2\right), \qquad q = 0.} \tag{48}$$

This is an analytical pure-state realization, not a numerical optimization or a compatibility ansatz.

Saturation of the paired-spectral cap. For (48), direct simplification yields

$$\Delta = 10(2p-1)^2, \qquad 5y + 6z - 15x - 85 = 20(2p-1) = \sqrt{40\Delta}, \tag{49}$$

where the last equality uses $p \ge 1/2$. Thus the entire family saturates (6f). It is also on $q = 0$, and hence on the exposed upper rim of the proved outer region.

The auxiliary construction in Section 17 provides an independent spectral interpretation. In the magnetic basis $m = 2, 1, 0, -1, -2$, its reduced matrix is

$$R = \operatorname{diag}\left(\frac{p}{2}, \frac{p}{2}, \frac{1-p}{2}, 0, \frac{1-p}{2}\right).$$

For $p \ge 1/2$ its two paired eigenvalues are $r = p/2$ and $s = (1-p)/2$, and time reversal reverses this diagonal. Consequently $\operatorname{Tr}(R\widetilde{R}) = 2rs + s^2$: the trace-rearrangement step in the curved-cap proof is an equality, not just a condition suggested by sector coordinates.

The rank-four cancellation and the formerly gray junction. Rank-four tensors also cannot connect levels separated by five. Only the $M = 0$ component can therefore be nonzero. In the normalized multipole convention (53),

$$T_{40}^{(6)} = \frac{35J_z^4 - 335J_z^2 + 360I}{60\sqrt{154}}, \qquad \rho_{40} = \frac{3p - 7(1-p)}{\sqrt{154}}.$$

Its diagonal entries at $m = 3, -2$ are $3/\sqrt{154}$ and $-7/\sqrt{154}$, respectively. It follows that

$$\boxed{a_4 = \frac{(10p-7)^2}{154}.} \tag{50}$$

At $p = 7/10$ the whole rank-four multipole vanishes, and

$$\boxed{|\chi_{7/10}\rangle = \frac{\sqrt{7}D_0 + \sqrt{3}D_5}{\sqrt{10}}, \qquad (x, y, z) = \left(\frac{3}{2}, \frac{99}{10}, 11\right).} \tag{51}$$

Its first multipole lengths and full sector tuple are

$$(a_1, a_2, a_3, a_4) = \left(\frac{9}{112}, \frac{7}{48}, \frac{1}{6}, 0\right),$$

$$(S_1, S_2, S_3, S_4, S_5, S_6) = \left(\frac{3}{2}, \frac{99}{10}, 11, \frac{36}{5}, \frac{39}{2}, \frac{139}{10}\right).$$

In particular $S_4 = \Phi(99/10) = 36/5$ and $\Delta = 8/5$, so all the bounds defining that junction are saturated.

Endpoints, known states and figure annotations. The lower endpoint is the attained state $\chi_{1/2} = (D_0 + D_5)/\sqrt{2}$, with tuple $(1/6, 15/2, 25/3)$. The previously labelled state χ_6 is exactly $\chi_{4/7}$, and $\chi_1 = D_0$ is the product vertex. For $7/10 \le p \le 1$,

$$S_4 = \frac{65 + 140p^2 - 160p}{3} = \Phi(5 + 10p^2),$$

so this part also saturates the curved orange bound. For $1/2 \le p \le 7/10$ the family traces the new paired-spectral rim. Both portions are drawn in violet, with labels χ_p, $\chi_{1/2}$ and $\chi_{7/10}$; the existing shared ψ_6, χ_6 label is retained. The different crease descending from $\chi_{7/10}$ toward GHZ is not part of this family and is not claimed attained. No inequality or outer-body geometry is changed by this new attainment result.

4 Origin of the manuscript's linear constraints

The linear starting system is Eqs. (42)–(43) of the local manuscript version [1]. Its reduction and the extra proved constraints are derived below so that the present note can be read independently.

4.1 Pure-state elimination

Let $r_t = \mathrm{Tr}(\rho_t^2)$ denote the purity of any t-qubit reduction, with $r_0 = 1$. Symmetry and the Pauli expansion give

$$r_t = 2^{-t} \sum_{k=0}^{t} \frac{\binom{t}{k}}{\binom{6}{k}} S_k. \tag{52}$$

For a pure global state $r_t = r_{6-t}$. Substituting (52) and solving the three linear identities yields (2).

4.2 Nonnegative multipole and two-copy weights

Use the manuscript's Hilbert–Schmidt orthonormal spin multipoles,

$$\rho = \sum_{L=0}^{6} \sum_{M=-L}^{L} \rho_{LM} T^{(6)}_{LM}, \qquad a_L = \sum_M |\rho_{LM}|^2 \geq 0, \qquad a_0 = \frac{1}{7}. \tag{53}$$

The collective spin is $J_i = \frac{1}{2}\sum_{a=1}^{6} \sigma_i^{(a)}$. Its dipole and Cartesian traceless quadrupole are

$$\begin{aligned} F_i &= \langle J_i \rangle, \qquad f = |F|, \\ \widehat{Q}_{ij} &= \frac{J_i J_j + J_j J_i}{2} - 4\delta_{ij} I, \qquad Q_{ij} = \langle \widehat{Q}_{ij} \rangle, \qquad \tau = \operatorname{Tr} Q^2. \end{aligned} \tag{54}$$

Here $\|X\|_F^2 = \operatorname{Tr}(X^\dagger X)$ is the Frobenius norm, and $X_{\mathrm{tf}} := X - \operatorname{Tr}(X) I_3/3$ denotes the traceless part of a 3×3 matrix. State-vector norms are Hilbert-space norms. The normalizations used below are

$$x = \frac{2}{3} f^2 = \frac{56}{3} a_1, \qquad y = 5 + \frac{4}{15}\tau = 5 + \frac{168}{5} a_2. \tag{55}$$

One explicit convention is

$$T^{(6)}_{LM} = \sqrt{\frac{2L+1}{7}} \sum_{m,m'=-3}^{3} \langle 3, m; L, M | 3, m' \rangle |3, m\rangle\langle 3, m'|.$$

For two identical copies define the nonnegative probabilities

$$p_J = \|P_J(|\psi\rangle \otimes |\psi\rangle)\|^2, \qquad J = 0, 2, 4, 6, \tag{56}$$

where P_J projects $V_3 \otimes V_3$ onto total spin J. Only even J occur because the two-copy vector is exchange symmetric. The sector–multipole conversion and Clebsch–Gordan expansion give the following exact formulas after (2):

Nonnegative quantity	Resulting inequality
$a_1 = 3x/56$	$x \geq 0$
$a_2 = 5(y-5)/168$	$y \geq 5$
$p_0 = (8 + 2x - z)/56$	$3z \leq 6x + 24$
$p_2 = 5(y + z - 5x - 5)/168$	$5x + 5 \leq y + z$
$p_4 = 3(162 + 8x - 10y - 3z)/616$	$10y + 3z \leq 8x + 162$
$a_4 = 3(70x - 20y - 21z + 324)/1232$	$20y + 21z \leq 70x + 324$

For example, all these formulas can be checked directly by inserting the displayed normalized tensors and the projectors $P_J = \sum_M |J, M\rangle\langle J, M|$ with $|J, M\rangle = \sum_{m,n} \langle 3, m; 3, n | J, M\rangle |3, m\rangle |3, n\rangle$. They are positivity statements, not fitted cuts.

4.3 The two remaining inequalities

Define time reversal by

$$\mathsf{T}_j \sum_m c_m |j, m\rangle = \sum_m (-1)^{j-m} \overline{c_m} |j, -m\rangle, \qquad \Theta := \langle \mathsf{T}_3 \psi | \psi \rangle, \qquad q = |\Theta|^2. \tag{57}$$

The Pauli spin-flip identity and (2) imply

$$q = \frac{1}{64} \sum_{k=0}^{6} (-1)^k S_k = 1 + \frac{x}{4} - \frac{z}{8} = 7 p_0. \tag{58}$$

In a one-versus-five Schmidt decomposition with eigenvalues $\lambda, 1-\lambda$, odd-five-qubit time reversal is antisymmetric. Its overlap formula is $|\Theta| = 2\sqrt{\lambda(1-\lambda)}\,|\langle \mathsf{T}_{5/2} u|v\rangle|$ for normalized Schmidt vectors u, v. Consequently

$$q \le 4\lambda(1-\lambda) = 2(1-r_1) = 1 - \frac{x}{6}, \qquad \text{hence} \quad 10x \le 3z.$$

Notice that $S_5 \ge 0$ alone gives only $9x \le 3z$, which is weaker. The Schmidt bound is retained here as background; the defining system uses its stronger multipole-dependent replacement proved next.

Finally every pure three-qubit state has second sector length 3, by equality of complementary one- and two-qubit purities. Convexity of sector lengths therefore gives $S_2(\tau) \le 3$ for every mixed three-qubit state. The symmetric three-qubit marginal here has $S_2(\rho_3) = y/5$, proving $y \le 15$. This completes all eight baseline inequalities. For later reference, two of the manuscript's baseline rows are

$$20y + 21z \le 70x + 324, \tag{59}$$

$$10y + 3z \le 8x + 162. \tag{60}$$

They are not added separately in (6): the first is already the linear orange branch, and the second is a consequence proved in Section 22.

5 Attained boundary patches from the C_2–C_6 families

This section gives the C_2–C_6 state constructions behind the violet patches, curves and contacts in the figures. It is self-contained: the compatibility tests below include state recovery and require no numerical optimization to verify a supplied witness, nor a separate algebraic certificate. The object being colored is the boundary of the necessary-condition region $\mathcal{R}$ in (6), not a claimed exact boundary of all achievable tuples. An attained family can also pass through the interior of $\mathcal{R}$; only its intersection with $\partial\mathcal{R}$ colors a boundary patch.

5.1 Rotation characters and the spin-multipole calculation

Write $D_k = |D_6^{(k)}\rangle = |3, 3-k\rangle$, and let $|\psi\rangle = \sum_{k=0}^{6} c_k D_k$ be normalized. Here C_n means symmetry of the *ray*: after choosing the symmetry axis as z, $e^{-2\pi i J_z/n}|\psi\rangle$ differs from $|\psi\rangle$ only by a global phase. Since $J_z D_k = (3-k)D_k$, all occupied k must belong to *one* residue class modulo n. All rotation characters must be retained, not just the character with eigenvalue one. Collective rotations do not change sector lengths, so aligning the axis loses no sector tuples. The labels C_n mean *at least* this symmetry; the families can overlap.

For this section recall the manuscript's dipole and Cartesian quadrupole, and the pure-state spin-flip overlap:

$$\begin{aligned} F_i &= \langle J_i\rangle, \qquad Q_{ij} = \frac{1}{2}\langle J_iJ_j + J_jJ_i\rangle - 4\delta_{ij}, \\ \Theta &= \sum_{k=0}^{6} (-1)^k c_k c_{6-k}, \qquad q = |\Theta|^2, \\ x &= \frac{2}{3}|F|^2, \qquad y = 5 + \frac{4}{15}\operatorname{Tr} Q^2, \qquad z = 8 + 2x - 8q. \end{aligned} \tag{61}$$

These are identities for a pure symmetric six-qubit state. The first two sector conversions follow by expanding the one- and two-qubit Pauli correlations; the last follows from $q = 2^{-6}\sum_{k=0}^{6}(-1)^k S_k$ and the pure-state elimination identities (2). They are not relaxation inequalities. For direct calculation one needs only

$$J_z D_k = (3-k)D_k, \qquad J_+ D_k = \sqrt{k(7-k)}\,D_{k-1}, \qquad J_+ = J_x + iJ_y.$$

For a C_2 state, $F_x = F_y = Q_{xz} = Q_{yz} = 0$. Reverse the symmetry axis, if needed, so that $f = F_z = \sqrt{3x/2} \ge 0$, and put $C = \langle J_+^2\rangle$. Then

$$\boxed{x = \frac{2}{3}f^2, \qquad y = 5 + \frac{2}{5}Q_{zz}^2 + \frac{2}{15}|C|^2, \qquad z = 8 + 2x - 8q.} \tag{62}$$

Indeed $\operatorname{Tr} Q^2 = 3Q_{zz}^2/2 + |C|^2/2$. This is how the rank-one and rank-two spin multipoles enter all the constructions below.

5.2 The two-level C_4, C_5, C_6 families

For $n = 4, 5, 6$, every character contains at most two Dicke levels. Up to the collective half-turn $D_k \leftrightarrow D_{6-k}$, the nontrivial families are listed below. In every row the representative is $\sqrt{p}D_k + \sqrt{1-p}D_l$, with $0 \le p \le 1$. A relative phase is removable by an axial rotation.

Family	(k,l)	x	y	z
C_6	$(0,6)$	$6(2p-1)^2$	15	$20(2p-1)^2$
C_5	$(0,5)$	$\frac{2}{3}(5p-2)^2$	$5+10p^2$	$8+\frac{4}{3}(5p-2)^2$
C_4	$(0,4)$	$\frac{2}{3}(4p-1)^2$	$5+\frac{2}{5}(8p-3)^2$	$8+\frac{4}{3}(4p-1)^2$
C_4	$(1,5)$	$\frac{8}{3}(2p-1)^2$	5	$\frac{40}{3}(2p-1)^2$

Analytical verification. Here $l-k \ge 4$, so no rank-one or rank-two spin operator connects the two occupied levels. Consequently

$$F = (0,0,f_z), \quad Q = \operatorname{diag}(-h/2, -h/2, h), \quad \begin{cases} f_z = p(3-k) + (1-p)(3-l), \\ h = p(3-k)^2 + (1-p)(3-l)^2 - 4. \end{cases}$$

Time reversal sends D_k to $(-1)^k D_{6-k}$, giving $q = 4p(1-p)$ if $k+l = 6$, and $q = 0$ otherwise. Substitution in (61) proves every table entry.

Which boundary pieces are attained? The C_6 family attains the entire edge $y = 15$, $z = 10x/3$, $0 \le x \le 6$. The middle C_4 character attains the entire edge $y = 5$, $z = 5x$, $0 \le x \le 8/3$, from the octahedral state to $W(6)$. Both saturate the lower boundary in (6a). The other C_4 family and all C_5 states in the table lie on the boundary plane $q = 0$, or $z = 8 + 2x$. They are attained curves, not two-dimensional faces.

The specifically labeled C_4 arc is obtained by $p = (1+f)/4$, $3/8 \le f \le 1/2$, and is precisely Ω_f in (39). Its equality with the appropriate member of the full supporting-plane family follows from the exact square (43). This does not mean that one state saturates every supporting plane. For the C_5 family,

$$\Delta = 10(2p-1)^2, \qquad 5y + 6z - 15x - 85 = 20(2p-1).$$

Thus the paired-spectral cap is saturated exactly for $p \ge 1/2$. For $p \ge 7/10$, $S_4 = \Phi(y)$ as well, giving the labeled rim and junction in Section 3.6. The continuation $p < 1/2$ remains attained on $q = 0$ but is not on that cap's equality rim.

Every singleton D_k is also ray-C_n for every n. In particular,

$$(x,y,z)(D_2) = \left(\frac{2}{3}, \frac{43}{5}, \frac{28}{3}\right) = (x,y,z)(\chi_{3/5}). \tag{63}$$

This is the labeled $D_6^{(2)}$ attainment-color contact in the figures, not an additional vertex or a geometric spike. Equal sector tuples do not imply collective-rotation equivalence of the states. The other singleton representatives are D_0, D_1, D_3, with tuples $(6, 15, 20)$, $(8/3, 5, 40/3)$ and $(0, 57/5, 0)$, respectively.

5.3 The C_3 family: exact phase intervals and side patches

The three-level character has the normalized representative

$$|\psi_{s,d,\phi}\rangle = \sqrt{\frac{s+d}{2}}D_0 + e^{i\phi}\sqrt{1-s}\,D_3 + \sqrt{\frac{s-d}{2}}D_6, \qquad 0 \le d \le s \le 1. \tag{64}$$

Let $h = \sqrt{s^2-d^2}$ and $b_3 = 1-s$. Direct spin-multipole evaluation gives

$$F = (0,0,3d), \qquad Q = \mathrm{diag}\left(-\frac{9s-4}{2}, -\frac{9s-4}{2}, 9s-4\right), \qquad \Theta = h - b_3 e^{2i\phi}.$$

There are no dipole or quadrupole coherences across a separation of three Dicke levels, but the time-reversal overlap retains the phase. Hence

$$x = 6d^2, \qquad y = 5 + \frac{2}{5}(9s-4)^2, \qquad z = 8 + 2x - 8\big(h^2 + b_3^2 - 2hb_3\cos 2\phi\big). \tag{65}$$

For a prescribed (x,y) put

$$v = \sqrt{\frac{5}{2}(y-5)}, \qquad s = \frac{4+v}{9}, \qquad d = \sqrt{\frac{x}{6}}. \quad \text{Require } 5 \le y \le 15, \quad 0 \le x \le 6s^2. \tag{66}$$

Then the entire interval

$$\boxed{L_3(x,y) := 8 + 2x - 8(h+b_3)^2 \;\le z \le U_3(x,y) := 8 + 2x - 8(h-b_3)^2} \tag{67}$$

is attained. For $hb_3 > 0$, an explicit state is recovered by

$$\cos 2\phi = \frac{h^2 + b_3^2 - q}{2hb_3}, \qquad q = 1 + \frac{x}{4} - \frac{z}{8}. \tag{68}$$

When $hb_3 = 0$ the interval collapses and the phase is immaterial. The endpoints correspond to $\phi = \pi/2$ and 0, respectively. No convex mixture is used to fill the interval: the phase varies in a single pure state.

For completeness, the second solution $s_- = (4-v)/9$, physical when $0 \le v \le 4$ and $d \le s_-$, does not enlarge the interval. Writing $s_+ = (4+v)/9$ and $h_\pm = \sqrt{s_\pm^2 - d^2}$, one has $h_+ + h_- \le s_+ + s_- = 8/9$ and $h_+ - h_- \ge s_+ - s_-$. Therefore $h_+ + 1 - s_+ \ge h_- + 1 - s_-$ and $|h_+ - (1-s_+)| \le |h_- - (1-s_-)|$. These inequalities show that the s_- phase interval is contained in (67).

The other two characters are related by a collective half-turn and add the curve

$$\begin{aligned} |\psi\rangle &= \sqrt{p}D_1 + \sqrt{1-p}D_4, \qquad p = 1 - v/3, \quad 0 \le v \le 3, \\ (x,y,z) &= \left(\frac{2}{3}(2-v)^2,\; 5 + \frac{2}{5}v^2,\; 8 + 2x\right). \end{aligned} \tag{69}$$

Again this follows from the diagonal moments and $\Theta = 0$. The complete C_3 fixed-pair set is the *union* of (67), when feasible, and the value in (69), when its pair matches. This union can have a gap: at $(x,y) = (0, 33/5)$ it is $[0, 64/9] \cup \{8\}$, not $[0,8]$.

Explicit two-dimensional boundary patches. Since $x \ge 0$ and $y \ge 5$ are global constraints, every state with $x = 0$ or $y = 5$ is on the outer boundary. Formula (67) proves attainment of

$$\boxed{x = 0, \quad 5 \le y \le 15, \quad 0 \le z \le 8 - \frac{8}{81}\left(2\sqrt{\frac{5}{2}(y-5)} - 1\right)^2.} \tag{70}$$

Its short upper rim $5 \le y \le 51/10$ is exactly the already labeled ψ_r family, with $r = (1-2v)/9$. On the other side face, it gives

$$y = 5, \qquad 0 \le x \le \frac{32}{27}, \qquad L_3(x,5) \le z \le U_3(x,5). \tag{71}$$

The same main character also has a $q = 0$ curve:

$$|\psi\rangle = \frac{1+t}{2} D_0 + \sqrt{\frac{1-t^2}{2}} D_3 + \frac{1-t}{2} D_6, \quad 0 \le t \le 1,$$

with $(x, y, z) = (6t^2,\, 5 + (1+9t^2)^2/10,\, 8 + 12t^2)$. Together with (69), this supplies attained curves on the $q = 0$ face. At $y = 15$ the phase interval reduces to the C_6 edge. On any other sheet of $\partial\mathcal{R}$, intersection with the exact interval test, and with the additional character curve, specifies the C_3-attained portion. These are not assertions that the C_3 extrema are unrestricted global extrema away from a saturated global bound.

5.4 The three-level C_2 character and the paired-purity patch

The two C_2 character spaces are $\mathrm{span}\{D_1, D_3, D_5\}$ and $\mathrm{span}\{D_0, D_2, D_4, D_6\}$. They must be united, not mixed coherently or replaced by their convex hull. In the first space set

$$\delta = \frac{f}{2} = \sqrt{\frac{3x}{8}}, \qquad w = \frac{y-5}{16}, \qquad r_\beta = \sqrt{(1-\beta)^2 - \delta^2}.$$

Every state has, up to phase and axial rotation, the form

$$|\psi_{\beta,\phi}\rangle = \sqrt{\frac{1-\beta+\delta}{2}} D_1 + e^{i\phi}\sqrt{\beta} D_3 + \sqrt{\frac{1-\beta-\delta}{2}} D_5, \qquad 0 \le \beta \le 1-\delta. \tag{72}$$

Evaluation of $Q_{zz} = -4\beta$, C and Θ gives

$$w = \beta - \frac{3}{5}\beta^2 + \beta r_\beta \cos 2\phi, \tag{73}$$

$$z = 15 + 5x - y - \frac{128}{5}\left(\beta - \frac{5}{8}\right)^2. \tag{74}$$

For example $C = 2\sqrt{30\beta}(Ae^{i\phi} + De^{-i\phi})$ and $\Theta = -(2AD + \beta e^{2i\phi})$, where A, D are the first and last amplitudes in (72). These expressions, inserted in (62), prove the two identities directly.

Thus a target is attained by this character if and only if one of

$$\beta = \frac{5}{8} \pm \sqrt{\frac{5}{128}(15 + 5x - y - z)} \tag{75}$$

is real, belongs to $[0, 1-\delta]$, and obeys

$$\left| w - \beta + \frac{3}{5}\beta^2 \right| \le \beta r_\beta.$$

Recover the phase from (73); when its denominator vanishes use the unsquared identity. This is an exact quartic phase-feasibility test, not interpolation between extrema.

In particular choose $\beta = 5/8$. The last condition reduces exactly to

$$\boxed{(4y-45)^2 + 600x \le 225, \quad x \ge 0, \qquad z = 15 + 5x - y.} \tag{76}$$

The ellipse implies the population domain and gives an explicit pure state for every point in this closed two-dimensional patch. Since $y + z \le 15 + 5x$ holds for all pure symmetric states, this patch attains the unrestricted upper boundary, not merely a C_2-restricted optimum.

5.5 The four-level C_2 character: two exact phase triangles

Use the representative

$$|\psi\rangle = \sqrt{p_0}e^{i\alpha}D_0 + \sqrt{p_2}D_2 + \sqrt{p_4}D_4 + \sqrt{p_6}e^{i\gamma}D_6. \tag{77}$$

The inner phases can be removed by a global phase and an axial rotation. To avoid confusion with two-copy weights, the p_k in this subsection are *Dicke populations*. Introduce two real coordinates s, u:

$$\begin{aligned} p_0 &= (s+u)/2, & p_6 &= (s-u)/2, \\ p_2 &= (1-s+f-3u)/2, & p_4 &= (1-s-f+3u)/2. \end{aligned} \tag{78}$$

Their exact domain is $0 \le s \le 1$, $|u| \le s$, $|f-3u| \le 1-s$. These formulas impose normalization and $F_z = f$ identically. Define

$$\begin{aligned} H &= \frac{15}{2}(y-5), & R &= s^2 - u^2 = 4p_0p_6, \\ W &= (1-s)^2 - (f-3u)^2 = 4p_2p_4, & T &= H - 3(8s-3)^2, \\ E &= (f-2u)^2 - (2s-1)^2. \end{aligned} \tag{79}$$

Here $Q_{zz} = 8s-3$ and $T = |C|^2$, so require $T \ge 0$.

Proposition 3 (Exact four-level compatibility and state recovery)**.** *At fixed (x, y) and physical (s, u), a pure state* (77) *with overlap q exists if and only if $0 \le q \le 1$ and*

$$\boxed{\begin{aligned} q \in\ & [(\sqrt{R}-\sqrt{W})^2, (\sqrt{R}+\sqrt{W})^2] \\ & \cap \left[\frac{(\sqrt{T}-6\sqrt{W})^2}{15} - E, \frac{(\sqrt{T}+6\sqrt{W})^2}{15} - E\right]. \end{aligned}} \tag{80}$$

Every value in this intersection is attained by a single pure state.

Proof. Put $\ell = \alpha + \gamma$, $U = \sqrt{p_0p_2} + \sqrt{p_4p_6}e^{i\ell}$ and $K = 12\sqrt{p_2p_4} = 6\sqrt{W}$. Direct evaluation yields

$$q = R + W + 2\sqrt{RW}\cos\ell, \qquad 4|U|^2 = q + E, \qquad C = 2\sqrt{15}\,e^{-i\alpha}U + K. \tag{81}$$

The first phase gives exactly the first interval. After it is chosen, the free phase α makes $|C| = \sqrt{T}$ if and only if lengths $\sqrt{15(q+E)}$, $6\sqrt{W}$ and $\sqrt{T}$ form a triangle. Equivalently, $|\sqrt{T} - 6\sqrt{W}| \le \sqrt{15(q+E)} \le \sqrt{T} + 6\sqrt{W}$, which is the second interval. No independent phase condition remains. For explicit recovery, choose

$$\begin{aligned} \ell &= \pm\arccos\frac{q-R-W}{2\sqrt{RW}}, \\ \alpha &= \arg U \ \pm \arccos\frac{T - 60|U|^2 - K^2}{4\sqrt{15}K|U|}, \qquad \gamma = \ell - \alpha. \end{aligned} \tag{82}$$

At a zero denominator use the corresponding vector-sum identity in (81): the phase is then free or fixed without division. Thus zero populations and collapsed triangles remain included. □

This provides an exact two-real-variable description of the entire four-level attainable set: substitute $q = 1 + x/4 - z/8$ and ask whether some physical (s, u) satisfies (80). Together with (75), it describes both C_2 characters. It does not presume that their union, or a fixed-pair fiber, is convex.

5.6 The specific four-level boundary patches in the figures

Further patches of the paired-purity plane. Let $q_{\mathrm{pp}} = (y - 7 - 3x)/8$, so that $z = 15 + 5x - y$, and put $q_- = (\sqrt{T} - 6\sqrt{W})^2/15 - E$. A direct completion of two squares gives

$$q_- - q_{\mathrm{pp}} = \frac{1}{20}(\sqrt{T} - 8\sqrt{W})^2 + \frac{16}{5}\left(u - \frac{f}{8}\right)^2. \tag{83}$$

Because compatibility requires $q \geq q_-$, equality on this plane is possible precisely on the branch

$$u = \frac{f}{8}, \qquad s = \frac{17}{32} \pm \frac{1}{16}\sqrt{H + 25f^2 - \frac{75}{4}}, \qquad q = q_{\mathrm{pp}}, \tag{84}$$

provided the populations and both phase triangles are physical. Indeed these equations give $T = 64W$ and $q_- = q_{\mathrm{pp}}$; state recovery then proves attainment. Retaining both physical roots gives the additional even-character pieces of the paired-purity face beyond the simple odd ellipse (76). The root formula alone is not a feasibility certificate.

Two-dimensional patches on the null-overlap plane. Every even-character state with $q = 0$ can, in a suitable gauge, be written

$$\begin{aligned} |\psi\rangle = {} & \cos a \cos b\, D_0 + \cos a \sin b\, D_2 \\ & + e^{i\chi} \sin a \cos b\, D_4 - e^{i\chi} \sin a \sin b\, D_6, \qquad 0 \leq a, b \leq \frac{\pi}{2}. \end{aligned} \tag{85}$$

Normalization is immediate and $\Theta = 2(c_0 c_6 + c_2 c_4) = 0$. Completeness follows by factorizing the rank-one matrix $\left(\begin{smallmatrix} c_0 & c_2 \\ c_4 & -c_6 \end{smallmatrix}\right)$. Set $\eta = \cos 2a$, $\nu = \cos 2b$. Direct calculation gives

$$\begin{aligned} f_z &= 2\eta + \nu, \qquad Q_{zz} = 1 + 4\eta\nu, \\ |C|^2 &= (1 - \nu^2)\left[9 + 6\eta^2 + 6\sqrt{15}\,\eta\sqrt{1 - \eta^2}\cos\chi\right]. \end{aligned} \tag{86}$$

These formulas and (62) parametrize attained patches on $z = 8 + 2x$. Equivalently, for a target point set $\nu = f - 2\eta$ and

$$Z = H - 3(1 + 4\eta\nu)^2 - (1 - \nu^2)(9 + 6\eta^2).$$

A phase exists exactly when some $|\eta| \leq 1$, $|\nu| \leq 1$ obeys

$$Z^2 \leq 540\eta^2(1 - \eta^2)(1 - \nu^2)^2. \tag{87}$$

Recover $\cos\chi$ from (86); a zero coefficient requires $Z = 0$. This proves the null-plane construction without a phase optimization. It does not assert that every point of the outer $q = 0$ face is attainable.

Patches on the side face $y = 5$. Here $Q = 0$, so $s = 3/8$ and $T = 0$. The second phase triangle fixes

$$q(u) = 1 - \frac{12}{5}(f - 3u)^2 - (f - 2u)^2 = 1 - \frac{3f^2}{32} - \frac{128}{5}\left(u - \frac{23f}{64}\right)^2. \tag{88}$$

For a target $q = 1 + x/4 - z/8$, test at most two candidates,

$$u = \frac{23f}{64} \pm \sqrt{\frac{5}{128}\left(1 - \frac{3f^2}{32} - q\right)}.$$

Keep only real candidates with physical populations and the first triangle in (80). Formula (82) then constructs the state. This is an analytical description of the even-character violet portions of the $y = 5$ face, supplementing the C_3 strip and C_4 edge above. The unconstrained vertex of the parabola need not be physical.

General four-level patches on the side face $x = 0$. Put $f = 0$ and $v_2 = u^2$. The physical domain and invariants become

$$0 \le v_2 \le \min\{s^2, (1-s)^2/9\}, \qquad R = s^2 - v_2, \quad W = (1-s)^2 - 9v_2, \quad E = 4v_2 - (2s-1)^2,$$

with $0 \le s \le 1$ and $T = H - 3(8s-3)^2 \ge 0$. For $q = 1 - z/8$, the exact two-triangle test is equivalently

$$\begin{aligned} 4RW - (q - R - W)^2 &\ge 0, \\ 144TW - [15(q+E) - T - 36W]^2 &\ge 0. \end{aligned} \tag{89}$$

For fixed s, y, q, these are concave quadratic polynomials in v_2, with leading coefficients -64 and -147456. Intersect their nonnegative intervals with the physical interval; any accepted v_2 gives $u = \pm\sqrt{v_2}$ and an explicit state by (82). Existence of such an s is a necessary and sufficient condition within the even character. This describes the additional four-level violet patches on the $x = 0$ face, including those located using the general resultant solver. The analytical verification is the two original phase triangles, not the resultant or a filled interval between its optimized endpoints.

5.7 Reading the attainment colors: scope and summary

The following is a compact guide to the actual boundary certificates. Each entry denotes the stated family image *intersected with the existing outer boundary*; no C_n-restricted inequality is added to the unrestricted region (6).

Boundary piece	Attaining construction	Exact test here
$x = 0$ side patches	C_3 phase intervals and even C_2 four-level states	(70), (89)
$y = 5$ side patches	C_3 strip, even C_2 states, and middle C_4 edge	(71), (88)
$z = 15 + 5x - y$	Odd C_2 ellipse and physical even C_2 branches	(76), (84)
$z = 8 + 2x$	Even C_2 null states; C_3, C_4 and C_5 curves	(85)–(87); Secs. 5.2, 5.3
$y = 15$, $z = 10x/3$	Complete C_6 edge	Sec. 5.2

The C_4 supporting-family arc, C_5 paired-spectral/orange rim and D_2 contact remain labeled as before. Two-dimensional patches come from the C_2 and C_3 constructions; the higher-order character families supply curves, edges and isolated Dicke points. Different symmetry classes can realize the same tuple. Other separately known attained curves retained in the figures need not have nontrivial cyclic symmetry; no classification of those curves is asserted here.

Family-resolved three-view figure. The separate main, alternative and rear views on pages 148, 149 and 150 each include a compact color key. Their complete plot regions retain the original scale and coordinates; the legends are added below, without moving or resizing the body, nodes, labels or leaders. Only true body edges are stroked. The keys distinguish surface fills from edge/node colors and label gray as excluded or uncertified; construction colors do not assert exclusive symmetry classes. The additional full-size fold-out on page 151 shows the same body from the main, alternative and rear perspectives, with a common color key. After its main view it includes two magnified details: the neighborhood of $|E_*\rangle$ and $|\omega\rangle$ from Fig. 4, and the small-dipole null roof showing the complete constructive C_2 band and the four unrestricted certified strips, with the second at its true enlarged width. The blue band is an attained inner image outside the certified strips, not an assertion of unrestricted sharpness. In the first detail both nodes retain C_2 blue and the surrounding supporting face remains gray. The interior equality trace is omitted, because it is not a body edge. Exact node positions, uniform sizes, and the separately routed border-connected leaders are unchanged. Blue and teal fills identify C_2 and C_3 surface constructions, respectively; gold, red and purple strokes show the genuine C_4, C_5 and C_6 body-edge portions. The family-colored views use the same edge-only selection as the violet views: full character curves are not overlaid where they lie inside a face. Rose identifies the additional rotated-W superposition image proved in Section 6; it is a separate state

construction, not another cyclic-symmetry class. Cyan identifies the GHZ–product superposition image of Section 7. Orange identifies the octahedral–product image of Section 8. The aligned three-state construction of Section 9 remains proved, but its interior roof trace is not stroked. Indigo identifies the complete rank-one roof image of Section 10; olive green identifies the filtered-octahedral lower-plane image of Section 11. Magenta identifies points reconstructed from the five-level null family of Section 12. Its complete image is not known: the magenta display is a positive-witness subset, not an inferred hull. Turquoise identifies the exact seven-level roof band and its quadrupole-free edge in Section 13. The complete image and inverse are explicit. The parabola bounding that attained band is not imposed as another universal inequality or drawn as a new geometric ridge of the body. Ochre identifies the additional four-level small-dipole roof image of Section 14. This color denotes that explicit state construction, not a further cyclic-symmetry class. A distinct deep-orange curve identifies the explicit global paired-rim family of Theorem 8, attaining $y = 7 + 3x$, $z = 8 + 2x$ throughout $2/27 \le x \le 1/6$. Its lower endpoint is now labeled $|\eta_6\rangle$: the interval is sharp and this endpoint is one C_2 orbit by Theorem 9. It is a one-dimensional locus, not an additional filled surface. The local strip existence theorem does not assign an unquantified finite display width below that rim. The complete sharp-plane equality segment of Section 23.4 retains the callout for the explicit real-radical family Ψ_t. Its endpoints keep their existing state colors and labels. The exceptional interior state $|E_*\rangle$ of Proposition 8 is marked in C_2 blue, as is $|\omega\rangle$; the intervening segment is not stroked. This construction is not assigned a fictitious cyclic-symmetry class, and its one-dimensional image does not fill the surrounding gray facet. Its interior stroke is omitted from both the single-color and family-colored body views, their magnified nearby-state details and combined figures, while its nodes, labels and leaders are retained. The full equality-state classification in Section 24 shows that this real family is one achiral boundary branch, not the whole set of equality states. Its sector image nevertheless is the complete attained segment. The exceptional $|E_*\rangle$ landmark adds no edge or filled surface. In all three views it stays at its exact sector coordinates, with a separately routed leader that avoids neighboring nodes.

The classes overlap, so the colors do not classify maximal state symmetry. A C_4 or C_6 state is also C_2, and every main-character C_3 state with $x = 0$ is also C_2 about a transverse axis: its coefficients obey $c_k = c_{6-k}$, hence $X^{\otimes 6}|\psi\rangle = |\psi\rangle$. For legibility the surface-fill priority is small-dipole four-level, seven-level, filtered octahedron, rank-one, five-level, octahedral–product, GHZ–product, rotated-W, C_3, then C_2 wherever their verified images overlap. The three-state locus is an attained curve, not a physical ridge; it is an interior trace of the larger indigo patch. Likewise the filtered-octahedral C_3 curve lies inside the flat lower-plane face. Neither construction receives a curve overlay in the edge-only views; the olive-green attained surface fill is retained. At coincident cyclic curves or nodes the displayed priority is C_6, C_5, C_4, C_3, C_2, not a claim of exclusive symmetry. Black marks other known attained paths; gray still means excluded or not certified by these constructions. Uniform flat nodes, corrected label leaders, lighter dashed rear edges and the exact necessary-condition geometry are retained. All attained fills are violet in the single-color views and their combined fold-out. All body views draw only genuine body edges, omitting attained construction traces lying inside faces: the three-state roof curve, filtered-octahedral lower-plane curve and sharp-plane equality segment. Their nodes, labels and leaders remain. The smooth part $\beta \le f < 3/8$ of the C_4 null-roof join is likewise not stroked; the genuine C_4 crease for $3/8 \le f \le 1/2$ is retained. This display convention does not change attainment of the full C_4 family. The family comparison now uses this same edge-only convention, including in its nearby-state detail. Its distinct surface colors, nodes, labels and leaders are unchanged. Omitting an interior stroke does not remove the state family or its attainment proof. The refined four-copy lower patch and its two open crease arcs are gray with *proved exclusion*, by Theorem 2; only their common product endpoint is attained. This stronger status is not asserted for unrelated gray pieces of older boundaries.

The equations above are exact analytical attainment criteria. The drawing is nevertheless a finite mesh: numerical population searches or resultant candidates propose states, which are reconstructed and checked against the original compatibility conditions. An independent check uses the Dicke

Schmidt matrix

$$B^{(t)}_{ar} = c_{a+r}\sqrt{\frac{\binom{t}{a}\binom{6-t}{r}}{\binom{6}{a+r}}}, \quad 0 \le a \le t, \quad 0 \le r \le 6-t, \qquad \rho_t = B^{(t)}B^{(t)\dagger}, \quad t = 1,2,3,$$

and (52). A numerical residual check at mesh samples is not an exact certificate for every point of a displayed triangle. The rigorous continuum statement is the family image, or equivalently the exact phase-feasibility sets just proved, on $\partial\mathcal{R}$. Attainment-color transitions are color seams, not additional geometric edges. Failure of these constructions means *not certified by these families*, not unattainable, unless a separate exclusion proof applies.

6 A further attained roof patch from rotated W states

The previous section restricts the states to cyclic-symmetry characters. We now superpose two different collective orientations of the same vertex state. The resulting sector tuples form a genuine surface, not the straight segment or convex hull of the two vertex tuples. In fact, an open part of this surface cannot be realized by *any* state with C_2, C_3, C_4, C_5 or C_6 symmetry, even after changing its collective axis.

6.1 The states and their spin-multipole verification

Keep $D_k = |D_6^{(k)}\rangle = |3, 3-k\rangle$ and set

$$\begin{aligned} &u := D_1 = |W(6)\rangle, \qquad v := e^{-i\beta J_y}u, \qquad \cos\beta = -\frac{2}{3}, \quad 0 < \beta < \pi, \\ &v = \frac{1}{216}\big(-\sqrt{30}D_0 - 24D_1 - 45\sqrt{2}D_2 - 20\sqrt{30}D_3 - 75\sqrt{2}D_4 + 25\sqrt{30}D_6\big). \end{aligned} \tag{90}$$

For example, the last line follows by rotating the Dicke polynomial $\sqrt{6}X^5Y$: replace (X, Y) by $(cX + sY, -sX + cY)$, with $c = 1/\sqrt{6}$, $s = \sqrt{5/6}$, and divide the coefficient of $X^{6-k}Y^k$ by $\sqrt{\binom{6}{k}}$. Both vectors are normalized, have sector tuple $(8/3, 5, 40/3)$, and obey $\langle u|v\rangle = -1/9$.

For real parameters a, b, define

$$\boxed{|\psi_{a,b}\rangle = \frac{u + (a+ib)v}{\sqrt{\mathcal{N}}}, \qquad \mathcal{N} := 1 + a^2 + b^2 - \frac{2a}{9}.} \tag{91}$$

Here $\mathcal{N} = (a - 1/9)^2 + b^2 + 80/81 > 0$. This is a coherent superposition of state vectors, not a mixture of their density matrices.

First, its time-reversal overlap vanishes identically. The symmetric bilinear form associated with integer-spin time reversal is

$$\mathcal{B}(s,t) := \sum_{k=0}^{6}(-1)^k s_k t_{6-k}.$$

Thus $\Theta_\psi = \mathcal{B}(\psi,\psi)$ in the convention of (61). We have

$$\mathcal{B}(u,u) = 0, \qquad \mathcal{B}(u,v) = -v_5 = 0, \qquad \mathcal{B}(v,v) = \mathcal{B}(u,u) = 0.$$

The last equality uses invariance under collective rotations, or follows from the displayed coefficients. Bilinearity gives $\Theta_{\psi_{a,b}} = 0$. Consequently *every* state in (91) lies on the globally exposed roof

$$q = 0, \qquad z = 8 + 2x. \tag{92}$$

The dipole and quadrupole give the remaining coordinates with little calculation. In the notation (54), both u and v have $Q = 0$: u has $\langle J_z^2 \rangle = 4$ and equal transverse second moments, and v is a rotation of u. Their dipoles are $(0, 0, 2)$ and $(2\sqrt{5}/3, 0, -4/3)$. The cross matrix elements are

$$\langle u|\mathbf{J}|v\rangle = \left(-\frac{2\sqrt{5}}{9}, \frac{7i\sqrt{5}}{36}, -\frac{2}{9}\right), \qquad \langle u|\widehat{Q}|v\rangle = R + iI,$$

where the two real symmetric matrices are

$$R = -\frac{25}{72}\begin{pmatrix} 4 & 0 & \sqrt{5} \\ 0 & -4 & 0 \\ \sqrt{5} & 0 & 0 \end{pmatrix}, \qquad I = \frac{5}{18}\begin{pmatrix} 0 & 5 & 0 \\ 5 & 0 & \sqrt{5} \\ 0 & \sqrt{5} & 0 \end{pmatrix}. \tag{93}$$

These entries follow directly from the displayed action of J_z, J_+ in Section 5. Writing $t = a^2 + b^2$, we obtain

$$\begin{aligned} \mathcal{N}F &= \left(\frac{2\sqrt{5}}{3}t - \frac{4\sqrt{5}}{9}a, -\frac{7\sqrt{5}}{18}b, 2 - \frac{4}{3}t - \frac{4}{9}a\right), \\ Q &= \frac{2}{\mathcal{N}}(aR - bI), \\ \operatorname{Tr} RI = 0, &\qquad \operatorname{Tr} R^2 = \frac{4375}{864}, \qquad \operatorname{Tr} I^2 = \frac{125}{27}. \end{aligned} \tag{94}$$

Insert these into $x = 2|F|^2/3$ and $y = 5 + 4\operatorname{Tr} Q^2/15$. The result is the exact parametrization

$$\boxed{\begin{aligned} x &= \frac{8}{3} - \frac{25(160a^2 + 163b^2)}{486\mathcal{N}^2}, \\ y &= 5 + \frac{25(35a^2 + 32b^2)}{162\mathcal{N}^2}, \\ z &= 8 + 2x. \end{aligned}} \tag{95}$$

This proves attainment analytically. In particular,

$$\frac{9u + v}{4\sqrt{5}} \quad \longmapsto \quad (x, y, z) = \left(\frac{41}{16}, \frac{2595}{512}, \frac{105}{8}\right). \tag{96}$$

Both component states have exactly the $W(6)$ vertex tuple, whereas their superposition has a different tuple. This illustrates why interpolating sector tuples would miss the patch.

6.2 The complete patch and an explicit inverse

For a proposed roof point define $d_x = 8/3 - x$, $d_y = y - 5$ and

$$A_W := \frac{18(163d_y - 96d_x)}{1625}, \qquad B_W := \frac{18(21d_x - 32d_y)}{325}. \tag{97}$$

Then the *complete* sector image of (91) is

$$\boxed{z = 8 + 2x, \quad A_W \geq 0, \quad B_W \geq 0, \quad 4(A_W + B_W) \leq \left(1 + \frac{2\sqrt{A_W}}{9}\right)^2.} \tag{98}$$

Indeed, inversion of the two linear combinations in (95) gives $A_W = a^2/\mathcal{N}^2$ and $B_W = b^2/\mathcal{N}^2$. Choosing the nonnegative signs for a, b, normalization becomes

$$(A_W + B_W)\mathcal{N}^2 - \left(1 + \frac{2\sqrt{A_W}}{9}\right)\mathcal{N} + 1 = 0.$$

Its discriminant is precisely the last condition in (98). If an original state had $a < 0$, its normalization quadratic has the smaller positive linear coefficient $1 - 2\sqrt{A_W}/9$, so the same condition with the plus sign remains necessary. Conversely, whenever (98) holds, put

$$\begin{aligned} k_W &:= 1 + \frac{2\sqrt{A_W}}{9}, \qquad D_W := k_W^2 - 4(A_W + B_W), \\ \mathcal{N} &:= \frac{2}{k_W + \sqrt{D_W}}, \qquad a := \mathcal{N}\sqrt{A_W}, \qquad b := \mathcal{N}\sqrt{B_W}. \end{aligned} \tag{99}$$

The rationalized root works also at $A_W = B_W = 0$ and recovers a normalized pure-state witness for every point of the patch. No optimization or convexification enters this test.

6.3 An open part is genuinely beyond all cyclic families

It is important to distinguish a state without an obvious symmetry from a tuple that *no* cyclic-symmetric state realizes. We prove the latter for the open parameter rectangle

$$0 < a < \frac{1}{9}, \qquad 0 < b < \frac{1}{18}. \tag{100}$$

First, the image is two-dimensional, since direct differentiation of (95) gives

$$\det \frac{\partial(x, y)}{\partial(a, b)} = \frac{40625}{2187} \frac{ab(1 - a^2 - b^2)}{\mathcal{N}^5} > 0 \tag{101}$$

throughout this rectangle.

Here are useful elementary bounds. Since $\mathcal{N} \geq 80/81$, (95) implies

$$\begin{gathered} 0 < d_y < \frac{43}{512} < \frac{1}{10}, \qquad 0 < d_x < \frac{803}{6144} < \frac{1}{3}, \qquad f = \sqrt{\frac{3x}{2}} > \frac{15}{8}, \\ \frac{96}{163} \leq \frac{d_y}{d_x} \leq \frac{21}{32}. \end{gathered} \tag{102}$$

For the ratio, divide the two expressions in (95): it is $3(35a^2 + 32b^2)/(160a^2 + 163b^2)$, between the two coefficient ratios. All these estimates are deliberately simple, not optimized.

Excluding both C_2 characters. For a C_2 state with nonzero dipole, its symmetry axis must be parallel to F, because a half-turn fixes no nonzero transverse vector. Choose its sign so that $\langle J_z \rangle = f > 0$. In the even character $\mathrm{span}\{D_0, D_2, D_4, D_6\}$, the available magnetic numbers are $3, 1, -1, -3$. On this subspace the diagonal operator inequality

$$J_z^2 - 4I \geq 4J_z - 7I$$

holds: its difference is $(J_z - I)(J_z - 3I)$, nonnegative at all four magnetic numbers. Hence $Q_{zz} \geq 4f - 7 > 1/2$. Formula (62) would force $d_y \geq 2Q_{zz}^2/5 > 1/10$, contrary to (102).

For the odd character $\mathrm{span}\{D_1, D_3, D_5\}$, write $\beta = |c_3|^2$. The null condition $\Theta = 0$ gives $\beta^2 = 4|c_1|^2|c_5|^2$; normalization and $f = 2(|c_1|^2 - |c_5|^2)$ then give $\beta = (4 - f^2)/8$. Substitution in (74), with $z = 8 + 2x$, yields

$$d_y = \frac{2}{5}(4 - f^2)(1 + f^2), \qquad \frac{d_y}{d_x} = 3 - \frac{9}{10} d_x > \frac{27}{10}.$$

This contradicts $d_y/d_x \leq 21/32$. Thus neither C_2 character can realize any tuple in the open patch, about *any* collective axis.

Excluding C_3, C_4, C_5, C_6. In the main C_3 character (64), the null condition is $h = 1 - s$, so $s^2 - d^2 = (1 - s)^2$ and $s = (1 + d^2)/2 \geq 1/2$. Consequently $d_y = 2(9s - 4)^2/5 \geq 1/10$, again impossible. The other C_3 characters in (69), and both orientations of the C_5 family, obey

$$d_y = \frac{2}{5}(2 \pm f)^2.$$

The plus sign gives $d_y > 6$. The minus sign gives instead

$$\frac{d_y}{d_x} = \frac{3}{5}\frac{2 - f}{2 + f} < \frac{3}{155} < \frac{96}{163},$$

where $f < 2$ follows from $d_x > 0$. Both contradict (102). Finally, every C_4 or C_6 state is also a C_2 state: its occupied Dicke levels all have the same parity. Singleton Dicke characters are already included in these exclusions. This covers every rotation character for $2 \leq n \leq 6$.

Thus (100) supplies a rigorously attained open roof patch beyond the *union* of all C_2–C_6 sector images. The full image (98) can overlap the earlier cyclic patches; only the open subpatch has been proved disjoint from all of them. The figures color the whole constructively attained image, violet in the single-color views and rose in the family-resolved comparison. Its perimeter is an attainment-color seam, not a newly proved physical ridge, and no extra inequality or artificial convex-hull face has been added to $\mathcal{R}$.

7 A further attained roof patch from GHZ–product superpositions

The GHZ and product vertices give a different construction. Their coherent superpositions do not, in general, have either vertex tuple or a convex interpolation of the two tuples. We give an explicitly normalized two-parameter family, calculate its sector lengths using only F, Q, Θ, and recover a state from every point of its image. An elementary compatibility argument then proves that an open part of this image is new even after including the preceding rotated-W patch.

7.1 A normalized null-overlap family

Set

$$\begin{aligned} g &:= \frac{D_0 + D_6}{\sqrt{2}} = |\mathrm{GHZ}(6)\rangle, \\ p &:= (c|0\rangle + e^{i\phi}s|1\rangle)^{\otimes 6}, \qquad c = \sqrt{\xi}, \quad s = \sqrt{1 - \xi}, \quad 0 \leq \xi \leq 1, \\ h &:= \langle g|p\rangle = \frac{c^6 + e^{6i\phi}s^6}{\sqrt{2}}. \end{aligned} \tag{103}$$

The new state is the following coherent superposition:

$$\boxed{|\Phi_{\xi,\phi}\rangle = 2hg - p = (2|g\rangle\langle g| - I)p.} \tag{104}$$

The operator in parentheses is a fixed unitary reflection: it is $+1$ on g and -1 on its orthogonal complement. Thus this formula is normalized for *all* parameters, including $h = 0$. More explicitly, $\|2hg - p\|^2 = 4|h|^2 + 1 - 4|h|^2 = 1$. The Dicke coefficients are

$$\Phi_0 = e^{6i\phi}s^6, \qquad \Phi_6 = c^6, \qquad \Phi_k = -\sqrt{\binom{6}{k}}\, c^{6-k}s^k e^{ik\phi} \quad (1 \leq k \leq 5). \tag{105}$$

In other words, the reflection exchanges the two endpoint coefficients of the product state and changes the signs of the five inner ones.

Recall the time-reversal bilinear form $\mathcal{B}(u, v) = \sum_k (-1)^k u_k v_{6-k}$, so that $\Theta_\Phi = \mathcal{B}(\Phi, \Phi)$ in (61). Here

$$\mathcal{B}(g, g) = 1, \qquad \mathcal{B}(g, p) = h, \qquad \mathcal{B}(p, p) = e^{6i\phi}(cs)^6 \sum_{k=0}^{6} (-1)^k \binom{6}{k} = 0.$$

Consequently

$$\Theta_\Phi = 4h^2 - 4h^2 = 0, \qquad q = 0, \qquad z = 8 + 2x. \tag{106}$$

When $h \neq 0$, this is precisely the nontrivial null-overlap branch of $g + \alpha p$: its bilinear norm is $1 + 2\alpha h$, hence it vanishes at $\alpha = -1/(2h)$. The product state itself is the other null line in this two-dimensional span. Formula (104) avoids a singular normalization at $h = 0$.

7.2 Direct verification with the dipole and quadrupole

The calculation uses exactly the manuscript's moments (54), not an optimization. Write $w = c^2 - s^2$ and $J_+ = J_x + iJ_y$. From (105) and the ladder-operator matrix elements one obtains

$$\begin{aligned} F_z &= 3w - 6(c^{12} - s^{12}), \\ F_x + iF_y &= A_1 e^{i\phi} - A_5 e^{-5i\phi}, \\ Q_{zz} &= \frac{15w^2 - 5}{2}, \\ Q_{xz} + iQ_{yz} &= E_1 e^{i\phi} + E_5 e^{-5i\phi}, \\ \langle J_+^2 \rangle &= M_2 e^{2i\phi} - M_4 e^{-4i\phi}, \end{aligned} \tag{107}$$

where all coefficients below are real:

$$\begin{aligned} A_1 &= 6cs(1 - c^{10} - s^{10}), & A_5 &= 6(cs)^5, \\ E_1 &= 15cs\,[w - (c^{10} - s^{10})], & E_5 &= 15w(cs)^5, \\ M_2 &= 30c^2s^2(1 - c^8 - s^8), & M_4 &= 30c^4s^4(c^4 + s^4). \end{aligned} \tag{108}$$

For example, J_z^2 does not distinguish D_0 from D_6; exchanging their populations therefore leaves the product state's Q_{zz} unchanged. Each of the other lines follows by retaining the matrix elements with Dicke-index differences one or two. The quadrupole norm can be evaluated without listing its nine entries:

$$\operatorname{Tr} Q^2 = \frac{3}{2} Q_{zz}^2 + 2|Q_{xz} + iQ_{yz}|^2 + \frac{1}{2}|\langle J_+^2 \rangle|^2.$$

This identity uses only symmetry and tracelessness of Q. Insert these expressions into $x = 2|F|^2/3$ and $y = 5 + 4\operatorname{Tr} Q^2/15$.

The result becomes particularly simple in the variables

$$\begin{aligned} &t := \xi(1 - \xi) = c^2 s^2 \in [0, 1/4], \qquad C_\phi := \cos 6\phi, \\ &\Lambda := 120t\big[(1 - 3t)^2 - 2(1 - C_\phi)t^3\big]. \end{aligned} \tag{109}$$

Namely, the exact sector map is

$$\boxed{x = 6 - (1 - t)\Lambda, \qquad y = 15 - (1 + 3t)\Lambda, \qquad z = 8 + 2x.} \tag{110}$$

Equations (107)–(108) provide a direct spin-multipole check of this compact parametrization; the third coordinate also follows independently from (106). Every displayed tuple is thus attained by an explicit pure state. Since $q \geq 0$ is a universal inequality, these tuples lie on an actual supporting plane of the unrestricted set.

7.3 The complete image and a constructive inverse

At fixed t, the exact range of Λ is the interval

$$\boxed{120t(1 - t)^2(1 - 4t) \leq \Lambda \leq 120t(1 - 3t)^2, \qquad 0 \leq t \leq \frac{1}{4}.} \tag{111}$$

Indeed (109) is affine in $C_\phi \in [-1, 1]$; at $C_\phi = -1$ use $(1 - 3t)^2 - 4t^3 = (1 - t)^2(1 - 4t)$. This is an exact attained interval, not an interpolation assumption.

For clarity, there is also a closed membership test with state recovery. The tuple $(6, 15, 20)$ belongs to the patch. For every other proposed tuple, put

$$U := 6 - x, \qquad V := 15 - y, \qquad H := V + 3U. \tag{112}$$

It belongs to the full image if and only if $z = 8 + 2x$, $H > 0$, and

$$t := \frac{V - U}{H} \in (0, 1/4], \qquad \Lambda := \frac{H}{4} \quad \text{satisfies (111).} \tag{113}$$

Necessity follows at once from (110). Conversely, recover

$$\xi = \frac{1 + \sqrt{1 - 4t}}{2}, \qquad C_\phi = 1 - \frac{(1 - 3t)^2 - \Lambda/(120t)}{2t^3}, \qquad \phi = \frac{1}{6} \arccos C_\phi. \tag{114}$$

The interval test ensures $C_\phi \in [-1, 1]$, and (104) then gives a normalized witness for the proposed point. If $\Lambda = 0$, (110) is the exceptional product tuple already listed. Thus the test covers the whole image, including its degeneracies, without a numerical optimization.

Finally, this is genuinely a two-dimensional image. Differentiating (110) gives

$$\det \frac{\partial(x, y)}{\partial(t, C_\phi)} = -960 \Lambda t^4. \tag{115}$$

It is nonzero whenever $t > 0$ and $\Lambda > 0$. In particular an interior parameter point with these properties has an open image in the roof plane, not just a curve.

7.4 An open part beyond all earlier attaining constructions

Consider the following open window on the null roof:

$$q = 0, \qquad \frac{23}{10} < f = \sqrt{\frac{3x}{2}} < \frac{7}{3}, \qquad 12 < y < \frac{1231}{100}. \tag{116}$$

We first exclude all C_2–C_6 sector images from this window, about *any* collective axis. Because $F \neq 0$, the axis of any nontrivial cyclic symmetry is parallel to F; choose its sign so that $F_z = f$. This reduces the question to the complete characters of Section 5, rather than testing only the displayed state's apparent symmetry.

The two C_2 characters. The odd character has magnetic support $\{2, 0, -2\}$, so $f \leq 2$ and is impossible. In the even character, $\Theta = 0$ is $c_0 c_6 + c_2 c_4 = 0$. Equivalently a two-by-two coefficient matrix has rank one. After a global phase and an axial rotation every such normalized state can therefore be written

$$\begin{gathered} c_0 = c_a c_b, \quad c_2 = c_a s_b, \quad c_4 = e^{i\chi} s_a c_b, \quad c_6 = -e^{i\chi} s_a s_b, \\ c_a^2 + s_a^2 = c_b^2 + s_b^2 = 1, \end{gathered} \tag{117}$$

with the four sine/cosine factors nonnegative. Put $\eta = c_a^2 - s_a^2$ and $\nu = c_b^2 - s_b^2$. Directly from these coefficients,

$$\begin{gathered} f = 2\eta + \nu, \qquad Q_{zz} = 1 + 4\eta\nu = 1 + 2f\nu - 2\nu^2, \\ |\langle J_+^2 \rangle|^2 = (1 - \nu^2)\big[9 + 6\eta^2 + 6\sqrt{15}\eta\sqrt{1 - \eta^2}\cos\chi\big]. \end{gathered} \tag{118}$$

Since $f > 2$, we have $\eta > 0$ and $0 < \nu \leq 1$. Cauchy–Schwarz bounds the bracket by $(\sqrt{15}\eta + 3\sqrt{1 - \eta^2})^2 \leq 24$. Moreover, $f < 7/3$ gives

$$0 < Q_{zz} \leq 1 + \frac{14}{3}\nu - 2\nu^2 \leq \frac{17}{8} + \frac{5}{3}\nu,$$

where the difference in the last inequality is $2(\nu - 3/4)^2$. A second Cauchy–Schwarz estimate yields

$$Q_{zz}^2 \le \left(\frac{289}{64} + \frac{25}{9}\right)(1+\nu^2) = \frac{4201}{576}(1+\nu^2) < \frac{15}{2}(1+\nu^2).$$

Thus

$$3Q_{zz}^2 + |\langle J_+^2\rangle|^2 < \frac{45}{2}(1+\nu^2) + 24(1-\nu^2) \le \frac{93}{2}.$$

Using (62), we obtain $y < 5 + (2/15)(93/2) = 56/5 < 12$, a contradiction. This excludes both C_2 characters, and hence also every C_4 or C_6 state, whose occupied Dicke levels have one parity.

The C_3 and C_5 characters. For the main C_3 character, null compatibility fixes

$$y = 5 + \frac{(1+f^2)^2}{10} \le 5 + \frac{1682}{405} < 12,$$

as follows from (61) or the C_3 formulas above. Its other characters have $f \le 2$. For C_5, the only physical two-level branch with $f > 2$ has

$$y = 5 + \frac{2}{5}(2+f)^2 > 5 + \frac{2}{5}\left(\frac{43}{10}\right)^2 = \frac{3099}{250} > \frac{1231}{100}.$$

These contradict the two sides of the window. Singleton Dicke characters are included in the preceding exclusions.

The rotated-W image is excluded even more simply: (95) gives $x \le 8/3$, whereas (116) requires $x > 529/150 > 8/3$.

An exact witness and a finite new patch. Take

$$\xi = \frac{39}{40}, \qquad \phi = \frac{\pi}{12},$$
$$p = \left(\frac{\sqrt{39}|0\rangle + e^{i\pi/12}|1\rangle}{\sqrt{40}}\right)^{\otimes 6}, \qquad \Phi = (2|g\rangle\langle g| - I)p. \tag{119}$$

Here $t = 39/1600$ and $C_\phi = 0$. Its exact tuple is

$$\boxed{(x,y,z) = \left(\frac{465105597769803}{131072000000000}, \frac{1612641542197791}{131072000000000}, \frac{989393597769803}{65536000000000}\right).} \tag{120}$$

Numerically this is $(3.54847410408, 12.30347856291, 15.09694820816)$, with $f = 2.30710016170$. The strict window conditions can also be checked directly from the rational coordinates, by squaring the positive bounds on f. They place this point outside all earlier cyclic and rotated-W images. Since $t \in (0, 1/4)$, $C_\phi \in (-1, 1)$ and (115) is nonzero, continuity and the inverse function theorem give a finite open roof patch with the same exclusion property.

The full null-family image (104) can overlap earlier attained sets; only the open part just proved is asserted to be new. The figures color its entire explicit image violet in the single-color views and cyan in the family comparison, with the surface-fill priority stated in Section 5.7. No extra inequality is assumed, no convex hull is filled, and a color seam is not an additional geometric edge. The known-state labels, leaders and attained cyclic arcs continue to represent exactly the same states as before.

8 An attained roof patch from octahedral–product superpositions

The reflection construction has another useful choice of reference state. Put

$$O := \frac{D_1 + D_5}{\sqrt{2}}, \qquad p := (c|0\rangle + e^{i\phi} s|1\rangle)^{\otimes 6}, \qquad c = \sqrt{\xi}, \quad s = \sqrt{1-\xi}. \tag{121}$$

Here O is the octahedral state: its dipole and quadrupole vanish. The product coefficients are $p_k = \sqrt{\binom{6}{k}}\, c^{6-k} s^k e^{ik\phi}$. With $h = \langle O|p\rangle = (p_1 + p_5)/\sqrt{2}$, define

$$\boxed{\Phi^{OP}_{\xi,\phi} = 2hO - p = (2|O\rangle\langle O| - I)p, \qquad 0 \le \xi \le 1, \quad \phi \in \mathbb{R}.} \tag{122}$$

This is normalized because the operator is unitary. Its particularly simple Dicke coefficients are

$$\Phi^{OP}_1 = p_5, \qquad \Phi^{OP}_5 = p_1, \qquad \Phi^{OP}_k = -p_k \quad (k \neq 1, 5). \tag{123}$$

Thus the reflection exchanges the two octahedral levels and changes the signs of all other coefficients.

The time-reversal form gives $\mathcal{B}(O, O) = -1$, $\mathcal{B}(O, p) = -h$, and $\mathcal{B}(p, p) = 0$. Consequently

$$\Theta_{\Phi^{OP}} = -4h^2 + 4h^2 = 0, \qquad q = 0, \qquad z = 8 + 2x. \tag{124}$$

This also proves completeness within each two-state span on this roof: $\mathcal{B}(aO + bp, aO + bp) = -a^2 - 2abh$, so its null lines are the product line $a = 0$ and $a = -2bh$. The second is exactly (122), up to normalization and phase. At $h = 0$ the two lines coincide, and the normalized reflection remains well defined.

8.1 The exact spin-multipole map

The following formulas provide direct analytical verification, using the same F, Q, Θ as (54). Write

$$t = \xi(1-\xi) \in [0, 1/4], \quad w = 2\xi - 1, \quad C_\phi = \cos 4\phi, \qquad a_t = 1 - 4t + 8t^2, \quad b_t = 4t(1-2t).$$

Applying J_z, J_+, J_+^2 to (123) gives

$$\begin{aligned} F_z &= 3w^5, & F_x + iF_y &= 6t^{5/2} e^{i\phi}(10 - 5e^{-4i\phi} - e^{4i\phi}), \\ Q_{zz} &= 5(1-6t), & Q_{xz} + iQ_{yz} &= 15wt^{5/2} e^{i\phi}(2 + 3e^{-4i\phi} - e^{4i\phi}), \\ \langle J_+^2 \rangle &= 30te^{2i\phi}(a_t - b_t e^{-4i\phi}). \end{aligned} \tag{125}$$

For example, the exchanged populations give $F_z = 3w^5$, whereas J_z^2 is unchanged by that exchange. The index-one and index-two matrix elements give the remaining lines. Using $\operatorname{Tr} Q^2 = 3Q_{zz}^2/2 + 2|Q_{xz} + iQ_{yz}|^2 + |\langle J_+^2\rangle|^2/2$ gives a useful factorization of the answer. Let α, β, γ be the squared components of the product Bloch direction along the octahedral axes $(\boldsymbol{e}_x + \boldsymbol{e}_y)/\sqrt{2}$, $(\boldsymbol{e}_x - \boldsymbol{e}_y)/\sqrt{2}$, and $\boldsymbol{e}_z$, respectively, and put

$$u = \alpha\beta + \beta\gamma + \gamma\alpha, \qquad v = \alpha\beta\gamma, \qquad \ell = u - v. \tag{126}$$

Here $\alpha, \beta, \gamma \ge 0$ and their sum is one. In the original parameters these invariants are simply

$$u = 4t + 2(C_\phi - 7)t^2, \qquad v = 2(1 + C_\phi)t^2(1 - 4t).$$

Substitution of the moments then gives the compact exact sector map

$$\boxed{x = 6 - 30\ell(1-u), \qquad y = 15 - 30\ell(1+3v), \qquad z = 8 + 2x.} \tag{127}$$

Every point of the full rectangle $0 \le t \le 1/4$, $-1 \le C_\phi \le 1$ is realized: choose $\xi = (1 + \sqrt{1-4t})/2$ and $\phi = \arccos(C_\phi)/4$, then use (122). Equivalently, every point of the simplex $\alpha + \beta + \gamma = 1$ is

a product direction. Conversely every octahedral–product null superposition has this form after a common collective rotation. Thus this is a complete two-state roof image, not a hull of sampled points.

There is also a constructive inverse. Besides the product tuple $(6, 15, 20)$, put

$$X = \frac{6-x}{30}, \qquad Y = \frac{15-y}{30}, \quad U = 3X + Y,$$
$$\ell = \frac{U}{2+\sqrt{4-3U}}, \quad v = \frac{Y/\ell - 1}{3}, \quad u = \ell + v.$$

A tuple belongs to the image exactly when $z = 8 + 2x$, these expressions are real with $\ell > 0$, and

$$\lambda^3 - \lambda^2 + u\lambda - v \tag{128}$$

has three nonnegative real roots. Indeed those roots recover α, β, γ and hence a product direction and its state. The inverse follows from $U = \ell(4 - 3\ell)$, with the physical root $0 < \ell \le 1/3$. For a purely algebraic test one may equivalently require $0 \le u \le 1/3$, $0 \le v \le 1/27$, and $u^2 - 4u^3 - 4v - 27v^2 + 18uv \ge 0$, the cubic discriminant.

8.2 A genuinely new two-dimensional patch

An exact witness is $t = 1/40$, $C_\phi = 0$, for example $\xi = (1 + 3/\sqrt{10})/2$, $\phi = \pi/8$. The corresponding invariants are $(u, v) = (73/800, 9/8000)$, and its tuple is

$$\boxed{(x, y, z) = \left(\frac{2267499}{640000}, \frac{78637599}{6400000}, \frac{4827499}{320000}\right).} \tag{129}$$

These rational coordinates satisfy the strict window (116). The character-by-character proof there therefore excludes every C_2–C_6 image and the rotated-W image at this point, regardless of collective orientation. It is outside the GHZ–product image as well. Its proposed inverse (112)–(113) gives

$$t_{GP} = \frac{757}{29837}, \qquad \Lambda_{GP} = \frac{64537431}{25600000},$$

but violates the necessary lower endpoint in (111). Explicitly,

$$120 t_{GP}(1 - t_{GP})^2(1 - 4t_{GP}) - \Lambda_{GP} = \frac{157292045200381114721 4009}{202889972125765570816000 00} > 0.$$

Finally direct differentiation of (127) gives

$$\det \frac{\partial(x, y)}{\partial(t, C_\phi)}\bigg|_{(1/40,0)} = \frac{51807845061}{51200000000} > 0. \tag{130}$$

The parameters are interior, all exclusions are strict, and the Jacobian is nonzero. The inverse function theorem therefore supplies a finite open attained roof patch beyond the union of all previous constructions. Only this open part is asserted to be new; the full image can overlap them. It is violet in the single-color views and orange in the family comparison. No additional inequality or geometric ridge is introduced by the edge of its color fill.

9 An attained roof curve from three vertex-state components

Adding a third state gives a transparent new boundary construction, even without relative rotations. Keep O from (121) and set

$$G = \frac{D_0 + D_6}{\sqrt{2}}, \qquad P = D_0.$$

Notice that $\langle G|P\rangle = 1/\sqrt{2}$: the squared amplitudes of O, G, P cannot simply be added to normalize a superposition. For $0 \le \sigma \le 1/2$, the following coefficients already include that overlap:

$$\boxed{\Psi_\sigma = \sqrt{\sigma}\, O + \frac{1-\sqrt{1-2\sigma}}{\sqrt{2}}\, G + \sqrt{1-2\sigma}\, P.} \tag{131}$$

In the orthonormal Dicke basis this is $A_\sigma D_0 + B_\sigma D_6 + C_\sigma O$, where

$$A_\sigma = \frac{1+\sqrt{1-2\sigma}}{2}, \qquad B_\sigma = \frac{1-\sqrt{1-2\sigma}}{2}, \qquad C_\sigma = \sqrt{\sigma}. \tag{132}$$

Indeed $A_\sigma^2 + B_\sigma^2 + C_\sigma^2 = 1$ and $2A_\sigma B_\sigma = C_\sigma^2$, so $\Theta = 0$.

9.1 Verification and the exact roof curve

For clarity, the entire calculation follows from five elementary moments. For a normalized state $AD_0 + BD_6 + \sqrt{\sigma}\, O$, where a global phase makes the last coefficient real, put $d = |A|^2 - |B|^2$. The Dicke ladder elements give

$$\begin{aligned} F_z &= 3d, & F_x + iF_y &= \sqrt{3\sigma}(\overline{A} + B), \\ Q_{zz} &= 5(1-\sigma), & Q_{xz} + iQ_{yz} &= \frac{5\sqrt{3\sigma}}{2}(\overline{A} - B), \\ \langle J_+^2 \rangle &= 0, & \Theta &= 2AB - \sigma. \end{aligned} \tag{133}$$

For (132), $A + B = 1$ and $A - B = d = \sqrt{1-2\sigma}$. Substituting into (54) gives the exact attained curve

$$\boxed{x = 6 - 10\sigma, \qquad y = 15 - 10\sigma - 10\sigma^2, \qquad z = 20 - 20\sigma = 8 + 2x.} \tag{134}$$

Equivalently,

$$y = \frac{27}{5} + \frac{11}{5}x - \frac{x^2}{10}, \qquad 1 \le x \le 6, \qquad z = 8 + 2x. \tag{135}$$

These states lie on the universal supporting plane $q = 0$, so the curve is on the actual pure-state boundary. It is not asserted to be an edge of $\mathcal{R}$ or the boundary of an attained surface. Its redundant interior stroke is omitted from both the violet-only and family-colored solid views; the state construction and its proof are unchanged. In particular,

$$\Psi_{1/4} = \frac{1}{2}O + \frac{\sqrt{2}-1}{2}G + \frac{1}{\sqrt{2}}P \quad \longmapsto \quad \boxed{(x, y, z) = \left(\frac{7}{2}, \frac{95}{8}, 15\right).} \tag{136}$$

9.2 A finite arc beyond the preceding families

We prove that every point with

$$\frac{249}{1000} \le \sigma \le \frac{251}{1000} \tag{137}$$

is outside all the preceding cyclic, rotated-W, GHZ–product and octahedral–product images. This is a statement about sector tuples, not merely the apparent symmetry of (131). Throughout this interval,

$$\frac{9}{4} < f = \sqrt{3x/2} < \frac{7}{3}, \qquad \frac{56}{5} < y < 12, \qquad x > \frac{8}{3}.$$

The even-C_2 null-state estimate derived from (118) gives $y < 56/5$ whenever $2 < f < 7/3$; the odd character has $f \le 2$. This also excludes C_4 and C_6. The main C_3 null character gives $y = 5 + (1+f^2)^2/10 \le 5 + 1682/405 < 56/5$; its other characters have $f \le 2$. The C_5 branch with $f > 2$ instead has $y = 5 + 2(2+f)^2/5 > 489/40 > 12$. These are all characters, since a cyclic axis must be parallel to the nonzero dipole. Finally, (95) gives $x \le 8/3$. All cyclic and rotated-W images are therefore excluded.

In fact the entire curve with $0 < \sigma \le 1/2$ is outside the GHZ–product image. Its proposed inverse is

$$t_{GP} = \frac{\sigma}{4+\sigma}, \qquad \Lambda_{GP} = \frac{5}{2}\sigma(4+\sigma).$$

It violates the lower bound in (111), because after cancelling positive factors that violation is exactly

$$(4+\sigma)^5 < 768(4-3\sigma).$$

The left side increases and the right side decreases, and even at $\sigma = 1/2$ one has $(9/2)^5 = 59049/32 < 1920$.

There is also a short global exclusion of the octahedral–product image near (136). The three antipodal Majorana axes of O are $\boldsymbol{e}_z$, $(\boldsymbol{e}_x + \boldsymbol{e}_y)/\sqrt{2}$ and $(\boldsymbol{e}_x - \boldsymbol{e}_y)/\sqrt{2}$. Its rotational stabilizer permutes these axes and commutes with $2|O\rangle\langle O| - I$. We may therefore rotate the largest component of the product Bloch vector onto z, without changing the sector tuple. Its squared component is at least $1/3$, so the representative in (127) has $t \le 1/6$. Expanding the first line of (127) in t, C_ϕ gives $x = 6(1-4t)^5 + 96t^5(29 - 30C_\phi + 5C_\phi^2)$. Since the bracket is at most 64 on $[-1,1]$, this implies

$$x \le f_0(t) := 6(1-4t)^5 + 6144t^5.$$

Both summands are convex for $0 \le t \le 1/6$. Hence, for $13/500 \le t \le 1/6$,

$$x \le \max\{f_0(13/500), f_0(1/6)\} = \frac{169188366}{48828125} < \frac{349}{100}.$$

But the arc (137) has $x \ge 349/100$. It must therefore have $t < 13/500$ if it belongs to the octahedral–product image. The quadrupole then gives the opposite contradiction:

$$y \ge 5 + \frac{2}{5}Q_{zz}^2 = 5 + 10(1-6t)^2 > \frac{75771}{6250} > 12.$$

Thus the whole finite arc, not just one sampled point, is new beyond every preceding construction. The curve can overlap earlier images elsewhere: for example $\sigma = 1/2$ is the already attained C_2 tuple $(1, 15/2, 10)$.

9.3 Why two of these vertex classes cannot supply this arc

The previous arguments exclude GHZ–product and octahedral–product pairs even when the two states have arbitrary independent collective orientations and complex amplitudes: a common rotation fixes the GHZ or octahedral reference, and the complete null branch was explicitly derived. To complete the comparison we exclude an octahedral–GHZ pair.

Choose phases so that $\mathsf{T}O = -O$ and $\mathsf{T}V = V$, where V is any collectively rotated GHZ state. Then $\langle O|V\rangle = i\rho$ with ρ real. Put $\delta = 1 - \rho^2 > 0$; equality would make the two rays identical, which is impossible since their quadrupoles differ. The two null roots in their span are proportional to

$$(-i\rho \pm \sqrt{\delta})O + V.$$

Their squared norm is 2δ. Because both reference states have zero dipole, their normalized dipole is $F_i = \pm h_i/\sqrt{\delta}$, where $h_i = \langle J_i O|V\rangle$ is real by time reversal. The vanishing octahedral dipole and quadrupole give

$$\langle O|J_i O\rangle = 0, \qquad \langle J_i O|J_j O\rangle = 4\delta_{ij}.$$

Bessel's inequality applied to the four orthonormal vectors $O, J_x O/2, J_y O/2, J_z O/2$ yields $\sum_i |h_i|^2 \le 4(1 - |\langle O|V\rangle|^2) = 4\delta$. Consequently every null octahedral–GHZ superposition satisfies

$$x = \frac{2}{3}|F|^2 \le \frac{8}{3}. \tag{138}$$

This excludes (137). Thus three distinct reference classes give strictly more boundary coverage than any pair among O, G, P, even allowing relative rotations. No assertion is made about pairs of other vertex classes or a classification of arbitrary pure states. Its redundant interior stroke is omitted from both the single-color and family-colored solid views. It remains an attained locus, not an added physical ridge or a new outer inequality.

10 A complete attained roof band from four Dicke levels

The earlier three-state curve can be enlarged to a whole surface without adding rotations or taking convex combinations of sector tuples. The essential observation is that the null-overlap condition is a rank-one condition on a small matrix of Dicke coefficients.

For $0 \le u, v \le 1$ define

$$\boxed{|\psi_{u,v}\rangle = \sqrt{uv}\, D_0 + \sqrt{u(1-v)}\, D_1 + \sqrt{(1-u)v}\, D_5 + \sqrt{(1-u)(1-v)}\, D_6.} \tag{139}$$

Its norm is one because the four populations sum to $(u+1-u)(v+1-v) = 1$. Moreover

$$\begin{pmatrix} c_0 & c_1 \\ c_5 & c_6 \end{pmatrix} = \begin{pmatrix} \sqrt{u} \\ \sqrt{1-u} \end{pmatrix} \begin{pmatrix} \sqrt{v} & \sqrt{1-v} \end{pmatrix}, \qquad \Theta = 2(c_0 c_6 - c_1 c_5) = 0.$$

Thus every state lies on $q = 0$, or $z = 8 + 2x$. In fact every complex null state supported on D_0, D_1, D_5, D_6 has a rank-one coefficient matrix. Its row and column phases do not alter the sector map below, so nonnegative representatives suffice for that entire four-level null image.

10.1 Multipoles and the exact image

Put $d = 2u - 1$, $e = 2v - 1$, so $-1 \le d, e \le 1$. The ladder elements give

$$\begin{aligned} F_z &= \frac{5d + e}{2}, & F_x + iF_y &= \frac{\sqrt{6}}{2}\sqrt{1-e^2}, \\ Q_{zz} &= \frac{5}{2}(1 + de), & Q_{xz} + iQ_{yz} &= \frac{5\sqrt{6}}{4} d\sqrt{1-e^2}, \\ \langle J_+^2 \rangle &= 0. \end{aligned} \tag{140}$$

For example, the two index-one coherences have the same phase and add in $F_x + iF_y$, whereas their opposite magnetic midpoints give their difference in $Q_{xz} + iQ_{yz}$. Using

$$\operatorname{Tr} Q^2 = \frac{3}{2} Q_{zz}^2 + 2|Q_{xz} + iQ_{yz}|^2 + \frac{1}{2}|\langle J_+^2 \rangle|^2$$

and (55) yields

$$\boxed{\begin{aligned} x &= \frac{25d^2 + 10de - 5e^2 + 6}{6}, \\ y &= \frac{5}{2}(3 + 2d^2 + 2de - d^2 e^2), \qquad z = 8 + 2x. \end{aligned}} \tag{141}$$

Theorem 3 (Complete four-level null image)**.** *The image of* (139) *is exactly*

$$\boxed{\begin{aligned} &0 \le x \le 6, \qquad z = 8 + 2x, \qquad f = \sqrt{3x/2}, \\ &\max\left\{5 + \frac{2}{5}(2-f)^2,\ 3x - 3\right\} \le y \le 5 + \frac{2}{5}(2+f)^2. \end{aligned}} \tag{142}$$

In particular every point of this two-dimensional band is attained by one pure symmetric state, not by a mixture.

Proof. First we prove the bounds, then prove that no holes remain between them. Write $K = 5(y-5)/2 - f^2 - 4$. Direct substitution in (141) gives the elementary factorization

$$\begin{aligned} 16f^2 - K^2 &= \frac{25}{16}(1-e^2)P(d,e), \\ P(d,e) &= [(5d^2 - 1)e - 8d]^2 + 15 - 6d^2 - 25d^4. \end{aligned} \tag{143}$$

The last expression is nonnegative on the square, although its displayed remainder need not be nonnegative everywhere. If $d^2 \le 1/5$, that remainder is at least $64/5$. Otherwise simultaneous reversal of d, e permits $d > 1/\sqrt{5}$. As a quadratic in e, P has its vertex at $8d/(5d^2-1) > 1$; indeed $8d > 5d^2 - 1$ for $0 \le d \le 1$. It is therefore minimized on $[-1,1]$ at $e = 1$, where

$$P(d,1) = 16(1-d)(5d^2+2d+1) \ge 0.$$

Hence $|K| \le 4f$, or

$$(f-2)^2 \le \frac{5}{2}(y-5) \le (f+2)^2.$$

The other lower bound follows directly from

$$y - (3x-3) = \frac{5}{2}(1-d^2)(e^2+3) \ge 0. \tag{144}$$

The larger lower bound switches at $f = 2$, equivalently $x = 8/3$. Finally $0 \le x \le 6$ follows from $|F| \le 3$ for a spin-three state.

For sufficiency fix x and hence $f \in [0,3]$. Its dipole equation is solved by the two branches

$$d_\pm(e) = \frac{-e \pm \sqrt{4f^2 - 6 + 6e^2}}{5}. \tag{145}$$

The following continuous paths remain inside $[-1,1]^2$ and join the two sharp y endpoints:

- If $0 \le f \le \sqrt{3/2}$, take both branches for $\sqrt{1-2f^2/3} \le e \le 1$. They meet where the square root vanishes. Their two endpoints at $e = 1$ are $d = (-1 \pm 2f)/5$.
- If $\sqrt{3/2} \le f \le 2$, take $d_+(e)$ for $-1 \le e \le 1$. Its two endpoints give the same two C_5 values, using simultaneous reversal of d, e at $e = -1$.
- If $2 \le f \le 3$, take $d_+(e)$ for $1 - 2\sqrt{9-f^2}/\sqrt{5} \le e \le 1$. The first endpoint has $d = 1$; the second has $d = (2f-1)/5$. The bound $d_+(e) \le 1$ is precisely $(e-1)^2 \le 4(9-f^2)/5$, so all parameters are physical.

The first two paths join the lower and upper C_5 endpoints. The third joins $y = 3x-3$ to the upper C_5 endpoint. Since y is continuous along each connected path, the intermediate value theorem supplies every intervening value. This is a pure-state coverage proof, with no convexity or monotonicity assumption on the fixed-x fiber. □

The upper and, for $f \le 2$, lower curves have the explicit states

$$|\chi_f^\pm\rangle = \sqrt{\frac{2 \pm f}{5}}\, D_0 + \sqrt{\frac{3 \mp f}{5}}\, D_5. \tag{146}$$

They meet at $(0, 33/5, 8)$. The upper branch ends at the product vertex; the lower branch ends at a collectively reversed W state. For $f \ge 2$ the lower line is the known product–W edge, obtained from (139) at $d = 1$. Thus all three boundaries of the band are constructively attained.

10.2 The whole high-dipole roof is settled

For $8/3 \le x \le 6$, (142) is not merely a subpatch: it is the entire exposed $q = 0$ part of the existing outer region. To see this, $p_2 \ge 0$ and $z = 8+2x$ imply $y \ge 3x-3$. If $y \ge 99/10$, put $\omega = \sqrt{(y-5)/10}$. The universal orange bound (6g), with $S_4 = 21 + 4x - 2y$ on this roof, rearranges to

$$(5\omega - 2)^2 \le \frac{3}{2}x = f^2.$$

It gives $y \le 5 + 2(2+f)^2/5$. If $y < 99/10$, the latter bound is automatic because $f \ge 2$ makes its right-hand side at least $57/5$. These are precisely the two boundaries filled by Theorem 3. Thus the

proved outer roof and actual pure-state roof coincide for $x \geq 8/3$. No corresponding completeness claim is made for $x < 8/3$ or for the other sheets.

The old three-state curve is the diagonal $d = e$, for which $x = 1 + 5d^2$ and (135) follows. Its Jacobian is

$$\det \frac{\partial(x, y)}{\partial(d, e)}\bigg|_{e=d} = 50d^2(1 - d^2).$$

It is therefore an interior trace of the enlarged patch whenever $0 < |d| < 1$, not its physical edge. For a target in (142), a scalar bracketed search along (145) gives a constructive state inverse; the continuous-path proof guarantees a root. The image is indigo in the family comparison and violet in the single-color figures.

11 A filled lower-plane patch from filtered octahedral states

Rotations alone leave sector lengths unchanged, but the same invertible nonunitary filter applied to each qubit can move a state along an exposed face. Here it preserves a particularly useful pure-state compatibility equation.

Let $O = (D_1 + D_5)/\sqrt{2}$. For $k \geq 0$ and $1/3 \leq c \leq 1$, set

$$\boxed{|\phi_{k,c}\rangle = \frac{e^{hJ_z} e^{-i\beta J_y} O}{\sqrt{Z(k,c)}}, \qquad h = \operatorname{arsinh} \sqrt{k}, \qquad \beta = \arccos \sqrt{c},} \tag{147}$$

where the exact squared norm is

$$Z(k, c) = 1 + 8k + [8 + 10(1 - c)(1 + 3c)]k^2 + 12(1 - c)(1 + c)^2 k^3. \tag{148}$$

The factor e^{hJ_z} is the action on six qubits of $\operatorname{diag}(e^{h/2}, e^{-h/2})^{\otimes 6}$. It is a filter, not a collective rotation; division by $\sqrt{Z}$ normalizes the resulting pure symmetric state.

11.1 Why the lower-plane equality is preserved

Let $\Pi_2 : V_3 \otimes V_3 \longrightarrow V_2$ be the normalized Clebsch–Gordan map, so $\Pi_2^\dagger \Pi_2 = P_2$ and $p_2 = \|\Pi_2(\psi \otimes \psi)\|^2$. For $O \otimes O$, the terms with magnetic sums $4, -4$ cannot contribute to spin two; the remaining coefficient is $\langle 3, 2; 3, -2 | 2, 0\rangle = 0$. Hence

$$\Pi_2(O \otimes O) = 0.$$

The Clebsch–Gordan map intertwines $SU(2)$ and therefore its complexified Lie algebra and exponentials. In particular, for $g \in SL(2, \mathbb{C})$,

$$\Pi_2\big(D^{(3)}(g)O \otimes D^{(3)}(g)O\big) = D^{(2)}(g)\Pi_2(O \otimes O) = 0.$$

This applies to the rotation and determinant-one filter in (147). The exact two-copy identity $p_2 = 5(y + z - 5x - 5)/168$ thus gives

$$\boxed{z = 5x + 5 - y.} \tag{149}$$

This plane equality is a pure-state compatibility statement, not an interpolation between the octahedral, W, and product vertices.

11.2 Norm and spin-multipole verification

Here is an explicit way to verify (148). Write the positive single-qubit matrix $H = e^{h\boldsymbol{n}\cdot\boldsymbol{\sigma}}$ as $\left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$, where $ad - |b|^2 = 1$. The required Dicke matrix elements are

$$\begin{aligned} \langle D_1 | H^{\otimes 6} | D_1 \rangle &= a^4(ad + 5|b|^2), \\ \langle D_5 | H^{\otimes 6} | D_5 \rangle &= d^4(ad + 5|b|^2), \\ \langle D_1 | H^{\otimes 6} | D_5 \rangle &= b^4(5ad + |b|^2). \end{aligned}$$

Half the sum of the first two entries plus the real part of the third is $\langle O|H^{\otimes 6}|O\rangle$. Substituting the boost direction with $n_z^2 = c$, $n_y = 0$, gives (148).

Let $Z_k = \partial_k Z$ and $Z_c = \partial_c Z$. Differentiating the norm gives $\partial_h Z = 2F_z Z$ and $\partial_\beta Z = -2\sinh(h)F_x Z$. Consequently, up to the irrelevant sign of F_x,

$$F_z = \frac{\sqrt{k(1+k)}Z_k}{Z}, \qquad F_x = \frac{\sqrt{c(1-c)}Z_c}{\sqrt{k}Z}, \qquad F_y = 0. \tag{150}$$

The $k = 0$ formula is read by continuity. The spin-flip bilinear form is invariant under determinant-one collective filters. Its unnormalized value is therefore still $\mathcal{B}(O, O) = -1$, so $\Theta = -1/Z$ after normalization. Together with (149), this proves the exact map

$$\boxed{\begin{aligned} &x = \frac{2}{3Z^2}\left[k(1+k)Z_k^2 + \frac{c(1-c)}{k}Z_c^2\right], \\ &y = 3x - 3 + \frac{8}{Z^2}, \qquad z = 2x + 8 - \frac{8}{Z^2} = 5x + 5 - y. \end{aligned}} \tag{151}$$

11.3 Complete image and a constructive coverage proof

Theorem 4 (Filtered-octahedral lower-plane patch)**.** *The family* (147)*, together with its already attained limiting product–W edge, fills exactly the closed region*

$$\boxed{\begin{aligned} &z = 5x + 5 - y, \qquad 5 \le y \le 15, \qquad t = \sqrt{\frac{y-5}{10}}, \\ &\frac{2}{3}t(t+8) \le x \le \frac{y+3}{3}. \end{aligned}} \tag{152}$$

This is a complete image theorem for this family, not a classification of all pure states with $p_2 = 0$.

Proof. The relevant Jacobian has an elementary factorization:

$$\begin{aligned} \det\frac{\partial(x,y)}{\partial(k,c)} &= -\frac{20480k^2(3c-1)[(3-c)k+2][(c+1)k+1]}{Z^5}\mathcal{H}(k,c), \\ \mathcal{H}(k,c) &= (9c^2 - 14c + 9)k^2 + (6-2c)k + 1. \end{aligned} \tag{153}$$

The identity follows by differentiating (151) and expanding. Every displayed factor apart from the leading minus sign is positive for $k > 0$, $1/3 < c < 1$: the quadratic $9c^2 - 14c + 9$ has negative discriminant -128, and $6 - 2c \ge 4$.

Fix any norm $Z_0 > 1$ and write $r = 1/Z_0$. All nonconstant coefficients of (148) are nonnegative on the parameter interval, and its linear coefficient is positive. Thus $Z_k > 0$, and for each c there is a unique continuous $k(c) > 0$ with $Z(k(c), c) = Z_0$. Along this norm level,

$$\frac{dx}{dc} = \frac{Z_0^3}{16Z_k}\det\frac{\partial(x,y)}{\partial(k,c)} < 0. \tag{154}$$

The endpoint $c = 1$ is the biased two-Dicke octa–W family: $Z = \cosh 4h$, $F_z = 2\tanh 4h$, and

$$x_W = \frac{8}{3}(1 - r^2), \qquad y_W = 5.$$

At $c = 1/3$ the boost is along an octahedral threefold axis. In that frame one may use

$$O_3 = \frac{\sqrt{2}D_0 + \sqrt{5}D_3 - \sqrt{2}D_6}{3}, \qquad Z = \frac{5 + 4\cosh 6h}{9}.$$

Its dipole and axial quadrupole give

$$x_{C_3} = \frac{2}{3}(1-r)(9-r), \qquad y_{C_3} = 5 + 10(1-r)^2.$$

Therefore continuity, and in fact strict monotonicity, fills the whole fixed-norm line interval

$$5 \le y \le 5 + 10(1-r)^2, \qquad x = \frac{y+3-8r^2}{3}, \qquad z = 5x + 5 - y. \tag{155}$$

Varying $0 < r \le 1$ gives every finite-filter point, with $r = 1$ the octahedral vertex. The case $r = 0$ is the known normalized edge $\sqrt{p}D_0 + \sqrt{1-p}D_1$, where $p = 1 - \sqrt{(15-y)/10}$; no infinite vector is assigned as a state. Finally, putting $y = 5 + 10t^2$ in (155) gives $r \le 1 - t$. Since $r^2 = (y - 3x + 3)/8$, this is exactly the lower bound in (152); $r^2 \ge 0$ is its upper bound. This proves both necessity and coverage of the stated image. □

The inverse is constructive: compute r from a proposed point, solve $Z(k, c) = 1/r$ uniquely for k at each c, then use the monotone equation (154) to recover c. Both are scalar bracketed inversions. The endpoints use the states just given.

The patch has three attained boundaries: octa–W, product–W, and the following C_3 octa–product curve:

$$(x, y, z) = \left(\frac{2}{3}t(t+8),\ 5 + 10t^2,\ \frac{20}{3}t(4-t)\right), \quad 0 \le t \le 1. \tag{156}$$

A direct Dicke representative, useful without applying a filter, is

$$\begin{aligned} |\zeta_t\rangle &= \sqrt{\frac{s+d}{2}}D_0 + \sqrt{\frac{5r}{9}}D_3 - \sqrt{\frac{s-d}{2}}D_6, \\ r &= 1 - t, \qquad s = 1 - \frac{5r}{9}, \qquad d = \sqrt{s^2 - \frac{16r^2}{81}}. \end{aligned} \tag{157}$$

It has $5c_0c_6 + 2c_3^2 = 0$ and $\Theta = -r$, giving precisely (156). Its interior curve overlay is omitted in the edge-only views; the filled triangle remains olive green in the family comparison and violet in the standard views.

This does not fill the whole old lower-plane outer face. Its intersection with the lower-Gram sheet has $x_{\mathrm{Gram}} = 2t(5t+4)/3$, whereas

$$x_{C_3} - x_{\mathrm{Gram}} = \frac{8}{3}t(1-t) > 0 \qquad (0 < t < 1).$$

No exclusion is claimed for every point of that intervening strip. The strict lower-Gram interior itself is excluded by the equality analysis in Section 15; attained bounding edges therefore do not imply an attained filling of every outer sheet.

12 A five-level null family beyond the rank-one band

The complete roof band leaves a small low-S_2 region to investigate. A different exact null construction reaches below its lower C_5 boundary. Unlike the previous two families, its complete sector image is not asserted here.

For real a, b and $0 \le p \le 1$, let

$$\begin{aligned} |\chi_{a,b}\rangle &= \cos a \cos b\, D_0 + \cos a \sin b\, D_2 + \sin a \cos b\, D_4 - \sin a \sin b\, D_6, \\ |\eta_{p,a,b}\rangle &= i\sqrt{p}\, D_1 + \sqrt{1-p}\, |\chi_{a,b}\rangle. \end{aligned} \tag{158}$$

The even-character vector is normalized and satisfies $\Theta_\chi = 2(c_0c_6 + c_2c_4) = 0$. It is orthogonal to D_1 and has zero bilinear overlap with D_1, because its D_5 coefficient vanishes. Hence (158) is normalized and $\Theta_\eta = 0$ for every parameter, not merely at fitted targets.

12.1 Spin-multipole map and an exact open-patch witness

Let c_0, c_2, c_4, c_6 denote the real coefficients of $\chi_{a,b}$. Define

$$\begin{aligned}
&\eta = \cos 2a, \qquad \nu = \cos 2b, \qquad f_\chi = 2\eta + \nu, \qquad q_\chi = 1 + 4\eta\nu,\\
&t_\chi = \frac{\sin 2b}{2}(\sqrt{60}\eta + 6\sin 2a), \quad h_\chi = \sqrt{6}c_0 - \sqrt{10}c_2,\\
&g_\chi = \frac{5\sqrt{6}}{2}c_0 - \frac{3\sqrt{10}}{2}c_2.
\end{aligned}$$

Here f_χ is a signed z component, not the length $|F|$. The same ladder calculation as before gives

$$\begin{aligned}
F_z &= 2p + (1-p)f_\chi, \qquad & F_x + iF_y &= i\sqrt{p(1-p)}h_\chi,\\
Q_{zz} &= (1-p)q_\chi, & Q_{xz} + iQ_{yz} &= i\sqrt{p(1-p)}g_\chi,\\
\langle J_+^2\rangle &= (1-p)t_\chi.
\end{aligned} \tag{159}$$

Consequently its exact sector map is

$$\begin{aligned}
x &= \frac{2}{3}\big([2p + (1-p)f_\chi]^2 + p(1-p)h_\chi^2\big),\\
y &= 5 + \frac{2}{5}(1-p)^2 q_\chi^2 + \frac{8}{15}p(1-p)g_\chi^2 + \frac{2}{15}(1-p)^2 t_\chi^2,\\
z &= 8 + 2x.
\end{aligned} \tag{160}$$

At $p = 1/2$, $a = -\pi/6$, $b = \pi/3$, this yields the exact state

$$|\eta_*\rangle = \frac{\sqrt{6}D_0 + 3\sqrt{2}D_2 - \sqrt{2}D_4 + \sqrt{6}D_6}{8} + \frac{iD_1}{\sqrt{2}}, \tag{161}$$

with

$$x_* = \frac{52 - 9\sqrt{5}}{24}, \qquad y_* = \frac{631 - 99\sqrt{5}}{80}, \qquad z_* = 8 + 2x_*. \tag{162}$$

Holding $a = -\pi/6$, direct differentiation of (160) gives

$$\det \frac{\partial(x,y)}{\partial(p,b)}\bigg|_{(1/2,\pi/3)} = \frac{111\sqrt{15} - 157\sqrt{3}}{40} > 0. \tag{163}$$

The inverse function theorem proves that even this fixed-a two-parameter family fills an open roof neighborhood of the example.

The neighborhood is genuinely beyond the complete rank-one image. For a short exact check put

$$A_* = \frac{3x_*}{2} = \frac{52 - 9\sqrt{5}}{16}, \qquad B_* = \frac{5}{2}(y_* - 5) = \frac{231 - 99\sqrt{5}}{32}.$$

$$\delta_* = 4 - A_* - B_* = \frac{117\sqrt{5} - 207}{32} > 0.$$

One obtains

$$\delta_*^2 - 4A_*B_* = \frac{4689\sqrt{5} - 10221}{512} > 0.$$

Both displayed signs follow already from $\sqrt{5} > 11/5$. Thus $\sqrt{A_*} + \sqrt{B_*} < 2$, or $y_* < 5 + 2(2 - \sqrt{3x_*/2})^2/5$. Since $x_* < 8/3$, this is strictly below the active lower boundary of (142). Continuity and (163) give a whole new open patch, not only an isolated state. This comparison excludes the rank-one image, not every other possible construction.

12.2 What the magenta display does and does not certify

The preceding local patch theorem is analytical. A complete boundary or inverse theorem for all of (158) has not been proved. In the figures this family therefore contributes magenta display points only after its parameters have been reconstructed and the actual normalized state has passed an independent marginal-purity check. The displayed search is restricted to the low-S_2 window $0.15 \le x \le 2.7$, $5 \le y \le 5.25$ on $q = 0$; it is a positively witnessed subset, not a claim that this window is wholly attained or that the family cannot extend outside it. Failure to reconstruct a state remains an unresolved point, not an exclusion proof. No hull of successful samples is used as an attainment criterion.

The exact defining inequalities of the outer body have not changed. These new families change only which parts have a constructive attainment explanation. In particular the lower-Gram exclusion and the separate exclusions of gray nodes remain in force.

13 An exact seven-level roof band and its quadrupole-free edge

The five-level family reaches very low S_2, but it does not fill the whole null-overlap roof. A different construction closes a genuine part of that gap. We first explain the particularly simple $Q = 0$ edge, then complexify its parametrization to obtain a complete two-dimensional image with an explicit inverse. Throughout, "seven-level family" means the specific states below, not every state with seven Dicke coefficients.

13.1 A real subspace with vanishing quadrupole

Use the three orthonormal spin-three vectors

$$\begin{aligned} e_0 &= \frac{\sqrt{2}}{3}(D_0 + D_6) + \frac{i\sqrt{5}}{3}D_3, \\ e_1 &= \frac{\sqrt{3}}{4}(D_0 - D_6) + \frac{\sqrt{5}}{4}(D_2 - D_4), \\ e_2 &= -\frac{i}{\sqrt{2}}(D_1 - D_5). \end{aligned} \tag{164}$$

Their disjoint or symmetric/antisymmetric supports immediately give $\langle e_a | e_b \rangle = \delta_{ab}$. In this basis the spin-flip bilinear form is

$$\Theta(\alpha e_0 + \beta e_1 + \gamma e_2) = \alpha^2 - \beta^2 - \gamma^2. \tag{165}$$

Moreover every *real* linear combination of these three vectors has zero unnormalized quadrupole:

$$\langle v | \tfrac{1}{2}\{J_i, J_j\} | v \rangle = 4\delta_{ij}\langle v | v \rangle \qquad (v \in \mathrm{span}_{\mathbb{R}}\{e_0, e_1, e_2\}).$$

One direct way to check the last statement is to use $J_+ D_k = \sqrt{k(7-k)}D_{k-1}$. The compressed quadrupole matrices have zero real part: their diagonal entries vanish and the remaining entries are purely imaginary and antisymmetric. Their expectation therefore vanishes on real coefficient vectors. The explicit moments in (172) below also verify all five quadrupole components. This real-span property must not be extended to arbitrary complex coefficients; that distinction is precisely what will produce the roof band.

For real r, u with $r^2 + u^2 = 1$, define

$$|\zeta_{r,u}\rangle = \frac{e_0 + r e_1 + u e_2}{\sqrt{2}}. \tag{166}$$

It has norm one, $\Theta = 0$ by (165), and $Q = 0$. Its seven Dicke coefficients are explicitly

$$\begin{aligned} (c_0, c_1, c_2, c_3) &= \left(\frac{1}{3} + \frac{\sqrt{6}r}{8}, -\frac{iu}{2}, \frac{\sqrt{10}r}{8}, \frac{i\sqrt{10}}{6} \right), \\ (c_4, c_5, c_6) &= \left(-\frac{\sqrt{10}r}{8}, \frac{iu}{2}, \frac{1}{3} - \frac{\sqrt{6}r}{8} \right). \end{aligned} \tag{167}$$

The dipole is

$$F_z = \frac{\sqrt{6}}{2} r, \qquad F_x + iF_y = i\left(\frac{5\sqrt{3}}{6} r - \frac{\sqrt{6}}{3} u\right),$$

so its sector map is

$$x = \frac{43r^2 + 8u^2 - 20\sqrt{2}ru}{18}, \qquad y = 5, \qquad z = 8 + 2x. \tag{168}$$

The real quadratic form for x has matrix

$$\begin{pmatrix} 43/18 & -5\sqrt{2}/9 \\ -5\sqrt{2}/9 & 4/9 \end{pmatrix}$$

and eigenvalues $1/6$ and $8/3$. Continuity on the unit circle, or its orthogonal eigenbasis, proves that the image is exactly the full segment

$$\boxed{\frac{1}{6} \le x \le \frac{8}{3}, \qquad y = 5, \qquad z = 8 + 2x.} \tag{169}$$

The lower endpoint has $(r, u) = (1/3, 2\sqrt{2}/3)$, and the upper endpoint has $(r, u) = (2\sqrt{2}/3, -1/3)$. They are the already labeled sector tuples of Ω_4 and $W(6)$, respectively; no duplicate endpoint node is required. Equation (169) proves attainment of the segment, not a classification of every unrestricted state satisfying $Q = \Theta = 0$.

13.2 Complexifying the null circle

The bilinear null condition is a conic. For $w \in \mathbb{C}$, its normalized parametrization in the same three-dimensional subspace is

$$\boxed{|\zeta_w\rangle = \frac{(1 + w^2)e_0 + (1 - w^2)e_1 + 2we_2}{\sqrt{2}(1 + |w|^2)}.} \tag{170}$$

The missing parameter value $w = \infty$ means $(e_0 - e_1)/\sqrt{2}$, up to an irrelevant global phase. Normalization and nullness follow from the elementary identities

$$|1 + w^2|^2 + |1 - w^2|^2 + 4|w|^2 = 2(1 + |w|^2)^2, \qquad (1 + w^2)^2 - (1 - w^2)^2 - (2w)^2 = 0.$$

Thus every member is one normalized pure state with $\Theta = 0$, without solving a purity constraint numerically.

Introduce the real unit-sphere coordinates

$$r = \frac{1 - |w|^2}{1 + |w|^2}, \qquad u = \frac{2\,\mathrm{Re}\, w}{1 + |w|^2}, \qquad v = \frac{2\,\mathrm{Im}\, w}{1 + |w|^2}, \qquad r^2 + u^2 + v^2 = 1. \tag{171}$$

For $r \neq -1$ the inverse is $w = (u + iv)/(1 + r)$; the remaining sphere point is $w = \infty$. Real w gives $v = 0$ and recovers the edge family (166).

For completeness, all nonzero first- and second-rank moments needed for the sector calculation are

$$\begin{gathered} F_x = 2v, \quad F_y = \frac{5\sqrt{3}r - 2\sqrt{6}u}{6}, \quad F_z = \frac{\sqrt{6}r}{2}, \\ Q_{xy} = \frac{5\sqrt{6}}{12} v(2\sqrt{2}r - u), \quad Q_{xz} = \frac{5\sqrt{3}}{12} v(2\sqrt{2}r - u), \\ Q_{xx} = Q_{yy} = Q_{zz} = Q_{yz} = 0. \end{gathered} \tag{172}$$

The unlisted entries follow by symmetry. These formulas follow by inserting (164) into the displayed Dicke ladder matrix elements. In particular

$$\|Q\|_F^2 = \frac{25}{8}v^2(2\sqrt{2}r - u)^2.$$

Using the same multipole-to-sector conversion as the rest of the document gives

$$\begin{aligned} x &= \frac{43r^2 + 8u^2 - 20\sqrt{2}ru + 48v^2}{18}, \\ y &= 5 + \frac{5}{6}v^2(2\sqrt{2}r - u)^2, \qquad z = 8 + 2x. \end{aligned} \tag{173}$$

13.3 The complete image and a constructive inverse

An orthogonal change of the two equatorial coordinates makes the image particularly transparent. Write

$$r = \frac{\ell + 2\sqrt{2}h}{3}, \qquad u = \frac{2\sqrt{2}\ell - h}{3}, \qquad \ell^2 + h^2 + v^2 = 1. \tag{174}$$

Then (173) reduces to

$$\boxed{x = \frac{1}{6} + \frac{5}{2}(h^2 + v^2), \qquad y = 5 + \frac{15}{2}h^2v^2, \qquad z = 8 + 2x.} \tag{175}$$

Theorem 5 (Complete image of the seven-level null conic)**.** *The family* (170) *has exactly the image*

$$\boxed{\frac{1}{6} \le x \le \frac{8}{3}, \qquad 5 \le y \le 5 + \frac{(6x-1)^2}{120}, \qquad z = 8 + 2x.} \tag{176}$$

Every point in this two-dimensional band is attained by an individual pure symmetric six-qubit state.

Proof. Put $H = h^2 + v^2 = (6x - 1)/15$. The sphere condition gives $0 \le H \le 1$. For fixed H, the two nonnegative numbers h^2, v^2 have sum H, so their product ranges over exactly $[0, H^2/4]$. Substituting into (175) proves both bounds and filling between them. This is an elementary compatibility argument, not a convexity assumption on sector tuples.

Here is an explicit state inverse. Given a point satisfying (176), define

$$\begin{aligned} H &= \frac{6x - 1}{15}, \qquad d = \sqrt{H^2 - \frac{8}{15}(y - 5)}, \\ h &= \sqrt{\frac{H + d}{2}}, \quad v = \sqrt{\frac{H - d}{2}}, \qquad \ell = \sqrt{1 - H}. \end{aligned} \tag{177}$$

All radicals are real by the stated bounds. Obtain r, u from (174), take $w = (u + iv)/(1 + r)$, and use (170). This chosen inverse has $r \ge 0$, so its denominator never vanishes. Direct substitution recovers the target (x, y, z) exactly, including both edge cases $y = 5$ and the upper parabola. □

The lower edge can be chosen with $v = 0$; it is precisely (169). The upper parabola has $h^2 = v^2 = H/2$. It is a sharp boundary of this *family image*, not a universal upper bound on arbitrary symmetric pure states. Both the band and its edge lie on the genuine supporting plane $q = 0$ of the full state set. Their edges need not be geometric ridges of the outer body.

13.4 Why this fills previously uncertified roof points

For comparison, (160) implies the exact restriction

$$\text{image}(\eta_{p,a,b}) \cap \{y = 5\} = \{(1/6, 5, 25/3), (4/9, 5, 80/9), (8/3, 5, 40/3)\}. \tag{178}$$

Indeed if $0 < p < 1$, the three nonnegative terms of $y - 5$ force $q_\chi = g_\chi = t_\chi = 0$. If $\cos a = 0$, then $q_\chi = 0$ requires $\nu = 1/4$ but $t_\chi \neq 0$. Otherwise $g_\chi = 0$ requires $\tan b = \sqrt{15}/3$, hence $\nu = -1/4$; then $q_\chi = 0$ forces $\eta = 1$, again contradicting $t_\chi = 0$. For $p = 0$, $q_\chi = t_\chi = 0$ gives either $\sin 2b = 0$ and $x = 1/6$, or $\sqrt{60}\eta + 6\sin 2a = 0$, whence $\eta^2 = 3/8$ and $x = 4/9$. Finally $p = 1$ is D_1 with $x = 8/3$. All three alternatives occur.

A particularly simple new witness is

$$|\zeta_1\rangle = \frac{7D_0 - 2i\sqrt{3}D_1 + \sqrt{15}D_2 + 2i\sqrt{10}D_3 - \sqrt{15}D_4 + 2i\sqrt{3}D_5 + D_6}{12}. \tag{179}$$

It is (166) at $(r, u) = (\sqrt{2/3}, 1/\sqrt{3})$, with $\Theta = 0$, $Q = 0$ and $F = (0, 1/\sqrt{2}, 1)$, so

$$(x, y, z) = (1, 5, 10). \tag{180}$$

Another useful edge representative is $|\zeta_*\rangle = |\zeta_{1,0}\rangle$, with $(x, y, z) = (43/18, 5, 115/9)$. Neither tuple belongs to the five-level image by (178). Its parameter domain is compact, so each omitted tuple has a relative open neighborhood disjoint from that image. The new band (176) has strictly positive height above both points. Its intersection with a sufficiently small such neighborhood therefore has positive area. Consequently the new construction adds a genuine roof patch, not merely isolated boundary points or a numerically interpolated chord.

The seven-level band is violet in the uniform views and turquoise in the family-resolved comparison, with the fill priority stated in Section 5. Its complete inverse permits direct pure-state checks at every displayed target. All existing known-state labels and leaders are retained. No extra outer inequality, ridge, or endpoint node is introduced by this change of attainment color.

14 The small-dipole null roof: a new attained patch and stricter compatibility

The condition $q = |\Theta|^2 = 0$ fixes $z = 8 + 2x$. For $0 \le x \le 1/6$, the previous necessary roof band was

$$L(x) \le y \le \min\{7 + 3x,\ 5 + (b + 3\sqrt{x})^2\}, \qquad L(x) = \begin{cases} 51/10 - 4x/5, & 0 \le x \le 3/32, \\ 5 + \frac{2}{5}(1 - 2\sqrt{3x/2})^2, & 3/32 \le x \le 1/6. \end{cases} \tag{181}$$

Here the lower branches follow from the full Gram family, the upper line from paired purity, and the upper curve from the earlier dipole-balancing estimate. A gray part of this band is not automatically unattainable: the following construction fills a genuine additional patch. Conversely, the linear lower rim is excluded at every interior point, and retaining the octupole improves the upper cap. These are unrestricted pure-state results, not an assumption of cyclic symmetry. The explicit construction in Section 14.4 also resolves attainment of the complete paired rim from $x = 2/27$ to $x = 1/6$.

14.1 An explicit four-level family and its sector image

Write $D_k = |D_6^{(k)}\rangle$ and set

$$\boxed{|\varphi_{t,k}\rangle = \frac{\sqrt{t}(1+k)}{2}D_0 + \sqrt{\frac{t(1-k^2)}{2}}D_3 - \sqrt{1-t}\,D_5 + \frac{\sqrt{t}(1-k)}{2}D_6, \quad 0 \le t \le 1,\ -1 \le k \le 1.} \tag{182}$$

The squared amplitudes on D_0, D_3, D_6 sum to t, so the state is normalized. Also $\Theta = 2c_0c_6 - c_3^2 = 0$ identically. For clarity, denote these four real amplitudes by a, d, e, h in that order. Direct angular-momentum action gives

$$F_z = t(2+3k) - 2, \quad F_+ = \sqrt{6}\,eh, \quad Q_{zz} = \frac{t}{2}(1+9k^2), \quad \langle J_+^2\rangle = \sqrt{120}\,de, \quad Q_{xz} + iQ_{yz} = -\frac{5\sqrt{6}}{2}eh.$$

Using (55), or contracting these Cartesian components, proves the exact sector map

$$\boxed{\begin{aligned} x(t,k) &= \frac{2}{3}[t(2+3k)-2]^2 + t(1-t)(1-k)^2, \\ y(t,k) &= 5 + \frac{t^2(1+9k^2)^2}{10} + t(1-t)(13-10k-3k^2), \\ z(t,k) &= 8 + 2x(t,k). \end{aligned}} \tag{183}$$

In particular $(t,k) = (5/8, 1/3)$ gives

$$\boxed{|\varphi_*\rangle = \frac{\sqrt{10}(2D_0 + 2D_3 + D_6) - 3\sqrt{6}D_5}{12}, \qquad (x,y,z) = \left(\frac{11}{96}, \frac{235}{32}, \frac{395}{48}\right).} \tag{184}$$

This point lies on the paired rim $y = 7 + 3x$.

This is a two-dimensional attained image, not a collection of samples. Indeed, its exact Jacobian satisfies

$$\det \frac{\partial(x,y)}{\partial(t,k)}\bigg|_{k=1/3} = \frac{64}{9} t(7t-5)(8t-5). \tag{185}$$

It is nonzero arbitrarily close to $t = 5/8$; the inverse function theorem therefore gives open attained roof patches accumulating at (184). For an exact regular example, $t = 313/500, k = 1/3$ has

$$(x,y,z) = \left(\frac{5129}{45000}, \frac{13766077}{1875000}, \frac{185129}{22500}\right), \qquad 7 + 3x - y = \frac{2}{78125} > 0.$$

It is important not to extend this particular family by interpolation. Its paired deficit is exactly

$$\Delta_0(t,k) := 7 + 3x - y = A(k)t^2 + B(k)t + 10,$$

where

$$\begin{aligned} A(k) &= \frac{-81k^4 + 102k^2 + 200k + 179}{10}, \qquad B(k) = 2(k+1)(3k-13), \\ B(k)^2 - 40A(k) &= 40(k^2-1)(3k-1)^2. \end{aligned}$$

Since $A(k) > 0$ for $-1 < k \le 1$, the only contacts with the paired rim in the parameter rectangle are $(5/8, 1/3)$ and $(1/2, 1)$. The latter is the C_5 state $\chi_{1/2} = (D_0 + D_5)/\sqrt{2}$, with tuple $(1/6, 15/2, 25/3)$; it is not the distinct Ω_4 tuple $(1/6, 5, 25/3)$. Near the first contact the upper boundary of *this family image* has the fold expansion

$$y_{\max,\varphi}(x) = 7 + 3x - \frac{1536}{115}\left(x - \frac{11}{96}\right)^2 + O\big((x - 11/96)^3\big). \tag{186}$$

For example this follows from $\nabla x = (-11/18, -5/8)$ and $\nabla^2\Delta_0 = \left(\begin{smallmatrix} 256/5 & 16 \\ 16 & 45/4 \end{smallmatrix}\right)$: the coefficient is $[2\nabla x^{\mathsf{T}}(\nabla^2\Delta_0)^{-1}\nabla x]^{-1}$. Equation (186) is not a universal inequality.

14.2 Why the interior of the old linear lower rim is excluded

Theorem 6 (Strict null-roof compatibility)**.** *Every normalized pure symmetric six-qubit state satisfies*

$$\boxed{q = 0, \quad 0 < x < \frac{3}{32} \quad \Longrightarrow \quad y > \frac{51}{10} - \frac{4}{5}x.} \tag{187}$$

The two endpoints are attained by ψ_0 and $\Omega_{3/8}$.

Proof. Use the harmonic-cubic tensors $H = R + iI$, their slices, and the positive Gram matrices $\mathcal{G}, \mathcal{H}$ defined explicitly in Section 18. Their exact difference is (255), with $r = 0$ here. Both matrices commute with $(u, v) \mapsto (\overline{v}, \overline{u})$, so the singular matrix $\mathcal{G}$ has a unit kernel vector $w = (u, \overline{u})/\sqrt{2}$. Write $u = a + ib$, with real a, b and $|a|^2 + |b|^2 = 1$, and define

$$A = aa^{\mathsf{T}} + bb^{\mathsf{T}}, \quad c = a \times b, \qquad Q_0 = \frac{3}{4}(A - I/3), \quad F_0 = -\frac{3}{4}c.$$

Testing the Gram difference on w and completing one quadratic norm gives the useful exact identity

$$\tau + 2f^2 - \frac{3}{8} = \frac{45}{4} w^\dagger \mathcal{H} w + \|Q - Q_0\|_F^2 + 2|F - F_0|^2. \tag{188}$$

To verify its constants, use $\|A - I/3\|_F^2 = 2/3 - 2|c|^2$, $\|Q_0\|_F^2 + 2|F_0|^2 = 3/8$, and $w^\dagger \mathcal{K} w = 2\,\mathrm{Tr}(QA) - 4F \cdot c = 1 + 15 w^\dagger \mathcal{H} w$. The line in (187) is exactly $\tau + 2f^2 = 3/8$. Equality would therefore force $Q = Q_0, F = F_0$ and $w \in \ker \mathcal{G} \cap \ker \mathcal{H}$.

The phase of u can be compensated by the global phase of the state. Choose it to make $a \perp b$, and then choose a collective Cartesian frame with $a = A_0 e_X, b = B_0 e_Y$, $A_0 \geq B_0 > 0$. Writing $t = B_0/A_0$, equality requires

$$0 < t < 1, \quad x = \frac{3t^2}{8(1+t^2)^2}, \qquad Q_{ZZ} = -\frac{1}{4}. \tag{189}$$

The common kernels are much more restrictive than the scalar norm: in the homogeneous harmonic polynomials representing R, I they give

$$\partial_X R + t \partial_Y I = 0, \qquad t L_Y R - L_X I = 0, \tag{190}$$

where $L_X = Y\partial_Z - Z\partial_Y$ and $L_Y = Z\partial_X - X\partial_Z$.

For $0 < t < 1$, their complete real solution is

$$R = -\frac{s}{t} U_Y - dt V_Y, \qquad I = s U_X + d V_X,$$

where s, d are real, $U_X = X(X^2 - 3Z^2)$, $V_X = X(Y^2 + Z^2 - 2X^2/3)$, and U_Y, V_Y exchange X, Y. For an elementary check of completeness, expand both polynomials in

$$X^3 - 3XZ^2,\ Y^3 - 3YZ^2,\ 3X^2Z - Z^3,\ X(Y^2 - Z^2),\ Y(X^2 - Z^2),\ Z(X^2 - Y^2),\ XYZ.$$

Equating coefficients in (190) leaves only the first and fourth coefficients of I free and gives exactly the displayed solution. Thus no rotational symmetry of the state has been assumed.

For a cubic monomial the tensor norm is $\|X^a Y^b Z^c\|^2 = a!b!c!/3!$. Hence $\|U_X\|^2 = 4$, $\|V_X\|^2 = 10/9$ and $\langle U_X, V_X \rangle = -5/3$, with the same values for Y. The real and imaginary supports are orthogonal. Nullness therefore gives

$$0 = \|R\|^2 - \|I\|^2 = (1 - t^2)\left(\frac{4s^2}{t^2} - \frac{10d^2}{9}\right), \qquad s = \epsilon \frac{\sqrt{10}}{6} t d, \quad \epsilon = \pm 1.$$

Here $d \neq 0$ for a nonzero state. Set $d = 1$ before normalizing and put $D = 2(1 + t^2) - \epsilon\sqrt{10}t > 0$. The squared tensor norm is $10D/9$. The normalized ZZ slice then is

$$C_{ZZ} = \frac{14(1 + t^2) - 8\epsilon\sqrt{10}t}{20D}.$$

Finally (253) gives

$$Q_{ZZ} + \frac{1}{4} = \frac{\epsilon\, 3\sqrt{10}t}{4D} \neq 0,$$

contradicting (189). This proves strictness. □

The exclusion removes more than a line. The normalized pure-state space is compact and its sector map is continuous. For every closed interval $I \subset (0, 3/32)$ there is consequently a $\delta_I > 0$ such that $q = 0, x \in I$ implies $y \geq 51/10 - 4x/5 + \delta_I$. Equivalently, each interior point of the old rim has a finite excluded relative neighborhood on the roof. No numerical value of this sharp gap is asserted or used to draw an invented replacement curve.

14.3 An octupole estimate that improves the implicit cap globally

Define the Cartesian octupole, with Sym averaging all six orderings,

$$T_{ijk} = \langle \mathrm{Sym}(J_i J_j J_k) \rangle - \frac{7}{3}(\delta_{ij} F_k + \delta_{ik} F_j + \delta_{jk} F_i). \tag{191}$$

It is symmetric and tracefree: $\sum_i \mathrm{Sym}(J_i J_i J_k) = (35/3) J_k$. Rotational covariance and the rank-three identity $a_3 = (1-q)/6$ proved in Section 17.2 give

$$\|T\|_F^2 = 540 a_3 = 90(1-q), \qquad \|T_{\boldsymbol{n}}\|_F^2 \le \frac{3}{5}\|T\|_F^2, \quad (T_{\boldsymbol{n}})_{ij} = \sum_k n_k T_{kij}. \tag{192}$$

The normalization follows by evaluating $|3,3\rangle$, for which $T_{ZZZ} = 6$, $T_{XXZ} = T_{YYZ} = -3$, including their permutations. For the contraction bound rotate $\boldsymbol{n}$ to Z; the squared contraction ratios on harmonic cubics of magnetic order $|m| = 0, 1, 2, 3$ are respectively $3/5, 8/15, 1/3, 0$. These follow by applying $\partial_Z/3$ to the seven cubics displayed above and using their tensor scalar products. In particular on $q = 0$, $\|T_{\boldsymbol{n}}\|_F \le \sqrt{54}$.

Theorem 7 (Octupole-strengthened global implicit cap)**.** *Every pure symmetric six-qubit state satisfies*

$$\boxed{\sqrt{y-5} \le b(1-r) + \sqrt{10}\, r + \frac{14}{5}\sqrt{x}\sqrt{\frac{1-r}{1+r}}, \qquad r = |\Theta|.} \tag{193}$$

Thus the coefficient 3 *in* (264) *can be replaced globally by* 14/5*, not merely on the null roof.*

Proof. First take $q = 0$ and $0 < f \le 1/20$. Along the exact dipole-balancing flow (271),

$$\dot f = -2v, \qquad v = 4 + Q_{\boldsymbol{nn}} - f^2 \ge V(f) := 4 - \sqrt{5(1+f^2)} - f^2.$$

Here $\boldsymbol{n} = F/f$ and $q = 0$ remains exact. The elementary operator identity

$$\frac{1}{2}\{J_{\boldsymbol{n}}, \mathrm{Sym}(J_i J_j)\} = \mathrm{Sym}(J_{\boldsymbol{n}} J_i J_j) + \frac{\delta_{ij}}{6} J_{\boldsymbol{n}} - \frac{n_i J_j + n_j J_i}{12}$$

and (191) give the instantaneous Cartesian equation

$$\dot Q = -2T_{\boldsymbol{n}} - 9f(\boldsymbol{n}\boldsymbol{n}^{\mathsf{T}} - I/3) + 2fQ. \tag{194}$$

It remains valid when the dipole axis varies; no derivative of the coordinate frame has been introduced. Using (192), paired purity and $\dot f = -2v$ gives

$$\Big\|\frac{dQ}{df}\Big\|_F \le \frac{H(f)}{V(f)}, \qquad H(f) = 3\sqrt{6} + \frac{3\sqrt{6}}{2} f + f\sqrt{\frac{15}{2}(1+f^2)}. \tag{195}$$

The variance has a positive uniform lower bound on this interval, so the flow reaches a pure $F = \Theta = 0$ limit, as justified in Section 19. Integrating from that limit and applying Lemma 1 yields the sharper integral estimate

$$\sqrt{y-5} \le b + \frac{2}{\sqrt{15}} \int_0^{\sqrt{3x/2}} \frac{H(s)}{V(s)}\, ds. \tag{196}$$

For a short closed bound, $0 \le f \le 1/20$ implies

$$H(f) < \frac{6137}{800} < \frac{77}{10}, \qquad V(f) > \frac{703}{400} > \frac{7}{4}, \qquad H(f)/V(f) < \frac{22}{5}.$$

These follow from $\sqrt{6} < 49/20$, $\sqrt{(15/2)(401/400)} < 11/4$ and $\sqrt{401/80} < 56/25$. Since $f = \sqrt{3x/2}$,

$$\sqrt{y-5} \le b + \frac{22\sqrt{10}}{25}\sqrt{x} \le b + \frac{14}{5}\sqrt{x}.$$

This proves the new null bound for $x \le 1/600$.

The same bound is automatic for every larger null dipole, a useful globalization. Paired purity gives $\sqrt{y-5} \le \sqrt{2(1+f^2)}$. For $f \ge 1/20$, use $b > 4/3$ and $(14/5)\sqrt{2/3} > 2$. The increasing difference

$$(4/3+2f)^2 - 2(1+f^2) = -\frac{2}{9} + \frac{16}{3}f + 2f^2 \ge \frac{89}{1800} > 0$$

shows $\sqrt{2(1+f^2)} < b + (14/5)\sqrt{2/3}\, f$. Therefore $\sqrt{y-5} \le b + (14/5)\sqrt{x}$ holds globally on $\Theta = 0$.

Finally use the exact time-reversal decomposition (275): $F_\psi = \sqrt{1-r^2}F_\phi$ and $Q_\psi = (1-r)Q_\phi + rQ_u$, where $\Theta_\phi = 0$ and $\|Q_u\|_F \le \sqrt{75/2}$. Applying the global null bound to ϕ proves (193) for $r < 1$. At $r = 1$, $x = 0$ and it is simply $y \le 15$. □

The coefficient is a convenient rigorous estimate, not a sharpness claim for positive x. Already the rational roof point

$$(x, y, z) = \left(\frac{1}{10000}, \frac{3441}{500}, \frac{40001}{5000}\right)$$

satisfies the old conditions and violates the new cap: $1.3428 < b < 1.3429$ gives $5 + (1.3428 + .030)^2 > 6.882$ but $5 + (1.3429 + .028)^2 < 6.882$. Continuity of the global inequality also excludes a finite-volume part of the previous outer region just below this roof.

14.4 An explicit family attaining the whole paired rim

The four-level family (182) has only isolated paired-rim contacts. This restriction belongs to that family, not to the pure-state image. Allowing all seven Dicke coefficients gives an elementary global construction.

Theorem 8 (The full small-dipole paired rim is attained)**.** *Every point*

$$\boxed{\frac{2}{27} \le x \le \frac{1}{6}, \qquad y = 7 + 3x, \qquad z = 8 + 2x} \tag{197}$$

is attained by a pure symmetric six-qubit state.

Idea of the construction. In a real spin-flip eigenbasis, equal squared norms in the positive and negative eigenspaces enforce $\Theta = 0$. We choose their relative directions so that the dipole and quadrupole contractions satisfy the paired-purity equality. A single real parameter then runs monotonically over the entire stated interval. Thus the proof uses actual pure states, not convex combinations of sector tuples.

Proof. For $k = 0, 1, 2$, introduce the orthonormal vectors

$$E_k^{\pm} = \frac{D_k \pm (-1)^k D_{6-k}}{\sqrt{2}}.$$

Together with D_3, they form a real orthonormal basis. The spin-flip bilinear form is $+1$ on the three E_k^+ and -1 on the three E_k^- and D_3. In particular, for real coefficients,

$$\Theta\left(\sum_{k=0}^{2} \alpha_k E_k^+ + \sum_{k=0}^{2} \beta_k E_k^- + \beta_3 D_3\right) = \sum_{k=0}^{2} \alpha_k^2 - \sum_{k=0}^{2} \beta_k^2 - \beta_3^2.$$

Let $0 \le h \le 1/3$, and first choose the negative root

$$s = -\frac{1+\sqrt{1+4h}}{2}, \qquad W = 1-h, \qquad R = 1+h, \qquad H = \sqrt{(s^2+3s+3)(1-3h)}. \tag{198}$$

Here $s(s+1)=h$, $W\geq 2/3$, $R\geq 1$, and $s^2+3s+3=(s+3/2)^2+3/4>0$, so all expressions are real and the denominator below never vanishes. Define

$$\begin{aligned}
\alpha_0 &= 3\sqrt{RW}(2-s-2s^2), & \beta_0 &= \sqrt{3}(s+2)(1-2s)H,\\
\alpha_1 &= -2\sqrt{3W}(1-s)H, & \beta_1 &= 6(s+1)\sqrt{R}(1-2s-s^2),\\
\alpha_2 &= -s\sqrt{15RW}, & \beta_2 &= 3\sqrt{5}\,sH,\\
& & \beta_3 &= 2\sqrt{15}\,s(s+1)\sqrt{R},
\end{aligned} \tag{199}$$

and the pure symmetric state

$$\boxed{|\Psi_h\rangle = \frac{\sum_{k=0}^{2}(\alpha_k E_k^{+}+\beta_k E_k^{-})+\beta_3 D_3}{12W^{3/2}}.} \tag{200}$$

The relation $s(s+1)=h$ immediately gives the two scalar identities

$$\sum_{k=0}^{2}\alpha_k^2 = \sum_{k=0}^{2}\beta_k^2+\beta_3^2 = 72W^3. \tag{201}$$

Consequently $\langle\Psi_h|\Psi_h\rangle=1$ and $\Theta(\Psi_h)=0$ exactly.

It remains to compute the first two spin multipoles. We use the manuscript's Cartesian notation $F_i=\langle J_i\rangle$ and $Q_{ij}=\langle\{J_i,J_j\}/2\rangle-4\delta_{ij}$. All entries follow directly from

$$J_zD_k=(3-k)D_k, \qquad J_+D_k=\sqrt{k(7-k)}D_{k-1}.$$

For clarity, the nonzero components for (200) are displayed in full:

$$\begin{aligned}
F_x &= \frac{s(1+5s+3s^2)}{\sqrt{6W}}, & F_z &= \frac{\sqrt{R}\,H}{2\sqrt{3W}},\\
Q_{xx} &= \frac{5(s^4+2s^3+5s^2+4s-1)}{4W}, & Q_{yy} &= -\frac{5}{4}(s^2+3s+1),\\
Q_{zz} &= -\frac{5(s+1)(s^3+2s^2+2s-1)}{2W}, & Q_{xz} &= Q_{zx} = \frac{5s\sqrt{R}\,H}{2\sqrt{2}\,W}.
\end{aligned} \tag{202}$$

The remaining entries F_y, Q_{xy}, Q_{yz} vanish because the Dicke coefficients are real. Contracting the displayed components, again using only $s^2+s=h$, gives

$$|F|^2=\frac{1-5h^2}{4}, \qquad \|Q\|_F^2=\frac{75}{8}(1-h^2).$$

The sector conversions (55) and (58) therefore yield

$$\boxed{x=\frac{1-5h^2}{6}, \qquad y=\frac{15}{2}-\frac{5}{2}h^2=7+3x, \qquad z=\frac{25}{3}-\frac{5}{3}h^2=8+2x.} \tag{203}$$

As h runs from 0 to $1/3$, x decreases continuously from $1/6$ to $2/27$. Conversely, every target in (197) is obtained by the explicit inverse

$$\boxed{h=\sqrt{\frac{1-6x}{5}}.} \tag{204}$$

This proves coverage of the complete closed rim. □

Connection with the earlier local seed. The negative-root branch in (198) extends the four-level seed (184), up to one fixed collective rotation. Specifically, let $\mathcal{Z} = Z^{\otimes 6}$ and choose $\cos\theta = \sqrt{5}/3$, $\sin\theta = -2/3$. Direct substitution gives

$$|\Psi_{1/4}\rangle = e^{-i\theta J_y}\mathcal{Z}|\varphi_*\rangle.$$

Here $\mathcal{Z}$ is a collective z rotation by π, up to an irrelevant global phase. The equally valid family $|\widehat{\Psi}_h\rangle = \mathcal{Z}e^{i\theta J_y}|\Psi_h\rangle$ therefore passes through φ_* itself and has the same sector map (203). In particular the previously local paired-rim construction now has an explicit global continuation.

A second branch and familiar endpoint states. All identities (201)–(203) also hold with the positive root $s = (-1 + \sqrt{1+4h})/2$ in the same coefficients. At $h = 0$ this branch is exactly $\chi_{1/2} = (D_0 + D_5)/\sqrt{2}$, the C_5 endpoint at $(1/6, 15/2, 25/3)$. At $h = 1/3$ it is $e^{-i\pi J_y/2}|\eta_6\rangle$, where

$$\boxed{\begin{aligned} |\eta_6\rangle &= \frac{5D_0 + \sqrt{35}D_2 + \sqrt{15}D_4 - \sqrt{21}D_6}{\sqrt{96}}, \\ (x,y,z) &= \left(\frac{2}{27}, \frac{65}{9}, \frac{220}{27}\right). \end{aligned}} \tag{205}$$

Thus the global construction also connects the known C_5 and C_2 endpoint tuples. The two root branches need not represent the same rotation orbit; equality of their sector maps is all that is needed for attainment. At the lower endpoint, however, Theorem 9 below proves that there is only one orbit.

14.5 Sharpness of the paired-rim endpoint and its unique state orbit

The preceding construction proves attainment. The converse requires compatibility of the *same pure state*'s spin multipoles, not merely its marginal purities. In particular, a generic complex state cannot be assumed collectively rotatable to real Dicke coefficients. The following argument treats all complex states directly.

Theorem 9 (The exact null paired rim and its sharp C_2 endpoint)**.** *For a pure symmetric six-qubit state,*

$$q = 0, \quad y = 7 + 3x \quad \Longrightarrow \quad \boxed{\frac{2}{27} \le x \le \frac{1}{6}.} \tag{206}$$

Every point of this closed interval is attained, with $z = 8 + 2x$. At $x = 2/27$, every realizing state is exactly

$$\boxed{|\psi\rangle = e^{i\phi}U^{\otimes 6}|\eta_6\rangle, \qquad U \in \mathrm{SU}(2),} \tag{207}$$

where η_6 is (205)*. Its projective rotational stabilizer in* SO(3) *is C_2.*

Proof sketch. Paired-purity equality makes an auxiliary spin-two reduction flat on its four-dimensional support. Its missing one-dimensional kernel is an actual pure spin-two state. A transition-quadrupole identity imposes one extra complex cubic condition on this kernel. Two elementary 3×3 trace identities then fix the physical contractions $F^{\mathsf{T}}Q^2F$ and $\mathrm{Tr}\,Q^3$. A short positive-norm identity forces $|F|^2 \ge 1/9$, hence $x \ge 2/27$. At equality it aligns F with a principal quadrupole axis, and a fixed linear spin equation has the unique complex solution η_6.

Proof. **Step 1: the flat auxiliary kernel and physical stationarity.** Set

$$u = |F|^2, \qquad \tau = \mathrm{Tr}\,Q^2, \qquad \alpha = F^{\mathsf{T}}QF, \qquad \beta = F^{\mathsf{T}}Q^2F, \qquad \omega = \mathrm{Tr}\,Q^3. \tag{208}$$

The sector conversion gives $x = 2u/3$ and, on this rim,

$$\tau = \frac{15}{2}(1+u). \tag{209}$$

Write $|\psi\rangle = \sum_{k=0}^{6} c_k D_k$. Its isometric embedding into the antisymmetric part of two auxiliary spin-two representations has coefficient matrix

$$A_{ij} = \langle 2, 2-i; 2, 2-j | 3, 4-i-j \rangle c_{i+j-1}, \qquad i, j = 0, \ldots, 4,$$

where out-of-range amplitudes are zero. For $R = AA^\dagger$, finite Clebsch–Gordan contractions give

$$\begin{aligned} \operatorname{Tr} R = 1, \qquad P := \operatorname{Tr} R^2 = \frac{25 + 5x - y - z}{40}, \\ \operatorname{Tr}(R \mathsf{T}_2 R \mathsf{T}_2^{-1}) = \frac{100 + 10x - 4y - 5z}{160}. \end{aligned} \tag{210}$$

The nonzero singular values of a complex antisymmetric 5×5 matrix occur in pairs. Thus $P \geq 1/4$, with equality precisely when

$$R = \frac{I_5 - |n\rangle\langle n|}{4}, \qquad \|n\| = 1. \tag{211}$$

Since $q = 0$ gives $z = 8 + 2x$, the stated rim has $P = 1/4$. The spin-two kernel n is therefore necessary, not an ansatz. The rank-one and rank-two intertwining factors give its multipoles

$$F_n = -2F, \qquad Q_n = -\frac{2}{5} Q. \tag{212}$$

The same global lower bound on P makes every rim state stationary on the physical unit sphere. Indeed $P = (17 + 3x - y + 8q)/40$, and the first derivative of $q = |\Theta|^2$ vanishes at $\Theta = 0$. Differentiating x and y in terms of F, Q gives the exact linear equation for this state:

$$H|\psi\rangle = \frac{15}{2}|\psi\rangle, \qquad H = \sum_{ij} Q_{ij} \frac{J_i J_j + J_j J_i}{2} - \frac{15}{2} \sum_i F_i J_i. \tag{213}$$

The eigenvalue follows from $\langle H \rangle = \tau - 15u/2 = 15/2$.

Step 2: the extra complex cubic compatibility. Represent n by a complex symmetric traceless Cartesian 3×3 matrix M, normalized by $\operatorname{Tr}(M^\dagger M) = 1$. For an explicit convention let e_x, e_y, e_z be the real coordinate vectors and put $e_\pm = e_x \pm i e_y$. The orthonormal spherical basis is

$$\begin{aligned} \mathcal{T}_2 &= e_+ e_+^{\mathsf{T}}/2, & \mathcal{T}_1 &= -(e_+ e_z^{\mathsf{T}} + e_z e_+^{\mathsf{T}})/2, \\ \mathcal{T}_0 &= (2 e_z e_z^{\mathsf{T}} - e_x e_x^{\mathsf{T}} - e_y e_y^{\mathsf{T}})/\sqrt{6}, & \mathcal{T}_{-1} &= (e_- e_z^{\mathsf{T}} + e_z e_-^{\mathsf{T}})/2, \\ \mathcal{T}_{-2} &= e_- e_-^{\mathsf{T}}/2, & M &= \sum_{k=0}^{4} n_k \mathcal{T}_{2-k}. \end{aligned}$$

Time reversal is $M \mapsto \overline{M}$. Define

$$\begin{aligned} M = X + iY, \qquad S = X^2 + Y^2, \qquad \vartheta = \operatorname{Tr} M^2, \\ \kappa = \operatorname{Tr}(MS) = \operatorname{Tr}(M^2 M^\dagger), \qquad \zeta = \operatorname{Tr} M^3, \end{aligned} \tag{214}$$

where X, Y are real symmetric traceless. The equality defining κ follows by trace cyclicity. The Cartesian spin-two formula $Q_n = 2I_3 - 6S$ and (212) yield

$$Q = 15S - 5I_3. \tag{215}$$

If c is the axial vector of $[X, Y]$, its dipole formula also gives $u = 4|c|^2$, $\alpha = 4c^{\mathsf{T}} Q c$ and $\beta = 4c^{\mathsf{T}} Q^2 c$; the axial-vector sign convention does not affect these scalars. Equation (211) gives $\operatorname{Tr}(R \mathsf{T}_2 R \mathsf{T}_2^{-1}) = (3 + |\vartheta|^2)/16$. Substitution into (210) proves

$$\mathcal{E} := 4u + 5|\vartheta|^2 - 1 = 0. \tag{216}$$

The additional compatibility, beyond these purities, is

$$\boxed{\kappa = 0.} \tag{217}$$

To prove it, use the transition quadrupole

$$W_{ij} = \langle \mathsf{T}_3\psi|(J_iJ_j + J_jJ_i)/2 - 4\delta_{ij}I|\psi\rangle.$$

The odd transition dipole $\langle \mathsf{T}_3\psi|J_i|\psi\rangle$ vanishes identically: its bilinear matrix B_7J_i is antisymmetric, where $(B_7)_{k,6-k} = (-1)^k$. Multiplying (213) by $\langle \mathsf{T}_3\psi|$ therefore gives $Q : W = 0$.

For clarity, the precise link between W and the kernel is elementary. Let $p_i = (-1)^i \operatorname{Pf}(A_{\widehat{i}})$, where $A_{\widehat{i}}$ deletes row and column i. Then $Ap = 0$; flatness gives $\|p\| = 1/4$, so choose $n = 4\overline{p}$. The quadratic transition identity is

$$W = \frac{5\sqrt{6}}{2}\overline{M}. \tag{218}$$

Both sides, after the displayed Pfaffian substitution, are the unique spin-two covariant in $\mathrm{Sym}^2(V_3)$. Its coefficient is fixed by $\psi = (D_0 + D_6)/\sqrt{2}$: $p_2 = 1/4$, $M = \operatorname{diag}(-1,-1,2)/\sqrt{6}$ and $W = \operatorname{diag}(-5/2,-5/2,5)$. Alternatively, expand the five 4×4 Pfaffians directly in the given embedding. Consequently $Q : \overline{M} = 0$. Since Q is real and M is traceless, (215) gives $0 = \operatorname{Tr}(MQ) = 15\kappa$, proving (217) without any real-state reduction.

Step 3: two small Cartesian trace identities. We give the algebraic identities explicitly so that no separate certificate is needed. For *arbitrary* real symmetric traceless X, Y, put $\nu = \operatorname{Tr} S$, replace Q by $15S - 5\nu I_3$, and use the same definitions of u, α, β, ω above. With $\mathcal{E} = 4u + 5|\vartheta|^2 - \nu^2$, one has

$$\omega - \frac{75}{4}(\alpha + 4\nu u - \nu^3) = -\frac{75}{4}\nu\mathcal{E} + 1125|\kappa|^2, \tag{219}$$

$$\beta - 4u^2 - \frac{37}{4}\nu^2 u + \nu^4 = \mathcal{E}\left(-\nu^2 + \frac{21}{4}u + \frac{5}{4}|\vartheta|^2\right) - 75\nu|\kappa|^2 + 75\Re(\vartheta\overline{\zeta}\kappa). \tag{220}$$

These are ordinary 3×3 trace identities. They follow by polarizing the traceless Cayley–Hamilton equation

$$Z^3 - \tfrac{1}{2}(\operatorname{Tr} Z^2)Z - \tfrac{1}{3}(\operatorname{Tr} Z^3)I_3 = 0$$

at $Z = X + tY$ and using the axial vector of $[X, Y]$. For example, two intermediate contractions are

$$3\operatorname{Tr} S^2 - 2|c|^2 = \nu^2 + \tfrac{1}{2}|\vartheta|^2,$$
$$6\operatorname{Tr} S^3 - 5\nu\operatorname{Tr} S^2 + \nu^3 - 2c^{\mathsf{T}}Sc = 2|\kappa|^2.$$

They give (219); the next polarized contraction with two commutators gives (220). Each identity can also be verified by direct substitution of arbitrary symmetric traceless X, Y and trace cyclicity. Their statements require neither an optimizing ansatz nor polynomial elimination.

For the physical kernel, $\nu = 1$ and (216)–(217) reduce them to the pure-state compatibility identities

$$\boxed{\beta = 4u^2 + \frac{37}{4}u - 1, \qquad \omega = \frac{75}{4}(\alpha + 4u - 1).} \tag{221}$$

Step 4: a short positive-norm identity. Every real symmetric traceless 3×3 matrix obeys $Q^3 = (\tau/2)Q + (\omega/3)I_3$. Substituting (209) and (221) gives

$$\begin{aligned}\frac{5}{6}&(9u-1)(82u^2 - 26u + 17)\\ &= \|(Q^2 + \tfrac{5}{2}Q - 25uI_3)F\|^2 + \frac{25}{6}\|(Q + (6u - \tfrac{3}{2})I_3)F\|^2\\ &\quad + \frac{25}{4}\|F \times (QF)\|^2.\end{aligned} \tag{222}$$

This is a short analytical identity in the physical dipole and quadrupole, not the earlier large amplitude-level certificate. For a hand check, expand the three Euclidean norms and use

$$F^{\mathsf{T}}Q^3F = \tfrac{\tau}{2}\alpha + \tfrac{\omega}{3}u, \quad F^{\mathsf{T}}Q^4F = \tfrac{\tau}{2}\beta + \tfrac{\omega}{3}\alpha, \quad |F \times QF|^2 = u\beta - \alpha^2.$$

All α-dependent terms cancel. The right side of (222) is nonnegative, while

$$82u^2 - 26u + 17 = 82\left(u - \frac{13}{82}\right)^2 + \frac{1225}{82} > 0.$$

Thus $u \geq 1/9$ and $x = 2u/3 \geq 2/27$. Equation (216) also gives $u \leq 1/4$, hence $x \leq 1/6$. The preceding explicit family proves attainment throughout this interval.

Step 5: complete equality classification. At $x = 2/27$, $u = 1/9$, and the second norm in (222) forces

$$QF = \frac{5}{6}F. \tag{223}$$

In particular $F \neq 0$. A collective rotation puts $F = (0, 0, 1/3)$; a further rotation about z diagonalizes the transverse quadrupole. Since $\operatorname{Tr} Q = 0$ and $\operatorname{Tr} Q^2 = 25/3$, we may order its two transverse eigenvalues as

$$Q = \operatorname{diag}\left(\frac{-5 + 5\sqrt{21}}{12}, \frac{-5 - 5\sqrt{21}}{12}, \frac{5}{6}\right). \tag{224}$$

Interchanging the first two axes is another collective rotation. In this frame (213), multiplied by 4/5, becomes the fixed linear equation

$$\left[J_z^2 - 2J_z - 10I + \frac{\sqrt{21}}{6}(J_+^2 + J_-^2)\right]|\psi\rangle = 0. \tag{225}$$

Its even-Dicke block on D_0, D_2, D_4, D_6 is tridiagonal with nonzero adjacent entries; its kernel therefore has dimension at most one. Direct substitution gives η_6. Its odd-Dicke block is

$$\begin{pmatrix} -10 & \sqrt{70} & 0 \\ \sqrt{70} & -10 & \sqrt{70} \\ 0 & \sqrt{70} & -2 \end{pmatrix}, \qquad \det = 640 \neq 0.$$

The full *complex* kernel is consequently $\mathbb{C}|\eta_6\rangle$, proving (207) and excluding additional chiral or time-reversed endpoint orbits.

Finally direct evaluation gives $\|\eta_6\| = 1$, $\Theta(\eta_6) = 0$, $F = (0, 0, 1/3)$ and (224); hence its sectors are $(2/27, 65/9, 220/27)$. A rotation preserving this state projectively must preserve its nonzero F and three distinct quadrupole axes. Only the identity and a π rotation about F do so. The latter sends η_6 to $-\eta_6$, so the projective stabilizer in SO(3) is exactly C_2. □

Geometric scope. The sharp threshold excludes the null paired line for $x < 2/27$; it does not exclude states with small S_1 away from that line. Accordingly the three-dimensional outer body's geometry is retained: the endpoint η_6 is marked in violet and the nonattained part of this rim is gray, without inventing a neighboring finite-width exclusion or a new surface facet.

14.6 A locally filled strip adjoining the paired rim

The global theorem concerns a one-dimensional rim, not a two-dimensional roof area. A transverse argument gives the following additional local area statement.

Theorem 10 (A whole one-sided attained roof strip)**.** *Some $\epsilon, \delta > 0$ exist such that every tuple*

$$\left|x - \frac{11}{96}\right| < \epsilon, \qquad 7 + 3x - \delta \leq y \leq 7 + 3x, \qquad z = 8 + 2x \tag{226}$$

is attained by a pure symmetric six-qubit state. Neither constant is quantified here. In particular the theorem does not justify coloring an arbitrary finite-width strip below the rim. Attainment of the rim itself throughout $2/27 \leq x \leq 1/6$ is independently established by Theorem 8.

Proof. Use the rotated real rim family $\widehat{\Psi}_h$ just constructed, with $\widehat{\Psi}_{1/4} = \varphi_*$ and $dx/dh = -5h/3 = -5/12$ at that point. Let $B_{i,6-i} = (-1)^i$ be the real spin-flip matrix, and work on the smooth real manifold $\mathcal{M} = \{p \in \mathbb{R}^7 : p^{\mathsf{T}}p = 1, p^{\mathsf{T}}Bp = 0\}$. The explicit rim curve has nonzero x derivative, so the three gradients of normalization, overlap and x are independent nearby. At its seed an explicit tangent preserving all three constraints is

$$w = \frac{D_0}{9\sqrt{10}} + \frac{7\sqrt{10}}{36}D_3 + \frac{\sqrt{6}}{3}D_5 + \frac{17}{9\sqrt{10}}D_6.$$

It is the tangent $(dt, dk) = (1, -44/45)$ of (182). The displayed Hessian of Δ_0 gives $d^2\Delta_0(w, w) = 92/3 > 0$. Let $G(p)$ have the three constraint gradients as rows and define

$$X(p) = \Big[I - G(p)^{\mathsf{T}}(G(p)G(p)^{\mathsf{T}})^{-1}G(p)\Big]\, w.$$

Its local analytical flow starting from each rim state preserves normalization, $\Theta = 0$ and x exactly. The paired deficit is nonnegative on $\mathcal{M}$ and vanishes on the rim, so its first derivative there is zero. Its second derivative along this flow is $92/3$ at the seed and remains uniformly positive on a sufficiently small closed parameter rectangle. The deficit therefore increases continuously from zero through a uniform positive interval at each allowed x. The intermediate value theorem fills all its values. Finally the nonzero derivative dx/dh makes x a local coordinate, proving (226). □

What is and is not settled. The explicit four-level image, the full paired rim $2/27 \le x \le 1/6$, the strict lower-rim exclusion, and the global octupole cap are proved analytical statements. The local strip theorem does not quantify a plotting width below the rim. Unrestricted Theorem 22 now determines the entire complex-C_2 null image for $0 \le f \le b_c$, including a monotone state inverse. The full physical Hessian has the strict local signs of Theorem 23. Neither fact by itself proves unrestricted global sharpness between the finite certified strips. The enlarged strip in Theorem 24 is globally complete; the remaining gaps concern possible additional states outside the proved inner band, not a failure to construct that band. In contrast, Theorem 9 settles the paired rim completely: it is attained exactly for $2/27 \le x \le 1/6$, with the unique C_2 endpoint orbit η_6 at $x = 2/27$. Remaining gray points are unresolved unless a stated exclusion applies.

15 A stronger lower boundary from a three-vector Gram matrix

The essential extra information is that the moments belong to one pure vector, not merely that the reduced density matrices have certain purities. Choose its global phase so that $\langle\psi|\mathsf{T}\psi\rangle = r \ge 0$. If $f > 0$, choose the collective axis $\boldsymbol{n} = F/f$ and put $J_{\boldsymbol{n}} = \boldsymbol{n}\cdot\mathbf{J}$. In a time-reversal-real basis, $J_{\boldsymbol{n}}$ is an imaginary antisymmetric matrix. Consequently $\langle\mathsf{T}\psi|J_{\boldsymbol{n}}\psi\rangle = 0$. The Gram matrix of $\psi, \mathsf{T}\psi, J_{\boldsymbol{n}}\psi$ is therefore

$$\begin{pmatrix} 1 & r & f \\ r & 1 & 0 \\ f & 0 & 4 + Q_{\boldsymbol{nn}} \end{pmatrix} \succeq 0. \tag{227}$$

Its determinant is nonnegative, giving

$$f^2 \le (1-q)(4 + Q_{\boldsymbol{nn}}) \le (1-q)\left(4 + \sqrt{\frac{2\tau}{3}}\right).$$

For the last step, a real traceless 3×3 quadrupole satisfies $Q_{\boldsymbol{nn}} \le \sqrt{2/3}\,\|Q\|_F$: the two transverse diagonal entries have sum $-Q_{\boldsymbol{nn}}$ and squared sum at least $Q_{\boldsymbol{nn}}^2/2$. The moment conversions now give

$$\boxed{12x \le (z-2x)\left(4 + \sqrt{\frac{5}{2}(y-5)}\right),} \tag{228}$$

which is precisely (6a). For $f = 0$, the same conclusion follows from $q \le 1$, so no axis choice is needed.

Attained edges and equality. Two continuous pure-state families attain the new lower boundary:

$$\psi_p^{(2)} = \sqrt{p}\, D_1 + e^{i\phi}\sqrt{1-p}\, D_5, \qquad (x,y,z) = \left(\frac{8}{3}(2p-1)^2, 5, \frac{40}{3}(2p-1)^2\right), \tag{229}$$

$$\psi_p^{(3)} = \sqrt{p}\, D_0 + e^{i\phi}\sqrt{1-p}\, D_6, \qquad (x,y,z) = \left(6(2p-1)^2, 15, 20(2p-1)^2\right), \tag{230}$$

where $0 \le p \le 1$. They contain the octahedral/W edge and the balanced/biased GHZ edge, respectively; ϕ is removable by a collective rotation about the chosen axis. For completeness these are all equality possibilities when $f > 0$. Equality in the Gram determinant makes $\mathrm{span}(\psi, \mathsf{T}\psi)$ invariant under $J_{\boldsymbol{n}}$, whose eigenvalues on that plane are $\pm h$. Equality in the quadrupole estimate also gives $h^2 = 4 + \sqrt{2\tau/3} \ge 4$. The spin-three spectrum forces $h = 2$ or 3, yielding exactly the two displayed families, up to rotations and global phase. At $f = 0$, equality means $z = 0$, equivalently that ψ is time-reversal real up to phase.

A strict cut of the previous outer region. The exact rational tuple

$$(x,y,z) = \left(1, \frac{33}{5}, \frac{7}{2}\right) \tag{231}$$

satisfies the previous plotted inequalities, including the stronger implicit red cap, but (6a) requires $z \ge 4$. Indeed $\sqrt{5(y-5)/2} = 2$. For the red cap, already $y - 5 = 8/5 < b^2$ makes its inequality automatic. The witness satisfies the previous conditions strictly and violates the new bound strictly, so continuity gives a finite-volume improvement, not just one point.

16 Origin and proof of the quantitative curved cap

The cap comes from the proved even-correlation theorem in Ref. [4], not from an optimization over six-qubit states. In fact it holds for every mixed symmetric state with at least four qubits, after the appropriate binomial rescaling.

Let $C^{(k)}$ be the common k-body Pauli correlation tensor, so $S_k = \binom{6}{k}\|C^{(k)}\|_F^2$. Set

$$u = \|C^{(2)}\|_F^2 = \frac{y}{15}, \qquad v = \|C^{(4)}\|_F^2 = \frac{S_4}{15}, \qquad t = \sqrt{\frac{3u-1}{2}} = \sqrt{\frac{y-5}{10}} \in [0,1]. \tag{232}$$

Take the physical four-qubit marginal and average it with its time reverse. This leaves the even tensors unchanged and eliminates the odd tensors. The resulting spin-two density matrix ρ_{e} has purity

$$P = \mathrm{Tr}\,\rho_{\mathrm{e}}^2 = \frac{1 + 6u + v}{16}. \tag{233}$$

The real symmetric tensor $C^{(2)}$ has trace one. With $\beta = \sqrt{2/3}$ (distinct from the constant b in (3)), define

$$B = \frac{C^{(2)} - I_3/3}{\beta t}, \qquad L_B = -\frac{1}{3}\sum_{ij} B_{ij}\frac{J_iJ_j + J_jJ_i}{2} \quad \text{on } V_2.$$

At $t = 0$, choose any real symmetric traceless B of unit Frobenius norm. For all t, $\mathrm{Tr}\,B = 0$, $\|B\|_F = 1$, and $\mathrm{Tr}(\rho_{\mathrm{e}}L_B) = -\beta t$. If b_1, b_2, b_3 are the eigenvalues of B, direct spin-two diagonalization gives

$$\mathrm{spec}\, L_B = \{-b_1, -b_2, -b_3, -\beta, \beta\}. \tag{234}$$

Equivalently its characteristic polynomial is $\prod_i(\xi + b_i)(\xi^2 - 2/3)$.

Dephase ρ_{e} in this eigenbasis. Its expectation of L_B is unchanged and its purity cannot increase. Let $p \ge 0$ be the dephased weight on one $+\beta$ eigenvector. The other four eigenvalues sum to $-\beta$, with squared deviations from their mean $-\beta/4$ summing to $3/2$. Cauchy–Schwarz on their four weights gives

$$P \ge p^2 + \frac{(1-p)^2}{4} + \frac{2}{3}\left(\beta t - \frac{\beta}{4} + \frac{5\beta}{4}p\right)^2. \tag{235}$$

The unconstrained minimizer is $p = 1/5 - 2t/7$. Minimizing over $p \geq 0$ therefore gives

$$P \geq \begin{cases} \dfrac{1}{5} + \dfrac{2t^2}{7}, & 0 \leq t \leq 7/10, \\ \dfrac{8t^2 - 4t + 5}{18}, & 7/10 \leq t \leq 1. \end{cases} \tag{236}$$

Insert (232) into (233). The result is exactly $S_4 \geq \Phi(S_2)$ in (7); the switch occurs at $y = 5 + 10(7/10)^2 = 99/10$. The lower branch is already (59), because

$$S_4 - \frac{6y - 9}{7} = \frac{324 + 70x - 20y - 21z}{7}. \tag{237}$$

The word "exact" refers to this analytical boundary formula. It does *not* assert that every point on the cap is realized by a pure state.

17 Auxiliary spin-multipole construction and spectral bounds

The previously drawn flat spectral half-space is the sector-coordinate form of the proved inequality [5]

$$\boxed{S_4 \geq \frac{45}{7} + \frac{20}{3} q.} \tag{238}$$

Substitution of (2) and (58) gives

$$-350x + 84y + 91z \leq 1340. \tag{239}$$

This remains a useful consequence, but is no longer a separate defining or plotted boundary. Section 17.3 retains the eigenvalue gap that was minimized out in this linear corollary.

17.1 A short spin-multipole derivation

Embed the physical spin-three vector into two auxiliary spin-two spaces:

$$\mathcal{V}|3, m\rangle = \sum_{m_1, m_2 = -2}^{2} \langle 2, m_1; 2, m_2 | 3, m\rangle |2, m_1\rangle |2, m_2\rangle.$$

This is an isometry, not a physical bipartition. Its coefficient matrix A is antisymmetric, since exchanging two spin-two factors in total spin three has sign -1. Thus $R = AA^\dagger$ has paired spectrum

$$\operatorname{spec} R = (r, r, s, s, 0), \qquad r \geq s \geq 0, \qquad r + s = \frac{1}{2}. \tag{240}$$

Indeed the skew-matrix singular values occur in pairs. On an eigenspace of R with eigenvalue $\ell > 0$, the antilinear map $u \mapsto A\overline{u}/\sqrt{\ell}$ is norm preserving and squares to $-I$; this proves even multiplicity without a nondegeneracy assumption.

Set $\widetilde{R} = \mathsf{T}_2 R \mathsf{T}_2^{-1}$, $P = \operatorname{Tr} R^2$ and $W = \operatorname{Tr}(R\widetilde{R})$. Rotation covariance of the auxiliary reduction multiplies each normalized multipole of rank L by a scalar κ_L. The squared gains and resulting traces are

L	0	1	2	3	4
κ_L^2	7/5	7/10	3/50	3/20	11/20

$$P = \frac{1}{5} + \frac{7}{10} a_1 + \frac{3}{50} a_2 + \frac{3}{20} a_3 + \frac{11}{20} a_4, \tag{241}$$

$$W = \frac{1}{5} - \frac{7}{10} a_1 + \frac{3}{50} a_2 - \frac{3}{20} a_3 + \frac{11}{20} a_4. \tag{242}$$

These gains follow directly by applying the embedding to $|3,3\rangle \mapsto (|2,2\rangle|2,1\rangle - |2,1\rangle|2,2\rangle)/\sqrt{2}$. The sector conversion and pure-state multipole compatibility are

$$S_4 = 3 + \frac{144}{5}a_2 + \frac{176}{3}a_4, \qquad 18a_2 + 11a_4 - 14a_1 - 3a_3 = 1 + 5q.$$

The first follows from (52) and normalized multipoles; the second follows from $r_2 = r_4$, $r_1 = r_5$, together with $q = \sum_L (-1)^L a_L$ and $\sum_L a_L = 1$. Substitution into (241)–(242) gives

$$S_4 = \frac{160}{3}P + \frac{80}{3}W + \frac{20}{3}q - \frac{35}{3}. \tag{243}$$

The trace rearrangement inequality for two positive matrices with spectrum (240) gives $W \geq 2rs + s^2$. For completeness, in eigenbases the trace is $\sum_{ij} \lambda_i \lambda_j |U_{ij}|^2$; the overlap matrix is doubly stochastic, and opposite ordering minimizes each permutation sum. With $\delta = r - s$, put $\varepsilon = W - (2rs + s^2) \geq 0$. Equation (243) becomes the exact remainder identity

$$\boxed{S_4 = \frac{45}{7} + \frac{20}{3}q + \frac{140}{3}\left(\delta - \frac{1}{14}\right)^2 + \frac{80}{3}\varepsilon.} \tag{244}$$

This proves (238) without optimization or a large polynomial certificate. In manuscript multipole lengths it reads

$$378a_2 + 770a_4 \geq 45 + \frac{175}{2}q. \tag{245}$$

17.2 The paired-purity plane

The same auxiliary spectrum gives a further independent bound:

$$P = 2(r^2 + s^2) \geq \frac{1}{4}.$$

Here r, s denote only the auxiliary eigenvalues in (240); later $r = |\Theta|$ will denote the overlap magnitude. The rank-three pure-state identity is $a_3 = (1 - q)/6$. One short verification uses the invariant operator $\sum_{M=-3}^{3} T_{3M}^{(6)} \otimes T_{3M}^{(6)\dagger}$: on the two-copy total-spin sectors $J = 0, 2, 4, 6$, its eigenvalues are $-1, 1/6, 1/6, 1/6$, respectively. Hence its expectation is $(1 - 7p_0)/6 = (1 - q)/6$. Eliminating a_4 from (241) with the displayed pure-state compatibility relation, and then using (55), gives

$$\boxed{P = \frac{3}{10} + \frac{f^2}{20} + \frac{q}{5} - \frac{\tau}{150} = \frac{25 + 5x - y - z}{40}.} \tag{246}$$

Consequently $P \geq 1/4$ proves (6e). In spin moments, the same inequality is

$$\tau \leq \frac{15}{2}(1 + f^2) + 30q. \tag{247}$$

The purity P is that of an auxiliary spin-two reduction. It must not be confused with a physical two- or three-qubit marginal purity.

17.3 The stronger paired-spectral curved cap

Theorem 11 (Paired-spectral compatibility)**.** *Every normalized pure symmetric six-qubit state satisfies*

$$\boxed{\Delta := 15 + 5x - y - z \geq 0, \qquad 5y + 6z - 15x - 85 \leq \sqrt{40\Delta}.} \tag{248}$$

The radical is nonnegative, but the left side need not be. The unsquared inequality is the one used to define and render the outer region.

Proof sketch. The same auxiliary matrix R is used throughout. Its paired spectrum determines $\delta = r - s$ from its purity, while pure-state multipole compatibility determines both $\operatorname{Tr} R^2$ and $\operatorname{Tr}(R\widetilde{R})$ from (x, y, z). Apply trace rearrangement without minimizing over δ. This retains the missing compatibility between the old spectral and paired-purity inequalities.

Proof. From (241)–(242),

$$P - W = \frac{7}{5}a_1 + \frac{3}{10}a_3 = \frac{2 + 3x - 2q}{40},$$

where $a_1 = 3x/56$, $a_3 = (1 - q)/6$, and $q = 1 + x/4 - z/8$. Together with (246), this gives the exact trace identities

$$\boxed{P = \frac{25 + 5x - y - z}{40}, \qquad W = \frac{100 + 10x - 4y - 5z}{160}.} \tag{249}$$

The paired spectrum (240) yields

$$P = \frac{1}{4} + \delta^2, \qquad \delta = r - s \in [0, 1/2], \qquad \Delta = 40\delta^2.$$

The trace rearrangement inequality already proved above gives

$$W \geq 2rs + s^2 = \frac{3}{16} - \frac{\delta}{4} - \frac{\delta^2}{4}.$$

Substitute (249) and multiply by 160:

$$0 \leq 160W - 30 + 40\delta + 40\delta^2 = 85 + 15x - 5y - 6z + 40\delta.$$

Finally $40\delta = \sqrt{40\Delta}$ proves (248). No optimization, state-support assumption, polynomial elimination or SOS certificate is used. □

Why the old flat facet is redundant. Equivalently, the exact remainder identity (244) gives

$$S_4 \geq \frac{45}{7} + \frac{20}{3}q + \frac{140}{3}\left(\sqrt{\frac{\Delta}{40}} - \frac{1}{14}\right)^2. \tag{250}$$

Thus (6f) implies (239), with a strictly stronger restriction whenever $\Delta \neq 10/49$. The separate spectral plane must not be drawn underneath the new cap. The paired plane $\Delta = 0$ remains part of the boundary only where the new condition allows it.

Excluded former corners and finite neighborhoods. Put $L = 5y + 6z - 15x - 85$. The two former corners give

(x, y, z)	Δ	L
$(2/7, 55/7, 60/7)$	0	$10/7$
$(6/7, 9, 68/7)$	$4/7$	$38/7$

and both have $L > \sqrt{40\Delta}$. For the second point, $L^2 - 40\Delta = 324/49 > 0$ with $L > 0$; the first would require $10/7 \leq 0$. By continuity, a relative open neighborhood of each point in the domain $\Delta \geq 0$ is excluded. These points are therefore removed from the three-dimensional boundary, not kept as gray nodes on a superseded body.

A unique sector tuple at the sharp fourth-sector minimum. If $S_4 = 45/7$, (250) forces $q = 0$ and $\Delta = 10/49$. Using $z = 8 + 2x$ and $S_4 = 45 + 10x - 2y - 3z$ then gives

$$S_4 = 45/7 \quad \Longrightarrow \quad (x, y, z) = \frac{1}{49}(24, 405, 440). \tag{251}$$

This is a classification of the sector tuple, not of all state orbits. The two known representatives ψ_6, χ_6 retain their shared violet marker and saturate the new curved cap.

17.4 What the spectral saturation marker means

Equality in (238) requires $r = 2/7$, $s = 3/14$ and equality in the trace rearrangement inequality. In this case R and $\widetilde{R}$ commute and their joint eigenvalue pairs are $(0,r),(r,0),(r,s),(s,r),(s,s)$. In a time-reversal-real basis choose orthonormal vectors $u, \mathsf{T}_2u, w, \mathsf{T}_2w, t$ with $\mathsf{T}_2t = t$ such that

$$\ker R = \operatorname{span}(u), \quad E_r = \operatorname{span}(\mathsf{T}_2u, w), \quad E_s = \operatorname{span}(\mathsf{T}_2w, t).$$

The skew structure forces $A = \alpha(\mathsf{T}_2u \wedge w) + \beta(\mathsf{T}_2w \wedge t)$. Its time reverse is in the orthogonal wedge span generated by $u \wedge \mathsf{T}_2w$ and $w \wedge t$. Since the embedding intertwines time reversal, $\Theta = 0$. Thus any physical equality state has $S_4 = 45/7$; the entire former flat facet is not achievable. Equation (251) fixes its only possible sector tuple.

The two labelled states ψ_6, χ_6 satisfy these spectral conditions, and both have the full sector tuple

$$(S_1, S_2, S_3, S_4, S_5, S_6) = \frac{1}{49}(24, 405, 440, 315, 1104, 799). \tag{252}$$

No exhaustive equality-orbit classification is needed for the defining inequalities: the spectral argument above and the explicit representatives already justify the bound and the violet marker.

18 The full teal family from pure-state multipole compatibility

We retain the full information in two compatible Gram matrices, rather than reducing it immediately to the older minimum-eigenvalue estimate. This gives all the supporting inequalities in (6i). The proof uses only the spin-three harmonic realization and elementary spectral duality; it does not use an amplitude-level sum-of-squares certificate or computer-assisted polynomial elimination. Group the six qubits into three spin-one pairs. A pure spin-three vector is a symmetric harmonic Cartesian cubic H_{ijk} with $\sum_i H_{iik} = 0$ and $\sum_{ijk} |H_{ijk}|^2 = 1$. In a Cartesian time-reversal-real basis, put

$$C_{ij} = \sum_{k\ell} H_{ik\ell}\overline{H_{jk\ell}}, \qquad B_{ij} = \sum_{k\ell} H_{ik\ell}H_{jk\ell}.$$

Then C is the physical two-qubit marginal on its spin-one space, $\operatorname{Tr} C = 1$, and $\operatorname{Tr} B = \Theta$. Spin contraction gives

$$C = \frac{I}{3} - \frac{Q}{15} - \frac{i}{6}E(F), \qquad M_{ij} := \langle J_iJ_j\rangle = 4\delta_{ij} + Q_{ij} + \frac{i}{2}E(F)_{ij}, \tag{253}$$

where $E(F)_{ij} = \epsilon_{ijk}F_k$. Its bilinear counterpart is

$$\psi^{\mathsf{T}}J_iJ_j\psi = 9\Theta\delta_{ij} - 15B_{ij}. \tag{254}$$

These identities follow by applying $(J_iH)_{abc} = -i(\epsilon_{iad}H_{dbc} + \epsilon_{ibd}H_{adc} + \epsilon_{icd}H_{abd})$. For example, the three same-index and six different-index terms in $(J_iH)\cdot(J_jH)$ contribute $-3(\Theta\delta_{ij} - B_{ij})$ and $-6(\Theta\delta_{ij} - 2B_{ij})$; using $J_i^{\mathsf{T}} = -J_i$ proves (254).

Each slice H_i is a symmetric traceless 3×3 matrix, a space of complex dimension five. Hence the Gram matrix of its six slices and conjugate slices is positive and singular:

$$\mathcal{G} = \begin{pmatrix} C & B \\ \overline{B} & \overline{C} \end{pmatrix} \succeq 0, \qquad \operatorname{rank}\mathcal{G} \le 5.$$

The Gram matrix of the six rotation vectors $J_i\psi, J_i\overline{\psi}$, after a block interchange and a sign change, gives

$$\mathcal{H} = \begin{pmatrix} K & B - \frac{3}{5}\Theta I \\ \overline{B} - \frac{3}{5}\overline{\Theta} I & \overline{K} \end{pmatrix} \succeq 0, \qquad K = \frac{\overline{M}}{15} = \frac{3}{5}I - \frac{2}{5}C - \frac{3}{5}\overline{C}.$$

Choose the global phase so that $\Theta = r \ge 0$. Substitution of (253) into this exact matrix difference gives

$$\mathcal{G} - \mathcal{H} = \frac{I_6 - \mathcal{K}}{15}, \qquad \mathcal{K} = \begin{pmatrix} 2Q + 2iE(F) & -9rI_3 \\ -9rI_3 & 2Q - 2iE(F) \end{pmatrix}. \tag{255}$$

Testing on a nonzero $w \in \ker\mathcal{G}$ gives $w^\dagger\mathcal{K}w \ge \|w\|^2$, because $\mathcal{H} \succeq 0$. Thus $\lambda_{\max}(\mathcal{K}) \ge 1$. This is where the pure-state compatibility and the singular slice Gram matrix enter the argument.

18.1 Reducing the six-dimensional spectral problem to one parameter

The Hermitian matrix $\mathcal{K}$ commutes with the conjugation $(u, v) \mapsto (\overline{v}, \overline{u})$. Its largest eigenspace therefore contains a vector fixed by this conjugation. Normalize it as $(u, \overline{u})/\sqrt{2}$ with $\|u\| = 1$, and write $u = a + ib$ for real vectors a, b. Define

$$p = 2|a \times b| \in [0, 1], \qquad A = aa^{\mathsf{T}} + bb^{\mathsf{T}}.$$

The elementary identities and estimates needed below are

$$\|A_{\mathrm{tf}}\|_F^2 = \frac{2}{3} - \frac{p^2}{2}, \qquad |u^{\mathsf{T}}u| = \sqrt{1 - p^2}, \qquad |u^\dagger iE(F)u| \le fp.$$

The first follows from $\operatorname{Tr} A = 1$ and $\operatorname{Tr} A^2 = 1 - p^2/2$; the second is $|u^{\mathsf{T}}u|^2 = (|a|^2 + |b|^2)^2 - 4|a \times b|^2$. For the third, $u^\dagger iE(F)u$ is, up to sign, $2F \cdot (a \times b)$. Evaluating the Rayleigh quotient and applying Cauchy–Schwarz to $\operatorname{Tr}(QA)$ now yields

$$1 \le \lambda_{\max}(\mathcal{K}) \le \sqrt{10(y-5)}\sqrt{1 - \frac{3}{4}p^2} + \sqrt{6x}\,p + 9r\sqrt{1 - p^2}. \tag{256}$$

The single real number p describes the possible relative orientation of the real and imaginary parts of the eigenvector; no rotational ansatz for the physical state has been imposed.

18.2 Weighted Cauchy–Schwarz produces every supporting plane

Fix any $t > 0$. Apply weighted Cauchy–Schwarz to the three terms on the right of (256). Put

$$U_t = 10(y-5) + \frac{24x}{3+4t} + \frac{81q}{t}.$$

The sum in (256) has square at most

$$U_t \left[1 - \frac{3}{4}p^2 + \left(\frac{3}{4} + t\right)p^2 + t(1 - p^2)\right] = (1+t)U_t.$$

Since the original Rayleigh quotient is at least one, this proves (6i) for every $t > 0$. For fixed t, $q = 1 + x/4 - z/8$ makes this an affine half-space in (x, y, z). All values of t are retained in the mathematical region and in the numerical envelope used for the displayed roof; finite selections of planes are used only for independent checks. Equivalently, define

$$\begin{aligned} Q_*(x, y) &= \max\left\{0, \sup_{t>0} \frac{t}{81}\left[\frac{1}{1+t} - 10(y-5) - \frac{24x}{3+4t}\right]\right\}, \\ Z_{\mathrm{teal}}(x, y) &= 8 + 2x - 8Q_*(x, y). \end{aligned} \tag{257}$$

The full teal condition together with $q \ge 0$ is exactly $z \le Z_{\mathrm{teal}}(x, y)$, not its convex hull or a weaker explicit root replacement. When $x = 0$, elementary one-variable maximization gives $Q_* = [1 - \sqrt{10(y-5)}]_+^2/81$, so the attained C_3 rim and every $S_1 = 0$ label are unchanged.

Attainment of every supporting plane. For each fixed $t > 0$, the known C_3 family in (36) attains equality at

$$x = 0, \qquad r = \frac{t}{9(1+t)}, \qquad y = 5 + \frac{1}{10(1+t)^2}, \qquad z = 8(1 - r^2).$$

Substitution makes the left side of (6i) equal to $1/(1+t)$. Thus the constant of every plane is attained. The same plane is also attained at nonzero x by the independently verified C_4 state $\Omega_{f(t)}$ in Section 3.5; the exact residual is (43). Thus both one-dimensional rims are attained. This does not assert sharpness of intermediate points on the two-dimensional envelope.

A strict upper-bound improvement. For an exact check, the tuple

$$(x, y, z) = \left(\frac{1}{100}, \frac{501}{100}, \frac{799}{100}\right), \qquad q = \frac{3}{800}, \tag{258}$$

satisfies every condition of the previous plotted region strictly. The historical teal cap has $D = 3/50$, while the red cap is automatic because $\sqrt{y-5} = 1/10 < b$. But at $t = 1$ the left side of (6i) is $2453/5600 < 1/2$. Thus even one member of the full family cuts a finite-volume region. That member can also be written as the simple supporting plane

$$567z - 1326x - 560y \le 1708. \tag{259}$$

It is not added separately, since it is already part of the full family.

Relation to Audenaert's inequality. Audenaert's theorem [3] compares Schatten norms of a bipartite state and its reductions. Applying it only to their purities does not supply the new cuts here. The extra information is the shared pure vector behind $\mathcal{G}$ and $\mathcal{H}$, their exact difference (255), and the forced zero eigenvalue of $\mathcal{G}$. Thus the present proof should be described as a pure-state multipole-compatibility argument, not as a consequence of Audenaert's theorem alone.

18.3 Historical spectral lemma and redundant root caps

For comparison, the earlier minimum-eigenvalue argument is reproduced here. It is useful for interpreting the known equality states, but its cap is no longer imposed as an additional boundary. If $\lambda = \lambda_{\min}(C)$ and $r = |\Theta|$, then

$$\mathcal{G} - \mathcal{H} \succeq \left[2\lambda - \frac{3}{5}(1+r)\right] I_6.$$

Testing on a nonzero vector in $\ker \mathcal{G}$ proves the spectral lemma

$$\lambda_{\min}(C) \le \frac{3}{10}(1+r). \tag{260}$$

Indeed its left quadratic form is $-w^\dagger \mathcal{H} w \le 0$.

Moreover (253) implies

$$D = 60\|C - I/3\|_F^2.$$

For $r < 1/9$, (260) and $\operatorname{Tr} C = 1$ give

$$D \ge 60\frac{3}{2}\left(\frac{1}{3} - \frac{3}{10}(1+r)\right)^2 = \frac{(1-9r)^2}{10}.$$

For $r \ge 1/9$ no estimate is needed. Thus every pure state obeys the historical root cap

$$\boxed{9\sqrt{q} + \sqrt{10(5x + y - 5)} \ge 1,} \tag{261}$$

or, equivalently,

$$z \le 8 + 2x - 8A(x, y)^2. \tag{262}$$

Its $x = 0$ curved rim is attained by (36), so the implication $x = 0, y = 5 \Rightarrow z \le 640/81$ is included globally, not merely overlaid. The full family implies the stronger simple corollary

$$9\sqrt{q} + \sqrt{8x + 10(y-5)} \ge 1. \tag{263}$$

Indeed $24x/(3+4t) \le 8x$, so (6i) implies $8x + 10(y-5) + 81q/t \ge 1/(1+t)$ for all $t > 0$; maximizing $t/(1+t) - [8x + 10(y-5)]t$ proves the claim. This also implies the old root cap because $8x \le 50x$. Neither root corollary is drawn as a second teal boundary: only the full family is used.

19 The earlier curved cap and its global extension mechanism

Theorem 12 (Global extension of the sharp endpoint)**.** *For every normalized pure symmetric six-qubit state, put* $r = \sqrt{q} = |\Theta|$*. Then*

$$\boxed{\sqrt{y-5} \le b(1-r) + \sqrt{10}\, r + 3\sqrt{x}\sqrt{\frac{1-r}{1+r}}.} \tag{264}$$

In particular it implies the simpler explicit inequality

$$\boxed{\sqrt{y-5} \le b + 3\sqrt{x} + d\sqrt{q},} \tag{265}$$

or, equivalently,

$$\boxed{z \le 8 + 2x - \frac{8}{d^2}\left[\sqrt{y-5} - b - 3\sqrt{x}\right]_+^2.} \tag{266}$$

Both inequalities are global necessary conditions. The base region uses their octupole-strengthened version (193)*, with* $\gamma = 14/5$ *instead of* 3*; the current figure uses the further unit-dipole strengthening* (377)*, with* $\gamma_{22} = \sqrt{3/2}$*. The earlier inequalities are retained here to explain the global time-reversal decomposition and historical comparisons. They are sharp at* $(x, y, z) = (0, s_*, 8)$*, but sharpness elsewhere is not asserted.*

Proof sketch. First prove the sharp quadrupole bound when both dipole and overlap vanish. Next, within $\Theta = 0$, use an imaginary collective rotation to decrease the dipole to zero; the paired-purity bound controls the path length, and hence the change in Q. Finally split an arbitrary state into two time-reversal-real vectors. This removes Θ exactly and expresses its quadrupole as a convex combination.

19.1 The sharp zero-dipole, zero-overlap input

Lemma 1. *If* $F = 0$ *and* $\Theta = 0$*, then*

$$\tau \le \tau_* := 30 - 6\sqrt{15}, \qquad y \le s_*. \tag{267}$$

Equality is attained by (37)*.*

Proof. This is the short multipole argument of Ref. [7], reproduced here with all tensors defined. Let $G = \sum_{M=-3}^{3} \rho_{3M} T^{(6)}_{3M}$ be the rank-three *density multipole*, not the physical three-qubit marginal. Define the real traceless rank-two contraction

$$W_{ij} = -36\,\mathrm{Tr}(G^2 \widehat{Q}_{ij}), \quad \kappa = \mathrm{Tr}\, Q^3, \quad e = \mathrm{Tr}(QW).$$

On $F = \Theta = 0$, the two scalar pure-state compatibility contractions are

$$\|W\|_F^2 + 2e = 48 - 8\tau, \qquad \mathrm{Tr}(Q^2 W) + e + 2\kappa = 6 - 3\tau. \tag{268}$$

For clarity, these are exact identities, not optimization assumptions. They are obtained by multiplying the normalized spin-three tensors in (53), using $\rho^2 = \rho$ and $F = \Theta = 0$. The scalar contraction table is

	1	τ	κ	e
$\|W\|_F^2$	48	-8	0	-2
$\mathrm{Tr}(Q^2 W)$	6	-3	-2	-1

.

This also provides a finite matrix-algebra verification prescription: form the seven-dimensional tensors displayed in Section 4, project ρ onto rank three, and take the indicated traces. Explicitly, write $\psi = \sum_{m=-3}^{3} c_m |3, m\rangle$ and substitute $\rho_{mn} = c_m \bar{c}_n$ in these matrices, subject to $\sum_m |c_m|^2 = 1$, $\langle J_i \rangle = 0$, and $\Theta = \sum_m (-1)^{3-m} c_m c_{-m} = 0$. The generators can be evaluated from $J_z |3, m\rangle = m |3, m\rangle$ and

$J_+|3,m\rangle = \sqrt{12 - m(m+1)}|3,m+1\rangle$. No large amplitude-level SOS certificate or stationary-state ansatz is used.

For a traceless 3×3 matrix, Cayley–Hamilton gives $\operatorname{Tr} Q^4 = \tau^2/2$ and $\|(Q^2)_{\mathrm{tf}}\|_F^2 = \tau^2/6$. Expanding one norm and using (268) gives

$$\tau^2 - 60\tau + 360 = 6\|(Q^2)_{\mathrm{tf}} + 2Q + W\|_F^2 \geq 0. \tag{269}$$

Also $\|Q + W\|_F^2 + 7\tau = 48$, excluding the upper branch $\tau \geq 30 + 6\sqrt{15}$. This proves (267). Direct spin-moment evaluation of (37) gives equality. □

19.2 A uniform quadrupole continuity estimate

The already proved $y \leq 15$ and (55) give $\|Q\|_F \leq R_0 := \sqrt{75/2}$ for every pure spin-three vector. For a real symmetric traceless matrix H with $\|H\|_F = 1$, the Hermitian operator $\widehat{Q}(H) = \sum H_{ij}\widehat{Q}_{ij}$ therefore has operator norm at most R_0. Duality and the pure-state trace-distance formula imply, for normalized vectors ψ, χ,

$$\|Q_\psi - Q_\chi\|_F \leq 2R_0\sqrt{1 - |\langle\psi|\chi\rangle|^2} = 2R_0 \sin d_{\mathrm{FS}}(\psi,\chi). \tag{270}$$

Here $d_{\mathrm{FS}} = \arccos|\langle\psi|\chi\rangle|$; a normalized horizontal path has speed $\|\dot\psi\|$. In particular a path of length L gives $\|Q_\psi - Q_\chi\|_F \leq 2R_0 L$.

19.3 Balancing the dipole while preserving zero overlap

Suppose first that $\Theta = 0$ and $0 < f \leq 1/5$. Let $\boldsymbol{n} = F/f$ and consider

$$\dot\psi = -(J_{\boldsymbol{n}} - f)\psi. \tag{271}$$

The flow preserves normalization and is horizontal. Integer-spin time reversal satisfies $\mathsf{T} J_i \mathsf{T}^{-1} = -J_i$; its bilinear form gives $\dot\Theta = 2f\Theta$. Hence the flow preserves $\Theta = 0$. Writing $v = \operatorname{Var}(J_{\boldsymbol{n}})$, differentiation gives

$$\dot f = -2v, \qquad \|\dot\psi\|^2 = v. \tag{272}$$

For a real traceless Q, its least eigenvalue is at least $-\sqrt{2\tau/3}$. By paired purity (247), throughout this decreasing-f path,

$$\begin{aligned} v &= 4 + \boldsymbol{n}^{\mathsf{T}} Q \boldsymbol{n} - f^2 \geq 4 - \sqrt{5(1+f^2)} - f^2 \\ &\geq \frac{99}{25} - \sqrt{\frac{26}{5}} > \frac{5}{3}. \end{aligned} \tag{273}$$

The last strict inequality follows from $(172/75)^2 - 26/5 = 334/5625 > 0$. The path reaches $f = 0$ in finite time. Although $\boldsymbol{n}$ is undefined at its endpoint, its finite length makes ψ Cauchy there; the endpoint is a normalized pure state with $F = \Theta = 0$. Equations (272)–(273) bound its length by

$$L = \int_0^{f_{\mathrm{initial}}} \frac{ds}{2\sqrt{v(s)}} \leq \sqrt{\frac{3}{20}}\, f_{\mathrm{initial}}.$$

Using Lemma 1 and (270) therefore yields

$$\|Q\|_F \leq \sqrt{\tau_*} + \sqrt{\frac{45}{2}}\, f. \tag{274}$$

This bound is also valid for $f \geq 1/5$, with no flow argument. Indeed paired purity gives $\tau \leq (15/2)(1+f^2)$, whereas

$$\left(\sqrt{\tau_*} + \sqrt{45/2}\, f\right)^2 - \frac{15}{2}(1+f^2) = \tau_* - \frac{15}{2} + 2\sqrt{\frac{45\tau_*}{2}} f + 15f^2 > 0.$$

The right side is increasing for $f \geq 0$ and is already positive at $f = 1/5$ (use $\tau_* > 6$). Thus (274) is global on the whole $\Theta = 0$ slice, including $f = 0$.

19.4 Removing the overlap exactly

Choose a global phase so that $\Theta = r \geq 0$. For $r < 1$ there exist orthonormal time-reversal-real vectors u, v such that

$$\psi = \sqrt{\frac{1+r}{2}}\, u + i\sqrt{\frac{1-r}{2}}\, v, \qquad \phi = \frac{u+iv}{\sqrt{2}}, \qquad \Theta_\phi = 0.$$

The spin generators are time-reversal odd and the quadrupole operators are even. Their real and imaginary cross terms therefore give exactly

$$F_\psi = \sqrt{1-r^2}\, F_\phi, \qquad Q_\psi = (1-r)Q_\phi + rQ_u. \tag{275}$$

Apply (274) to ϕ and $\|Q_u\|_F \leq R_0$:

$$\|Q_\psi\|_F \leq (1-r)\sqrt{\tau_*} + rR_0 + \sqrt{\frac{45}{2}}\, f_\psi \sqrt{\frac{1-r}{1+r}}.$$

Multiplication by $2/\sqrt{15}$, followed by $f_\psi = \sqrt{3x/2}$, proves (264). If $r = 1$, the state is time-reversal real up to phase, $F = 0$, and the same inequality reduces to the established $y \leq 15$. Finally $\sqrt{(1-r)/(1+r)} \leq 1$ proves (265); using $q = 1 + x/4 - z/8$ gives (266). This completes the proof of Theorem 12.

19.5 A genuine finite-volume improvement

Even the simpler consequence removes a finite-volume region. The rational point

$$(x, y, z) = (10^{-5},\, 139/20,\, 399997/50000) = (10^{-5}, 6.95, 7.99994), \qquad q = 10^{-5},$$

satisfies all previous conditions and paired purity strictly, but the explicit roof gives $z \leq 7.995321229\ldots$. For an elementary strict check, its left side in (265) exceeds 1.39, while the right side is less than 1.36. By continuity, an open three-dimensional neighborhood of this point is removed. The improvement is not confined to the line $x = 0, z = 8$.

The historical implicit condition is strictly better than its simpler root consequence whenever its active boundary has $x > 0$ and $r > 0$: then $\sqrt{(1-r)/(1+r)} < 1$, so the same point cannot saturate both versions. At $x = 0$ or $r = 0$ the two right sides agree. The violet $S_1 = 0$ families and the sharp endpoint Ξ_* are consequently unchanged.

For a strict comparison with the previously plotted simplified cap, put $x = 10^{-4}$, $z = 7.997$, so that $r = 1/50$, and define

$$t = b + \frac{d}{50} + \frac{3}{200}\left(1 + \sqrt{\frac{49}{51}}\right), \qquad y = 5 + t^2 = 6.98509610744\ldots.$$

This point satisfies the other defining conditions and the simplified red cap strictly, but violates the stronger cap by $\frac{3}{200}(1 - \sqrt{49/51}) > 0$ in square-root coordinates. The stronger roof at the same (x, y) is $z = 7.99694667472\ldots$. Continuity again gives a finite-volume improvement, now relative to the previously displayed region.

20 The $S_1 = 0$ slice and interpretation of the labels

Historical base slice. The following formula describes the base region $\mathcal{R}$ before the 22 October side update. It is retained to identify the older excluded gray junction and its proof, not as a currently plotted rim. On $x = 0$ this former upper boundary was

$$Z_0(y) = \begin{cases} 8 - \dfrac{8}{81}\big(1 - \sqrt{10(y-5)}\big)^2, & 5 \leq y \leq 51/10, \\ 8, & 51/10 \leq y \leq s_*, \\ 8 - k\big(\sqrt{y-5} - b\big)^2, & s_* \leq y \leq y_c, \\ 15 - y, & y_c \leq y \leq 15, \end{cases} \qquad k := \frac{8}{d^2}. \tag{276}$$

The crossing of the red cap with paired purity is

$$t_c = \frac{kb - \sqrt{kb^2 - 2(k-1)}}{k-1}, \qquad y_c = 5 + t_c^2 = 7.01408446898\ldots. \tag{277}$$

Thus the slice of the base $\mathcal{R}$ is exactly $5 \le y \le 15$, $0 \le z \le Z_0(y)$. The first curved rim is achieved by (36). At its left endpoint,

$$x = 0,\ y = 5 \quad \Longrightarrow \quad q \ge 1/81, \qquad z \le 640/81. \tag{278}$$

The entire violet vertical edge is achieved as well: the phase family

$$|\mathrm{AC}_2(\phi)\rangle = \frac{\sqrt{2}D_0 + e^{i\phi}\sqrt{5}D_3 + \sqrt{2}D_6}{3} \quad \text{has} \quad (x,y,z) = \left(0, 5, \frac{640}{81}\cos^2\phi\right).$$

The full horizontal segment

$$x = 0, \quad z = 8, \qquad \frac{51}{10} \le y \le s_* \tag{279}$$

is achieved by pure states; this is not an appeal to convex mixtures. For explicit continuous paths, write

$$(a_u, b_u, c_u, d_u) = \frac{1}{2}\big(\sqrt{(1+u)(1-2u)}, \sqrt{(1+u)(1+2u)}, \sqrt{(1-u)(1-2u)}, \sqrt{(1-u)(1+2u)}\big).$$

The normalized states

$$\Phi_u^- = a_u D_0 + b_u D_2 - c_u D_4 + d_u D_6, \qquad \Phi_u^+ = a_u D_0 + b_u D_2 + c_u D_4 - d_u D_6$$

have $F = \Theta = 0$. The first path, $\sqrt{10}/8 \le u \le 1/2$, connects the tuple of ψ_0 to C_4; the second, $u_* \le u \le 1/2$, connects Ξ_* to C_4. Their quadrupole lengths are

$$\tau_\pm(u) = 6 - 39u^2 + 84u^4 \pm 3\sqrt{15}\, u(1-4u^2)\sqrt{1-u^2}.$$

Continuity, together with the proved endpoint bounds, establishes (279). The C_4 state at $y = 27/5$ is an interior point. Among this horizontal segment, only ψ_0 and C_4 saturate the spectral lemma (260).

Violet therefore marks explicit attained states or families. Three former gray nodes are unattainable, as proved in Section 21. Gray curves still carry no blanket attainment or exclusion claim. In particular the red rim beyond Ξ_* was not claimed achievable; its gray junction with the paired-purity line was specifically excluded. Both are superseded by the necessary side envelope and explicit attained families of Section 27.2. Figure 5 uses that current envelope, not the historical red branch in (276). The formerly problematic point $(0, 5, 8)$ is no longer a corner of the displayed region: the global teal cap excludes it and a finite neighborhood.

21 Analytical exclusion of three former gray nodes

The following nodes belonged to the necessary-condition region before adding (6h), but not to the pure-state sector set. The new plane removes A and creates different junctions, settled in Section 23: its equality segment is fully attained, and the adjacent gray plane junctions are excluded. The other listed points remain useful geometric junctions. Their exclusion does not, by itself, exclude every adjacent gray curve or surface. No numerical optimization, polynomial elimination or large SOS certificate is used in the proofs below.

Using the constants b, d of (3), define

$$\eta := \frac{2 - b^2}{8 - d^2}, \qquad \varepsilon := \left(\frac{2 - b^2}{\sqrt{3b^2 + 2\gamma^2 - 6} + b\gamma}\right)^2. \tag{280}$$

The three former gray nodes, in sector coordinates (x, y, z), are

$$\begin{aligned} A &= \left(\frac{25}{54}, \frac{215}{18}, \frac{145}{27}\right), \\ B &= (0,\, 7+8\eta^2,\, 8-8\eta^2), \\ C &= (\varepsilon,\, 7+3\varepsilon,\, 8+2\varepsilon). \end{aligned} \tag{281}$$

For locating the two tightly packed nodes in the magnifications,

$$B \simeq (0, 7.014084469, 7.985915531), \qquad C = (\varepsilon, 7+3\varepsilon, 8+2\varepsilon).$$

These decimals are locators only; the proofs use the exact radicals.

Node	Obstruction to a pure-state realization	Proof
A	Required $q = 4/9$ contradicts auxiliary compatibility $q \le 1/4$	Sec. 21.1
B	Cap equality would require a quadrupole eigenvalue below -4	Sec. 21.2
C	The octupole-balancing estimate is strictly stronger there	Sec. 21.3

Recall the manuscript spin-multipole conventions:

$$F_i = \langle J_i \rangle, \quad f = |F|, \quad Q_{ij} = \frac{1}{2}\langle J_i J_j + J_j J_i \rangle - 4\delta_{ij}, \quad \tau = \|Q\|_F^2,$$

so that $x = 2f^2/3$, $y = 5 + 4\tau/15$ and $q = |\Theta|^2$. A useful elementary compatibility condition, valid for every normalized spin-three vector, is

$$Q \succeq -4I, \qquad \boldsymbol{n}^{\mathsf{T}} Q \boldsymbol{n} = \langle J_{\boldsymbol{n}}^2 \rangle - 4 \ge -4 \quad (|\boldsymbol{n}| = 1). \tag{282}$$

21.1 Node A: auxiliary support and overlap compatibility

Proposition 4. *The sector tuple A in* (281) *is unattainable.*

Proof. Suppose it were realized. The exact identities (4) and (249) give

$$q = \frac{4}{9}, \qquad P = \frac{1}{4}, \qquad W = \frac{3}{16}.$$

Use the isometry $\mathcal{V} : V_3 \longrightarrow V_2 \otimes V_2$ of Section 17, and set $|\Upsilon\rangle = \mathcal{V}|\psi\rangle$. It is a normalized antisymmetric vector; its one-factor reduction is R. The real Clebsch–Gordan embedding intertwines time reversal. Thus, for $|\widetilde{\Upsilon}\rangle = (\mathsf{T}_2 \otimes \mathsf{T}_2)|\Upsilon\rangle$,

$$|\langle \Upsilon | \widetilde{\Upsilon} \rangle| = |\Theta|.$$

The paired spectrum (240) and $P = 1/4$ force

$$R = \frac{1}{4}(I - |u\rangle\langle u|), \qquad \|u\| = 1.$$

Put $v = \mathsf{T}_2 u$; then $\widetilde{R} = (I - |v\rangle\langle v|)/4$ and

$$W = \frac{3 + |\langle u|v\rangle|^2}{16}.$$

Hence $u \perp v$. Let $K = \mathrm{span}\{u, v\}^{\perp}$, a three-dimensional subspace of V_2, and let Π project onto $\Lambda^2 K$. Since R has kernel u, the vector Υ lies in $\Lambda^2(u^{\perp})$; similarly $\widetilde{\Upsilon}$ lies in $\Lambda^2(v^{\perp})$. Their components outside $\Lambda^2 K$ lie in the mutually orthogonal spaces $v \wedge K$ and $u \wedge K$. Consequently

$$\langle \Upsilon | \widetilde{\Upsilon} \rangle = \langle \Pi\Upsilon | \Pi\widetilde{\Upsilon} \rangle.$$

Antisymmetry allows at most one factor in the direction v. Its expected occupation is $2\langle v|R|v\rangle = 1/2$, so

$$\|\Pi\Upsilon\|^2 = \frac{1}{2}, \qquad \|\Pi\widetilde{\Upsilon}\|^2 = \frac{1}{2}.$$

The second equality follows in the same way, using the direction u and $\widetilde{R}$. Cauchy–Schwarz now gives $|\Theta| \le 1/2$, hence $q \le 1/4$, contradicting $q = 4/9$. □

21.2 Node B: the quadrupole equality conditions are incompatible

We first extract the equality information already contained in the null-slice multipole identities; no classification of state amplitudes is needed.

Lemma 2 (Quadrupole spectrum at the sharp null endpoint)**.** *If $F = 0$, $\Theta = 0$ and $\tau = \tau_* = 30 - 6\sqrt{15}$, then*

$$\operatorname{spec} Q = \left\{ \sqrt{15} - 3, \quad \frac{3 - \sqrt{15} \pm \sqrt{6\sqrt{15} - 12}}{2} \right\}. \tag{283}$$

Proof. Equation (269) forces $W = -(Q^2)_{\mathrm{tf}} - 2Q$, where W is the rank-two octupole contraction in (268), not the auxiliary scalar trace W used in the preceding proof. Thus, with $\kappa = \operatorname{Tr} Q^3$,

$$\operatorname{Tr}(QW) = -\kappa - 2\tau, \qquad \|W\|_F^2 = \frac{\tau^2}{6} + 4\tau + 4\kappa.$$

The first identity in (268) then gives

$$\kappa = 24 - 4\tau - \frac{\tau^2}{12} = 54 - 9\tau = -216 + 54\sqrt{15}.$$

Since Q is traceless, its characteristic polynomial is $t^3 - (\tau/2)t - \kappa/3$. Substitution and factorization give (283). □

Proposition 5. *The implicit cap* (264) *is strict whenever $x = 0$ and $0 < r = |\Theta| < 1$. In particular the node B is unattainable.*

Proof. For $x = 0$ and $0 < r < 1$, the exact time-reversal decomposition (275) gives normalized vectors u, ϕ with $F_\phi = 0$, $\Theta_\phi = 0$ and

$$Q_\psi = (1 - r)Q_\phi + rQ_u.$$

The cap was obtained from

$$\|Q_\psi\|_F \le (1 - r)\|Q_\phi\|_F + r\|Q_u\|_F \le (1 - r)\sqrt{\tau_*} + rR_0, \qquad R_0 = \sqrt{75/2}. \tag{284}$$

Because both weights are positive, equality in the cap would force equality in both steps. Hence

$$\|Q_\phi\|_F = \sqrt{\tau_*}, \qquad \|Q_u\|_F = R_0, \qquad Q_u = \frac{R_0}{\sqrt{\tau_*}} Q_\phi.$$

The last equality is the positive-collinearity condition for equality in the Frobenius triangle inequality. Lemma 2 fixes the least eigenvalue of Q_ϕ as

$$\lambda_* := \frac{3 - \sqrt{15} - \sqrt{6\sqrt{15} - 12}}{2} < -2.$$

For an exact check of the strict inequality, square $\sqrt{6\sqrt{15} - 12} > 7 - \sqrt{15}$; it is equivalent to $\sqrt{15} > 19/5$. Also $\tau_* < 7$ since $\sqrt{15} > 23/6$. Therefore the required quadrupole Q_u would satisfy

$$\lambda_{\min}(Q_u) = \frac{R_0}{\sqrt{\tau_*}} \lambda_* < -\sqrt{\frac{150}{7}} < -4,$$

contradicting (282) for the normalized vector u. This proves strictness of the cap on the stated slice.

For the node B, $q = \eta^2$ and $0 < \eta < 1$. Its defining radical obeys

$$(b + d\eta)^2 = 2 + 8\eta^2 = y_B - 5.$$

Indeed $b + d = \sqrt{10}$ makes the quadratic factor as

$$(b + dr)^2 - 2 - 8r^2 = (d^2 - 8)(r - 1)(r - \eta).$$

Thus B lies exactly on the implicit-cap equality surface at $x = 0$, $r = \eta$, and is excluded by the strictness just proved. □

21.3 Node C: retain the strict octupole estimate

Proposition 6. *If $\Theta = 0$ and $f > 0$, then*

$$\boxed{\sqrt{y-5} < b + \gamma\sqrt{x}.} \tag{285}$$

In particular the node C is unattainable.

Proof. The proof of Theorem 7 is already strict for every positive null dipole: when $f \le 1/20$ its coefficient is at most $22\sqrt{10}/25 < \gamma$, and for $f \ge 1/20$ the paired-purity comparison is strict. This proves (285). At C the positive radical in (280) solves $(\gamma^2-3)u^2 + 2b\gamma u + b^2 - 2 = 0$. Thus

$$(b + \gamma\sqrt{\varepsilon})^2 = 2 + 3\varepsilon = y_C - 5,$$

and $q_C = 0$, $x_C = \varepsilon > 0$. Thus C would require equality in the strict inequality (285), a contradiction. □

What these exclusions change. All three listed nodes are certified unattainable, not merely unresolved numerical candidates. Their earlier exclusion remains valid after adding the sharp plane; A is now outside $\mathcal{R}$ rather than a current node. The defining inequalities have not been replaced by a claim of exact attainability. The quantitative octupole cap shifts C from its older coefficient-3 position; the updated junction remains rigorously excluded. Exclusion of a node alone does not classify all nearby states or all gray boundary curves.

22 Redundant consequences: stated, but not plotted

The following conditions add no restriction to (6); no separate facets, equality curves or special markers for them appear in Fig. 1.

Baseline rows absorbed into the defining system. First, $y \le 15$ and the historical lower bound $10x \le 3z$ really are consequences of the updated system, not extra domain assumptions. Put $v = \sqrt{5(y-5)/2} \ge 0$. Combining $q \ge 0$ with the new lower bound gives $x \le 2(v+4)/3$. If $v > 5$, the $t = 1$ orange plane $4y + 3z \le 60 + 10x$ and the same lower bound instead give $x \ge 2(v+5)(v+4)/5$. The two estimates would imply $v + 5 \le 5/3$, a contradiction. Hence $v \le 5$, or $y \le 15$. Now the new lower denominator is at most nine, proving $z \ge 10x/3$; combining this with $q \ge 0$ also gives $x \le 6$. Thus neither the old lower plane nor the upper domain bounds is separately imposed. The $t = 7/10$ plane is exactly (59). The explicit domain condition $q \ge 0$ gives $-6x + 3z \le 24$. Finally the identity

$$-8x + 10y + 3z = \frac{5}{2}(-10x + 4y + 3z) + \frac{1}{2}(-6x + 3z) + 2(10x - 3z) \le 162$$

proves (60). These are consequences, not extra defining inequalities. The historical teal root cap is redundant by (263); only the full supporting family is used as its replacement.

The older fourth-sector interval. The curved bound implies the old blue bound (239); together with $q \ge 0$ it implies $S_4 \ge 45/7$. Also

$$S_4 = 45 + 10x - 2y - 3z \le 45 - 2y \le 35.$$

Hence the previously imposed interval $45/7 \le S_4 \le 35$ is already contained in the displayed system. Neither endpoint is an extra assumption.

The sixth-sector spectral inequality. The analogous proved bound $S_6 \ge 1 + (46/3)q$ is equivalent to $70x - 12y - 11z \le 20$, and is already implied by the baseline system:

$$70x - 12y - 11z = \frac{5}{3}(-6x + 3z) + 4y + 16(5x - y - z) \le 40 + 60 - 80 = 20.$$

Central purity convexity and log-convexity. For pure six-qubit states, the central conditions $r_2 - 2r_3 + r_4 \geq 0$ and $r_2 r_4 - r_3^2 \geq 0$ are equivalent, by $r_4 = r_2$, to $r_2 \geq r_3$, namely

$$S_4 \geq 2y - 15 \quad \Longleftrightarrow \quad 4y + 3z \leq 60 + 10x. \tag{286}$$

The cap proves this directly, since

$$\Phi(y) - (2y - 15) = \begin{cases} 8(12 - y)/7, & 5 \leq y \leq 99/10, \\ 80(1 - t)^2/3, & 99/10 \leq y \leq 15, \end{cases}$$

where $t = \sqrt{(y - 5)/10}$. Both expressions are nonnegative on their domains.

The noncentral quantitative conditions previously called Conjecture 5. Their $N = 6$ instances are proved in Ref. [4], and are already implied by the baseline system:

$$36y \geq 15x^2 + 5(6 - x)^2, \tag{287}$$

$$5(6 + x)(4y + 3z) \geq 2(5x + 2y)^2 + 4(15 - y)^2. \tag{288}$$

For (287), the baseline gives $0 \leq x \leq 6$ and $y \geq \max\{5, 3x - 3\}$. Its gap is $36y - 20x^2 + 60x - 180$, bounded below by $20x(3 - x) \geq 0$ for $x \leq 3$ and by $4(6 - x)(5x - 12) \geq 0$ for $3 \leq x \leq 6$. For (288), the exact nonnegative factorization is

$$\begin{aligned} &5(6 + x)(4y + 3z) - 2(5x + 2y)^2 - 4(15 - y)^2 \\ &\quad = 4(15 - y)(5x + 3y - 15) + 5(6 + x)(3z - 10x) \geq 0. \end{aligned}$$

The ordinary noncentral log-convexity bounds $12y \geq 5x^2$ and $5(6 + x)(4y + 3z) \geq 2(5x + 2y)^2$ follow immediately by dropping the extra squared terms. Thus no pure-state conjecture is being assumed in the figure. Equality annotations for redundant conditions are omitted.

The former simplified red cap. The explicit conditions (265)–(266) are consequences of the stronger implicit condition (6j), which is now part of the defining system. They therefore add no restriction and are not plotted separately.

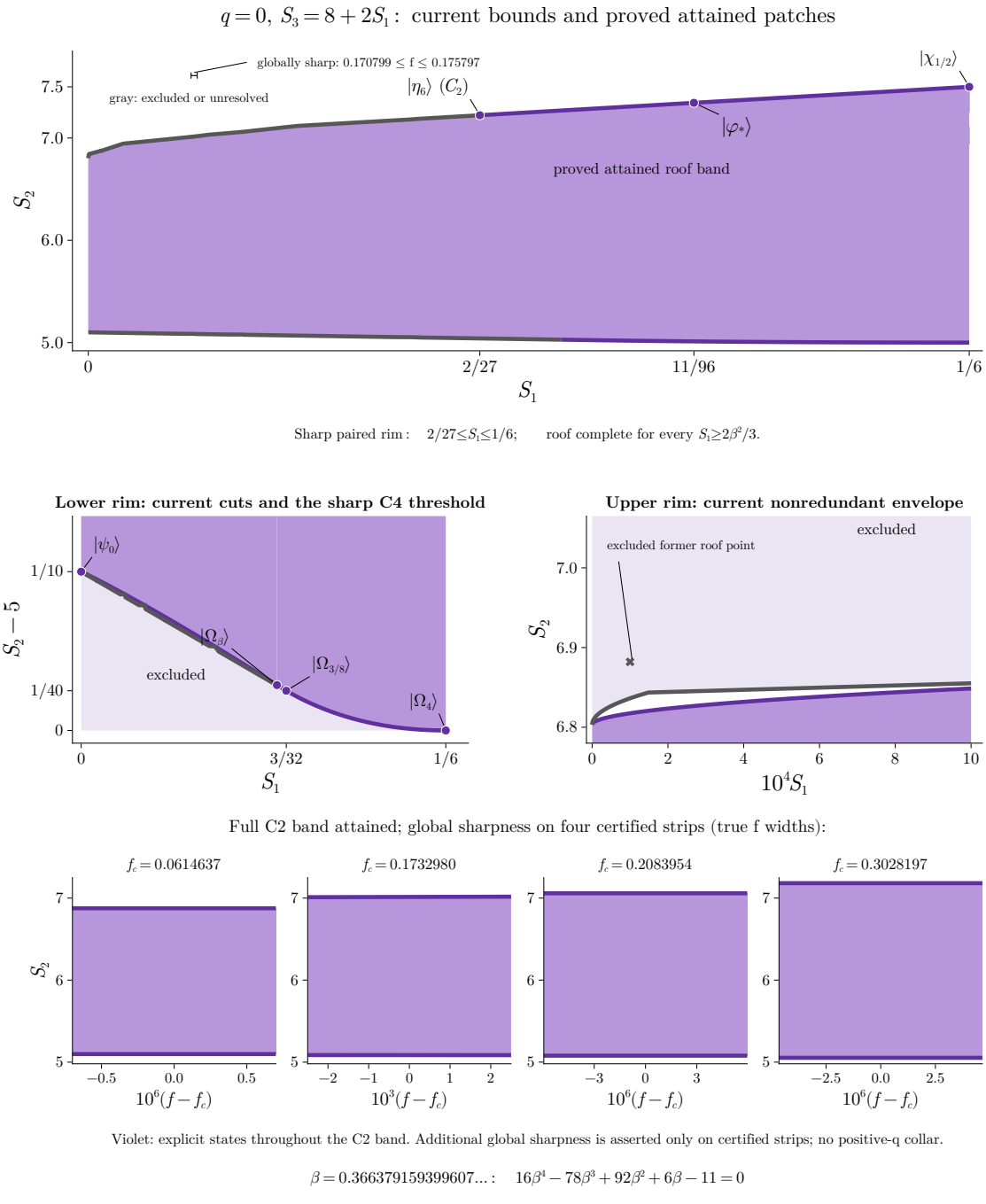


Figure 10: Resolving the small-dipole $q = 0$ roof. The full violet C_2 band is the complete restricted image in Theorem 22, drawn from the normalized family (438), not from a hull. Its sharp unrestricted status is proved on four finite strips; the enlarged second strip $[.170799, .175797]$ is resolved at its actual width in a detail panel. Outside the certified strips, constructive coverage is not a proof that no other state lies beyond the band. The full violet paired rim for $2/27 \le x \le 1/6$ is attained by (200), with exact inverse (204); its contact φ_* with the four-level family is (184). The newly labeled node $|\eta_6\rangle$ is the sharp C_2 lower endpoint at $x = 2/27$. Theorem 9 excludes the rest of the null paired line below it and classifies every state at equality, without a real-amplitude assumption. The interior of the old straight lower rim is unattainable by Theorem 6; no guessed finite-gap curve replaces it. The small-x inset resolves the current stronger cap from Theorem 7, with only nonredundant boundaries displayed. The unquantified strip-existence theorem 10 is not used to assign an arbitrary plotting width. Other gray roof points remain unresolved unless an explicit exclusion is stated.

23 A sharp singlet-dependent plane and its state ω

23.1 Multipole normalization and the proof in one paragraph

Here and below $\mathbf{F} = \langle \mathbf{J} \rangle$ and $Q_{ij} = \langle (J_i J_j + J_j J_i)/2 \rangle - 4\delta_{ij}$. Thus $x = 2|\mathbf{F}|^2/3$, $y = 5 + 4\|Q\|_F^2/15$ and $q = |\Theta|^2 = (8 + 2x - z)/8$. The rank-four multipole length a_4 uses orthonormal Hilbert–Schmidt spherical tensors, with $a_0 = 1/7$. The idea is to separate the singlet contribution from this nonnegative length. The remainder is not manifestly positive in $\mathbf{F}, Q$, but becomes a positive quadratic form in five irreducible blocks of quartic spin covariants. This gives a short logical proof, with longer exact coefficient data supplied below for reproducibility. It is a finite spin-projection Gram proof, not a certificate-free scalar proof.

Theorem 13 (Sharp rank-four bound)**.** *Every normalized pure symmetric six-qubit state satisfies*

$$\boxed{a_4 \geq \frac{9}{286}|\Theta|^2, \qquad S_4 \geq \frac{6S_2 - 9}{7} + \frac{24}{13}|\Theta|^2, \qquad 65S_2 + 63S_3 - 217S_1 \leq 1011.} \tag{289}$$

All three forms are equivalent; the coefficient $9/286$ *is optimal.*

Proof. Homogenize by writing $N = \langle \psi|\psi \rangle$, $F_i = \langle \psi|J_i|\psi \rangle$, $\Theta = \sum_{k=0}^{6} (-1)^k c_k c_{6-k}$, $u = |\mathbf{F}|^2$, $q = |\Theta|^2$, $\tau = \|Q\|_F^2$, where now $Q_{ij} = \langle \psi|\widehat{Q}_{ij}|\psi \rangle$ and $\widehat{Q}_{ij} = (J_i J_j + J_j J_i)/2 - 4\delta_{ij} I$. The exact pure-state multipole conversion is

$$L := 8N^2 + \frac{8}{3}u - \frac{16}{21}\tau + 24q = \frac{176}{3}a_4.$$

The projection-data subsection 25.6 defines explicit quartic covariants $B_{J,a}$ and positive rational matrices H_J such that

$$N^2 G = \sum_{J=2}^{6} \sum_{a,b} (H_J)_{ab} \operatorname{Re}\langle B_{J,a}, B_{J,b} \rangle, \qquad G := L - \frac{24}{13}q. \tag{290}$$

Every block is positive definite, so $G \geq 0$ for every nonzero state. For $N = 1$, substitution of $u = 3x/2$, $\tau = 15(y - 5)/4$ gives

$$G = \frac{4}{91}(1011 + 217x - 65y - 63z).$$

The remaining claims follow from $L = 176a_4/3$ and the explicit equality state below, whose overlap is nonzero. □

23.2 The new labeled equality state and the scope of the equality line

$$\boxed{|\omega\rangle = \frac{(\sqrt{21} + 3\sqrt{2})D_1 + 7\sqrt{2}D_3 + (\sqrt{21} - 3\sqrt{2})D_5}{4\sqrt{11}}.} \tag{291}$$

Direct spin-three ladder contractions give

$$\Theta(\omega) = -\frac{13}{22}, \qquad (S_1, S_2, S_3)_\omega = \left(\frac{63}{121}, \frac{618}{55}, \frac{756}{121} \right). \tag{292}$$

Inserting these fractions into (289) gives equality. The nonzero overlap proves optimality of the coefficient.

Proposition 7 (Exact equality segment and its maximal endpoint)**.** *Every equality state of Theorem 13 has*

$$x = \frac{3}{2} - \frac{474}{169}q, \quad y = \frac{99}{10} + \frac{3234}{845}q, \quad z = 11 - \frac{2300}{169}q, \qquad 0 \leq q \leq \frac{169}{484}. \tag{293}$$

At $q = 0$ *there is one rotation/phase orbit, represented by* $\sqrt{7/10}D_0 + \sqrt{3/10}D_5$*. At* $q = 169/484$ *there is one rotation/phase orbit, represented by* ω*, whose projective collective stabilizer is* C_2*. Conversely, every sector tuple in* (293) *is attained by a normalized pure symmetric state.*

The reduction below proves necessity and classifies the endpoint orbits. The explicit family in Theorem 14 proves sufficiency for every intermediate q. Thus this proposition determines the exact *sector image on the supporting plane*. Theorem 15 below gives the finer classification of all individual equality states: the displayed real family is not exhaustive. The full segment is colored attained; the rest of the two-dimensional facet remains gray.

A genuine cut and the new gray junctions. The plane removes finite volume from the previous plotted region. For example the rational point $(x, y, z) = (551/400, 157/16, 1067499/100000)$ satisfies all its previous conditions strictly but violates the new one by $41937/100000$. For direct verification, $q = 8001/800000 > 0$, $\Delta = 140001/100000 > 0$, the old orange slack is $3/100000$, and, for $L = 5y + 6z - 15x - 85 > 0$, $40\Delta - L^2 = 1246984991/2500000000 > 0$. The full Gram family is inactive since $10(y - 5) > 1$, and the old implicit cap is inactive since $(14/5)^2 x > 10$. The two lower sheets are strictly below z. Continuity therefore gives an open excluded neighborhood.

Every state on the new plane must lie on (293); its other points are excluded, not merely unresolved. In particular the geometric junction $(7/16, 185/16, 45/8)$ has $q = 13/32 > 169/484$ and is not attained. The orange/paired/new-plane junction is

$$t_0 = \frac{196 + 3\sqrt{91}}{262}, \quad y_0 = 5 + 10t_0^2, \quad x_0 = (33 - y_0)/49, \quad z_0 = 15 + 5x_0 - y_0.$$

Here $y_0 > 12 > 618/55$, whereas every equality state has $y \le 618/55$ by Proposition 7. Thus this new gray junction is excluded as well. These excluded junctions lie outside the fully attained segment.

Equality reduction. We give the contractions explicitly to fix the scope of this result. Let $P_L = [\psi \otimes \psi]_L$, $A_L = [P_L \otimes \mathsf{T}\psi]_3$, $B_L = [P_L \otimes \mathsf{T}\psi]_1$ ($L = 2, 4$ for B_L), and use $\mathrm{bil}_J(v, w) = \sum_k (-1)^k v_k w_{2J-k}$. Set

$$\begin{aligned}
\mathscr{D} &= \left[\frac{21}{2} NI - Q : \widehat{Q} + \frac{7}{2}\mathbf{F}\cdot\mathbf{J}\right]\psi + \frac{378}{13}\Theta\mathsf{T}\psi,\\
C_1 &= 13\sqrt{5}B_2 + 4\sqrt{22}B_4, \qquad C_4 = 507\,\mathrm{bil}_2(P_2, P_2) - 112\Theta^2,\\
V_{ij} &= \langle \mathsf{T}\psi | \widehat{Q}_{ij} | \psi\rangle,\\
K_e &= (Q^2)_{\mathrm{tf}} - \frac{49}{12}(\mathbf{F}\mathbf{F}^T)_{\mathrm{tf}} - \frac{49}{13}\,\mathrm{Re}(\overline{\Theta}V),\\
K_o &= \mathrm{Im}(\overline{\Theta}V) - \frac{13}{42}([\mathbf{F}]_\times Q - Q[\mathbf{F}]_\times).
\end{aligned}$$

Here $[\mathbf{F}]_\times v = \mathbf{F} \times v$ and tf means the symmetric tracefree part. At equality every $B_{J,a}$ in (290) vanishes, because every H_J is strictly positive. The derivative of $21G/16 \ge 0$ gives $\mathscr{D} = 0$. The following exact projection memberships give $C_1 = K_e = K_o = 0$:

$$\begin{aligned}
\frac{[C_1 \otimes \mathsf{T}\psi]_4}{35\sqrt{30}} &= \frac{4}{35}B_{4,1} + \frac{227}{539}B_{4,2} + \frac{4i}{35}B_{4,3} + \frac{i}{77}B_{4,4} + \frac{36}{77}B_{4,5},\\
(K_e)_{22} &= (0, -5700/49, 0, 275/7, -45240/121)\cdot \boldsymbol{B}_2,\\
(K_o)_{22} &= (195/7, 0, 325/7, 0, 0)\cdot \boldsymbol{B}_2.
\end{aligned}$$

Here $\boldsymbol{B}_2 = (B_{2,1}, \ldots, B_{2,5})^T$, not the cubic spin-one covariant B_2 used in the contractions. The highest spherical tensor component is $K_{22} = (K_{xx} - K_{yy} - 2iK_{xy})/2$; lowering gives the other components. The Cartan product $[C_1 \otimes \mathsf{T}\psi]_4$ is multiplication of binary spin polynomials and has no zero divisor, so its vanishing implies $C_1 = 0$. Two universal contractions, checked by the same finite

CG formula as the projection data, are as follows. Abbreviate $R_0 = 169(4u - 9N^2) + 2844q$.

$$\begin{aligned}\overline{\Theta}C_4 = {}& \frac{3679\sqrt{7}}{1029}\,\mathrm{bil}(\mathscr{D}, A_0) - \frac{845\sqrt{35}}{4116}\,\mathrm{bil}(\mathscr{D}, A_2) - \frac{9295\sqrt{7}}{686}\,\mathrm{bil}(\mathscr{D}, A_4)\\ &+ \frac{20449\sqrt{91}}{4116}\,\mathrm{bil}(\mathscr{D}, A_6) - \frac{507\sqrt{22}}{28}\,\mathrm{bil}(C_1, B_4),\\ \Theta R_0 = {}& -\frac{247299\sqrt{7}}{686}\,\mathrm{bil}(\mathscr{D}, A_0) - \frac{116103\sqrt{35}}{2744}\,\mathrm{bil}(\mathscr{D}, A_2)\\ &+ \frac{782301\sqrt{7}}{1372}\,\mathrm{bil}(\mathscr{D}, A_4) - \frac{417261\sqrt{91}}{2744}\,\mathrm{bil}(\mathscr{D}, A_6)\\ &- \frac{59319\sqrt{5}}{98}\,\mathrm{bil}(C_1, B_2) + \frac{150579\sqrt{22}}{392}\,\mathrm{bil}(C_1, B_4)\\ &- \frac{6591}{98}K_e : V + \frac{507i}{2}K_o : V.\end{aligned}$$

For $q > 0$ they give $C_4 = 0$ and $u = 9/4 - 711q/169$; $G = 0$ then gives $\tau = 147/8 + 4851q/338$. This is the displayed sector line. At $q = 0$ the rank-four-zero classification in Lemma 3 below gives the stated C_5 endpoint. The same contractions give

$$V : V = \frac{4704}{169}\Theta^2, \quad Q : V = \frac{1029}{26}\Theta, \quad \|V\|_F^2 = \tau + 30 - \frac{15}{2}u - 30q.$$

Choose phase so $\Theta = \sqrt{q}$. The Gram determinant of $Q, \mathrm{Re}\, V$ is therefore

$$\det \mathrm{Gram}(Q, \mathrm{Re}\, V) = \frac{9261(64q - 169)(484q - 169)}{913952} \geq 0.$$

Since $q \leq 1$, this proves $q \leq 169/484$.

At the upper endpoint $Q = \mathrm{Re}\, V$. Write $\psi = \sqrt{35/44}\, a + i\sqrt{9/44}\, b$ with a, b real under time reversal, unit and orthogonal. Then $Q(b) = 0$, so the zero-dipole case of the side-face classification below makes b octahedral. Choose $b = (D_1 - D_5)/\sqrt{2}$. An orthonormal time-real basis of its orthogonal complement is

$$e_i = \frac{i}{2}J_i b, \qquad f_z = iD_3, \quad f_x = e^{-i\pi J_y/2} f_z, \quad f_y = e^{i\pi J_x/2} f_z.$$

Write $a = \sum p_i e_i + \sum t_i f_i$ with real p_i, t_i. The dipole and quadrupole values force $P := \sum p_i^2 = 3/10$, $T := \sum t_i^2 = 7/10$ and $\|Q(a)\|_F^2 = 924/25$. To see the rigidity transparently, put $w_i = p_i^2 - 3t_i^2/7$ and $U_{ij} = (p_i p_j, t_i t_j, \sigma_{ij}(p_i t_j - p_j t_i))^T$, with $\sigma_{xy} = \sigma_{yz} = 1$, $\sigma_{xz} = -1$. Direct spin-three multiplication gives the small identity

$$120PT + 24T^2 - \|Q(a)\|_F^2 + \frac{385}{4}(P - 3T/7)^2 = \frac{385}{4}\sum_i w_i^2 + \sum_{i<j} U_{ij}^T K U_{ij},$$

$$K = \begin{pmatrix} 80 & 75 & 15\sqrt{15}/2 \\ 75 & 720/7 & 27\sqrt{15}/2 \\ 15\sqrt{15}/2 & 27\sqrt{15}/2 & 30 \end{pmatrix} = L\,\mathrm{diag}(80, 3645/112, 1/6)L^T,$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 15/16 & 1 & 0 \\ 3\sqrt{15}/32 & 161\sqrt{15}/810 & 1 \end{pmatrix}.$$

Its left side is zero for the required values. Positivity forces p, t to have the same single Cartesian support, with moduli $\sqrt{3/10}, \sqrt{7/10}$. Octahedral rotations and global phase identify all axes and sign pairs. Taking the z axis recovers (291). Its nonzero dipole fixes the stabilizer axis; the occupied Dicke gaps two and four give exactly C_2. □

23.3 Why the zero-overlap endpoint is a unique orbit

Lemma 3. *A normalized pure spin-three state with $a_4 = 0$ is, up to collective rotation and global phase, $\sqrt{7/10}D_0 + \sqrt{3/10}D_5$.*

Proof. Rotate a global maximum of the spin-coherent overlap to the north pole. Coherent-state resolution gives $|c_0|^2 \geq 1/7$; first-order stationarity gives $c_1 = 0$. Scale c_0 to one and use axial phase to put $s = \sqrt{15}c_2/c_0 \geq 0$. The second derivative of the coherent overlap gives $s \leq 3$: near the pole its amplitude is $(1 + s\zeta^2 + O(\zeta^3))/(1 + |\zeta|^2)^3$. Define

$$f(z) = z^6 + sz^4 + tz^3 + uz^2 + vz + w, \qquad (t, u, v, w) = (\sqrt{20}c_3, \sqrt{15}c_4, \sqrt{6}c_5, c_6)/c_0.$$

The antipodal pole and normalization give $|w| \leq 1$ and

$$P := |v|^2 \leq 36 - \frac{2}{5}s^2. \tag{294}$$

The spin-four coupling of ψ and $\mathsf{T}\psi$ is zero exactly when

$$3(f''g + fg'') - 5f'g' = 0, \qquad g(z) = \bar{w}z^6 - \bar{v}z^5 + \bar{u}z^4 - \bar{t}z^3 + sz^2 + 1.$$

This is simply the CG coupling $3 \otimes 3 \to 4$ written with ordinary derivatives, not an invariant-theory classification. Its five independent coefficients give

$$\begin{aligned}
&\bar{u} + s\bar{w} = 0, \qquad -18\bar{t} + 4s\bar{v} + 18t\bar{w} = 0,\\
&36s - 8s\bar{u} - 3t\bar{v} + 36u\bar{w} = 0,\\
&6s\bar{t} - 6t\bar{u} - 16u\bar{v} + 60v\bar{w} = 0,\\
&90 + 2s^2 + 9|t|^2 + 2|u|^2 - 35|v|^2 + 90|w|^2 = 0.
\end{aligned}$$

The last equation ensures $v \neq 0$. Substituting $u = -sw$ and $a = |w|^2$ reduces the middle equations to

$$\begin{aligned}
t - w\bar{t} &= (2s/9)v,\\
H := t\bar{v} &= 12s(1 - a) + (8/3)s^2\bar{w},\\
3s(\bar{t} + t\bar{w}) + 8sw\bar{v} + 30v\bar{w} &= 0.
\end{aligned}$$

If $s = 0$, these force $t = w = u = 0$, and the last coefficient gives $P = 18/7$. Normalizing proves the stated orbit.

Suppose instead $0 < s \leq 3$, so $P < 36$. If $a = 1$, the first equation and its conjugate give $v + w\bar{v} = 0$; the third then gives $3s(\bar{H} - H) + (8sw - 30)P = 0$. Its real part requires $\operatorname{Re} w = 15/(4s) \geq 5/4$, impossible. Thus $a < 1$. Solve the first equation for $t = (2s/9)(v + w\bar{v})/(1 - a)$. The last equation now becomes

$$A\bar{v} + Bv\bar{w} = 0, \qquad A = A_0 + A_1 w, \quad A_0 = s^2(1+a)/3, \quad A_1 = 4s(1-a), \quad B = 2s^2/3 + 15(1-a) > 0.$$

Here $w \neq 0$, since $w = 0$ would force $A_0\bar{v} = 0$. Taking modulus and combining with the displayed expression for H gives

$$|A|^2 = B^2 a, \qquad (1 - a)AH = (2s/9)P(A - Ba).$$

If $\operatorname{Im} w \neq 0$, their imaginary parts and the first modulus equation give, with $X = s^2$, $C = s \operatorname{Re} w$,

$$P = 54(1 - a)^2 - X(1 + a), \qquad C = \frac{B^2 a - A_0^2 - 16X(1 - a)^2 a}{8A_0(1 - a)}.$$

Insert these into the real part of the preceding equation. Elementary collection yields

$$0 = -\frac{2a(a - 1)}{27}(-2X + 45a - 45)(-X + 54a - 99),$$

impossible for $0 < a < 1, X > 0$. Hence w is real.

Then $v/\bar{v} = \varepsilon = \pm 1$. If $\varepsilon = 1$, the equation $A + Bw = 0$ forces $w = -h$, $0 < h < 1$, and $s^2(1-h) = 3h(1+h)(4s+15)$. Since $s^2 \le 9$, this gives $h < 1/6$. The first two reduced equations yield

$$P = (1+h)\{54(1-h^2) - 12sh\} > 93/2 > 36,$$

contradicting (294). If $\varepsilon = -1$, the equation $A - Bw = 0$ gives $w = h$, $0 < h < 1$, and $s^2(1-h) = 3h(1+h)(15-4s)$, so $s \ge 3\sqrt{h}$. Now

$$P = (1+h)\{54(1-h^2) + 12sh\} \ge (1+h)\{54(1-h^2) + 36h^{3/2}\} \ge 36(1+h) > 36.$$

This is the same contradiction. Consequently $s = 0$, the case already settled. The surviving phase of v is removed by an axial rotation. □

23.4 An explicit pure-state family attaining the full sharp-plane segment

The necessity argument above leaves a line segment rather than a two-dimensional attainable facet. We now construct a pure state at *every* point of that segment. The construction uses only real radicals; its verification is by direct spin-three moments and does not invoke the Gram identity that proves the supporting inequality.

Theorem 14 (Full attainment of the sharp-plane equality segment)**.** *For every $1 \le t \le \sqrt{7/5}$ the state Ψ_t defined below is a normalized pure symmetric six-qubit state. Put*

$$r(t) = \frac{13(t^2-1)}{4(5-2t^2)}. \tag{295}$$

Its exact sector tuple is

$$\boxed{\begin{aligned} S_1 &= \frac{3}{2} - \frac{474}{169} r(t)^2, \\ S_2 &= \frac{99}{10} + \frac{3234}{845} r(t)^2, \\ S_3 &= 11 - \frac{2300}{169} r(t)^2. \end{aligned}} \tag{296}$$

These tuples fill the entire closed segment joining $\chi_{7/10}$ to ω, and satisfy $65S_2 + 63S_3 - 217S_1 = 1011$ identically. At $t = 1$ the state is a collective rotation and phase of $\chi_{7/10}$; at $t = \sqrt{7/5}$ it is exactly $-\omega$.

Closed Dicke coefficients. The following Greek letters are auxiliary real amplitudes, not the multipole lengths a_L. Define

$$g = 13 - 7t^2, \qquad \mathcal{D} = 427 - 639t + 287t^2 - 35t^3, \tag{297}$$

$$\alpha = \sqrt{\frac{35(7-5t^2)}{16g}}, \qquad R = \frac{25(23t^2 - 84t + 91)}{108g}, \tag{298}$$

$$U = -\frac{3(7-3t)g}{\mathcal{D}}, \qquad V = -\frac{24\alpha(t-1)g}{5\mathcal{D}}, \tag{299}$$

$$P = \frac{5\alpha}{54g^2}(1985 - 3484t + 1594t^2 + 84t^3 - 119t^4). \tag{300}$$

All square roots in the following formula are the nonnegative roots:

$$\delta = \sqrt{\frac{R + \sqrt{R^2 - 4P^2}}{2}}, \qquad \epsilon = \frac{P}{\delta}, \tag{301}$$

$$\beta = U\delta + V\epsilon, \qquad \gamma = -V\delta - U\epsilon. \tag{302}$$

Define the polynomial-scaled Dicke amplitudes by

$$\begin{pmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \\ p_6 \end{pmatrix} = \begin{pmatrix} \alpha/3 - \gamma/4 - \epsilon/20 \\ \beta/2 - 1 + 3\delta/10 \\ \alpha - \gamma/4 + 3\epsilon/4 \\ \beta - \delta \\ \alpha + \gamma/4 - 3\epsilon/4 \\ \beta/2 + 1 + 3\delta/10 \\ \alpha/3 + \gamma/4 + \epsilon/20 \end{pmatrix}. \tag{303}$$

The normalized state is

$$\boxed{|\Psi_t\rangle = \frac{1}{\sqrt{\mathcal{N}}} \sum_{k=0}^{6} \frac{p_k}{\sqrt{\binom{6}{k}}} D_k, \qquad \mathcal{N} = \frac{64(5-2t^2)}{27g}.} \tag{304}$$

There is no implicit root choice, matrix-kernel solve, optimization or convex interpolation in this prescription.

Proof. We first establish that the formula exists on its *whole closed parameter interval*, then compute its multipoles, and finally prove that it reaches every sector tuple in the segment.

1. All radicals and denominators are valid. Put $s = t-1$. Since $0 \le s < 1/5$, one has $g \ge 16/5 > 0$ and

$$\mathcal{D} = 40 - 170s + 182s^2 - 35s^3 \ge \frac{143}{25} > 0,$$
$$23t^2 - 84t + 91 = 30 - 38s + 23s^2 \ge \frac{112}{5} > 0.$$

Consequently α is real, $R > 0$ and $\mathcal{N} > 0$. The remaining discriminant has the exact factorization

$$\begin{aligned} R^2 - 4P^2 &= \frac{125\mathcal{D}^2\mathcal{C}(t)}{729g^5}, \\ \mathcal{C}(t) &= -21t^4 - 42t^3 + 64t^2 + 70t - 35. \end{aligned} \tag{305}$$

Indeed

$$\mathcal{C}(1+s) = 36 - 12s - 188s^2 - 126s^3 - 21s^4 \ge \frac{15649}{625} > 0.$$

Thus $\delta > 0$ and $\epsilon = P/\delta$ are well defined, even at both endpoints. In particular their defining relations are

$$\alpha^2 = \frac{35(7-5t^2)}{16g}, \qquad \delta^2 + \epsilon^2 = R, \qquad \delta\epsilon = P. \tag{306}$$

2. Direct spin-multipole calculation. Let $C_k = \binom{6}{k}$, $\widetilde{c}_k = p_k/\sqrt{C_k}$ and $|\widetilde{\Psi}_t\rangle = \sum_k \widetilde{c}_k D_k$. Use the unnormalized moments

$$\widetilde{F}_i = \langle\widetilde{\Psi}_t|J_i|\widetilde{\Psi}_t\rangle, \quad \widetilde{Q}_{ij} = \langle\widetilde{\Psi}_t|(J_iJ_j + J_jJ_i)/2 - 4\delta_{ij}I|\widetilde{\Psi}_t\rangle,$$

and $\widetilde{\Theta} = \sum_{k=0}^{6}(-1)^k\widetilde{c}_k\widetilde{c}_{6-k}$. Substitution of (303), using only (306), gives the four exact identities

$$\sum_{k=0}^{6}\frac{p_k^2}{C_k} = \mathcal{N}, \qquad \widetilde{\Theta} = -r\mathcal{N}. \tag{307}$$

$$\begin{aligned} |\widetilde{\mathbf{F}}|^2 &= \left(\frac{9}{4} - \frac{711}{169}r^2\right)\mathcal{N}^2, \\ \|\widetilde{Q}\|_F^2 &= \left(\frac{147}{8} + \frac{4851}{338}r^2\right)\mathcal{N}^2. \end{aligned} \tag{308}$$

For a self-contained way of checking the expansion, set

$$n = \sum_{k=0}^{6}\frac{p_k^2}{C_k}, \qquad m_2 = \sum_{k=0}^{6}\frac{(3-k)^2p_k^2}{C_k},$$
$$\widetilde{F}_x = \sum_{k=0}^{5}\frac{(k+1)p_kp_{k+1}}{C_k}, \qquad \widetilde{F}_z = \sum_{k=0}^{6}\frac{(3-k)p_k^2}{C_k},$$
$$\mathcal{A} = \sum_{k=0}^{4}\frac{(k+1)(k+2)p_kp_{k+2}}{C_k}, \qquad \widetilde{Q}_{xz} = \sum_{k=0}^{5}\frac{(5-2k)(k+1)p_kp_{k+1}}{2C_k}.$$

The spin-three ladder matrices give

$$\widetilde{Q}_{xx} = 2n - m_2/2 + \mathcal{A}/2,$$
$$\widetilde{Q}_{yy} = 2n - m_2/2 - \mathcal{A}/2, \qquad \widetilde{Q}_{zz} = m_2 - 4n.$$

Reality of the amplitudes gives $\widetilde{F}_y = \widetilde{Q}_{xy} = \widetilde{Q}_{yz} = 0$. Hence $|\widetilde{\mathbf{F}}|^2 = \widetilde{F}_x^2 + \widetilde{F}_z^2$ and

$$\|\widetilde{Q}\|_F^2 = \widetilde{Q}_{xx}^2 + \widetilde{Q}_{yy}^2 + \widetilde{Q}_{zz}^2 + 2\widetilde{Q}_{xz}^2.$$

These formulas and the three quadratic relations verify (307)–(308) by ordinary polynomial algebra; they are not sampled-state tests or positivity certificates.

The state is normalized by division by $\sqrt{\mathcal{N}}$; its moments are divided by $\mathcal{N}$ and therefore obey

$$\Theta = -r, \qquad |\mathbf{F}|^2 = \frac{9}{4} - \frac{711}{169}r^2, \qquad \|Q\|_F^2 = \frac{147}{8} + \frac{4851}{338}r^2.$$

The manuscript's conversions $S_1 = 2|\mathbf{F}|^2/3$, $S_2 = 5 + 4\|Q\|_F^2/15$ and $S_3 = 8 + 2S_1 - 8|\Theta|^2$ now give (296) directly.

3. The entire interval and both endpoints. The overlap amplitude is continuous and strictly increasing:

$$r'(t) = \frac{39t}{2(5-2t^2)^2} > 0, \qquad r(1) = 0, \qquad r\Big(\sqrt{7/5}\Big) = \frac{13}{22}.$$

Thus $q = r^2$ covers every value in $[0, 169/484]$. Since the three sector lengths are affine in q, every point of (293) is attained by an actual pure state, not by a convex mixture of endpoint states. Direct endpoint substitution gives $\Psi_{\sqrt{7/5}} = -\omega$. At $t = 1$ the tuple is $(3/2, 99/10, 11)$ and $q = a_4 = 0$; the rank-four-zero classification in Lemma 3 makes this state a collective rotation and phase of $\chi_{7/10}$. □

Meaning of the new segment in the figures. The single-color views show this entire one-dimensional locus in violet, with the same attained-curve thickness as the other families. The family-colored comparison uses a separate dark-wine stroke and retains the endpoint state labels and leaders. The complete classification in Theorem 15 identifies this family as the upper achiral boundary of a larger state-orbit space; its non-endpoint members have trivial projective collective stabilizer. It does not fill any two-dimensional part of the sharp plane. Together with Proposition 7, it determines the *exact sector image on that plane*, although generic equality states are not rotations and phases of these real representatives.

24 Complete equality-state classification on the sharp plane

The family in Section 23.4 proves attainment of the whole sector-length segment, but does not exhaust the states above it. We now classify those states up to a collective rotation $U^{\otimes 6}$ and a global phase. The result is a two-real-parameter family, with a chirality sign. At a fixed non-endpoint sector tuple, the space of inequivalent equality states is a circle.

Proof strategy and exact-arithmetic status. There are three distinct tasks. First, equality fixes four scalar spin-multipole invariants and leaves one further invariant free. Second, a 3×3 Gram matrix gives its exact domain and puts the dipole and quadrupole into a canonical collective frame. Finally, an explicit spin-three stationarity matrix has a unique kernel state, and its determinant derivatives prove that this state has precisely the prescribed moments. The last step uses finite, exact rational polynomial identities. Their matrix definitions, degree bounds, unisolvent verification grid and calculation procedure are given below. Thus the classification is an exact algebraic, computer-assisted result; it is not inferred from numerical optimization or a sampled orbit census.

24.1 The classification and its explicit canonical family

Throughout this section $D_k = |D_6^{(k)}\rangle$, $0 \le k \le 6$, and

$$\mathbf{F} = \langle \mathbf{J} \rangle, \qquad Q_{ij} = \left\langle \frac{J_i J_j + J_j J_i}{2} \right\rangle - 4\delta_{ij}, \qquad q = |\Theta|^2. \tag{309}$$

The five real rotational invariants used below are

$$u = |\mathbf{F}|^2, \quad \tau = \operatorname{Tr} Q^2, \quad h = \mathbf{F}^T Q \mathbf{F}, \quad b = \mathbf{F}^T Q^2 \mathbf{F}, \quad \delta = \det Q. \tag{310}$$

The symbol h is a scalar invariant, not a Dicke amplitude or an entry of Q. The chirality invariant is

$$\chi = \det(\mathbf{F}, Q\mathbf{F}, Q^2\mathbf{F}). \tag{311}$$

It is invariant under proper collective rotations and changes sign under complex conjugation in the standard Dicke basis.

Theorem 15 (All pure-state equality orbits)**.** *Every normalized pure symmetric six-qubit state satisfying* $65S_2 + 63S_3 - 217S_1 = 1011$ *belongs to exactly one of the following classes, subject only to the sign identification stated below.*

For $q = 0$ *there is one orbit,*

$$\sqrt{\frac{7}{10}} D_0 + \sqrt{\frac{3}{10}} D_5. \tag{312}$$

For $0 < q \le 169/484$*, define*

$$u(q) = \frac{9}{4} - \frac{711}{169} q, \qquad \tau(q) = \frac{147}{8} + \frac{4851}{338} q, \tag{313}$$

$$b(q) = \frac{441}{16} - \frac{9702}{169} q - \frac{853335}{28561} q^2, \qquad \delta(q,h) = \frac{49}{36} h + \frac{16807}{338} q. \tag{314}$$

Choose exactly those $h \in \mathbb{R}$ *for which*

$$\boxed{M(q,h) := \begin{pmatrix} u & h & b \\ h & b & \tau h/2 + \delta u \\ b & \tau h/2 + \delta u & \tau b/2 + \delta h \end{pmatrix} \succeq 0.} \tag{315}$$

Let $\lambda_1 < \lambda_2 < \lambda_3$ *be the three real roots of*

$$\lambda^3 - \frac{\tau}{2}\lambda - \delta = 0, \tag{316}$$

and put

$$w_i = \frac{b + \lambda_i h + (\lambda_i^2 - \tau/2)u}{3\lambda_i^2 - \tau/2} \quad (i = 1, 2, 3). \tag{317}$$

These roots are distinct and $w_i \geq 0$ throughout the domain. In a canonical collective frame choose

$$Q = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3), \qquad \mathbf{F} = (\sqrt{w_1}, \sqrt{w_2}, \varepsilon\sqrt{w_3}), \quad \varepsilon \in \{+1, -1\}. \tag{318}$$

Define the following explicit seven-dimensional matrices:

$$\begin{aligned} \widehat{Q}_{ij} &= \frac{J_i J_j + J_j J_i}{2} - 4\delta_{ij} I_7, \\ A_\pm &= \frac{21}{2} I_7 - Q : \widehat{Q} \pm \frac{7}{2} \mathbf{F} \cdot \mathbf{J}, \\ L &= A_- A_+ - \left(\frac{378}{13}\right)^2 q I_7. \end{aligned} \tag{319}$$

Then $\mathrm{rank}_{\mathbb{C}}\, L = 6$. *With $C = \mathrm{adj}\, L$, the canonical pure state, specified up to its irrelevant global phase, is*

$$\boxed{\rho(q, h, \varepsilon) = \frac{CC^\dagger}{\mathrm{Tr}(CC^\dagger)}.} \tag{320}$$

Equivalently, normalize any nonzero column of C and use its seven entries as Dicke coefficients. If $\det M > 0$, *the two signs give inequivalent orbits and $\varepsilon = \mathrm{sgn}\,\chi$. If* $\det M = 0$, *the two signs give the same orbit. There are no further identifications or equality states.*

Formula (320) is a finite algebraic family: it uses ordered cubic roots, nonnegative square roots and 7×7 cofactors. There is no unknown nonlinear compatibility condition or optimization left to perform. Its sector tuple is

$$(S_1, S_2, S_3) = \left(\frac{3}{2} - \frac{474}{169}q, \frac{99}{10} + \frac{3234}{845}q, 11 - \frac{2300}{169}q\right). \tag{321}$$

The additional parameter h and the chirality sign are invisible to all sector lengths: for pure symmetric states the remaining sectors are already determined by this tuple.

24.2 Necessity: the equality contractions fix the scalar data

We derive the extra invariants in (314), rather than assume a real Dicke frame. The equality reduction in Section 23 already gives (313), $0 \leq q \leq 169/484$, and the $q = 0$ classification of Lemma 3. For $q > 0$, fix phase so $\Theta = r = \sqrt{q} > 0$ and set

$$V_{ij} = \langle \mathsf{T}\psi | \widehat{Q}_{ij} | \psi \rangle, \qquad [\mathbf{F}]_\times v = \mathbf{F} \times v.$$

The vanishing tensors K_e, K_o from that reduction read

$$\mathrm{Re}\, V = \frac{13}{49r}\left[(Q^2)_{\mathrm{tf}} - \frac{49}{12}(\mathbf{F}\mathbf{F}^T)_{\mathrm{tf}}\right], \tag{322}$$

$$\mathrm{Im}\, V = \frac{13}{42r}\left([\mathbf{F}]_\times Q - Q[\mathbf{F}]_\times\right). \tag{323}$$

Their scalar contractions, also established there, are

$$Q : V = \frac{1029}{26} r, \qquad V : V = \frac{4704}{169} q, \qquad \|V\|_F^2 = \tau + 30 - \frac{15}{2}u - 30q. \tag{324}$$

Contracting (322) with Q and using $\mathrm{Tr}\, Q^3 = 3 \det Q$ gives

$$3\delta - \frac{49}{12} h = \frac{50421}{338} q,$$

which is exactly the formula for δ.

To obtain b, diagonalize Q temporarily. For a traceless symmetric 3×3 matrix, a direct entrywise computation gives

$$\big\|[\mathbf{F}]_\times Q - Q[\mathbf{F}]_\times\big\|_F^2 = 4u\tau - 6b.$$

Thus (323) and (324) imply

$$\frac{169}{1764q}(4u\tau - 6b) = \frac{1}{2}\left(\tau + 30 - \frac{15}{2}u - 30q - \frac{4704}{169}q\right).$$

Solving and substituting (313) yields (314). These deductions apply to arbitrary complex equality spinors.

24.3 The exact moment domain and the canonical collective frame

Cayley–Hamilton gives

$$Q^3 = \frac{\tau}{2}Q + \delta I_3. \tag{325}$$

Consequently M in (315) is precisely the Gram matrix of $\mathbf{F}, Q\mathbf{F}, Q^2\mathbf{F}$, and

$$\chi^2 = \det M. \tag{326}$$

This proves necessity of its positive semidefiniteness. We next prove the exact domain and its sufficiency at the level of real moment pairs.

Closed radical endpoints. For $r = \sqrt{q}$, define

$$\alpha_\pm(r) = -\frac{3087r^2(10r+13)}{4394} \pm \frac{63(22r+13)}{17576}\sqrt{(13-8r)^3(13-20r)}, \tag{327}$$

$$\beta_\pm(r) = \frac{3087r^2(10r-13)}{4394} \pm \frac{63(13-22r)}{17576}\sqrt{(13+8r)^3(13+20r)}. \tag{328}$$

All roots are real for $0 \le r \le 13/22$. Direct factorization gives

$$\det M = -\frac{49}{36}(h-\alpha_-)(h-\alpha_+)(h-\beta_-)(h-\beta_+). \tag{329}$$

For $q > 0$ the exact domain is the single interval

$$\boxed{\max\{\alpha_+(\sqrt{q}), \beta_-(\sqrt{q})\} \le h \le \beta_+(\sqrt{q}).} \tag{330}$$

The lower endpoint switches at

$$q_* := \frac{14027}{5375} - \frac{10647\sqrt{21}}{21500} = 0.340340305851403\ldots. \tag{331}$$

Here is an exact sign-chart verification of the interval, including the point where its lower branch changes. For $0 < q < q_*$ the root ordering is $\alpha_- < \beta_- < \alpha_+ < \beta_+$; for $q_* < q < 169/484$ it is $\alpha_- < \alpha_+ < \beta_- < \beta_+$. The discriminant in h of the quartic in (329), up to a nonzero rational constant, is

$$q^8(64q-169)^3(400q-169)(484q-169)^2(86000q^2-448864q+142805)^2.$$

Only q_* is an interior collision in our range. The indicated ordering is fixed by direct comparisons of the explicit radicals at $r = 1/2$ and $r = 59/100$, respectively. The resultant of that quartic and $ub - h^2$, again up to a nonzero rational constant, is

$$q^4(64q-169)^2(484q-169)^2(86000q^2-448864q+142805)^2.$$

There is therefore no other crossing of a quartic root and the Cauchy boundary $|h| = \sqrt{ub}$. At the same two test values, comparison of the explicit radicals shows that the lower quartic interval lies outside $|h| \le \sqrt{ub}$, whereas the upper interval lies inside it. At q_* the common root is $-\sqrt{ub}$. This proves (330) for $ub - h^2 > 0$ and its continuous endpoints; the next calculation treats the degenerate case directly.

Construction, including the aligned strata. If $ub - h^2 > 0$, set

$$\begin{aligned} &z = h/u, && H = b/u - z^2 > 0, && C_0 = \tau/2 - b/u, \\ &a = (\delta + C_0 z)/H - z, && d = \varepsilon\sqrt{C_0 - a^2 - az}. \end{aligned} \tag{332}$$

The square root is real since $C_0 - a^2 - az = \det M/(u^3 H^2)$. Then

$$\mathbf{F} = (0, 0, \sqrt{u}), \qquad Q = \begin{pmatrix} a & d & \sqrt{H} \\ d & -a - z & 0 \\ \sqrt{H} & 0 & z \end{pmatrix} \tag{333}$$

has all the required scalar invariants and $\chi = u^{3/2} H d$.

If $ub - h^2 = 0$, positivity forces the second row of M to equal (h/u) times its first row. Using (314), this forces

$$h = h_{\mathrm{al}}(q) := -\frac{441(316q - 169)^2}{2704(440q + 169)}. \tag{334}$$

Indeed the coefficient of h in the row relation is $C_0 + 49u/36 = -686q(440q+169)/[169(316q-169)] \neq 0$. The remaining equality factors as

$$\begin{aligned} ub - h_{\mathrm{al}}^2 = \frac{35721(64q - 169)(316q - 169)(484q - 169)}{1235663104(440q + 169)^2} \\ \times (86000q^2 - 448864q + 142805). \end{aligned} \tag{335}$$

Thus the only possibilities for $q > 0$ are $q = q_*$ and $q = 169/484$. At either point, with $z = h/u$, take

$$\mathbf{F} = (0, 0, \sqrt{u}), \qquad Q = \operatorname{diag}\left(\frac{-z + \sqrt{2\tau - 3z^2}}{2}, \frac{-z - \sqrt{2\tau - 3z^2}}{2}, z \right). \tag{336}$$

The transverse gap squared is $(1040319 + 52479\sqrt{21})/43000 > 0$ at q_*, and $15435/484 > 0$ at $169/484$. In both cases $M = u(1, z, z^2)^T(1, z, z^2) \succeq 0$. This completes moment-pair sufficiency without excluding a singular chart by a generic-rank assumption. At $q = 0$ the matrix test would also allow the spurious isolated value $h = -63/8$; this is why that endpoint is classified separately by Lemma 3.

Why the ordered quadrupole frame is nonsingular. The cubic discriminant

$$\Delta_Q = \frac{\tau^3}{2} - 27\delta^2$$

is a concave quadratic in h. Its resultant with $\det M$, up to a nonzero constant, is

$$\begin{aligned} &q^6(64q - 169)^2 R(q), \\ R(q) = 1020930304q^4 - 22446877440q^3 - 35772995232q^2 \\ - 5174339248q + 7341576489. \end{aligned}$$

On $[0, 169/484]$ one has

$$\begin{aligned} R'(q) &= 16\{q^2(255232576q - 4208789520) - 4471624404q - 323396203\} < 0, \\ R(169/484) &= \frac{49920018425052048}{214358881} > 0. \end{aligned}$$

Consequently Δ_Q never vanishes on either continuous $\det M = 0$ boundary branch for $q > 0$. Both branches meet at

$$q = \frac{169}{484}, \qquad h = -\frac{9261}{5324}, \qquad \Delta_Q = \frac{147088897515}{453519616} > 0.$$

Continuity proves positivity on both branches, and concavity in h proves it throughout the interval. Thus all three roots in (316) are real and distinct.

The spectral projector of Q at λ_i is

$$\Pi_i(Q) = \frac{Q^2 + \lambda_i Q + (\lambda_i^2 - \tau/2)I_3}{3\lambda_i^2 - \tau/2}.$$

Hence $\mathbf{F}^T \Pi_i(Q)\mathbf{F} = w_i$, proving (317) and $w_i \geq 0$. Equivalently, $M = V\,\mathrm{diag}(w_1, w_2, w_3)V^T$, where the ith column of V is $(1, \lambda_i, \lambda_i^2)^T$; its invertible Vandermonde matrix makes positivity of the weights explicit. A proper ordered eigenframe can flip any two coordinate signs. For all $w_i > 0$, this leaves only the product sign, which is $\operatorname{sgn}\chi$ because the ordered Vandermonde product is positive. If one $w_i = 0$, that unused sign removes the distinction between $\varepsilon = \pm 1$. This proves the canonical moment-pair classification globally.

24.4 The spectral kernel proves existence and uniqueness of the state

The spin-three matrices in the orthonormal Dicke basis are completely specified by

$$\begin{aligned} &(J_z)_{kk} = 3 - k, \quad (J_+)_{k,k+1} = \sqrt{(k+1)(6-k)}, \\ &J_- = J_+^T, \qquad\qquad J_x = (J_+ + J_-)/2, \qquad J_y = (J_+ - J_-)/(2i). \end{aligned} \tag{337}$$

The real matrix $T_{k,6-k} = (-1)^k$ represents time reversal as $\mathsf{T}c = T\overline{c}$, so $\Theta = c^T T c$. With $r = \sqrt{q}$, define the real symmetric matrix

$$K = \begin{pmatrix} \operatorname{Re} A_+ + (378/13)rT & -\operatorname{Im} A_+ \\ \operatorname{Im} A_+ & \operatorname{Re} A_+ - (378/13)rT \end{pmatrix}. \tag{338}$$

The equality Euler equation $\mathscr{D} = 0$ already obtained in Section 23 is exactly

$$A_+ c + \frac{378}{13} rT\overline{c} = 0, \quad \text{or} \quad K \begin{pmatrix} \operatorname{Re} c \\ \operatorname{Im} c \end{pmatrix} = 0, \tag{339}$$

after the phase choice $\Theta = r > 0$.

Lemma 4 (Exact spectral identities)**.** *Temporarily regard the six scalar invariants $(u, \tau, \delta, h, b, q)$ as independent formal variables and write $P = \det K$, $\kappa = \operatorname{Tr}(\operatorname{adj} K)$. On the substitutions* (313)–(314)*, the following are global polynomial identities:*

$$\boxed{\begin{array}{lll} P = 0, & P_u = \frac{7}{4}\kappa, & P_\tau = -\frac{1}{2}\kappa, \\ P_\delta = P_h = P_b = 0, & P_q = \frac{189}{13}\kappa. & \end{array}} \tag{340}$$

Moreover

$$\kappa = -\frac{875693751355552500}{13^{12}} q^3 \mathcal{S}(q, h), \tag{341}$$

where

$$\begin{aligned} \mathcal{S}(q,h) = {} & 4942652416h^2 + (157996710144q + 119202875064)h \\ & + 106662334464q^3 + 417664093392q^2 + 4136455642410q - 1245244319865. \end{aligned} \tag{342}$$

An exact finite-arithmetic proof of all these identities is supplied in Subsection 24.7.

Lemma 5 (No rank drop on the entire physical domain)**.** *For every $0 < q \leq 169/484$ and every h satisfying* (315)*, $\kappa > 0$. Consequently* $\operatorname{rank}_{\mathbb{R}} K = 13$.

Proof. Both u and b are positive and decreasing on the closed q interval, so $ub \le (63/8)^2$ and $h \ge -63/8$. Therefore

$$\partial_h \mathcal{S} \ge 14394744(10976q + 2873) > 0.$$

It is enough to prove $\mathcal{S}(q, \beta_+(\sqrt{q})) < 0$. Put $r = \sqrt{q}$, $D = (13 + 8r)^3(13 + 20r)$. Direct substitution in (342) gives $\mathcal{S}(q, \beta_+) = S_0 + S_1\sqrt{D}$, where

$$\begin{aligned} S_0 &= 2528253(8r + 13)^2 F(r), \\ F(r) &= 4112r^4 - 1976r^3 + 1326r^2 + 2704r - 2197, \\ S_1 &= 194481(8r + 13)(13 - 22r)(160r^2 - 104r + 169) \ge 0. \end{aligned}$$

For $0 \le r \le 13/22$,

$$F'(r) \ge 2704 - 5928(13/22)^2 > 0, \qquad F(13/22) = -\frac{1245699}{29282} < 0,$$

so $S_0 < 0$. The exact quadratic-field norm is

$$S_0^2 - DS_1^2 = 992623121070084r^4(8r + 13)^4 G(r),$$

with

$$G(r) = 33344r^4 - 89024r^3 + 108160r^2 - 87880r + 28561.$$

Here $G''(r) = 128(3126r^2 - 4173r + 1690) > 0$, since the inner quadratic has discriminant -3717831. Nevertheless $G'(13/22) = -34325928/1331 < 0$, so G is decreasing on our interval, and $G(r) \ge G(13/22) = 1399489/14641 > 0$. For $r > 0$ the norm is strictly positive. Since $S_0 < 0$ and $S_1 \ge 0$, this proves $S_0 + S_1\sqrt{D} < 0$, hence $\mathcal{S} < 0$. Equation (341) now gives $\kappa > 0$. Finally $\det K = 0$ while its adjugate is nonzero, so K has rank 13. The proof includes both aligned points and does not omit a singular normalization chart. □

Completion of Theorem 15. Take any admissible canonical moment pair and let $v = (x, y)^T$ be a unit real kernel vector of K. Its unique rank-one projector is $vv^T = \operatorname{adj} K/\kappa$, and $c = x + iy$ has unit norm. The determinant derivative identity $dP = \operatorname{Tr}[(\operatorname{adj} K)dK] = \kappa v^T(dK)v$ gives

$$\nabla_{\mathbf{F}} P = \frac{7}{2}\kappa\, \mathbf{F}_c, \qquad \nabla_Q P = -\kappa Q_c,$$

where $\mathbf{F}_c, Q_c$ are the actual moments of c and the second gradient is in the real tracefree symmetric matrix space. On the other hand, the invariant derivatives in (340) give $\nabla_{\mathbf{F}} P = (7/2)\kappa\mathbf{F}$ and $\nabla_Q P = -\kappa Q$. Thus $\mathbf{F}_c = \mathbf{F}$ and $Q_c = Q$. Also $\partial_r P = 2rP_q = (378/13)\kappa r$; differentiation of (338) identifies this with $(378/13)\kappa \operatorname{Re}\Theta_c$. Hence $\operatorname{Re}\Theta_c = r$. Taking the imaginary part of $c^\dagger(A_+c + (378/13)rT\overline{c}) = 0$ gives $\operatorname{Im}\Theta_c = 0$, since A_+ is Hermitian and $r > 0$. The reconstructed state therefore has every prescribed moment and overlap. Substitution in the pure-state sector conversions gives (321), so it is an equality state.

There is exactly one such state up to phase: any other equality state with these moment data can have its phase fixed by $\Theta = r$, and then its realification lies in the same one-dimensional kernel of K. This proves existence and exhaustion, not just a count of invariants.

For the shorter formula (320), let $s = (378/13)r$. The unitary change of coordinates $(x, y) \mapsto (c, T\overline{c})/\sqrt{2}$ changes K into

$$\begin{pmatrix} A_+ & sI_7 \\ sI_7 & A_- \end{pmatrix}.$$

The identity $T\overline{A_+}T = A_-$ follows from the even time parity of $\widehat{Q}$ and odd parity of $\mathbf{J}$. Since $s \ne 0$, the first block row determines the second block component as $-A_+c/s$; the second then becomes $Lc = 0$. Thus $\dim_{\mathbb{C}} \ker L = 1$. Its adjugate has rank one and nonzero columns spanning that right kernel, proving (320). The doubled kernel possesses the time-reversal real structure just displayed, so this reduction introduces no spurious complex solution. The moment-frame classification proved above now supplies precisely the stated orbit identifications. The independent $q = 0$ rigidity result finishes the classification at the remaining endpoint. □

24.5 Chirality, the exceptional C_2 orbit and the circle of states

The matrix M is positive definite for h strictly between its two endpoints, so $\chi \neq 0$ there. Complex conjugation changes the sign of χ and interchanges the two inequivalent canonical orbits. On either boundary, $\chi = 0$ and the signs coincide. Such boundary states admit real Dicke coefficients in a collective frame: put the zero component of $\mathbf{F}$ on the y axis while Q stays diagonal. Then A_+ is real and K is block diagonal between real and imaginary spinors. Its simple real kernel lies entirely in one block, so a global phase makes the Dicke coefficients real. Conversely a real Dicke spinor has $F_y = Q_{xy} = Q_{yz} = 0$, hence $\chi = 0$.

The real family Ψ_t of Section 23.4 has $h = \beta_+(\sqrt{q})$. It covers only the upper achiral boundary, not the other real boundary or the intervening chiral states. For every $0 < q < 169/484$, doubling the nondegenerate h interval and identifying its two endpoint signs gives a circle of rotation/phase orbits. All states on that circle share the same sector tuple (321). The circles collapse to the unique C_5 orbit at $q = 0$ and the unique ω orbit at $q = 169/484$.

For $q > 0$, Q has distinct eigenvalues. A projective collective stabilizer must therefore preserve its ordered eigenaxes and the dipole. If at least two components of $\mathbf{F}$ in that frame are nonzero, the only proper rotation doing so is the identity. If only one is nonzero, the moment-pair stabilizer is exactly the half-turn group C_2. By uniqueness of the reconstructed state, every symmetry of its moment pair is a projective state symmetry as well. One nonzero component is equivalent to $ub - h^2 = 0$, already classified by (335). Thus the only nontrivial projective collective stabilizers for $q > 0$ are the two C_2 orbits: the known state ω and the additional state at

$$q = q_*, \qquad h = h_{\mathrm{al}}(q_*), \qquad \mathbf{F} = (0, 0, \sqrt{u(q_*)}), \qquad Q \text{ as in (336)}. \tag{343}$$

Equations (319)–(320) already give a closed algebraic representative of this exceptional orbit. In the aligned frame it has even Dicke parity, in contrast to the odd-parity representative of ω.

24.6 The exceptional aligned C_2 orbit

Besides ω, the classification contains one further equality orbit with a nontrivial collective rotational stabilizer. We denote it by $|E_*\rangle$. The following formula gives its Dicke coefficients by ordinary radicals and four explicit cubic expressions; it requires no kernel solve or choice of a vanishing denominator.

Proposition 8 (Explicit exceptional equality state)**.** *Put*

$$\ell = \sqrt{21}, \qquad q_* = \frac{14027}{5375} - \frac{10647\ell}{21500}, \qquad r_* = \sqrt{q_*}, \tag{344}$$

$$f_* = \sqrt{\frac{-187677 + 44793\ell}{21500}}, \qquad z_* = \frac{49 - 21\ell}{20}, \qquad g_* = \frac{21(7+\ell)}{26} r_*. \tag{345}$$

Thus $f_* > 0$, $g_* > 0$. Define the following real numbers

$$\alpha_\pm = \frac{21}{2} - \frac{15}{2} z_* \pm \frac{21}{2} f_*, \qquad \beta_\pm = \frac{21}{2} + \frac{9}{2} z_* \pm \frac{7}{2} f_*, \tag{346}$$

$$s_* = \frac{378}{13} r_*, \qquad v_* = -\frac{\sqrt{15}}{2} g_*, \qquad w_* = s_* - 3g_*. \tag{347}$$

The four unnormalized coefficients are

$$\begin{aligned} k_0 &= \beta_+(\beta_-\alpha_- - v_*^2) - w_*^2\alpha_-, \\ k_2 &= -v_*(\beta_-\alpha_- - v_*^2 + w_* s_*), \\ k_4 &= v_*(w_*\alpha_- + \beta_+ s_*), \\ k_6 &= s_*(w_*^2 - \beta_+\beta_-) - w_* v_*^2. \end{aligned} \tag{348}$$

With $\mathcal{N}_* = k_0^2 + k_2^2 + k_4^2 + k_6^2$, set

$$\boxed{|E_*\rangle = -\frac{k_0 D_0 + k_2 D_2 + k_4 D_4 + k_6 D_6}{\sqrt{\mathcal{N}_*}}.} \tag{349}$$

Here $D_k = |D_6^{(k)}\rangle$; the overall minus sign is only a convenient phase convention. The state has

$$\Theta(E_*) = r_*, \qquad \mathbf{F}(E_*) = (0, 0, f_*), \qquad Q(E_*) = \operatorname{diag}\left(\frac{-z_* + g_*}{2}, \frac{-z_* - g_*}{2}, z_*\right). \tag{350}$$

It saturates $65S_2 + 63S_3 - 217S_1 = 1011$ at $q = q_*$ and $h = -\sqrt{u(q_*)b(q_*)}$. It is the unique orbit there up to collective rotations and phase, and its projective collective rotational stabilizer is exactly C_2.

Proof. The formulas are first seen from an elementary four-dimensional reduction. In the even Dicke subspace (D_0, D_2, D_4, D_6), the real phase-gauged equality equation becomes

$$\begin{pmatrix} \alpha_+ & v_* & 0 & s_* \\ v_* & \beta_+ & w_* & 0 \\ 0 & w_* & \beta_- & v_* \\ s_* & 0 & v_* & \alpha_- \end{pmatrix} \begin{pmatrix} k_0 \\ k_2 \\ k_4 \\ k_6 \end{pmatrix} = 0. \tag{351}$$

The four expressions in (348) are simply the first adjugate column of this displayed matrix. Substitution of (344)–(347) gives zero determinant and verifies (351) directly. They cannot all vanish: expansion of the last expression gives

$$k_6 = \frac{83349}{223600}\, r_*(19229 - 4086\sqrt{21}) > 0,$$

because $19229^2 - 21 \cdot 4086^2 = 19151125 > 0$. Consequently $\mathcal{N}_* > 0$ and the state is well defined.

For completeness, its multipoles can be checked without using the general reconstruction theorem. Write $k = (k_0, k_2, k_4, k_6)$ and $N = \sum k_j^2$. Direct substitution in the four cubic formulas gives

$$2(k_0k_6 + k_2k_4) = r_*N, \tag{352}$$

$$3k_0^2 + k_2^2 - k_4^2 - 3k_6^2 = f_*N, \tag{353}$$

$$5k_0^2 - 3k_2^2 - 3k_4^2 + 5k_6^2 = z_*N, \tag{354}$$

$$2\sqrt{15}(k_0k_2 + k_4k_6) + 12k_2k_4 = g_*N. \tag{355}$$

These identities follow by expansion and the three relations

$$\ell^2 = 21, \qquad f_*^2 = \frac{-187677 + 44793\ell}{21500}, \qquad r_*^2 = \frac{14027}{5375} - \frac{10647\ell}{21500}.$$

The left sides are respectively the spin-flip bilinear, J_z, $J_z^2 - 4I$, and $J_x^2 - J_y^2$ ladder contractions. Even Dicke parity and real coefficients make all other dipole components and all off-diagonal quadrupole components zero. This proves (350). The same elementary substitutions give

$$f_*^2 = u(q_*), \quad \frac{3}{2}z_*^2 + \frac{1}{2}g_*^2 = \tau(q_*), \quad f_*^2 z_* = -\sqrt{u(q_*)b(q_*)}.$$

The sector conversions therefore give precisely

$$(S_1, S_2, S_3)_{E_*} = \left(\frac{3}{2} - \frac{474}{169}q_*,\ \frac{99}{10} + \frac{3234}{845}q_*,\ 11 - \frac{2300}{169}q_*\right),$$

which proves saturation. The uniqueness part of Theorem 15 identifies every state at this aligned moment pair with (349), up to rotation and phase.

Finally $f_* > 0$ fixes the z axis, while the quadrupole has three distinct eigenvalues. Indeed $g_* > 0$ and

$$g_*^2 - 9z_*^2 = \frac{3087}{21500}(331\sqrt{21} - 1659) < 0,$$

where $1659^2 - 21 \cdot 331^2 = 451500 > 0$. The only possible nonidentity proper rotation preserving these moments is a half-turn about z. It fixes the state projectively, because all four occupied levels have odd J_z quantum number. The projective stabilizer is therefore exactly C_2. □

As a numerical orientation only, the phase convention in (349) gives

$$|E_*\rangle \simeq 0.276666534D_0 + 0.895775185D_2 + 0.343219868D_4 - 0.056943560D_6.$$

The exact formulas and verification above, not these rounded coefficients, define the orbit.

24.7 Self-contained exact verification of the spectral identities

For clarity, we record a finite procedure proving Lemma 4 from the displayed matrices. It does not assume a precomputed polynomial or an external certificate. The finite arithmetic is substantial, but the reason it proves an all-parameter identity is elementary.

A rational matrix before any equality substitution. In the polynomial-scaled Dicke basis $p_k = \sqrt{\binom{6}{k}}\, c_k$, put

$$(J_+)_{k,k+1} = k + 1, \quad (J_-)_{k+1,k} = 6 - k, \quad J_z = \mathrm{diag}(3, 2, 1, 0, -1, -2, -3).$$

This is a similarity transform of (337); it leaves determinants and adjugate traces unchanged. Use independent variables a, z, f, ℓ, d, r and set

$$A_0 = \frac{21}{2}I_7 - a\widehat{Q}_{xx} + (a + z)\widehat{Q}_{yy} - z\widehat{Q}_{zz} - 2\ell\widehat{Q}_{xz} + \frac{7}{2}fJ_z,$$

$$A_1 = i(J_xJ_y + J_yJ_x),$$

$$K_{\mathrm{pol}} = \begin{pmatrix} A_0 + (378/13)rT & -dA_1 \\ dA_1 & A_0 - (378/13)rT \end{pmatrix}.$$

All entries are rational polynomials. Its determinant is even in f, ℓ, d, r. Compute that determinant exactly, then replace $f^2 = U$, $\ell^2 = H$, $d^2 = D$, $r^2 = q$. Denote the resulting polynomial by $\mathcal{P}(a, z, U, H, D, q)$; it has 3604 monomials. Define, also before any substitution,

$$\mathcal{K} = \frac{2}{21}\left(14\mathcal{P} - a\mathcal{P}_a - z\mathcal{P}_z - 2U\mathcal{P}_U - 2H\mathcal{P}_H - 2D\mathcal{P}_D - 2q\mathcal{P}_q\right). \tag{356}$$

Euler's identity with the scalar diagonal term $21/2$ shows that this is exactly $\mathrm{Tr}(\mathrm{adj}\, K)$, not an additional conjectured coefficient formula.

For an explicit determinant procedure one may use fraction-free elimination. Starting from $B = K_{\mathrm{pol}}$ and $p = 1$, choose a nonzero pivot $v = B_{kk}$ (swap rows if necessary) and update, for $i, j > k$,

$$B_{ij} \longleftarrow \frac{vB_{ij} - B_{ik}B_{kj}}{p}.$$

Every entry uses the old denominator p; only after completing the entire trailing block set $p \leftarrow v$. After the last step, the final diagonal entry, corrected for the row swap sign, is the determinant. All divisions are exact in the rational polynomial ring. Thus the displayed matrix definitions determine the calculation without supplying any expanded polynomial file.

Why forty rational evaluations are conclusive. The determinant is invariant under proper rotations and under Dicke complex conjugation, hence is an $O(3)$ scalar. A scalar polynomial in one real vector and one symmetric tracefree matrix is generated by u, τ, δ, h, b: pairwise index contractions form closed chains $\mathrm{Tr}\, Q^j$ and open chains $\mathbf{F}^T Q^j \mathbf{F}$; pairs of Levi-Civita tensors reduce to Kronecker tensors, and the odd pseudoscalar is excluded by reflection invariance. Cayley–Hamilton (325) reduces all chains to the five listed generators. The determinant is even in r, so its sixth generator is $q = r^2$.

Give these six invariants weights $(2, 2, 3, 3, 4, 2)$ in the order $(u, \tau, \delta, h, b, q)$. Since K is 14×14 and each nonconstant entry is linear in $Q, \mathbf{F}, r$, P has weighted degree at most 14. The substitutions (313)–(314) do not increase weight. Hence its restriction is a polynomial in q, h of separate degrees at most 7 and 4. The invariant derivatives and the residuals in (340) and (341) obey the same bounds (in fact at most 13 weighted degree). It is therefore sufficient to verify them on

$$(q, h) \in \{1, 2, \ldots, 8\} \times \{0, 1, 2, 3, 4\}. \tag{357}$$

These are formal rational evaluations, not physical states; they need not lie inside the admissible moment domain. The evaluation matrix on $q^j h^k$, $0 \le j \le 7$, $0 \le k \le 4$, is the tensor product of two Vandermonde matrices and is invertible. Exact zero values at all 40 grid points therefore prove an identity everywhere.

The entirely rational evaluation and derivative recipe. At each grid point use the rational functions (313)–(314) and put

$$z = h/u, \quad H = b/u - z^2, \quad C_0 = \tau/2 - b/u, \quad a = (\delta + C_0 z)/H - z, \quad U = u, \quad D = C_0 - a^2 - az.$$

None of the denominators vanishes on (357). After evaluating the formal derivatives of $\mathcal{P}$, recover its invariant partial derivatives by the explicit chain rule

$$g_\delta = \frac{\mathcal{P}_a - (2a + z)\mathcal{P}_D}{H}, \qquad g_\tau = \frac{\mathcal{P}_D + z g_\delta}{2},$$
$$g_b = \frac{\mathcal{P}_H - 2g_\tau - (a + z)g_\delta}{u},$$
$$g_h = \frac{\mathcal{P}_z - (2a + 4z)g_\tau - (-a^2 - 2az - D + H)g_\delta - 2uz g_b}{u},$$
$$g_u = \mathcal{P}_U - z g_h - (H + z^2) g_b.$$

The formal $\mathcal{P}_q$ already means differentiation at fixed other invariants, since those five invariants do not involve the independent variable q. The eight residuals to check are

$$\mathcal{R} = \Bigg(\mathcal{P}, \quad g_u - \frac{7}{4}\mathcal{K}, \quad g_\tau + \frac{1}{2}\mathcal{K}, \quad g_\delta, \quad g_h, \quad g_b, \\ \mathcal{P}_q - \frac{189}{13}\mathcal{K}, \quad \mathcal{K} + \frac{875693751355552500}{13^{12}} q^3 \mathcal{S}(q,h)\Bigg). \tag{358}$$

Here is the complete zero-residual result; each vector applies at *all five* indicated h values, with exact rational arithmetic.

q	h values checked	$\mathcal{R}$ at every listed h
1	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$
2	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$
3	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$
4	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$
5	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$
6	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$
7	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$
8	$0,1,2,3,4$	$(0,0,0,0,0,0,0,0)$

Together with the degree and Vandermonde argument, this finite exact calculation proves Lemma 4 globally. No numerical tolerance, unproved generic-rank assertion, external spin-projection coefficient file or optimization result enters this verification. The positive-adjugate proof then establishes existence and uniqueness on the whole physical domain, including the exceptional C_2 stratum.

25 Completed pure-state boundary patches

The statements in this section determine exact images on specified equality faces. Except for (6h), they are not additional unconditional three-dimensional inequalities in the plotted system. Violet means an actual normalized pure-state construction, not interpolation of sector tuples or mixtures of density matrices.

25.1 The complete lower spin-two plane

Theorem 16. *On $y+z=5x+5$, the complete sector image is*

$$0\le q\le 1,\qquad 4(1-q)\le f^2\le 9-10\sqrt{q}+q,\quad x=\frac{2}{3}f^2,\quad y=3x-3+8q,\quad z=8+2x-8q. \tag{359}$$

Every point is attained by a filtered octahedral state or a limiting W–product state. No other part of this plane is attained.

Proof. The plane is $p_2=\|[\psi\otimes\psi]_2\|^2=0$. Write $\psi=\sum c_kD_k$ and $b_k=\sqrt{\binom{6}{k}}\,c_k$. Its five spin-two projection amplitudes, up to nonzero CG factors, are

$$\begin{aligned}&10b_0b_4-5b_1b_3+2b_2^2, &&25b_0b_5-5b_1b_4+b_2b_3,\\ &50b_0b_6-2b_2b_4+b_3^2, &&25b_1b_6-5b_2b_5+b_3b_4,\\ &10b_2b_6-5b_3b_5+2b_4^2.\end{aligned}$$

These equations are preserved by collective $SL(2,\mathbb{C})$ filters. If the spin polynomial $\sum b_kX^{6-k}Y^k$ has a simple root, put it at $[1:0]$, so $b_0=0,b_1\neq0$, and shear $X\mapsto X+tY$ to set $b_2=0$. The equations successively force $b_3=b_4=b_6=0$. Thus the polynomial is $b_1X^5Y+b_5XY^5$: it is in the W orbit if $b_5=0$, otherwise a diagonal filter makes it octahedral. A root of multiplicity two, three or four contradicts, respectively, the terms $2b_2^2,b_3^2,2b_4^2$. If there is no simple root the only remaining partition is six, the product orbit. This proves the classification, including all multiple-root strata.

For $q>0$ only the octahedral orbit is possible. Polar decomposition gives, up to collective rotation,

$$\psi=Z^{-1/2}e^{h\mathbf{n}\cdot\mathbf{J}}O,\quad O=(D_1-D_5)/\sqrt{2},\quad |\mathbf{n}|=1,\quad h\ge0,\qquad q=Z^{-2}.$$

Put $k=\sinh^2h$, $a_i=n_i^2$, $\sigma_2=\sum_{i<j}a_ia_j$, $\sigma_3=a_1a_2a_3$. The spin-three moments give exactly

$$Z=1+8k+(8+40\sigma_2)k^2+48(\sigma_2-\sigma_3)k^3.$$

The full dipole is the polar gradient of this norm: $f^2Z^2=k(1+k)Z_k^2+|\nabla_{S^2}Z|^2/(4k)$, with its continuous limit at $k=0$. If $A=40k^2+48k^3$, $B=-48k^3$, direct differentiation gives

$$\frac{|\nabla_{S^2}Z|^2}{4}=A^2(\sigma_2+3\sigma_3-4\sigma_2^2)+AB(4\sigma_3-12\sigma_2\sigma_3)+B^2(\sigma_2\sigma_3-9\sigma_3^2).$$

Consequently, for $U=1-3\sigma_2\ge0$, $V=1-27\sigma_3\ge0$,

$$9Z^2-10Z+1-f^2Z^2=\frac{320k^2}{9}\{k^2U^2+(4k^2+15k+9)U+(4k^2+3k)V\}\ge0.$$

This proves the upper bound in (359); the lower one is $y\ge5$. At fixed $Z>1$, vary $\mathbf{n}=(\sqrt{(1-c)/2},\sqrt{(1-c)/2},\sqrt{c})$, $1/3\le c\le1$. The norm equation has a unique $k\ge0$ continuously depending on c. The dipole ranges continuously between the threefold upper endpoint and the fourfold lower endpoint, so every intermediate value occurs in an actual state. At $q=0$ use $\sqrt{p}D_0+\sqrt{1-p}D_1$, with $f^2=4+10p-5p^2$. At $q=1$ use O. This proves coverage of every limiting stratum as well. □

25.2 The complete quadrupole-free side face

Theorem 17. *For $y = 5$, put $f = \sqrt{3x/2}$. The exact pure-state image is*

$$0 \le f \le 2, \qquad q_{\min}(f) \le q \le 1 - f^2/4, \qquad z = 8 + 2x - 8q. \tag{360}$$

Here $q_{\min} = 0$ for $1/2 \le f \le 2$. For $0 \le f \le 1/2$ it is the sharp cubic parametrization

$$f = \frac{t(9t^2 - 8)}{2}, \qquad q_{\min} = \frac{(1 - t^2)(3t^2 - 2)^2}{4}, \qquad \frac{2\sqrt{2}}{3} \le t \le 1. \tag{361}$$

The inverse $t(f)$ is unique. The upper-S_3 boundary is attained by

$$\psi_\beta = \sqrt{\frac{3}{8}}(\cos 3\beta\, D_0 + \sin 3\beta\, D_6) + \sqrt{\frac{5}{8}}(-\sin\beta\, D_2 + \cos\beta\, D_4), \quad t = \cos 2\beta. \tag{362}$$

Each point of this small-dipole boundary has a single rotation/phase orbit, including its AC_2 and Ω_4 endpoints.

Spin-multipole reduction and sharp bound. The 3|3 Schmidt matrix is $A_{ij} = c_{i+j}\sqrt{\binom{3}{i}\binom{3}{j}/\binom{6}{i+j}}$. Let $J_{i,3-i} = (-1)^i$ be the real spin-3/2 time-reversal matrix and $B = AJ^\dagger$. The even part of the marginal has only scalar and quadrupole components; hence

$$Q = 0 \quad \Longleftrightarrow \quad \{B, B^\dagger\} = I_4/2.$$

The five Hermitian matrices

$$\Gamma_1 = (j_x^2 - j_y^2)/\sqrt{3}, \quad \Gamma_2 = j_z^2 - 5I/4, \quad (\Gamma_3, \Gamma_4, \Gamma_5) = (\{j_x, j_y\}, \{j_y, j_z\}, \{j_z, j_x\})/\sqrt{3}$$

satisfy the Clifford relations. The ten products $\Gamma_a\Gamma_b$ span $\mathfrak{sp}(4, \mathbb{C})$, which contains B. The displayed anticommutator makes B^2 normal. Its scalar-plus-vector Clifford expansion therefore supplies a nonzero real quadrupole H commuting with B. For completeness, if B^2 is scalar, the bivector of B is decomposable, $\xi \wedge \eta$, and the anticommutator also implies $(\xi \wedge \eta) \wedge \overline{(\xi \wedge \eta)} = 0$. The complex span of its plane and conjugate has real fixed dimension at most three in $\mathbb{R}^5$; a nonzero real vector perpendicular to it gives the same commuting H. Thus the singular case is included.

A physical rotation diagonalizes H. Choose its largest-absolute Cartesian eigenvalue positive, use that axis as z, and order the other axes. This gives $H = \Gamma_2 + v\Gamma_1$, $0 \le v \le 1/\sqrt{3}$, up to scalar. The complete physical commutant has the orthonormal time-real basis

$$\begin{aligned}
e_1 &= (3v + 2\sqrt{3}, 0, \sqrt{15}v, 0, \sqrt{15}v, 0, 3v + 2\sqrt{3})/\sqrt{24P},\\
e_2 &= i(2\sqrt{3} - 3v, 0, \sqrt{15}v, 0, -\sqrt{15}v, 0, 3v - 2\sqrt{3})/\sqrt{24M},\\
e_3 &= i(0, v, 0, -\sqrt{10}, 0, v, 0)/\sqrt{2O},\\
P &= 2v^2 + \sqrt{3}v + 1, \quad M = 2v^2 - \sqrt{3}v + 1, \quad O = v^2 + 5.
\end{aligned}$$

Write $\psi = \sum z_i e_i$, $z = a + ib$ with a, b real, and set $S = aa^T + bb^T$, $W = 2a \times b$. Then $|W|^2 = 1 - q$, $SW = 0$ and the two nonzero eigenvalues of S are $(1 \pm \sqrt{q})/2$. Direct projected spin contractions give

$$\begin{aligned}
Q_{zz} &= \mathrm{Tr}(\Lambda S), & \Lambda &= \mathrm{diag}\left(\frac{5(1 + \sqrt{3}v)}{P}, \frac{5(1 - \sqrt{3}v)}{M}, -\frac{20}{O}\right),\\
f^2 &= 4|W|^2 + W^T \,\mathrm{adj}(L) W, & L &= \sqrt{(1 + v^2)/5}\,\Lambda.
\end{aligned}$$

The other diagonal quadrupole entries are fixed multiples of Q_{zz} and the off-diagonal entries vanish. Assume $q < 1$, so $W \neq 0$. Let the compression of L to $W^\perp$ have eigenvalues $r, -s$. The condition $\mathrm{Tr}(LS) = 0$ gives, with $\kappa = (1 - \sqrt{q})/(1 + \sqrt{q})$, $\kappa r \le s$ and $\kappa s \le r$. This undivided form includes

the zero compression. If the eigenvalues of L are $a \ge b \ge 0 > -d$, interlacing and the trace condition imply

$$f^2 = (1-q)(4-rs), \qquad rs \le \min\{ad, d^2/\kappa\}.$$

On $0 \le v \le 1/\sqrt{3}$, d increases and ad decreases:

$$(ad)' = -\frac{40v(3-v^2)(1-\sqrt{3}v)(v^2+\sqrt{3}v+2)}{P^2O^2} \le 0, \qquad \frac{d'}{d} = \frac{v(3-v^2)}{(1+v^2)(v^2+5)} \ge 0.$$

For $q \le 1/81$ the maximum of the last minimum occurs at the unique v with $d/a = \kappa$. It is attained by the principal $a, -d$ plane and $S = \mathrm{diag}((1-\sqrt{q})/2, 0, (1+\sqrt{q})/2)$. Substitution of $t^2 = 8(1+\sqrt{3}v)/(3(v+\sqrt{3})^2)$ gives exactly (361). For $q > 1/81$ the same lower bound is automatic. Strict interlacing and monotonicity classify the interior equality states; the residual two choices are related by a physical half-turn. The endpoint degeneracies are axial rotations, not new state orbits. The triple-angle state verifies the equality representative directly.

The opposite bound is shorter. Align $\mathbf{F}$ with z and project off $\mathsf{T}\psi$; time-reversal oddness gives

$$f^2 \le (1-q)\langle J_z^2 \rangle = 4(1-q).$$

For $q < 1$, equality forces $J_z^2\psi = 4\psi$, so all equality states are the two-level C_4 family $aD_1 + bD_5$. At $q = 1$ the lower-plane classification and its norm formula force the unique unfiltered octahedral orbit. This also justifies the time-real octahedral step in Proposition 7. □

Explicit coverage, not merely extremal states. The following real span extends the null construction in Section 13 to nonzero overlap:

$$\begin{aligned} e_0 &= \sqrt{2}(D_0 + D_6)/3 + i\sqrt{5}D_3/3, \\ e_1 &= \sqrt{3}(D_0 - D_6)/4 + \sqrt{5}(D_2 - D_4)/4, \\ e_2 &= -i(D_1 - D_5)/\sqrt{2}, \qquad \psi = \alpha e_0 + \beta e_1 + \gamma e_2. \end{aligned}$$

For real unit (α, β, γ), compression gives $Q = 0$ and $\Theta = 2\alpha^2 - 1$. For a target with $q < 1$, put $A = (1+\sqrt{q})/2$, $\lambda = (4f^2/(1-q) - 1)/15$, and choose

$$\begin{aligned} \alpha &= \sqrt{A}, \\ \beta &= \sqrt{1-A}\big(\sqrt{1-\lambda} + 2\sqrt{2\lambda}\big)/3, \\ \gamma &= \sqrt{1-A}\big(2\sqrt{2(1-\lambda)} - \sqrt{\lambda}\big)/3. \end{aligned}$$

Then $f^2 = (1-q)(1+15\lambda)/4$. This exact inverse fills $\max(0, 1-4f^2) \le q \le 1 - f^2/4$; $q = 1$ uses e_0. For $f \le 1/2$ the remaining interval has an elementary four-level inverse. Set

$$\begin{aligned} A &= \frac{24 + 23f + \sqrt{160(1-q) - 15f^2}}{128}, \\ p_0 &= A, \qquad p_6 = 3/8 - A, \\ p_2 &= f/2 + 7/8 - 3A, \qquad p_4 = 3A - f/2 - 1/4. \end{aligned}$$

The quadrupole phase equation is the closed triangle

$$\sqrt{60p_0p_2}e^{-i\alpha} + 12\sqrt{p_2p_4} + \sqrt{60p_4p_6}e^{i\gamma} = 0.$$

Choose either of its orientations and use amplitudes $\sqrt{p_0}e^{i\alpha}D_0 + \sqrt{p_2}D_2 + \sqrt{p_4}D_4 + \sqrt{p_6}e^{i\gamma}D_6$. Its exact Heron sign is nonnegative for $A_L = 3/16 + 3f/8 \le A \le A_{\max}$, where $A_{\max}$ is the D_0 population in (362). Indeed the sign reduces to the cubic

$$\begin{aligned} C_f(A) = &-331776A^3 + (165888f + 186624)A^2 \\ &- (27648f^2 + 62208f + 31792)A + 1536f^3 + 5184f^2 + 5382f + 1587. \end{aligned}$$

It is strictly concave to the right of A_L, $C_f(A_L) = 750f(1-4f^2) \ge 0$, and its first zero strictly to the right of A_L is $A_{\max}$ (with the degenerate $f = 1/2$ limit). The resulting overlap $q = -(1024A^2 - 368Af - 384A + 34f^2 + 69f + 26)/10$ decreases continuously from $1 - 4f^2$ to $q_{\min}(f)$. Thus every interval point is a pure state. This establishes full coverage in (360).

25.3 A self-contained kernel construction for the paired face

Embed a physical vector $c = (c_0, \dots, c_6)$ in the antisymmetric auxiliary spin-two pair, as in (210). In normalized wedge coordinates, let $C_{(ij),k} = \sqrt{2}\langle 2, 2-i; 2, 2-j | 3, 3-k\rangle$ $(i < j)$, and let D be the analogous total-spin-one isometry. Thus $CC^T + DD^T = I_{10}$. For $n \in \mathbb{C}^5$, define $(L_n a)_{ij} = \sum_{k<l,m} \epsilon_{ijklm} n_m a_{kl}$ and $K(n) = C^T L_n C$. Explicitly, for $n = (a, b, c, d, e)$,

$$K(n) = \begin{pmatrix} 0 & 0 & 0 & 0 & \sqrt{10}e/5 & -d & c \\ 0 & 0 & 0 & -2\sqrt{5}e/5 & \sqrt{15}d/5 & 0 & -b \\ 0 & 0 & 2\sqrt{6}e/5 & -\sqrt{2}d/5 & -3c/5 & \sqrt{15}b/5 & \sqrt{10}a/5 \\ 0 & -2\sqrt{5}e/5 & -\sqrt{2}d/5 & 4c/5 & -\sqrt{2}b/5 & -2\sqrt{5}a/5 & 0 \\ \sqrt{10}e/5 & \sqrt{15}d/5 & -3c/5 & -\sqrt{2}b/5 & 2\sqrt{6}a/5 & 0 & 0 \\ -d & 0 & \sqrt{15}b/5 & -2\sqrt{5}a/5 & 0 & 0 & 0 \\ c & -b & \sqrt{10}a/5 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{363}$$

The antisymmetric-pair purity is minimal precisely on $\Delta = 0$, where $AA^\dagger = (I - nn^\dagger)/4$ for a unit kernel n.

Write the Cartesian spin-two kernel as $M = \sum_{m=-2}^{2} n_{2-m} T_m = X + iY$, using

$$T_2 = e_+ e_+^T/2, \qquad T_1 = -(e_+ e_z^T + e_z e_+^T)/2, \qquad T_0 = (3e_z e_z^T - I)/\sqrt{6},$$
$$T_{-1} = (e_- e_z^T + e_z e_-^T)/2, \qquad T_{-2} = e_- e_-^T/2, \qquad e_+ = (1, i, 0)^T, \quad e_- = \overline{e}_+.$$

Put $S = X^2 + Y^2$, $[X, Y] = [v]_\times$, $u = 4|v|^2$, $T = |\operatorname{Tr} M^2|^2$, $\kappa = \operatorname{Tr}(MS)$. For unit n the exact compatibility criterion is

$$\boxed{\mathfrak{f} := 4u + 5T - 1 - 24|\kappa|^2 = 0.} \tag{364}$$

It is equivalent to a unit Takagi singular value of $K(n)$. Every unit solution $K(n)c = \bar{c}$ is a physical pure state with

$$x = \frac{2}{3}u, \quad q = \frac{4u + 5T - 1}{4}, \quad y = 7 + 3x + 8q, \quad z = 15 + 5x - y. \tag{365}$$

These auxiliary variables are not the physical quadrupole Q.

Proof of the criterion and state recovery. Define $(B_{\bar{n}})_{r,(ij)} = \delta_{ri}\bar{n}_j - \delta_{rj}\bar{n}_i$ and $\mathscr{G} = (B_{\bar{n}} C; D^T L_n C)$. Epsilon contraction and completeness give

$$\nu I - K^\dagger K = \mathscr{G}^\dagger \mathscr{G}, \qquad \nu = \|n\|^2.$$

The row relation $(\bar{n}_0, \dots, \bar{n}_4, 0, 0, 0)\mathscr{G} = 0$ and one determinant expansion give $\det \mathscr{G}_{\widehat{0}} = -\bar{n}_0 \mathfrak{f}/125$, where homogeneously $\mathfrak{f} = \nu(4u + 5T - \nu^2) - 24|\kappa|^2$. All other maximal minors are the corresponding $\bar{n}_i$ multiples or zero. Cauchy–Binet yields

$$\det(\nu I - K^\dagger K) = \nu \mathfrak{f}^2/5^6.$$

These finite epsilon/CG identities also follow by 3×3 Cayley–Hamilton in the displayed Cartesian basis. Positivity proves the singular-value criterion, including singular kernels. A Takagi vector $Kc = \bar{c}$ saturates the positive matrix identity, giving $A\bar{n} = 0$ and no spin-one leakage; hence $L_n a = \bar{a}$. Send n to the last axis by a unitary frame change. The remaining four-dimensional relation is $*A = \bar{A}$. Its real self-dual and imaginary anti-self-dual summands commute and have scalar squares. Thus $AA^\dagger = (I - nn^\dagger)/4$. Conversely the Pfaffian/Hodge identity gives this Takagi equation for every flat physical state. This proves necessity and sufficiency, not merely a test on a marginal.

25.4 The complete paired face and its sharp lower edge

Set $u = 3x/2$, $x_* = (15\sqrt{2} - 21)/4$, and $q_\pm(u) = (17 - 8u \pm 5\sqrt{9 - 16u})/32$.

Theorem 18. *The entire pure-state image on $\Delta = 0$ is*

$$0 \le x \le \frac{3}{8}, \qquad q_{\rm low}(x) \le q \le q_+(3x/2), \qquad y = 7 + 3x + 8q, \quad z = 8 + 2x - 8q, \tag{366}$$

where

$$q_{\rm low}(x) = \begin{cases} 1/16 - x, & 0 \le x \le x_*, \\ q_{\rm arc}(3x/2), & x_* \le x \le 2/27, \\ 0, & 2/27 \le x \le 1/6, \\ (6x-1)/4, & 1/6 \le x \le 1/3, \\ q_-(3x/2), & 1/3 \le x \le 3/8. \end{cases}$$

The smooth function is specified without elimination by

$$1 \le R \le 3\sqrt{2} - 3, \qquad u = \frac{(3-R)^3(3+R)}{288}, \qquad q_{\rm arc} = \frac{(R-1)^2(R^2+6R+21)}{192}. \tag{367}$$

Every point between these boundaries is attained by a pure state.

Sharp upper boundary: Bessel's inequality. Choose kernel phase so $\operatorname{Tr} M^2 = t \ge 0$. Then $\operatorname{Tr} XY = 0$, $\|X\|^2 = (1+t)/2$, $\|Y\|^2 = (1-t)/2$ and

$$\|S - I/3\|^2 = (t^2 + u)/6, \qquad |\kappa|^2 \le (1+t)(t^2+u)/12.$$

The criterion gives $u \le (1-t)(1+2t)/2 = 9/16 - (t - 1/4)^2$. At $t = 1$ this follows directly from $Y = 0, u = 0$. It proves $x \le 3/8$ and $q \le q_+(u)$. The old odd-C_2 family fills the ellipse

$$(-17 + 8u + 32q)^2 + 400u \le 225, \tag{368}$$

attaining the upper boundary and the lower boundary for $1/3 \le x \le 3/8$. The next two lower branches are $q \ge 0$ and the already proved paired-spectral bound $q \ge (6x-1)/4$.

Sharp straight lower edge. Let $p = \operatorname{Tr} S^2$, $P = \det S$, $L = \lambda_{\max}(S) \ge 1/3$. The elementary trace identities

$$3p - 2|v|^2 = 1 + T/2, \qquad 3\operatorname{Tr} S^3 - 5p/2 + 1/2 - v^T S v = |\kappa|^2$$

and Cayley–Hamilton give, for $D = 16u + 12T - 3$,

$$(6L-1)\{D - (12L-5)^2\} = 96 v^T (LI - S) v + 4\mathfrak{f}. \tag{369}$$

Compatibility implies $D \ge 0$, exactly $q + x \ge 1/16$. Equality forces $L = 5/12$, $Sv = (5/12)v$. For $u > 0$, the eigenvalue of $M\bar{M}$ in direction v is simple. Since $(M\bar{M})M = M(\bar{M}M)$, a rotation and kernel phase give

$$M = \begin{pmatrix} -m/2 + a & b & 0 \\ b & -m/2 - a & 0 \\ 0 & 0 & m \end{pmatrix}, \quad m = \sqrt{5/12}, \quad a = Ae^{i\chi}, \quad b = iBe^{i\chi},$$

$A^2 = (3 + \sqrt{9 - 64u})/32$, $B^2 = (3 - \sqrt{9 - 64u})/32$. The remaining condition is $\cos 2\chi = -(27 + 32u)/(15\sqrt{9 - 64u})$. Its feasibility is exactly $64u^2 + 1008u - 81 \le 0$, giving $0 \le x \le x_*$. The two conjugate interior orbits are distinct since $I = \operatorname{Im}[\overline{\operatorname{Tr} M^2}\,\kappa^2]$ changes sign and $I^2 = u^2(81 - 1008u - 64u^2)/46656 > 0$ there; they merge at both endpoints. At $u = 0$, commuting X, Y diagonalize with squared diagonal moduli $\{5/12, 5/12, 1/6\}$; the tracefree triangle gives a unique orbit. Physical Takagi uniqueness follows from the odd-block characteristic $\lambda(10\lambda - 9)(10\lambda - 1)/100$ and the positive even-block principal minor $(16b_0^2 + 31)(256b_0^4 - 96b_0^2 + 81)/160000$, $b_0 = A - B$.

Proof of the smooth lower edge. This spectral-dual argument is for arbitrary complex kernels, not a restriction to the displayed equality states. Put $h = \det(M\bar{M})$, $e = v^T S^2 v$, $E = 4u+5T-1 = 4q$. Polarized Cayley–Hamilton gives

$$\begin{aligned} &\operatorname{Tr} S^2 = 1/3 + (T+u)/6, && \det S = (7-3T-4u-24h)/192,\\ &v^T S v = \det S - h, && (e-e_0(h))^2 \le hTE/384,\\ &e_0(h) = (T+u-1)(T+4u-4)/144 - 3h/4. \end{aligned}$$

Indeed $e - e_0 = \operatorname{Re}[\operatorname{Tr}(M^2)\overline{\operatorname{Tr}(M^3)}\kappa]/12$, $|\operatorname{Tr} M^3|^2 = 9h$ and $|\kappa|^2 = E/24$ prove the last inequality. Suppose $u_* < u < 1/9$ and $q < q_{\mathrm{arc}}(u)$, where $u_* = (45\sqrt{2}-63)/8$. Define

$$\begin{aligned} &\lambda = (5-\sqrt{16u+12T-3})/12, && h_0 = \lambda(5-6\lambda-8u)/36, && e_C = u\lambda^2/4,\\ &C_g = -(72\lambda^2-42\lambda+u+5)/108, && g_h = -(-72\lambda^3+114\lambda^2+2\lambda u-47\lambda+5)/96. \end{aligned}$$

The straight-edge classification handles the zero radicand. Otherwise $7/18 \le \lambda < 5/12$, $h_0 > 0$. The matrix $W = (T+u)I/9 - (S-I/3)^2 \succeq 0$, because $S - I/3$ is tracefree. Two spectral norms reduce to

$$\begin{aligned} N_0 &:= \|(S-\lambda I)v\|^2 = (e-e_C) + (9\lambda/4)(h-h_0),\\ G_0 &:= v^T(S-\lambda I)^2 W v = C_g(e-e_C) + g_h(h-h_0) - 9(h-h_0)^2/64. \end{aligned}$$

Put $r_0 = \sqrt{TE/(384h_0)}$, $k_0 = -3/4 + r_0/2$, $\alpha = (-g_h - C_g k_0)/(k_0 + 9\lambda/4)$. The concave upper bound $e_H(h) = e_0(h) + \sqrt{hTE/384}$ lies below its tangent at h_0, yielding

$$0 \le G_0 + \alpha N_0 \le C_{\mathrm{tot}}\{e_H(h_0) - e_C\} - 9(h-h_0)^2/64, \quad C_{\mathrm{tot}} = \frac{-(3\lambda-1)(6\lambda-1)(12\lambda-5)}{96(k_0+9\lambda/4)} > 0. \tag{370}$$

The coefficients have the stated signs: $k_0 + 9\lambda/4 \ge 1/8$, $r_0^2 \le 45/6496 < 1/64$ using $T < 1/6$, $E \le 5/18$, $h_0 \ge 203/11664$. If $C_g > 0$, use $-g_h + 3C_g/4 = (3\lambda-1)(36C_g+u)/96 \ge C_g/16$; otherwise $\alpha > 0$ is immediate. Finally, for $A_0 = e_C - e_0(h_0)$,

$$A_0^2 - h_0 TE/384 = \frac{(-72\lambda^2+42\lambda+8u-5)^2}{1679616}\mathscr{Q}(\lambda,u),$$

$$\mathscr{Q} = 324\lambda^4 - 702\lambda^3 + 36\lambda^2 u + 540\lambda^2 + 42\lambda u - 168\lambda + u^2 - 8u + 16.$$

Its λ derivative is positive on this interval: its u-independent part is at least $175/18$, and the other term is $(72\lambda+42)u$. On the asserted arc, $\lambda = (R^2+6R+21)/72$, $\mathscr{Q} = 0$ and $A_0 = \sqrt{h_0 TE/384}$. Decreasing T at fixed u increases λ, so below the arc $A_0 > \sqrt{h_0 TE/384}$. The right side of (370) is then negative, a contradiction. The endpoints follow by continuity.

Explicit states on the smooth edge. For (367) put

$$\begin{aligned} &s = (5R^2+30R+57)/192, && f = \sqrt{u},\\ &p_0 = (s+f/8)/2, && p_6 = (s-f/8)/2,\\ &p_2 = (1-s+5f/8)/2, && p_4 = (1-s-5f/8)/2. \end{aligned}$$

Then

$$\psi_R = \sqrt{p_0}D_0 + \sqrt{p_2}D_2 + \sqrt{p_4}D_4 - \sqrt{p_6}D_6 \tag{371}$$

has the required sectors by direct multipole contraction. At $R = 1$ it is η_6. Equality in the dual proof forces $(S-\lambda I)v = 0$ and a collapsed purification triangle. The kernel has the same axial block form as on the straight edge, now with $m = \sqrt{\lambda}$. Here put $A^2 = (9-R^2)/48$, $B^2 = (3-R)^2/96$, $b_0 = A - B$. Its remaining phase satisfies $\cos 2\chi = -1$, so the two conjugate charts are related by transverse quarter- and half-turns. This gives one kernel rotation/phase orbit. Its physical unit Takagi space is simple, as seen from the positive even-block principal minor $(R-3)^2(R+5)(96b_0^2+11R^2+66R+87)/240000$ and nonzero odd-block determinant $(4k_0^2-1)^2/25$, where $k_0^2 = 3\lambda/2$. Thus (371) classifies all equality states. All four coefficients are nonzero and $f > 0$, so the stabilizer is C_2.

Every point at small x: an explicit four-level path. For $0 \le u \le 1/9$ use the same populations with $1/4 \le s \le s_{\min}(u) \le 17/32$, and set

$$q(s) = \frac{64s^2 - 68s + 19 - 10u}{15}, \quad R_0 = 4p_0p_6 = s^2 - u/64, \quad W_0 = 4p_2p_4 = (1-s)^2 - 25u/64.$$

Recover actual phases by

$$\ell = \pm \arccos \frac{q - R_0 - W_0}{2\sqrt{R_0 W_0}}, \qquad U_0 = \sqrt{p_0p_2} + \sqrt{p_4p_6}e^{i\ell},$$
$$\alpha = \arg U_0, \qquad \gamma = \ell - \alpha.$$

The state is $\sqrt{p_0}e^{i\alpha}D_0 + \sqrt{p_2}D_2 + \sqrt{p_4}D_4 + \sqrt{p_6}e^{i\gamma}D_6$. The second phase triangle is identically collapsed; the first is physical exactly when $P(s,u) \le 0$, with

$$P = 625u^2 + 6400us^2 + 17200us - 1100u + 16384s^4 - 50176s^3 + 52224s^2 - 19456s + 1024.$$

Take $s_{\min} = 17/32$ for $u \le u_*$ and the smooth-edge value above for $u \ge u_*$. Both endpoints of $[1/4, s_{\min}]$ have $P \le 0$, while $P_{ss} = 196608(s - 17/32)(s-1) + 12800u \ge 0$. Convexity therefore proves the entire path feasible. Since $q'(s) = (128s - 68)/15 \le 0$, its image is $[q_{\text{low}}, 2/5 - 2u/3]$. The latter endpoint lies strictly in the odd-C_2 ellipse, so the two pure-state intervals overlap and fill up to $q_+(u)$, without taking mixtures.

Every remaining point: a real-Hodge bridge. For $x \ge 2/27$ put $r = \sqrt{q}$ and $d = \sqrt{(1 + 4q - 6x)/5}$. The necessary conditions are $0 \le d \le 1$, $q \le (1+d)^2/4$ and $q \ge \max(0, (5d^2 - 5/9)/4)$. The odd ellipse fills $q \ge (1-d)^2/4$. Below it, choose

$$n = ((\ell - t)/\sqrt{6}, (\sqrt{V} + \sqrt{U})/\sqrt{2}, t, (\sqrt{V} - \sqrt{U})/\sqrt{2}, (\ell - t)/\sqrt{6}),$$
$$V = (r_0 + 1 - w)/2, \quad t = (\ell - 1)(2r_0 - w)/(r_0 + 1 - w), \quad U = 1 - V - (\ell - t)^2/3 - t^2.$$

If $r \ge d - 1/4$, take $r_0 = r$, $w = r - d$, $\ell = (1 - \sqrt{1 + 4r - 4d})/2$. Then

$$U = \frac{(1 - d - 2r)\{4r^2 - 3d^2 + 2r + 5d + 2 + (3d + 2r + 1)\sqrt{1 + 4r - 4d}\}}{6(1+d)^2} \ge 0.$$

Otherwise take $r_0 = -r$, $w = d - r$ and $\ell = (1 + \varepsilon\sqrt{1 + 4d - 4r})/2$, where $\varepsilon = 1$ if $3d + 2r - 1 \ge 0$ and $\varepsilon = -1$ otherwise. Either sign is valid at zero; the value zero itself is not used. Here $V = (1-d)/2 > 0$ and

$$U = \frac{(1 + d + 2r)(A + |B|\sqrt{H})}{6(1-d)^2}, \quad A = 4r^2 - 3d^2 - 2r - 5d + 2, \quad B = 3d + 2r - 1, \quad H = 1 + 4d - 4r.$$

The bracket is nonnegative. To check this without optimization, use the convex quadratic

$$A^2 - B^2H = 16q^2 - (24d^2 + 8d)q + 9d^4 - 6d^3 + 28d^2 - 18d + 3.$$

For $d \le 1/4$ one has $A \ge 5/16$. For $1/4 \le d \le 1/3$ its positive component is adjacent to $q = 0$, where $A \ge 0$; elsewhere $|B|\sqrt{H} \ge |A|$. For $1/3 \le d \le 5/6$ its values at $q = (5d^2 - 5/9)/4$ and $q = d(1+3d)/4$ are nonpositive; the former sign follows from the increasing negative cubic $27d^3 - 99d^2 + 141d - 67$ on this interval. This includes the whole second branch since $r < d - 1/4$ implies $q < d(1+3d)/4$; combining this with $q \ge (5d^2 - 5/9)/4$ gives $d \le 5/6$. Thus $U, V \ge 0$. Substitution gives $\mathfrak{f} = 0$ and, more specifically, the real $+1$ eigenvalue condition $(\ell - 2)(\ell^2 - 1) + 2V(2\ell - t - 2) = 0$ for the displayed $K(n)$. A real unit solution of $K(n)c = c$ gives its Dicke coefficients by finite linear algebra. On simple charts any nonzero adjugate column suffices; at a rank drop use the nullspace instead. No denominator-zero chart is silently discarded. This bridge and the small-x path complete the proof of (366).

25.5 The complete zero-overlap roof for $x \geq 3/32$

Theorem 19. *For $q = 0$ and $3/32 \leq x \leq 6$, the exact sector image is*

$$z = 8 + 2x, \qquad \ell(x) \leq y \leq h(x), \tag{372}$$

where $f = \sqrt{3x/2}$ and

$$\ell(x) = \begin{cases} 5 + \frac{2}{5}(1-2f)^2, & 3/32 \leq x \leq 1/6, \\ 5, & 1/6 \leq x \leq 8/3, \\ 3x - 3, & 8/3 \leq x \leq 6, \end{cases} \qquad h(x) = \begin{cases} 7 + 3x, & x \leq 1/6, \\ 5 + \frac{2}{5}(2+f)^2, & x \geq 1/6. \end{cases}$$

Every point is attained by explicit pure-state paths. Below $3/32$ the unrestricted lower endpoint is not asserted solved here.

The mechanism: a circle fixes the entire dipole. Use the real spin-flip bases $E_k^{\pm}$ of (199), with $E_3^- = D_3$, and their column matrices E_+, E_-. For real unit vectors $a \in \mathbb{R}^3$, $b \in \mathbb{R}^4$, the state

$$\psi(a,b) = (E_+ a + E_- b)/\sqrt{2} \tag{373}$$

is normalized and has $\Theta = 0$. Its dipole obeys $F_y = 0$ and $F_x = a^T A_x b$, $F_z = a^T A_z b$, where

$$A_x = \begin{pmatrix} 0 & \sqrt{6}/2 & 0 & 0 \\ \sqrt{6}/2 & 0 & \sqrt{10}/2 & 0 \\ 0 & \sqrt{10}/2 & 0 & \sqrt{6} \end{pmatrix}, \qquad A_z = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

For fixed a, let L have rows $a^T A_x, a^T A_z$ and $G = LL^T$. If $\det G > 0$, define $b_0 = L^T G^{-1} L b$, $w = b - b_0$, and choose a vector $v \in \ker L$ orthogonal to w with $|v| = |w|$. Then

$$b(\theta) = b_0 + \cos\theta\, w + \sin\theta\, v \tag{374}$$

is unit and has exactly the original dipole. The vector v is obtained by normalizing the four-dimensional cofactor vector of the three rows L_1, L_2, w^T. Thus these are algebraic states, not sector interpolation. The reflected point $\theta = \pi$ is $b' = 2b_0 - b$.

Filling the missing low-S_2 band for $1/6 \leq x \leq 8/3$. Put $T_0 = (4f^2 - 1)/15$, $l = \sqrt{1 - T_0}$, $h = \sqrt{T_0}$ and

$$r = (l + 2\sqrt{2}h)/3, \quad u = (2\sqrt{2}l - h)/3, \quad a = (\sqrt{6}r/4, u, -\sqrt{10}r/4), \quad b = (2/3, 0, 0, \sqrt{5}/3).$$

Here $r^2 + u^2 = 1$. Direct contraction gives $Q = 0$, $f^2 = 1/4 + 15h^2/4$ and

$$L = \begin{pmatrix} \sqrt{6}u/2 & -r/2 & \sqrt{10}u/2 & -\sqrt{15}r/2 \\ 3\sqrt{6}r/4 & 2u & -\sqrt{10}r/4 & 0 \end{pmatrix}, \qquad LL^T = 4I.$$

Take v equal to one quarter of the cofactor vector of L_1, L_2, b^T. It has $Lv = 0$, $v \cdot w = 0$, $|v|^2 = |w|^2 = 1 - f^2/4$; at $f = 2$ both vectors vanish and the circle contracts regularly. The reflected state's exact sector length is

$$Y_{\text{ref}} = 5 + \frac{(4 - f^2)(f^2 + 1)(17 - 3f^2)}{50}.$$

The old rank-one band starts at $Y_{\text{rank}} = 5 + 2(2 - f)^2/5$. Their difference is

$$Y_{\text{ref}} - Y_{\text{rank}} = \frac{2 - f}{50}(-3f^5 - 6f^4 + 14f^3 + 28f^2 + 37f - 6) \geq 0.$$

For $1/2 \leq f \leq 2$, the bracket is at least $-10f^3 + 28f^2 + 37f - 6 \geq 8f^2 + 37f - 6 \geq 29/2$. Continuity along the actual circle therefore fills $5 \leq y \leq Y_{\text{rank}}$. The rank-one family of Section 10 fills from there to $h(x)$, and already fills the entire roof for $x \geq 8/3$.

Filling the missing upper band for $3/32 \le x \le 1/6$. Start with the paired-rim state (200), but choose the positive root $s = (-1 + \sqrt{1 + 4h})/2$ in (198), with $h = \sqrt{(1 - 6x)/5}$. The same norm and multipole identities hold for either root of $s(s + 1) = h$. Set $a = \sqrt{2}E_+^T \Psi_h$, $b = \sqrt{2}E_-^T \Psi_h$ in the general circle. In fact this construction is nonsingular on the larger interval $2/27 \le x \le 1/6$. Put

$$P(s) = 25s^4 + 50s^3 + 20s^2 - 5s + 34.$$

Direct multiplication of the displayed matrices gives

$$\det G = P(s)/4 > 0, \quad |w|^2 = \frac{5(s^6 - 13s^5 + 2s^4 + 35s^3 + 10s^2 - 5s + 25)}{4P(s)} > 0.$$

Here $0 \le s < 1/3$; the positive constant terms dominate the indicated negative ones, so no endpoint chart is lost. The reflected state has

$$6 - Y_{\mathrm{ref}} = \frac{N(s)}{8P(s)^4} > 0.$$

For clarity, the entire polynomial is specified by its nonzero degree/coefficient pairs:

Degree	Coefficient	Degree	Coefficient
20	3353125	10	−385448875
19	62106250	9	−405679710
18	278314375	8	−315058875
17	630628750	7	−129806740
16	999224375	6	16940830
15	1262987500	5	145643540
14	1231961250	4	189358835
13	994259500	3	88087690
12	629239805	2	22940595
11	19716470	1	9491510
		0	3512093

All unlisted coefficients are zero. On $0 \le s \le 1/3$, $N(s) \ge 3512093 - 7942799860/59049 > 0$. The circle starts at $y = 7 + 3x$ and ends below six, and thus attains every $6 \le y \le 7 + 3x$.

An elementary even-C_2 path joins the lower endpoint to this band. For $(f - 1)/2 \le e \le 0$, put $\nu = f - 2e$ and

$$c_a = \sqrt{(1 + e)/2}, \quad s_a = \sqrt{(1 - e)/2}, \quad c_b = \sqrt{(1 + \nu)/2}, \quad s_b = \sqrt{(1 - \nu)/2},$$

and use

$$c_a c_b D_0 + c_a s_b D_2 + s_a c_b D_4 - s_a s_b D_6.$$

It has $\Theta = 0$ and $\mathbf{F} = (0, 0, f)$ identically. Its endpoint values are $5 + 2(1 - 2f)^2/5 < 6$ and $33/5 - 6f^2/5 \ge 63/10 > 6$. Its continuous image therefore contains the interval from the lower endpoint through six. Together with the circle it fills the entire claimed roof interval. The universal lower and upper bounds in (372) are the existing Gram, paired-purity and C_5 spectral bounds, so no unrestricted sharpness is inferred merely from these constructions.

The same two explicit paths also prove the attained sub-band

$$\frac{2}{27} \le x < \frac{3}{32}, \qquad 5 + \frac{2}{5}(1 - 2\sqrt{3x/2})^2 \le y \le 7 + 3x, \qquad z = 8 + 2x.$$

This extension is used in the attainment colors. Its lower endpoint is not asserted to be the unrestricted minimum on that smaller-x interval.

A finite algebraic inverse. Set $t = \tan(\theta/2)$. Then $b(t) = b_0 + (1 - t^2)w/(1 + t^2) + 2tv/(1 + t^2)$. For a target y between the proved endpoint values,

$$(1 + t^2)^4 \{S_2(\psi(a, b(t))) - y\} = 0$$

is a real polynomial of degree at most eight with a nonnegative root, by the opposite endpoint signs. This supplies an exact algebraic state prescription, independently of numerical optimization.

What remains gray. The preceding results settle complete equality faces, not the full three-dimensional boundary. Small-dipole null-roof extrema below 3/32 remain separate problems. The sharp-plane sector segment, in contrast, is completely attained by Theorem 14; no surrounding facet area is inferred from this one-dimensional construction. A failed restricted-family reconstruction is never used as an exclusion proof. The extra conditional face bounds above are used to determine attainment colors; they have not silently been imposed as unconditional finite-width cuts on $\mathcal{R}$.

25.6 Exact projection data for the new plane

This appendix specifies every coefficient of the identity used in Theorem 13; no external data file is needed. Write $[v \otimes w]_{JM} = \sum_{m+n=M} C^{JM}_{jm,kn} v_m w_n$. The real Condon–Shortley coefficients used here are

$$\begin{aligned} C^{JM}_{jm,kn} = \delta_{M,m+n} &\sqrt{\frac{(2J+1)(J+j-k)!(J-j+k)!(j+k-J)!}{(j+k+J+1)!}} \\ &\times \sqrt{(J+M)!(J-M)!(j-m)!(j+m)!(k-n)!(k+n)!} \\ &\times \sum_t \frac{(-1)^t}{t!(j+k-J-t)!(j-m-t)!(k+n-t)!(J-k+m+t)!(J-j-n+t)!}, \end{aligned}$$

with only nonnegative factorial arguments included. Index k means $M = J - k$. Put $P_L = [\psi \otimes \psi]_L$ and $A^J_{LK} = [P_L(\psi) \otimes P_K(\mathsf{T}\psi)]_J$, and define

$$R^J_{LK} = (A^J_{LK} + \mathsf{T}_J A^J_{LK})/2, \quad I^J_{LK} = (A^J_{LK} - \mathsf{T}_J A^J_{LK})/(2i), \quad (\mathsf{T}_J v)_k = (-1)^k \overline{v_{2J-k}}.$$

For each displayed column X_J set $B_{J,a} = \sum_r (C_J)_{ra} (X_J)_r$. The symmetric matrices H_J are specified by their nonzero upper-triangle entries. The usual unitary component inner product, without binomial weights, is used.

Spin $J = 2$

$$X_{2,1} = \frac{14}{5\sqrt{2}} R^2_{2,0}, \qquad X_{2,2} = \frac{14}{5\sqrt{2}} I^2_{2,0}, \qquad X_{2,3} = \frac{91}{2\sqrt{3003}} R^2_{6,6}$$

$$X_{2,4} = \frac{143}{15\sqrt{143}} R^2_{6,4}, \qquad X_{2,5} = \frac{143}{15\sqrt{143}} I^2_{4,6}, \qquad X_{2,6} = \frac{49}{25\sqrt{21}} R^2_{2,2}$$

$$X_{2,7} = \frac{154}{25\sqrt{154}} I^2_{2,4}, \qquad X_{2,8} = \frac{154}{25\sqrt{154}} R^2_{2,4}, \qquad X_{2,9} = \frac{121}{30\sqrt{462}} R^2_{4,4}$$

$$C_2 = \begin{pmatrix} 0 & \frac{38371}{253110} & 0 & -\frac{3243}{42185} & \frac{15917}{901992} \\ -\frac{11}{273} & 0 & \frac{17}{65} & 0 & 0 \\ 0 & \frac{196}{1475} & 0 & -\frac{72}{295} & \frac{121}{4130} \\ 0 & \frac{3423}{6490} & 0 & -\frac{84}{3245} & -\frac{33}{3304} \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

a	b	$(H_2)_{ab}$
1	1	$\frac{95137887410219959802263 1}{17401224094796903961600}$
1	3	$-\frac{12993609267779101941600 4343}{10231919767740579529420800}$
2	2	$\frac{51538456173887123112468494 29}{22397118506981316421779456}$
2	4	$\frac{10444024790391015380876867 3207}{11518518089304677016915148 80}$
2	5	$-\frac{22097462929141201900731274 571}{43208726711491378938700800}$
3	3	$\frac{39576524135715017173571985 7}{98226429770309563482439680}$
4	4	$\frac{57070672010227787489000842 91}{10695766797211485801421209 6}$
4	5	$-\frac{58133206389402048233725168287 59}{46220992230806777927570227200}$
5	5	$\frac{62626580694851540100628790 2819}{42032979045872565906355200 0}$

For $D_k = \det H_2[1{:}k, 1{:}k]$, exact rational comparison gives

$$10^1 \le D_1 < 10^2, \quad 10^4 \le D_2 < 10^5, \quad 10^4 \le D_3 < 10^5, \quad 10^5 \le D_4 < 10^6, \quad 10^6 \le D_5 < 10^7$$

Sylvester's criterion therefore gives $H_2 \succ 0$.

Spin $J = 3$

$$X_{3,1} = \frac{13}{\sqrt{715}} I^3_{6,6}, \qquad X_{3,2} = \frac{14}{25\sqrt{2}} I^3_{2,2}, \qquad X_{3,3} = \frac{143}{6\sqrt{4290}} I^3_{4,6}$$

$$X_{3,4} = \frac{143}{6\sqrt{4290}} R^3_{6,4}, \qquad X_{3,5} = \frac{11}{5\sqrt{66}} I^3_{2,4}, \qquad X_{3,6} = \frac{11}{5\sqrt{66}} R^3_{2,4}$$

$$X_{3,7} = \frac{121}{210\sqrt{22}} I^3_{4,4}$$

$$C_3 = \begin{pmatrix} -\frac{792}{10235} & 0 & -\frac{3509}{71645} \\ \frac{1065}{28658} & 0 & -\frac{56265}{200606} \\ -\frac{13452}{10235} & 0 & -\frac{6666}{14329} \\ 0 & \frac{12}{35} & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

a	b	$(H_3)_{ab}$
1	1	$\frac{898816787635009351151075701}{331480375946636956655616 0}$
1	3	$\frac{139330778405064162543345566 1071}{462209922308067779275702 2720}$
2	2	$\frac{136838901751983103671359 81}{556839171033500926771 20}$
3	3	$\frac{175798558247063825282913365 77}{100879149710094158175252 48}$

For $D_k = \det H_3[1{:}k, 1{:}k]$, exact rational comparison gives

$$10^2 \le D_1 < 10^3, \quad 10^4 \le D_2 < 10^5, \quad 10^7 \le D_3 < 10^8$$

Sylvester's criterion therefore gives $H_3 \succ 0$.

Spin $J = 4$

$$X_{4,1} = \frac{77}{3\sqrt{385}} R^4_{0,4}, \qquad X_{4,2} = \frac{77}{3\sqrt{385}} I^4_{4,0}, \qquad X_{4,3} = \frac{182}{15\sqrt{182}} R^4_{2,6}$$

$$X_{4,4} = \frac{182}{15\sqrt{182}} I^4_{2,6}, \qquad X_{4,5} = \frac{14}{25} R^4_{2,2}, \qquad X_{4,6} = \frac{442}{9\sqrt{24310}} R^4_{6,6}$$

$$X_{4,7} = \frac{143}{12\sqrt{3003}} R^4_{4,6}, \qquad X_{4,8} = \frac{11}{30\sqrt{3}} R^4_{4,2}, \qquad X_{4,9} = \frac{143}{12\sqrt{3003}} I^4_{4,6}$$

$$X_{4,10} = \frac{11}{30\sqrt{3}} I^4_{2,4}, \qquad X_{4,11} = \frac{1573}{630\sqrt{715}} R^4_{4,4}$$

$$C_4 = \begin{pmatrix} \frac{1235953}{5777772} & -\frac{47}{2535} & 0 & 0 & -\frac{439}{656565} \\ 0 & 0 & \frac{437}{5460} & \frac{539}{975} & 0 \\ \frac{1756}{30303} & \frac{253}{1170} & 0 & 0 & \frac{2266}{30303} \\ 0 & 0 & \frac{11}{14} & \frac{33}{10} & 0 \\ -\frac{2084305}{7999992} & \frac{233}{1404} & 0 & 0 & -\frac{66215}{1272726} \\ -\frac{3157}{14430} & \frac{242}{975} & 0 & 0 & -\frac{8591}{50505} \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

a	b	$(H_4)_{ab}$
1	1	$\frac{460065005715398001522949}{820686052056911040000}$
1	2	$-\frac{773598286712943637457652 1}{7222037258100817152000}$
1	5	$-\frac{2587}{9}$
2	2	$\frac{1307535172142762758055680 09}{63553927871287190937600}$
2	5	551
3	3	$\frac{7639}{8}$
3	4	-264
4	4	75
5	5	$\frac{1489}{10}$

For $D_k = \det H_4[1{:}k, 1{:}k]$, exact rational comparison gives

$$10^2 \le D_1 < 10^3, \quad 10^3 \le D_2 < 10^4, \quad 10^6 \le D_3 < 10^7, \quad 10^7 \le D_4 < 10^8, \quad 10^7 \le D_5 < 10^8$$

Sylvester's criterion therefore gives $H_4 \succ 0$.

Spin $J = 5$

$$X_{5,1} = \frac{91}{5\sqrt{1430}} I^5_{6,2}, \qquad X_{5,2} = \frac{91}{5\sqrt{1430}} R^5_{2,6}, \qquad X_{5,3} = \frac{221}{66\sqrt{221}} I^5_{6,6}$$

$$X_{5,4} = \frac{77}{10\sqrt{1155}} I^5_{2,4}, \qquad X_{5,5} = \frac{77}{10\sqrt{1155}} R^5_{4,2}, \qquad X_{5,6} = \frac{13}{6\sqrt{195}} I^5_{4,6}$$

$$X_{5,7} = \frac{13}{6\sqrt{195}} R^5_{6,4}, \qquad X_{5,8} = \frac{143}{180\sqrt{91}} I^5_{4,4}$$

$$C_5 = \begin{pmatrix} 0 & \frac{3025}{4074} \\ \frac{15}{56} & 0 \\ 0 & -\frac{121}{970} \\ 0 & -\frac{220}{6111} \\ \frac{35}{33} & 0 \\ 0 & -\frac{1166}{291} \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

a	b	$(H_5)_{ab}$
1	1	$\frac{23951}{9}$
2	2	$\frac{6}{5}$

For $D_k = \det H_5[1{:}k, 1{:}k]$, exact rational comparison gives

$$10^3 \le D_1 < 10^4, \quad 10^3 \le D_2 < 10^4$$

Sylvester's criterion therefore gives $H_5 \succ 0$.

Spin $J = 6$

$$X_{6,1} = \frac{7}{\sqrt{7}} R^6_{6,0}, \qquad X_{6,2} = \frac{7}{\sqrt{7}} I^6_{0,6}, \qquad X_{6,3} = \frac{7}{\sqrt{330}} R^6_{2,6}$$

$$X_{6,4} = \frac{154}{15\sqrt{770}} R^6_{4,2}, \qquad X_{6,5} = \frac{7}{\sqrt{330}} I^6_{2,6}, \qquad X_{6,6} = \frac{154}{15\sqrt{770}} I^6_{2,4}$$

$$X_{6,7} = \frac{323}{22\sqrt{6783}} R^6_{6,6}, \qquad X_{6,8} = \frac{34}{45\sqrt{34}} R^6_{6,4}, \qquad X_{6,9} = \frac{11}{12\sqrt{105}} R^6_{4,4}$$

$$X_{6,10} = \frac{34}{45\sqrt{34}} I^6_{6,4}$$

$$C_6 = \begin{pmatrix} \frac{3025}{79443} & 0 \\ 0 & -\frac{4}{65} \\ -\frac{3025}{12222} & 0 \\ -\frac{55}{36666} & 0 \\ 0 & \frac{15}{28} \\ 0 & \frac{35}{132} \\ \frac{121}{8730} & 0 \\ \frac{583}{873} & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

a	b	$(H_6)_{ab}$
1	1	$\frac{31443}{7}$
2	2	3371

For $D_k = \det H_6[1{:}k, 1{:}k]$, exact rational comparison gives

$$10^3 \le D_1 < 10^4, \quad 10^7 \le D_2 < 10^8$$

Sylvester's criterion therefore gives $H_6 \succ 0$.

Finite verification of the identity. Insert these arrays and the displayed factorial formula, replace $(\mathsf{T}\psi)_k$ by $(-1)^k\overline{c_{6-k}}$, and collect monomials in the seven amplitudes and their conjugates. The exact identity is

$$N^2\left(8N^2+\frac{8}{3}|\mathbf{F}|^2-\frac{16}{21}\|Q\|_F^2+\frac{288}{13}|\Theta|^2\right)=\sum_{J=2}^{6}\sum_{a,b}(H_J)_{ab}\,\mathrm{Re}\langle B_{J,a},B_{J,b}\rangle.$$

This is polynomial coefficient equality, not a finite-state sample test. There are $5,3,5,2,2$ multiplets of spins $2,3,4,5,6$: 17 multiplets, or 139 real scalar components, not 17 scalar squares.

26 Additional necessary cuts from pure-state compatibility

This section gives the strengthened defining system used in the updated figures. Sections 27 and 28 separate actual state constructions from the finite, exact computations needed for certain unrestricted sharpness claims. No failed optimization is an exclusion argument. In particular, a necessary equality sheet need not be a physical face, and an existence theorem with an unquantified neighborhood does not justify coloring a finite ribbon violet.

We retain the manuscript's density multipoles $\rho=\sum_{L=0}^{6}\rho_L$, $a_L=\mathrm{Tr}\,\rho_L^2$, and write

$$\boldsymbol{F}=\langle\boldsymbol{J}\rangle,\quad f=|\boldsymbol{F}|,\quad Q_{ij}=\langle\{J_i,J_j\}/2\rangle-4\delta_{ij},\quad \tau=\mathrm{Tr}\,Q^2=126a_2,\quad r=|\Theta|,\quad q=r^2. \tag{375}$$

The sector conversion, valid for every pure symmetric six-qubit state, is

$$x=\frac{2}{3}f^2,\qquad y=5+\frac{4}{15}\tau,\qquad z=8+2x-8q. \tag{376}$$

Thus every inequality below is an inequality in S_1,S_2,S_3 alone. The auxiliary variables are explicit functions of those sectors, not additional independent data.

26.1 The previously proved unit-dipole cap adopted in the current figure

The coefficient 14/5 belongs to the historical base system, not to the latest rendering. The current region uses the stronger, previously proved global condition

$$\boxed{\sqrt{y-5}\le b(1-r)+\sqrt{10}\,r+\sqrt{\frac{3}{2}}\sqrt{x}\sqrt{\frac{1-r}{1+r}}.} \tag{377}$$

At $r=1$ the last term is zero. Thus in (6j) the current coefficient is $\gamma_{22}=\sqrt{3/2}$. This is a necessary cap, not a claim of sharpness at nonzero dipole. For clarity, a direct proof of the adopted coefficient follows.

Finite-dipole compatibility on the null slice. Assume first $\Theta=0$. Use the Cartesian octupole T of (191), so $\|T\|_F^2=90$. For a vector v and a symmetric matrix A, write $\mathsf{T}(v)_{ij}=\sum_k T_{ijk}v_k$ and $\mathsf{t}(A)_i=\sum_{jk}T_{ijk}A_{jk}$. Let tf mean the tracefree part and define

$$W_{ij}=5\delta_{ij}-\tfrac{1}{6}\sum_{kl}T_{ikl}T_{jkl},\qquad K=(Q^2)_{\mathrm{tf}}+2Q+W,$$
$$L=\mathsf{T}(F)+\tfrac{9}{2}(FF^{\mathsf{T}})_{\mathrm{tf}},\qquad V=4\mathsf{t}(Q)+\tfrac{1}{3}\mathsf{t}((Q^2)_{\mathrm{tf}})-\tfrac{2}{3}Q\mathsf{t}(Q).$$

Pure-state compatibility gives the exact scalar identity

$$\Phi(\tau):=\tau^2-60\tau+360=6\|K\|_F^2-6K:L-6R_0, \tag{378}$$

where, with $u=f^2$, $a=F^{\mathsf{T}}QF$ and $c=F^{\mathsf{T}}Q^2F$,

$$R_0=-45u+12a-\tfrac{9}{2}c+\tfrac{3}{2}\tau u+F\cdot V-3T(F,F,F).$$

Here are two small contraction identities specifying this compatibility input completely; put $\kappa = \operatorname{Tr} Q^3$:

$$\begin{aligned}\|W\|_F^2 + 2Q : W &= 48 - 8\tau - 24u + 4T(F,Q) + T(F,W),\\ Q^2 : W + Q : W + 2\kappa &= 6 - 3\tau - \tfrac{21}{2}u + \tfrac{21}{2}a + \tfrac{9}{4}F^{\mathsf{T}}WF + T(F,Q)\\ &\quad - \tfrac{3}{2}T(F,F,F) + \tfrac{2}{3}T(F,Q^2) - \tfrac{1}{3}T(QF,Q).\end{aligned}$$

We use $T(v,A) = v \cdot \mathsf{t}(A)$. Expanding K and using these identities, $\operatorname{Tr} Q^4 = \tau^2/2$ and $W = 5I - (T : T)/6$ proves (378). The two contractions themselves can be checked directly from $J_z D_k = (3-k)D_k$ and $J_+ D_k = \sqrt{k(7-k)}D_{k-1}$. One finite coefficient verification uses $c_k = \sqrt{\binom{6}{k}}z_k$, a simple Majorana zero $z_0 = 0$, $z_1 = 1$, z_2 real and $z_5 = 5z_2z_4/2 - 5z_3^2/3$; nullness holds identically. Use independent conjugate variables and homogenize by $N = \langle\psi|\psi\rangle$ before expansion. Both residuals are the zero polynomial in the seven free coordinates. Rotational invariance, homogeneity and continuity extend this dense chart to all physical null states. The supplied `prior_20261020/roof/general_null_compatibility_exact_fast.py` performs precisely this direct spin-matrix expansion, not elimination.

The elementary STF contraction estimates give $\|\mathsf{T}(F)\|_F \le \sqrt{54}f$ and $|T(F,F,F)| \le 6f^3$. Also $c \ge a^2/u$ (by continuity at $u = 0$), so $12a - 9c/2 \le 8u$. Consequently

$$R_0 \le F \cdot V - (37 - 3\tau/2)f^2 + 18f^3, \qquad \|L\|_F \le \sqrt{54}f + \tfrac{3\sqrt{6}}{2}f^2. \tag{379}$$

A directional estimate and one scalar inequality. Suppose $\tau \le 39/5$ and put $k = \|K\|_F$, $U_0 = 4I - 2Q/3$, $P = (30 - 2\tau)I + 12Q + 6Q^2$. The exact definition of K gives $T : T = P - 6K \ge 0$. Moreover $F \cdot V = \mathsf{T}(U_0F) : Q + \mathsf{T}(F) : (Q^2)_{\mathrm{tf}}/3$. For an eigenvalue s of Q, $|s| \le \sqrt{2\tau/3} < 23/10$ and $\tau \ge 3s^2/2$. Thus

$$(4 - 2s/3)^2(30 - 2\tau + 12s + 6s^2) \le g(s) := \tfrac{4}{3}(6-s)^2[(s+2)^2 + 6] < 560.$$

Indeed $g'(s) = \frac{16}{3}(s-6)(s-1)(s+1)$, so the possible maxima on this interval are its endpoints and $s = 1$; $g(-23/10) = 1398467/2500 < 560$ and $g(1) = 500$. Also $P = 6(Q+I)^2 + (24 - 2\tau)I > 0$, $\|U_0\|_{\mathrm{op}}^2 < 31$ and $\|P\|_{\mathrm{op}} < 30 + 2\tau + 12\sqrt{2\tau/3} < 75$. Cauchy–Schwarz and $\|(Q^2)_{\mathrm{tf}}\|_F = \tau/\sqrt{6}$ now give

$$|F \cdot V| \le f\left\{\sqrt{\tau}\sqrt{560 + 186k} + \frac{\tau}{3\sqrt{6}}\sqrt{75 + 6k}\right\} \le f[M_0(\tau) + M_1(\tau)k],$$

where

$$M_0(\tau) = \sqrt{560\tau} + \frac{5\tau}{3\sqrt{2}}, \qquad M_1(\tau) = \frac{93\sqrt{\tau}}{\sqrt{560}} + \frac{\tau}{15\sqrt{2}}.$$

The last step is concavity of the square root at $k = 0$. Substitute this and (379) into (378), and minimize the quadratic in $k \ge 0$. It follows that

$$\begin{aligned}E_2(\tau,f) := \tau^2 - 60\tau + 360 + 6fM_0(\tau) - (222 - 9\tau)f^2 + 108f^3\\ + \tfrac{3}{2}f^2\left[\sqrt{54} + \tfrac{3\sqrt{6}}{2}f + M_1(\tau)\right]^2 \ge 0\end{aligned} \tag{380}$$

for every null state with $\tau \le 39/5$.

The unit-dipole coefficient. Set $\tau_* = 30 - 6\sqrt{15}$ and $\tau_c(f) = 15(b+f)^2/4$. On $0 < f \le f_* = 19/250$, a violation of $\sqrt{y-5} \le b + f$ would imply $6 < \tau_c < \tau \le 15(1+f^2)/2 < 39/5$ by paired purity. On $6 \le \tau \le 8$ one has $M_0' < 8$, $M_1' < 2$ and $\sqrt{54} + (3\sqrt{6}/2)f + M_1 < 22$. Hence

$$\partial_\tau E_2 < 16 - 60 + 6(2/25)8 + 9(2/25)^2 + 3(2/25)^2\,22 \cdot 2 < 0.$$

Direct substitution gives $E_2(\tau_c(f), f)/f = \sum_{j=0}^{5} c_j f^j$, with rational enclosures

$$\begin{gathered}-52 < c_0 < -51, \quad 592 < c_1 < 593, \quad 931 < c_2 < 932,\\ 264 < c_3 < 265, \quad 6 < c_4 < 7, \quad 0 < c_5 < 1.\end{gathered}$$

These are short exact radical estimates, not floating-point evidence. For reproducibility, substitute $\tau_c = t_0 + t_1 f + t_2 f^2$ and $\sqrt{\tau_c} = s_0 + s_1 f$ into (380), with

$$t_0 = 30 - 6\sqrt{15}, \quad t_1 = 15b/2, \quad t_2 = 15/4, \quad s_0 = \sqrt{6}\sqrt{5 - \sqrt{15}}, \quad s_1 = \sqrt{15}/2.$$

The bounds follow by interval arithmetic from $3.872983 < \sqrt{15} < 3.872984$, $1.061610 < \sqrt{5 - \sqrt{15}} < 1.061611$ and the enclosing pairs $(1.41421356, 1.41421357)$, $(2.44948974, 2.44948975)$, $(3.16227766, 3.16227767)$, $(23.66431913, 23.66431914)$ for $\sqrt{2}, \sqrt{6}, \sqrt{10}, \sqrt{560}$, respectively. Each enclosure is verified by rational squaring (twice for the nested radical); the supplement supplies the complete rational-arithmetic check. In particular, for $0 < f \le f_*$,

$$E_2(\tau_c(f), f)/f < -51 + 593 f_* + 932 f_*^2 + 265 f_*^3 + 7 f_*^4 + f_*^5 < -2/5.$$

Monotonicity contradicts (380). At $f = 0$ the established sharp null zero-dipole bound gives $\tau \le \tau_*$. For $f_* \le f \le 2$, the concave expression $(b + f)^2 - 2(1 + f^2) = b^2 - 2 + 2bf - f^2$ is positive at both endpoints (greater than $1/1000$ and $10/9$, respectively), so paired purity proves the same bound. For $f \ge 2$, use $y \le 15$ and $b + f > 10/3 > \sqrt{10}$. Thus every null state satisfies $\sqrt{y - 5} \le b + f$.

Finally the exact time-reversal decomposition of Section 19 gives a null state ϕ with $F_\psi = \sqrt{1 - r^2} F_\phi$ and $Q_\psi = (1 - r) Q_\phi + r Q_u$, where $\|Q_u\|_F \le \sqrt{75/2}$. Multiplying the quadrupole triangle inequality by $2/\sqrt{15}$ proves (377). The case $r = 1$ follows directly from $y \le 15$.

26.2 The retained paired-spectral inequalities

Put $\delta = \sqrt{\Delta/40}$ and use different letters for the *auxiliary* eigenvalues, to avoid confusing them with $r = |\Theta|$:

$$\alpha = \frac{1}{4} + \frac{\delta}{2}, \quad \beta = \frac{1}{4} - \frac{\delta}{2}, \quad H = \frac{1}{4} + \delta + 5\delta^2, \quad K = (\sqrt{2}\alpha + 2\beta)^2 + 8\alpha\delta. \tag{381}$$

For physical states $0 \le \Delta \le 10$, so $0 \le \delta \le 1/2$ and $\alpha, \beta \ge 0$. Define

$$G = 85 + 15x - 5y - 6z + \sqrt{40\Delta}. \tag{382}$$

The old paired-spectral condition was $G \ge 0$. Its useful retained strengthening is

$$\boxed{G \ge \max\left\{ \frac{4\Delta q}{K}, \quad \frac{16\Delta q}{\left(\sqrt{H + 4\alpha\sqrt{q}} + \sqrt{H}\right)^2} \right\}.} \tag{383}$$

Both denominators are positive, including $\Delta = 0$. The simpler corollary

$$\boxed{G \ge \frac{8}{5}\Delta q} \tag{384}$$

is not plotted separately: it is implied by (383).

Proof of the retained inequalities. Use the isometric spin-three summand of $\bigwedge^2 V_2$, with normalized antisymmetric coefficient matrix A, and set $R = AA^\dagger$. Section 17.3 gives

$$\operatorname{spec} R = (\alpha, \alpha, \beta, \beta, 0), \qquad \alpha + \beta = \tfrac{1}{2}, \quad \alpha - \beta = \delta, \quad \Delta = 40\delta^2.$$

The embedding preserves the spin-flip contraction. In a skew Takagi basis $Z = (U, V, n)$, where U, V are orthonormal two-frames and n is a kernel vector, write

$$A = \sqrt{\alpha}\,(u_1 \wedge u_2) + e^{i\phi}\sqrt{\beta}\,(v_1 \wedge v_2).$$

Let $P = UU^\dagger$, $N = nn^\dagger$ and put $t = \operatorname{Tr}(P\overline{P})$, $a = \operatorname{Tr}(P\overline{N})$. The exact projector expansion of the trace-rearrangement slack is

$$\epsilon := \operatorname{Tr}(R\overline{R}) - (2\alpha\beta + \beta^2) = \delta^2 t + 2\beta\delta(1 - a) + \beta^2 \operatorname{Tr}(N\overline{N}), \qquad G = 160\epsilon. \tag{385}$$

In particular $0 \le t \le 2$, $0 \le a \le 1$ and $\epsilon \ge \delta^2 t + 2\beta\delta(1-a)$.

The symmetric unitary matrix $Z^{\mathsf{T}}Z$ has two-by-two blocks $B = U^{\mathsf{T}}U$, $C = U^{\mathsf{T}}V$, $D = V^{\mathsf{T}}V$. Contracting the two wedges gives

$$\sqrt{q} \le 2\alpha|\det B| + 2\beta|\det D| + 4\sqrt{\alpha\beta}|\det C|. \tag{386}$$

Three elementary estimates are

$$2|\det B| \le t, \qquad |\det C|^2 \le 1-a, \qquad |\det D| \le \sqrt{t}.$$

The first uses the two singular values of B. For the second, $CC^{\dagger} \le I - U^{\mathsf{T}}N\overline{U}$, whose subtracted rank-one term has trace a; apply determinant monotonicity. For the third, Jacobi's complementary-minor identity identifies $|\det D|$ with the modulus of the complementary three-by-three minor $E = \left(\begin{smallmatrix} B & h \\ h^{\mathsf{T}} & j \end{smallmatrix}\right)$. It is a contraction. Replacing B by zero leaves rank at most two, so its smallest singular value is at most $\|B\|_{\mathrm{op}} \le \sqrt{t}$; the other singular values are at most one.

Using $t \le 2$ in (386) and weighted Cauchy–Schwarz yields

$$\sqrt{q} \le (\sqrt{2}\alpha + 2\beta)\sqrt{t} + 4\sqrt{\alpha\beta}\sqrt{1-a}, \qquad \delta^2 q \le K\epsilon.$$

Together with (385), this is the first bound in (383). Before using $t \le 2$, the sharper version gives

$$\sqrt{q} \le \alpha t + 2\beta\sqrt{t} + 4\sqrt{\alpha\beta}\sqrt{1-a} \le \frac{\alpha\epsilon}{\delta^2} + \frac{\sqrt{H\epsilon}}{\delta}, \qquad H = 4\beta^2 + 8\alpha\delta.$$

After substitution this becomes $4\Delta\sqrt{q} \le \alpha G + 2\sqrt{\Delta H G}$. Solving the quadratic inequality for $\sqrt{G}$ gives the second bound. The case $\delta = 0$ follows by continuity from $G \ge 0$. Finally $K = 1 + (2\sqrt{2} - 8)\alpha + (22 - 4\sqrt{2})\alpha^2$ is convex on $1/4 \le \alpha \le 1/2$ and at most $5/2$; this proves (384). □

Only the larger retained branch is active at a given point. Indeed, for $\Delta q > 0$, the second is at least the first exactly when

$$\sqrt{q} \le \frac{K - \sqrt{KH}}{\alpha}. \tag{387}$$

This follows by comparing positive denominators; $K - H > 0$. Thus the two formulas do not create two overlaid physical sheets. Their maximum is one necessary cap, whose exposed part alone is drawn.

26.3 Null-seed certificates lifted into a finite neighborhood

For $q < 1$ put

$$U = \frac{f^2}{1-q} = \frac{3x}{2(1-q)}, \qquad a_0 = \frac{15}{2}(1+U), \qquad c_0 = 30 - \frac{15}{2}U. \tag{388}$$

The following lifts are applied only when $U \le 2/3$. Let $L(U)$ and $B(U)$ be respectively any proved lower and upper bounds on τ for *arbitrary null states* with squared dipole U. Then

$$\boxed{\tau \ge \left[\sqrt{L(U)} - \sqrt{q\big(L(U) + c_0\big)}\right]_+^2,} \tag{389}$$

$$\boxed{\Delta \ge \frac{4}{15}\left[\sqrt{a_0 - \overline{B}(U)} - \sqrt{q\big(c_0 - \overline{B}(U)\big)}\right]_+^2,} \tag{390}$$

where $\overline{B} = \min(B, a_0)$ and L is replaced by $\max(L, 0)$ if necessary. Multiple lower inputs enter through their maximum; multiple upper inputs enter through their minimum.

The exact C_4 threshold in Section 27.4 gives one additional continuous null lower input, important near its new endpoint:

$$L_{C_4}(U) = \begin{cases} \dfrac{3}{2}(1-2b_c)^2 + \dfrac{1}{2}(b_c^2 - U), & 0 \le U < b_c^2, \\ \dfrac{3}{2}(1-2\sqrt{U})^2, & b_c^2 \le U \le 1/4, \\ 0, & U > 1/4. \end{cases} \tag{391}$$

The first branch is the safe lower multiplier enclosure from the critical null certificate; the second is the exact global C_4 bound. It enters the same maximum defining $L(U)$ in (389). Thus the newly exact null rim changes the geometry as well as its attainment color. The positive-q lifted sheet is conservative and is not asserted to be the sharp physical boundary.

Proof. Every pure state with $r < 1$ is a time-reversal bias of a null seed χ. Varying the bias and its phase gives an actual pure-state disk

$$Q(w) = Q_0 + w_1 D + w_2 E, \qquad |\boldsymbol{F}(w)|^2 = (1-|w|^2)U, \qquad |w| \le 1.$$

With $A_0 = \|Q_0\|_F^2$ and $C_{ij} = D_i : D_j$, $D_1 = D, D_2 = E$, pure-state compatibility gives

$$\operatorname{Tr} C = A_0 + c_0. \tag{392}$$

One direct derivation uses $\boldsymbol{J}^{(1)} \cdot \boldsymbol{J}^{(2)}$, whose eigenvalues on the symmetric two-copy channels $0, 2, 4, 6$ are $-12, -9, -2, 9$. The operator corresponding to $\|Q\|_F^2 + 30 - 30|\Theta|^2 - (15/2)|\boldsymbol{F}|^2$ is $126P_2$; apply it to the null seed and its entire bias disk. Triangle inequality now gives

$$\sqrt{\tau} \ge \sqrt{A_0} - \sqrt{q(A_0 + c_0)}.$$

Its right side increases with $A_0 \ge 0$, since $c_0 \ge 0$ and $q < 1$; substitution of $A_0 \ge L$ proves (389).

For the other direction, paired purity on every disk point gives $\|Q(w)\|_F^2 \le a_0 + c_0|w|^2$. Consequently the quadratic

$$p(w) = (a_0 - A_0) - 2(Q_0 : D, Q_0 : E) \cdot w + w^{\mathsf{T}}(c_0 I - C)w$$

is nonnegative on the unit disk. Its quadratic trace is $T = c_0 - A_0$, and $T \ge 2(a_0 - A_0)$ when $U \le 2/3$. The elementary disk lemma below therefore gives

$$p(w) \ge \left[\sqrt{a_0 - A_0} - |w|\sqrt{c_0 - A_0}\right]_+^2.$$

At the target, $|w|^2 = q$ and $p(w) = 15\Delta/4$. The displayed positive-part square is nonincreasing in $A_0 \le a_0$, so $A_0 \le \overline{B}$ proves (390). □

Lemma 6 (A nonnegative quadratic on the disk)**.** *If $p(w) = a - 2b \cdot w + w^{\mathsf{T}} M w \ge 0$ for $|w| \le 1$, $a \ge 0$ and $T = \operatorname{Tr} M \ge 2a$, then $p(w) \ge [\sqrt{a} - |w|\sqrt{T}]_+^2$.*

Proof. Rotate the target onto the first coordinate and call its quadratic coefficient m. If $m \le T$, nonnegativity on that diameter gives the bound by its vertex/endpoint test. More explicitly, when $m \le a$ one has $|b_1| \le (a+m)/2$; for $t \le \sqrt{a/T}$, $a - (a+m)t + mt^2 \ge (\sqrt{a} - \sqrt{T}t)^2$ follows by checking the endpoints of the resulting affine factor. If $m > a$, the vertex test gives $|b_1| \le \sqrt{am}$ and the conclusion follows from $m \le T$. For $t \ge \sqrt{a/T}$ the asserted bound is just nonnegativity. If $m = T + k > T$, the other coefficient is $-k$. Averaging the two unit-circle points with a prescribed first coordinate gives $k \le a$ and $b_1^2 \le (a-k)(T+2k) \le aT$, because $T \ge 2a$. Then $p(t, 0) \ge a - 2\sqrt{aT}t + Tt^2$. □

26.4 A dipole-balancing lift of any zero-dipole bound

Let $\mathcal{B}(r)$ be a nondecreasing rigorous upper bound on τ for all pure $\boldsymbol{F} = 0$ states. Define

$$d_0 = (1-r)^2 - 2f^2 > 0, \qquad R = \frac{r}{r+\sqrt{d_0}}, \qquad a = 1-R, \qquad c = a - 2f^2 > 0. \tag{393}$$

At $r = 0$ use $R = 0$. Retaining the decreasing dipole in the covariance estimate gives the following two explicit comparison integrals:

$$\mathcal{I}_1(f,a) = \frac{\operatorname{artanh}\Big(f\sqrt{2/a}\Big)}{\sqrt{2a}}, \qquad \mathcal{I}_2(f,a) = \frac{1}{4}\log\frac{a}{c}. \tag{394}$$

Both vanish at $f = 0$ and are real because $2f^2 < a$. The current figure uses the stronger bound

$$\boxed{\sqrt{\tau} \le \sqrt{\mathcal{B}(R)} + 6\sqrt{6}\sqrt{1-q}\,\mathcal{I}_1(f,a) + 3\sqrt{6}\,\mathcal{I}_2(f,a).} \tag{395}$$

For example, every proved support $\tau \le A + Bq$, $B \ge 0$, gives $\mathcal{B}(R) = A + BR^2$. Section 27.2 supplies a tighter piecewise envelope. A formula proved only on a conditional range is used only there; the two remaining tiny side tails use the rigorous upper bracket, not an unproved sharp extrapolation.

Proof. Evolve an actual normalized pure symmetric state by $\dot\psi = -(\boldsymbol{F}\cdot\boldsymbol{J} - f^2)\psi$. Direct spin differentiation gives

$$\dot{\boldsymbol{F}} = -2\,\mathrm{Cov}(\boldsymbol{J})\boldsymbol{F}, \qquad \dot r = 2f^2 r, \qquad \mathrm{Cov}(\boldsymbol{J}) \ge [(1-r)/2 - f^2]I.$$

For the last estimate, in any quantization axis the $m = 0$ population is at most $(1+r)/2$; hence $\langle J_n^2\rangle \ge (1-r)/2$. Set $u = f^2$. Covariance is positive semidefinite, so u decreases. For initial $r_0 > 0$, integration of $d(u/r^2)/dr \le (r-1)/r^3$ gives

$$u(r) \le \frac{u_0 + r_0 - 1/2}{r_0^2} r^2 - r + \frac{1}{2}.$$

The first zero beyond r_0 is exactly R in (393); the physical flow cannot cross a negative right side. Thus $r(t) \le R$, and covariance stays at least $cI/2$. More precisely, with the fixed number $a = 1 - R$ one has

$$\dot f(t) \le -\big[a - 2f(t)^2\big] f(t).$$

Compare this with the scalar solution $g(0) = f_0$, $\dot g = -g(a - 2g^2)$. Since $c > 0$, it decreases to zero and $0 \le f(t) \le g(t) \le f_0 e^{-ct}$. Substituting $dt = -dg/[g(a-2g^2)]$ gives

$$\int_0^\infty f(t)\,dt \le \int_0^{f_0} \frac{ds}{a - 2s^2} = \mathcal{I}_1(f_0, a),$$
$$\int_0^\infty f(t)^2\,dt \le \int_0^{f_0} \frac{s\,ds}{a - 2s^2} = \mathcal{I}_2(f_0, a).$$

The state speed is at most $3f$, so the flow converges to a pure zero-dipole state χ with $|\Theta_\chi| \le R$.

The Cartesian octupole compatibility identity, in the normalization of (191), gives the norm-speed bound

$$\frac{d}{dt}\|Q\|_F \ge -6\sqrt{6}\sqrt{1-q_0}\,f(t) - 3\sqrt{6} f(t)^2.$$

Here $q(t)$ increases, so replacing it by q_0 weakens the estimate. This follows by contracting the exact octupole norm $\|T\|_F^2 = 90(1-q)$ and using Cauchy–Schwarz, as in the balancing proof of Section 14. Integrating and using $\|Q_\chi\|_F \le \sqrt{\mathcal{B}(R)}$ proves (395). At $r_0 = 0$, the overlap remains zero and the same proof uses $R = 0$. Norm zeros are handled by absolute continuity. □

A useful algebraic improvement and the superseded sheet. The former exponential comparison gave the weaker right-hand side

$$H_{\mathrm{old}} := \sqrt{\mathcal{B}(R)} + \frac{6\sqrt{6}\sqrt{1-q}\,f + (3\sqrt{6}/2)f^2}{c}. \tag{396}$$

For $0 \le s \le f$, the identity

$$\frac{1}{c} - \frac{1}{a-2s^2} = \frac{2(f^2-s^2)}{c(a-2s^2)} \ge \frac{2(f^2-s^2)}{ac}$$

and its integral after multiplication by s give

$$\mathcal{I}_1 \le \frac{f}{c} - \frac{4f^3}{3ac}, \qquad \mathcal{I}_2 \le \frac{f^2}{2c} - \frac{f^4}{2ac}.$$

Consequently the exact comparison bound implies the simpler algebraic condition

$$\boxed{\sqrt{\tau} \le H_{\mathrm{old}} - \epsilon, \qquad \epsilon = \frac{8\sqrt{6}\sqrt{1-q}\,f^3 + (3\sqrt{6}/2)f^4}{ac}.} \tag{397}$$

In particular $\epsilon > 0$ whenever $f > 0$ in the guard domain. Neither (396) nor (397) is drawn as an additional surface: both are consequences of the stronger (395). Section 32 uses this strict improvement to exclude the former tiny gray junction. All quantities in these conditions are explicit functions of the three sector lengths through (376) and (393).

The updated defining system and its origins. The updated region $\mathcal{R}_{22}$ is obtained from (6) by *replacing* the old (6j) with (377), replacing (6f) with (383), and adding (389), (390) with the nine inputs in Table 1 and (391), and (395) on its explicit guard domain. On $x = 0$, use the necessary side envelope of Section 27.2. Conditional statements are not imposed outside their stated domains. Only the maximum lower graph and minimum upper graph are rendered; a dominated condition does not add a facet.

Current strengthening	Origin	Proof
Unit-dipole cap	Finite-dipole scalar compatibility, directional octupole estimate and time-reversal lift	Sec. 26.1
Retained paired cap	Skew Takagi decomposition, projector overlap and two-by-two determinant bounds	Sec. 26.2
Null-seed lifts	Pure-state bias disk, two-copy spin-two compatibility and Lemma 6	Sec. 26.3
Zero-dipole flow lift	Covariance barrier and octupole contraction along an actual pure-state flow	Sec. 26.4
Side middle support	Ten quintic multipole covariants in spin $1, 3, 4$; exact positive Gram identity	Secs. 27.2,28
Finite null supports	Exact harmonic-Gram identities and directed interval continuation	Secs. 27.3,28

These additions are genuine cuts, not an assumption of attainment. For example $(x, y, z) = (1/2, 42/5, 44471/5000)$ satisfies the preceding defining system but violates even (384). Selected lifted null caps have strict positive-q rational witnesses as well; their exact verification is included in the finite supplement.

27 Explicit state families and the newly certified boundary patches

Throughout this section $D_k = |D_6^{(k)}\rangle = |3, 3-k\rangle$. The families below are actual normalized pure states. A claim that they are *global extrema over all pure states* uses the additional compatibility arguments or exact computational lemmas stated explicitly. The already proved qualitative origin image is not counted as a new patch: the new finite strips have quantified intervals away from that unquantified origin neighborhood.

27.1 One four-level state and its elementary multipole calculation

Let $0 \le r < 1$, $g = \sqrt{1-r^2}$, $A_r = \sqrt{(1+g)/2}$ and $B_r = r/\sqrt{2(1+g)}$. For $e, \nu \in [-1,1]$ set

$$c_a = \sqrt{\frac{1+e}{2}}, \quad s_a = \sqrt{\frac{1-e}{2}}, \qquad c_b = \sqrt{\frac{1+\nu}{2}}, \quad s_b = \sqrt{\frac{1-\nu}{2}}.$$

Define the real even-C_2 state

$$\begin{aligned} |\psi(e,\nu,r)\rangle = {} & (A_r c_a c_b + B_r s_a s_b) D_0 + (A_r c_a s_b - B_r s_a c_b) D_2 \\ & + (A_r s_a c_b - B_r c_a s_b) D_4 - (A_r s_a s_b + B_r c_a c_b) D_6. \end{aligned} \tag{398}$$

It has, exactly,

$$\|\psi\| = 1, \quad \Theta = -r, \quad \boldsymbol{F} = (0, 0, g(2e+\nu)), \quad Q_{xy} = Q_{xz} = Q_{yz} = 0. \tag{399}$$

Put $s = \sqrt{1-e^2}$, $t = \sqrt{1-\nu^2}$ and

$$\begin{aligned} Z(e,\nu,r) &= 1 + 4e\nu + 4rst, \\ H(e,\nu,r) &= \sqrt{15}(et - rs\nu) + 3[st - r(1-e\nu)], \\ T(e,\nu,r) &= \frac{3}{2}Z^2 + \frac{1}{2}H^2. \end{aligned} \tag{400}$$

Then $Q_{zz} = Z$, $Q_{xx} - Q_{yy} = H$ and $\tau = T$.

Proof. Write the four real coefficients as a, b, c, d. Direct expansion gives $a^2 + b^2 + c^2 + d^2 = (A_r^2 + B_r^2)(c_a^2 + s_a^2)(c_b^2 + s_b^2) = 1$ and $2(ad + bc) = -2A_rB_r = -r$. The J_z eigenvalues are $3, 1, -1, -3$, so $3a^2 + b^2 - c^2 - 3d^2 = g(2e + \nu)$. Even support makes $F_x = F_y = 0$. The $J_z^2 - 4$ eigenvalues are $5, -3, -3, 5$, giving $Q_{zz} = Z$. The only J_+^2 links have coefficients $2\sqrt{15}, 12, 2\sqrt{15}$. Consequently

$$Q_{xx} - Q_{yy} = 2\sqrt{15}(ab + cd) + 12bc = H,$$

because $ab + cd = (et - rs\nu)/2$ and $bc = [st - r(1 - e\nu)]/4$. Realness and even support make the off-diagonal quadrupole entries zero. Finally $\operatorname{Tr} Q = 0$ gives $\tau = 3Z^2/2 + H^2/2$. This verifies the state and sectors without any positivity certificate. □

At fixed f, r, impose $\nu = f/g - 2e$. The allowed e interval is connected:

$$\max\{-1, (f/g - 1)/2\} \le e \le \min\{1, (f/g + 1)/2\}. \tag{401}$$

Whenever two proven global endpoint values occur on this path, continuity supplies *every* intermediate S_2 at the same f, r. This is the intermediate-value theorem, not convexity of the full sector-length image. It does not assert that this real path alone fills the entire zero-dipole side; additional phases are used below.

27.2 The almost complete zero-dipole side

Define

$$r_c = \sqrt{\frac{648\sqrt{5} - 1445}{931}}, \quad r_- = \frac{102734367}{1024000000000000}, \quad r_+ = \frac{5350769}{81920000}, \quad \mathcal{I} = [r_-, r_+] \cup [r_c, 1]. \tag{402}$$

Here $r_c = 0.065317940252779\ldots$, $r_- = 1.003265302734375 \times 10^{-7}$ and $r_+ = 0.06531700439453125$.

Theorem 20 (Complete side fibers on certified ranges)**.** *For every $r \in \mathcal{I}$, the complete pure-state sector image at $\boldsymbol{F} = 0$, $|\Theta| = r$ is*

$$x = 0, \quad z = 8 - 8r^2, \qquad 5 + \frac{[1-9r]_+^2}{10} \le y \le U(r). \tag{403}$$

For $r \ge r_c$ the sharp upper curve is

$$U(r) = \begin{cases} \dfrac{239}{35} + \dfrac{64}{5}r^2, & r_c \le r \le 1/7, \\ \dfrac{33}{5} + \dfrac{16}{5}r + \dfrac{8}{5}r^2, & 1/7 \le r \le 1/4, \\ 7 + 8r^2, & 1/4 \le r \le 1. \end{cases} \tag{404}$$

On $[r_-, r_+]$, $U(r) = 5 + 4T(e, -2e, r)/15$, with e given by the radical family (409) *below. The global sharpness on this low interval is an exact computer-assisted result, whose finite verification is specified in Section 28.*

The central new compatibility input is the unrestricted conditional inequality

$$\boxed{\boldsymbol{F} = 0 \quad \Longrightarrow \quad \tau \le \frac{48}{7} + 48q} \qquad \Longleftrightarrow \qquad \boxed{S_1 = 0 \quad \Longrightarrow \quad 35S_2 + 56S_3 \le 687.} \tag{405}$$

Its proof is the positive spin-multipole identity of Lemma 7; it is not inferred from a cyclic-state optimization. It is globally necessary even where it is not the sharp side bound.

Why the middle-to-high branch has a short analytical proof. For a normalized null zero-dipole seed ϕ, put

$$a_r = \frac{\sqrt{1+r} + \sqrt{1-r}}{2}, \quad b_r = \frac{\sqrt{1+r} - \sqrt{1-r}}{2}, \quad \psi_r = a_r\phi + b_r\mathsf{T}\phi.$$

Time reversal gives $\boldsymbol{F}(\psi_r) = \sqrt{1-r^2}\boldsymbol{F}(\phi)$ and

$$Q(\psi_r) = Q(\phi) + r\Re V(\phi), \qquad V_{ij}(\phi) = \langle \mathsf{T}\phi | \widehat{Q}_{ij} | \phi \rangle. \tag{406}$$

Conversely every zero-dipole state with $r < 1$ has a null seed obtained by the inverse bias. Thus

$$K_0(r) := \max_{\boldsymbol{F}=0, |\Theta|=r} \|Q\|_F = \sup_{\phi : \boldsymbol{F}=0, \Theta=0} \|Q(\phi) + r\Re V(\phi)\|_F \tag{407}$$

is convex. The seed phase $\phi \mapsto i\phi$ changes V to $-V$ without changing Q, so its even extension is convex as well; in particular K_0 is nondecreasing on $r \ge 0$. The sharp values at $r = 1/7$ and $r = 1/4$, supplied respectively by (405) and paired purity, are $8\sqrt{6}/7$ and $5\sqrt{6}/4$. Their chord is $\sqrt{6}(1+r)$. Convexity therefore proves $\tau \le 6(1+r)^2$ on $[1/7, 1/4]$. It is attained throughout by

$$H_r = \sqrt{\frac{1-r}{4}}(D_1 + D_5) + i\sqrt{\frac{1+r}{2}}D_3. \tag{408}$$

This proves the middle line of (404). The last line is paired purity. The first line is (405).

Explicit low-side radicals. For $k_c = (24\sqrt{5} - 45)/19 \le k \le k_0 = 4\sqrt{15} - 15$, define

$$\begin{aligned} D &= k^2 + 15, & H_0 &= 5 - k^2, \\ P &= 121k^4 - 180k^3 + 1510k^2 - 4500k + 2025, & & \\ B &= -9k^4 + 90k^3 + 360k^2 - 1050k + 225, & & \\ C &= 5(k-3)(k^2 + 30k - 15), & & \\ r(k) &= \frac{2C\sqrt{H_0/5}}{-B + D\sqrt{P}}, & e(k) &= \frac{k}{\sqrt{D}}, \qquad \nu = -2e(k). \end{aligned} \tag{409}$$

Insert these quantities into (398) to obtain L_k. The denominator is positive, $P > 0$ and $C \geq 0$ on this interval. The state exists on its *whole* parameter interval, not just where global sharpness is certified. Its quadrupole norm is the elementary polynomial in r

$$\begin{aligned} T &= A_0 + A_1 r + A_2 r^2, \\ A_0 &= \frac{3(17k^4 - 45k^3 - 135k^2 + 225k + 450)}{D^2}, \\ A_1 &= -\frac{27\sqrt{5H_0}(k-1)(k^2 + 10k + 5)}{D^2}, \\ A_2 &= \frac{9(9k^4 - 60k^3 - 50k^2 - 300k + 1425)}{2D^2}. \end{aligned} \tag{410}$$

For $A_s = 10k^3 + 90k^2 - 450k - 450$,

$$\partial_k T = \frac{9(3k+5)}{D^3}\left[A_s r^2 + B\sqrt{5/H_0}\, r + C\right].$$

Its discriminant is $D^2 P$, so (409) is exactly the stationary branch, not the outcome of an optimization. At k_0, $r = 0$ and $T = 30 - 6\sqrt{15}$; at k_c, $r = r_c$ and $T = 48/7 + 48r_c^2$. Directed bounds $T_{ee} < 0$, $T_{er} < 0$ throughout $0 \leq r \leq 0.066$, $0.116 \leq e \leq 0.130$ make this a single monotone branch. Its global sharpness on the rational interval $[r_-, r_+]$ uses the finite proof below, not this restricted stationarity.

State families filling every side fiber. The following formulas supply the constructive part of Theorem 20; phases may be chosen from either triangle orientation. All are normalized and have $\boldsymbol{F} = 0$, $|\Theta| = r$. First, for $(1-r)/4 \leq p \leq (1+r)/4$, use

$$A_{r,p} = \sqrt{p}(D_0 + D_6) + \sqrt{1-2p}\, e^{i\chi} D_3, \qquad \cos 2\chi = \frac{4p^2 + (1-2p)^2 - r^2}{4p(1-2p)}. \tag{411}$$

It has $\tau = \frac{3}{2}(18p - 4)^2$ and reaches the sharp spectral lower endpoint. The five-level bridge

$$B_{r,t} = \sqrt{\frac{1-r}{4}}(D_0 + D_6) + i\sqrt{\frac{(1+r)(1-t)}{4}}(D_2 + D_4) - \sqrt{\frac{(1+r)t}{2}} D_3, \qquad 0 \leq t \leq 1, \tag{412}$$

has

$$\tau = 6 - 3r + \frac{57}{2}r^2 - \frac{15}{2}(1+r)t + \frac{15}{8}(1+r)^2 t^2.$$

For $r < 1$ this decreases strictly with t. On the low branch, if $L_k = aD_0 + bD_2 + cD_4 + dD_6$, multiplying c, d by a common phase gives

$$\tau_\theta = \frac{3}{2}Q_{zz}^2 + \frac{1}{2}\left|2\sqrt{15}(ab + cd) + 12bc\, e^{i\theta}\right|^2. \tag{413}$$

Its interval overlaps the bridge uniformly by more than 0.264 in τ on the low chart. Hence it fills every value from the spectral lower endpoint to $T(r, e(r))$ on the certified band.

For $r_c \leq r \leq 1/7$, put

$$s = \frac{29}{56}, \quad d^2 = \frac{27}{128}(1/49 - r^2), \quad n_0 = \frac{s+d}{2}, \quad n_6 = \frac{s-d}{2}, \quad n_2 = \frac{1-s-3d}{2}, \quad n_4 = \frac{1-s+3d}{2}.$$

Let $v = 2\sqrt{n_0 n_6}$, $w = 2\sqrt{n_2 n_4}$ and choose $\cos\eta = (r^2 - v^2 - w^2)/(2vw)$. With $\alpha = \arg(\sqrt{n_0 n_2} + \sqrt{n_4 n_6} e^{i\eta})$, the states

$$D_{r,\theta} = \sqrt{n_0} D_0 + \sqrt{n_2} D_2 + \sqrt{n_4} e^{i(\alpha+\theta)} D_4 + \sqrt{n_6} e^{i(\alpha+\theta+\eta)} D_6 \tag{414}$$

have $Q_{zz} = 8/7$. At $\theta = 0$ they reach $\tau = 48/7 + 48r^2$; varying θ covers $[3351/784 + (363/16)r^2, 48/7 + 48r^2]$. Feasibility is exactly the triangle condition

$$4v^2 w^2 - (r^2 - v^2 - w^2)^2 = \frac{5(45619r^4 + 141610r^2 - 605)}{614656} \geq 0,$$

whose first nonnegative root is r_c. The bridge overlaps this interval because $1353/784-3r+(93/16)r^2 > 0$ on $[0,1/7]$.

For $r \ge 1/7$, the odd-C_2 family

$$C_{r,p} = \sqrt{\frac{1-p}{2}}(D_1+D_5) + \sqrt{p}\, e^{i\chi} D_3, \qquad \cos 2\chi = \frac{r^2-(1-p)^2-p^2}{2p(1-p)} \tag{415}$$

has $\tau = -96p^2+120p-30+30r^2$. For $r \le 1/4$, vary p from $(1-r)/2$ to $(1+r)/2$; for $r \ge 1/4$, vary it to $5/8$. It reaches the required upper endpoint and overlaps the bridge since $6-3r+(57/2)r^2-6(1-r)^2 = 9r+(45/2)r^2 \ge 0$. At $r=1$ it alone fills the full interval. At parameter endpoints where a displayed phase formula has a zero denominator, use its continuous state limit. These elementary interval overlaps prove complete coverage, without convex-hull interpolation or a connected-fiber assumption.

The two remaining tails and the necessary side envelope. Write $T_* = 30-6\sqrt{15}$, $m = \sqrt{42\sqrt{15}-162}$ and $t_* = m/(7\sqrt{T_*})$. Convexity of K_0 and the line from $K_0(0) = \sqrt{T_*}$ tangent to $\sqrt{48/7+48r^2}$ at t_* prove

$$\tau \le (\sqrt{T_*}+mr)^2, \qquad 0 \le r \le t_*. \tag{416}$$

Use this necessary bound below r_-, the sharp radical branch on $[r_-, r_+]$, the middle support on (r_+, r_c), and the sharp piecewise formula (404) thereafter. Call the resulting pointwise necessary τ envelope $\mathcal{B}_0(r)$. It may have tiny downward jumps where a certificate interval starts; no monotonicity of this stitched formula is assumed. For the flow lift, (407) nevertheless gives $\tau_\chi \le K_0(R)^2 \le \mathcal{B}_0(R)$ whenever $|\Theta_\chi| \le R$. Thus it is a valid input in (395). For convenient monotone inversion, the renderer uses the slightly weaker envelope $\widehat{\mathcal{B}}_0$: write $T_{\mathrm{low}}(r) = T(e,-2e,r)$ for the sharp radical value and, on $[r_-, r_+]$, replace it by

$$\widehat{\mathcal{B}}_0(r) = \max\left\{T_{\mathrm{low}}(r), (\sqrt{T_*}+mr_-)^2\right\}, \tag{417}$$

and otherwise use $\mathcal{B}_0(r)$ (with the middle support continued to $1/7$ and the sharp high branches beyond it). The constant only affects a microscopic initial part of the certified low branch. It does not mean that the exact sharp side curve is flat. This monotone closure is necessary for every physical state and is the choice of $\mathcal{B}$ in the plotted flow lift.

The explicit biased endpoint states and these necessary bounds give an outer-minus-attained S_2 gap at most

$$\frac{353}{10^{10}} = 3.53\times 10^{-8} \tag{418}$$

on the *entire* zero-dipole side. Near zero the gap is at most $(658/1875)r$. Near r_c, the exact residual is $(4/15)[48-A_2(k_c)](r-r_c)^2$, with $48-A_2(k_c) < 24$. These two estimates and the rational endpoints (402) prove (418). This is a rigorous uncertainty bracket: the tiny tails are not claimed globally sharp, excluded, or completely classified.

27.3 Nine null contacts, their global supports, and four complete strips

For $u > 0$ put $d = 1+15u^2$, $e = (1-15u^2)/d$ and define

$$\begin{aligned} A(u) &= (3u+1)(5u-1)d(15u^2-30u-1),\\ B(u) &= -12u(15u^2+10u-1)(45u^2-10u-3),\\ C(u) &= -d(225u^4-180u^3-90u^2+12u+1),\\ A(u)\nu^2+B(u)\nu+C(u) &= 0, \qquad \nu = \frac{15-k^2}{15+k^2}. \end{aligned} \tag{419}$$

Choose the positive k branch specified in Table 1. The exact algebraic null state is

$$\chi(u,k) = \frac{D_0+(k/\sqrt{15})D_2+\sqrt{15}uD_4-ukD_6}{\sqrt{(1+15u^2)(1+k^2/15)}}. \tag{420}$$

Its $f = 2e + \nu$ is positive at the listed contacts and

$$\tau = \frac{3}{2}(1 + 4e\nu)^2 + \frac{15}{2}(1 - \nu^2)(e + 6u/d)^2. \tag{421}$$

This gives an exact specification of each contact without rounding its amplitudes or sector coordinates. The decimal k column selects the isolated root; the finite supplement gives rational isolating intervals.

Extremum	u	k (root selector)	f	m	h
Lower	39/100	1.2842821304	.0208235808	−333/20	31/2
Lower	$38581311/10^8$	1.2025521793	.0614637090	−297/20	287/20
Upper	2/9	4.9282885716	.0614637279	38	4
Upper	$10643749/(5 \cdot 10^7)$	4.7833179314	.1732980136	583/40	13/8
Lower	3/8	.9483569982	.1732980713	−212/25	213/25
Upper	21/100	4.7388130275	.2083954442	62/5	8/5
Lower	$3718048909/10^{10}$	.8553697513	.2083954448	−13/2	13/2
Upper	$10120399/(5 \cdot 10^7)$	4.6205271031	.3028197911	359/40	63/40
Lower	4/11	.5382785491	.3028198094	−12/5	21/10

Table 1: Nine unrestricted global null contacts. The state and exact coordinates are (419)–(421). Each is the unique equality orbit up to collective rotations and phase. The last columns define the globally necessary null supports (422), not sharp finite sheets.

For each row let $u_* = f_*^2$ and τ_* be its exact values. The finite Gram certificates prove the following bounds for *all* null states, not just the family (420):

$$\begin{aligned} B_j(U) &= \tau_* + m(U - u_*) + h|U - u_*| && \text{(upper row)}, \\ L_j(U) &= \max\{0, \tau_* + m(U - u_*) - h|U - u_*|\} && \text{(lower row)}. \end{aligned} \tag{422}$$

Their minimum and maximum are the B, L used in (390)–(389). The rational numbers m, h are safe multiplier enclosures; the finite certificate specifies the sharper exact algebraic endpoint multipliers. The convenient lifted sheets are not asserted attained away from their null contacts.

Theorem 21 (Four finite complete roof strips)**.** *For every f in the following union of closed rational intervals,*

$$\begin{aligned} J_1 &= [.0614630, .0614644], \quad J_2 = [.170799, .175797], \\ J_3 &= [.2083895, .2084014], \quad J_4 = [.3028151, .3028244], \end{aligned} \tag{423}$$

the complete fixed-f null sector image is the interval between its unique real-C_2 global minimum and maximum from (420)*. The widened J_2 is proved in Theorem 24. The previously established positive-q continuation retains its original scope: put $J_2^{\mathrm{old}} = [.173293, .173303]$. There is one positive,* unquantified *ε such that on $J_1 \cup J_2^{\mathrm{old}} \cup J_3 \cup J_4$ and $0 \le q < \varepsilon$, the complete local three-dimensional image is*

$$x = 2f^2/3, \quad z = 8 + 2x - 8q, \qquad 5 + \frac{4}{15}T_-(f, \sqrt{q}) \le y \le 5 + \frac{4}{15}T_+(f, \sqrt{q}). \tag{424}$$

Here $T_\pm$ are the analytic stationary branches of (400) *with $\nu = f/\sqrt{1-q} - 2e$ through the certified null endpoints. Every intermediate value is attained by* (398)*.*

Proof. The exact identities and directed continuation in Lemma 8, with the widened null interval from Theorem 24, prove unrestricted global lower and upper arcs, not merely stationary C_2 arcs. Their strictly monotone f maps have the four intervals in (423) as *inner* overlaps; no outer enclosure is mistaken for an image. At $q = 0$ both endpoint states lie in the connected feasible interval (401). The intermediate-value theorem therefore fills precisely the interval between them.

Strict Gram positivity and the full feature derivative leave only normalization, phase and rotation zero modes. Thus the physical constrained Hessian has the strict minimum/maximum sign in every remaining direction. The singlet responses have opposite, nonzero signs. The implicit-function theorem selects the real singlet phase and continues both unique global extrema. To see globality, any sequence of competing global extrema with $q \to 0$ has a subsequence converging to a null extremum. Its unique orbit must enter this controlled local slice. A finite cover over the four original compact intervals gives one positive q width. The continuous feasible e path at exactly fixed f, r again fills every intermediate value. No width is inferred from sample points. Unrestricted sharpness between the certified intervals remains open; the constructive C_2 inner band in those gaps is now filled by Theorem 22. □

27.4 The exact C4 threshold and the critical cusp

Let $b_c = 0.366379159399607379\ldots$ be the unique root in $(0, 1/2)$ of

$$16b_c^4 - 78b_c^3 + 92b_c^2 + 6b_c - 11 = 0. \tag{425}$$

The null lower state

$$\Omega_f = \frac{\sqrt{1+f}D_0 + \sqrt{3-f}D_4}{2} \tag{426}$$

has $\tau = \frac{3}{2}(1-2f)^2$ and is globally minimal down to $f = b_c$. Together with the preceding complete-roof result, this determines the entire null roof for $f \geq b_c$; for $b_c \leq f \leq 1/2$ its interval is

$$5 + \frac{2}{5}(1-2f)^2 \leq y \leq 7 + 2f^2, \qquad z = 8 + 4f^2/3. \tag{427}$$

On the newly certified $b_c \leq f \leq 3/8$ the lower equality orbit is unique. The exact finite computation behind the endpoint and its continuum is stated in Lemma 9.

The threshold also follows locally from elementary spin algebra. For the null four-level family let $e_0 = (f-1)/2$, $\nu = f - 2e$ and $\mathcal{T}(e, f) = T(e, \nu, 0)$. The physical endpoint $e = e_0$ has $\mathcal{T} = \frac{3}{2}(1-2f)^2$. Its inward derivative vanishes precisely at (425) on the indicated interval. It changes sign there, while $\mathcal{T}_{ee}(e_0, b_c) > 0$. Consequently below b_c an interior real-C_2 state lowers the value: the C_4 state is then not even a local minimum.

For completeness, the full sharp critical boundary has a nonanalytic opening into $q > 0$. Put $w = \sqrt{3 + 2b_c - b_c^2}$ and

$$\begin{aligned} E &= 6(7b_c^2 - 42b_c + 29) + \frac{6\sqrt{15}(b_c^3 - 7b_c^2 + 7b_c + 7)}{w}, \\ C &= 2E/15, \qquad \sigma = \frac{4}{5}[\sqrt{15}b_c(3 - b_c) - (7b_c - 6)w], \\ \kappa &= (\sigma/(4C))^{1/3}, \qquad A = 3C\kappa^4 = 2.644953610896\ldots. \end{aligned} \tag{428}$$

All constants are positive. For sufficiently small positive q,

$$y_{\min}(f = b_c, q) = 5 + \frac{2}{5}(1 - 2b_c)^2 - Aq^{2/3} + O(q). \tag{429}$$

The actual equality state is (398) with $\nu = \cos 2b$, $e = (b_c/\sqrt{1-q} - \nu)/2$ and $b = \kappa q^{1/6} + O(q^{1/2})$ selected by the exact stationary equation. This expansion is not itself proposed as a truncated universal inequality.

Proof of the sharp cusp, given the exact critical lemma. At the unique critical orbit one real physical null mode has zero Hessian; all other five have positive Hessian. Eliminate them by an equivariant implicit-function theorem. In the remaining signed mode b and complex singlet coordinate u, $|u| = r$, residual C_4 and conjugation give

$$G(b, u) - G(0, 0) = Cb^4 - \sigma b \Re u + O(b^6 + |u||b|^3 + |u|^2).$$

Direct differentiation of (400) gives exactly (428); $e - e_0 = b^2 + O(b^4)$ explains $C = 2E/15$. With $b = r^{1/3}B$, division by $r^{4/3}$ gives the limiting minimum problem $CB^4 - \sigma B\cos\phi$. It has the two C_4-equivalent nondegenerate minima $(B,\phi) = (\kappa, 0), (-\kappa, \pi)$. Analytic continuation and conjugation keep the selected phase exactly real. Unique critical equality and compactness force every nearby global minimizer into this local slice; quartic coercivity gives $b = O(r^{1/3})$. Thus the local minimum is the global one and its value is $-3C\kappa^4 r^{4/3} + O(r^2)$, proving (429).

The same proof is uniform for $|f - b_c| \le Mq^{1/3}$, for each fixed $M < \infty$ and sufficiently small q. It adds $K_c(f - b_c)b^2$ to the normal form, where $K_c = 15.5183831231587\ldots > 0$ is obtained by differentiating the endpoint derivative. The rescaled minimum has the unique positive root $4CB^3 + 2K_c\eta B - \sigma = 0$, $\eta = (f - b_c)/q^{1/3}$. Its Hessian is positive since $12CB^2 + 2K_c\eta = 8CB^2 + \sigma/B > 0$. This proves a genuine two-parameter sharp wedge, but no numerical positive q width is supplied by this argument. □

27.5 A sharp surface away from both the roof and the side

The low-side contact $r_* = 0.01076063265637951975\ldots$ has exact

$$k_* = \frac{5324405030127873}{10947851158810243},$$
$$q_* = \frac{13035854269122306431146976122147619377793896158940371857 89}{11258068455785308792543637961491281111125099638943137341409875}.$$

Use $r_* = \sqrt{q_*}$, $e_* = k_*/\sqrt{k_*^2 + 15}$, $\nu_* = -2e_*$ in (398). The exact quintic certificate, complete equality reconstruction and full physical Hessian at this contact prove that there exist $\epsilon, \eta > 0$ such that for

$$0 \le f < \epsilon, \qquad |r - r_*| < \eta, \tag{430}$$

the global upper boundary is its unique real even-C_2 continuation. The two widths are unquantified. In particular an arbitrary sampled rectangle near this point is not automatically a sharp patch.

An explicit two-parameter radical selector removes any optimization. For $s = \sqrt{1 - e^2}$, $t = \sqrt{1 - \nu^2}$ set

$$z_0 = 1 + 4e\nu, \quad z_1 = 4st, \quad h_0 = \sqrt{15}et + 3st, \quad h_1 = -\sqrt{15}\nu s - 3(1 - e\nu), \quad D = \partial_e - 2\partial_\nu.$$

Define

$$\begin{aligned} A &= 3z_1 Dz_1 + h_1 Dh_1, \\ B &= 3(z_0 Dz_1 + z_1 Dz_0) + h_0 Dh_1 + h_1 Dh_0, \\ C &= 3z_0 Dz_0 + h_0 Dh_0, \qquad r(e,\nu) = \frac{2C}{-B + \sqrt{B^2 - 4AC}}. \end{aligned} \tag{431}$$

Choose the continuous root through the contact and use (398). Since at fixed f, r one has $d\nu/de = -2$, this is exactly the equation $D\tau = 0$. The map $(e,\nu) \mapsto (r, f)$ is locally invertible because T_{ee} and T_{er} are nonzero on the side branch. For the upper branch select $2e + \nu \ge 0$; the negative-dipole branch near the same representative is attained but is not uppermost at fixed $|\boldsymbol{F}|$.

Here is why this restricted-looking formula solves the unrestricted problem. The finite multipole-rigidity computation in Section 28 proves one global equality orbit at the contact, not just equality of a few scalar invariants. The full five-dimensional physical constrained Hessian is negative. In the principal quadrupole frame the dipole response is $\beta_z = 4.238896132006\ldots > 4$. Eliminating the internal modes gives an analytic energy $E(p, r)$ with $\nabla_p E(0, r_*) = \beta_z e_z$. On $|p| = f$ its unique maximum is near the north pole; C_2 symmetry makes that pole an exact critical point. Conjugation then makes the unique internal state real and even. A sequence of distant competing global maxima approaching the contact would limit to its unique equality orbit, hence enter the same local slice. This compactness argument proves the global statement (430). The leading sector response is

$$y_{\max}(x, r) = y_{\max}(0, r) + \frac{4}{15}\beta_z(r)\sqrt{\frac{3x}{2}} + O(x). \tag{432}$$

Thus it is a genuine nonzero-S_1, nonzero-q sharp surface, not a convex-hull face or merely a roof contact.

28 Finite exact verification data and the limits of the plotting claims

The analytical proofs above are self-contained. Some new unrestricted sharpness statements also use large but finite exact identities and directed interval computations. We state those finite lemmas precisely here instead of presenting a numerical fit as a proof or concealing hundreds of rational coefficients behind an informal "compatibility" claim. The supplied portable data supplement is

`assets/N6_gray_search_proof_data_20261022.zip`.

It contains the exact defining data, verifier sources, accepted interval reports and an integrity manifest. Large regenerated coefficient-operator caches, numerical search grids and redundant expanded-polynomial backups are not necessary mathematical input and are not included. The reader can reconstruct the operators from the stated finite covariants and spin matrices. These are *computer-assisted exact proofs*, not claims of short certificate-free analytical derivations.

28.1 The ten-multiplet proof of the middle side support

Lemma 7 (Finite multipole identity)**.** *Let $N = \langle\psi|\psi\rangle$. On the full physical $\boldsymbol{F} = 0$ locus, there are charge-one quintic spin multiplets $C_i^{(J)}$, of multiplicities $3, 4, 3$ for $J = 1, 3, 4$, and positive definite rational matrices $G^{(J)}$, such that*

$$\left[\frac{48}{7}N^2 + 48|\Theta|^2 - \tau\right] N^3 = \sum_{J=1,3,4} \sum_{i,j} G_{ij}^{(J)} \langle C_i^{(J)}, C_j^{(J)}\rangle. \tag{433}$$

Consequently (405) *holds for every pure symmetric six-qubit state with zero dipole.*

Exact finite specification and verification. Use the manuscript's normalized Condon–Shortley projections. With $r^{(L)} = (\psi \otimes \mathsf{T}\psi)_L$ define natural paths

$$B_{LMK}^{(J)} = [\,[r^{(L)} \otimes r^{(M)}]_K \otimes \psi\,]_J, \qquad L, M \in \{0, 2, 3, 4\}, \quad L \le M.$$

For $L = M$, take K even. Each $C_i^{(J)}$ is the finite linear combination of these paths in `caps/middle_ten_multipole_exact_channels.json`; that file gives every rational path coefficient, its exact square-root Condon–Shortley scale, and every rational entry of the three G. It specifies ten multiplets and 22 independent Gram entries, not an unspecified positive-matrix existence problem. Only $\rho_0, \rho_2, \rho_3, \rho_4$ enter. The scalar-times-spinor component of an earlier eleven-multiplet representation is absent; it is not silently discarded from a matrix whose positivity would then be uncertain.

Here is a complete finite algorithm for checking the identity. Write $c_k = \sqrt{\binom{6}{k}}z_k$ and treat its conjugates as independent $\sqrt{\binom{6}{k}}w_k$. Construct each highest-weight quintic from the listed coupled paths. The remaining spin-J components follow by ladder action; in binomial coordinates their squared norm is $\sum_{m=0}^{2J}\binom{2J}{m}P_{i,m}(z,w)P_{j,m}(w,z)$. Form the left side independently from the standard spin-three ladder matrices. Substitute the following seven-variable zero-dipole gauge, where $y_k = (-1)^k w_{6-k}$:

$$\begin{gathered} z = (0, 1, b, c, d, 1, 0), \qquad (y_2, y_3, y_4, y_5) = (u, v, w, t), \\ y_1 = (5du - 5bw + 4t)/4, \quad y_0 = 10bv - 10cu + 5dy_1 - 5w, \quad y_6 = 5bt - 10cw + 10dv - 5u. \end{gathered} \tag{434}$$

The three dipole polynomials vanish identically in these seven free variables. Every one of the 3901 rational coefficient residuals of (433) is exactly zero. Sylvester's test gives positive rational leading minors of all three matrices. The independent physical reconstructions and integer-arithmetic positivity tests are recorded in the supplement; floating SDP values are not used in this verification.

For completeness, the gauge covers the whole physical locus by polynomial continuity, not only a guessed generic branch. Complexify the two spin-three spinors. The zero-dipole incidence $\{(c, d) : d^{\mathsf{T}} J_i c = 0,\ i = 1, 2, 3\}$ is an irreducible rank-four vector bundle over the open rank-three set,

of dimension 11. Rank drops occur only over the scaled $m = 0$ and coherent orbits, of dimensions 3 and 2; their incidence dimensions are at most 8 and 7. The zero-spinor fiber has dimension 7. A set cut out by three equations in $\mathbb{C}^{14}$ has no component of dimension below 11, so none of these smaller strata is an additional component. A generic sextic spinor can be put into the displayed gauge by an SL_2 transformation and scale: choose two simple roots as zero and infinity and fix the adjacent nonzero coefficients. Invariance and homogeneity preserve the identity. Exact vanishing on this dense gauge therefore proves it on every physical rank-drop state as well. Nonnegativity of the complete right side and $N = 1$ finish the proof. □

28.2 Why the low-side band is a proved continuum

The low-side certificate has the same form as (433), with target $[a(r)N^2 + b(r)|\Theta|^2 - \tau]N^3$ and multiplicities $5, 9, 7$. Here $e(r)$ is the stationary root of (400) at $\nu = -2e$, $b(r) = T_r/(2r)$ and $a(r) = T - b(r)r^2$. It is enough to continue *exact identities*, not a list of optimized values:

1. The real time-even invariant quotient in balanced degree $(5, 5)$ has dimension 24. The exact character calculation gives total dimension $90 - 70 + 20 - 4 = 36$ and time-reversal trace $32 - 16 - 8 + 4 = 12$, hence $(36 + 12)/2 = 24$. Evaluation and the r, e jets are three independent functionals. For the third jet one can use $I = Q : W$, $W_{ij} = -36\,\mathrm{Tr}(\rho_3^2 \widehat{Q}_{ij})$; its exact formula is

$$I = (1 - r^2)\left[-24e^2 Z - 4\sqrt{15}e\sqrt{1 - 4e^2}H\right].$$

   Its e derivative stays nonzero across the whole low chart. Thus the uniaxial contact, where the alternative $\mathrm{Tr}\, Q^3$ jet degenerates, is not lost.

2. The annihilating quintic multiplet spaces in spins $1, 3, 4$ have dimensions $5, 9, 7$. Their symmetric norm image is contained in the 21-dimensional common jet kernel. An exact nonzero 21-minor makes it equal that kernel. The target belongs to it because $T_e = 0$ and the definitions of a, b kill the other two jets.

3. Fix the free Gram coefficients and solve the persisting 21 coordinates. This is an exact identity, not a small residual. Directed interval Neumann inversion and a strict rational positive base margin show that each resulting Gram matrix remains positive.

The supplement contains 1347 accepted closed interval reports. For each, the root boundary signs are opposite, $T_{ee} < 0$, all evaluation minors and the coefficient minor are invertible, and each Gram perturbation is smaller than its exact positive base bound. Their rational union is exactly $[r_-, r_+]$ in (402), with no gap. Fresh directed bounds also prove the phase/bridge overlap in every cell. This establishes a continuous unrestricted sharp boundary and complete fibers, not a finite collection of numerical witnesses.

The core input tables are `zero_dipole/global_tangent_exact_basis.json`, `zero_dipole/F0_tangent_24_whitened_frame.json`, the two `sharp_low_contact*_exact_basis.json` files and their exact Gram candidates. The interval catalog and independent union audit identify the accepted reports by name and hash. The analytical argument above specifies what every finite check establishes; the exact coefficients themselves are supplied rather than rounded into print.

28.3 Null contacts, finite arcs and the critical algebraic point

Lemma 8 (Finite null certificates)**.** *All nine rows of Table 1 are unrestricted global null extrema with one equality orbit. The rational supports* (422) *hold globally. Directed continuation gives the original four inner overlaps:* J_1, J_3, J_4 *in* (423) *and* $J_2^{\mathrm{old}} = [.173293, .173303]$. *The widened current* J_2 *is proved separately in Theorem 24.*

Exact finite identity and its equality argument. Every pure state admits the complete real harmonic-cubic kernel chart $\partial_X R + t\partial_Y I = 0$, $0 \le t \le 1$, with nine real chart coordinates z. The explicit

coordinate map is the one obtained by solving the linear kernel equation in the harmonic basis already used in Section 18; its exact coefficient table is included in each state-data file. No cyclic symmetry is assumed in this chart. Let h contain its quadratic monomials, split into the four principal half-turn characters, and use $v_3 = (h, th, t^2h, t^3h)$ and $v_2 = (h, th, t^2h)$. For a lower contact the supplied exact identity is

$$\frac{\tau_H - \tau_* N^2 + \mu(t)(|F_H|^2 - f_*^2 N^2)}{100} = \sum_b w_b(t)(Z_b^{\mathsf{T}} v_b)^{\mathsf{T}} P_b (Z_b^{\mathsf{T}} v_b) + \Re\Theta_H L_R + \Im\Theta_H L_I. \tag{435}$$

Here $N, \boldsymbol{F}_H, \Theta_H$ are the unnormalized quadratic physical moments, τ_H is quartic, μ is quadratic in t, and w_b is either 1 or $t(1-t)$. For an upper contact replace the first two terms in the numerator by $-\tau_H + \tau_* N^2$. The eight small P_b have dimensions $59, 44, 40, 30, 40, 30, 40, 30$ and are strictly positive definite. All entries lie in the explicitly isolated degree-four field $\mathbb{Q}(k)$. Every one of the 945 physical coefficient identities is exactly zero. Rational root isolation, rational congruences and integer diagonal-dominance or directed LDL bounds prove positivity; floating eigenvalues are not accepted.

On a null state at $f = f_*$ the multiplier terms vanish, proving the unrestricted extremum. Equality kills every unweighted facial feature. Its trivial character is proportional to the contact feature $(h_*, t_* h_*, t_*^2 h_*, t_*^3 h_*)$ and its other characters vanish. Since h contains coordinate squares and the state is nonzero, the first two levels force $t = t_*$; recombining characters gives $zz^{\mathsf{T}} = cz_* z_*^{\mathsf{T}}$. Norm fixes the scale and only the contact's rotation/phase orbit remains. Bounds on the quadratic $\mu(t)$ over $[0, 1]$ give the m, h supports in Table 1 with the signs in (422).

For finite continuation the coefficient operator has a fixed exact rank-870 part and a 75-dimensional quotient. A uniform invertible 73-minor solves the remaining coefficients. The last two equations are not discarded: evaluation at the contact and differentiation along its fixed-f physical null fiber are independent exact left syzygies. Their independence is the checked nonzero determinant $t_e z_0^8$. Stationarity (419) annihilates the target's derivative, so all 945 equations follow from the $870 + 73$ solve. Rational-congruence perturbation bounds prove all eight matrices positive throughout the recorded root rectangles. Uniform derivative bounds make the four original f intervals strict inner overlaps. The fourth-order replay of Section 31 enlarges the second null interval. This proves the finite arcs and completes the lemma. □

Lemma 9 (The exact critical C_4 endpoint)**.** *At $f = b_c$ the unique null minimum is Ω_{b_c}. The C_4 bound continues globally on $b_c \le f \le 3/8$. Its full physical null Hessian has one real zero mode at b_c; the other five modes are positive.*

Finite verification and the singular equality face. The same all-state harmonic chart has an exact certificate over $\mathbb{Q}(b_c)$, isolated by (425). Two Gram blocks are positive semidefinite with precisely one added kernel each; the other six are positive definite. All 945 coefficient equations and 99 kernel equations hold exactly, with total rank 1028. The supplement supplies the matrices, root interval and congruence bounds. A rational Schur/Bernstein continuation gives the whole interval up to $3/8$, with 148 positive matrix controls.

An enlarged linear Gram kernel does not by itself add physical states. In the trivial character it permits $t^k h = u + c(dh + k\kappa h_0)$, $k = 0, 1, 2, 3$, with $\kappa \neq 0$. Its second difference gives $(t-1)^2 h = 0$. Coordinate squares make $h \neq 0$, so $t = 1$; its first difference then gives $c = 0$. The old rank-one C_4 equality face therefore survives unchanged. It gives exactly Ω_{b_c}, not an unclassified larger equality set. Direct physical differentiation of the spin-three moments gives the one zero and five positive Hessian modes used in the cusp proof. The same critical multiplier audit gives $\tau \ge \tau_{b_c} + \frac{1}{2}(b_c^2 - f^2)$ for $f \le b_c$, which is the first branch of (391). □

28.4 The first side-anchor rigidity input

For the off-side theorem in Section 27.5, equality must be classified, not inferred from the quadrupole spectrum alone. The finite proof has the following complete logical chain.

The strict quintic support at r_* kills its $5, 9, 7$ multiplets. Couple these with ψ and $\mathsf{T}\psi$ to sextic multiplets and take time-even norms and cross-norms. The exact degree-$(6, 6)$ invariant space on

$\boldsymbol{F} = 0$ has dimension 76 and time-reversal trace 18, so its time-even part has dimension 47. The three evaluation/r/e jets leave dimension 44. Forty-four selected exact sextic scalars are independent: their rational-point evaluation determinant is nonzero modulo 1000000007 and independently modulo 1000000009, with residues 682680425 and 845874211. This is an exact rational determinant test, not sampled numerical rank.

Hence every equality evaluation annihilates the entire three-jet kernel. The nonnegative scalar $(q - q_* N^2)^2 N^2$ fixes $q = q_*$. Writing $\kappa = \operatorname{Tr} Q^3$, the values of κ and κ^2 then remove the remaining e-jet and fix every time-even invariant through degree six. With

$$V_{ij} = \langle \mathsf{T}\psi | \widehat{Q}_{ij} | \psi \rangle, \quad C = \Re(\overline{\Theta} V), \quad D = \Im(\overline{\Theta} V), \quad W_{ij} = -36 \operatorname{Tr}(\rho_3^2 \widehat{Q}_{ij}),$$

the positive commutator norms force Q, C, W to commute. τ, κ fix three distinct Q eigenvalues. Trace zero and the contractions with Q, Q^2 fix C, W by a Vandermonde system. The universal pure-state identity on $\boldsymbol{F} = 0$,

$$\Re(V V^\dagger)_{\mathrm{tf}} = (Q^2)_{\mathrm{tf}} + 5NQ + \frac{3}{2} W - 5C,$$

fixes D^2 to its canonical $\operatorname{diag}(d^2, d^2, 0)$. The square of the time-odd scalar $Q : D$ is among the fixed invariants, so $Q : D = 0$; consequently D has only its xy entry, whose sign is removed by a half-turn. This fixes the *full* transition pencil. It does not claim $[Q, V] = 0$, which would be false here.

Finally the physical $N, \Theta, \boldsymbol{F}, Q, V$ determine all 21 real quadratic entries of the six-by-six slice Gram matrix of the real and imaginary harmonic cubics. The exact change of basis is invertible. A positive parent matrix, made from eight of these fixed quadratic forms, has rank 13 and kernel precisely the canonical real 14-amplitude ray. Its 13 nonzero LDL pivots are proved positive by directed arithmetic; its kernel is exact. Any physical state with this pencil is therefore the canonical ray. This completes the unique equality-orbit input. The full five-mode negative Hessian and positive dipole response are independently checked, so the analytical continuation/compactness proof in Section 27.5 applies.

Portability, reproduction and honest display scope. The supplement's manifest lists the exact coefficient tables and accepted reports, not numerical optimizer outputs. Its verifier sources reconstruct direct spin operators and covariants independently. Research drivers retain the strict absolute time guards requested for the search; a later expressly requested replay should set a fresh bounded audit deadline in a copy, preserving the frozen inputs. The resulting mathematical statements distinguish:

- explicit attained states and continuous fiber coverage;
- unrestricted sharpness/exclusion from exact identities and rigorous intervals;
- numerical marginal-purity checks, which test implementation only.

The four current f strips are finite and may be displayed, with the enlarged second interval at $q = 0$. The previously proved positive-q widths on the original intervals, the off-side sharp patch widths and the critical cusp width are not quantified and are not assigned an invented plotting thickness. Two small zero-dipole tails retain the error bar (418). The remaining roof gaps have the full constructive C_2 band proved in Theorem 22, but their unrestricted completeness and the general nonzero-f, nonzero-q boundary are not characterized.

29 The complete complex-C_2 null-roof image

This section describes an *attained inner band* over the entire small-dipole range. It does not assume that an unrestricted extremizer has a cyclic symmetry. The separate unrestricted conclusion, on the finite intervals where it is proved, is stated in Section 31. All notation is that of the spin-multipole decomposition used above:

$$f = |F|, \quad \tau = \operatorname{Tr} Q^2 = 126 a_2, \quad q = |\Theta|^2, \qquad S_1 = \frac{2}{3} f^2, \quad S_2 = 5 + \frac{4}{15}\tau, \quad S_3 = 8 + \frac{4}{3} f^2 - 8q. \tag{436}$$

Here $F_i = \langle J_i\rangle$, $Q_{ij} = \langle\{J_i, J_j\}/2\rangle - 4\delta_{ij}$, and $D_k = |D_6^{(k)}\rangle$ satisfies $J_z D_k = (3-k)D_k$. The *null roof* means $\Theta = 0$, equivalently $q = 0$. Write $b_c = 0.366379159399607379\ldots$ for the root in $(0, 3/8)$ of (425).

29.1 One normalized family and an exact scalar selector

For a signed real parameter u, define

$$d_u = 1 + 15u^2, \qquad e = \frac{1-15u^2}{d_u}, \qquad \nu = f - 2e, \qquad H_u = \frac{1+6u-15u^2}{d_u},$$
$$c = \sqrt{\frac{1+\nu}{2}}, \qquad s = \sqrt{\frac{1-\nu}{2}}. \tag{437}$$

Whenever $|\nu| \le 1$, the state

$$\boxed{|\chi(f,u)\rangle = \frac{cD_0 + sD_2 + \sqrt{15}\,ucD_4 - \sqrt{15}\,usD_6}{\sqrt{d_u}}} \tag{438}$$

is normalized and has $\Theta = 0$, $F = (0, 0, f)$ and

$$\tau = \mathcal{T}(u,f) := \frac{3}{2}(1+4e\nu)^2 + \frac{15}{2}(1-\nu^2)H_u^2. \tag{439}$$

The following degree-eight polynomial selects its stationary states:

$$\begin{aligned} P_f(u) &= A(u)f^2 + B(u)f + C(u),\\ A(u) &= (3u+1)(5u-1)d_u^2(15u^2-30u-1),\\ B(u) &= 4d_u\big(3375u^6 - 8325u^5 - 2475u^4 + 1410u^3 + 165u^2 - 37u - 1\big),\\ C(u) &= (15u^2+6u-1)(45u^2-10u-3)\\ &\qquad \times (225u^4 - 900u^3 - 90u^2 + 60u + 1),\\ \partial_u\mathcal{T}(u,f) &= -\frac{90P_f(u)}{d_u^5}. \end{aligned} \tag{440}$$

Theorem 22 (Complete restricted image, including complex phases)**.** *For every $0 \le f \le b_c$, P_f has exactly one root $u_L(f)$ in $(7/20, 2/5)$ and exactly one root $u_U(f)$ in $(19/100, 23/100)$. Set*

$$\tau_L(f) = \mathcal{T}(u_L(f), f), \qquad \tau_U(f) = \mathcal{T}(u_U(f), f). \tag{441}$$

Among all complex C_2-symmetric pure states*, allowing both characters of the half-turn, the complete null-roof sector image at dipole length f is exactly*

$$\boxed{S_1 = \frac{2}{3}f^2, \quad S_3 = 8 + 2S_1, \qquad 5 + \frac{4}{15}\tau_L(f) \le S_2 \le 5 + \frac{4}{15}\tau_U(f).} \tag{442}$$

Every intermediate value is attained by (438) *with $u_U(f) \le u \le u_L(f)$. On that interval $\mathcal{T}$ is strictly decreasing, so there is a unique such u for every interior value of S_2. Each extremum has a single equality orbit up to collective rotations and a global phase, represented by $|\chi(f, u_L(f))\rangle$ or $|\chi(f, u_U(f))\rangle$, respectively.*

The branches are strictly decreasing in f. Their useful endpoint identifications are

$$u_L(0) = \frac{5+4\sqrt{10}}{45}, \qquad \tau_L(0) = \frac{3}{8}, \qquad \tau_U(0) = 30 - 6\sqrt{15},$$
$$u_U(1/3) = \frac{1}{5}, \qquad \tau_U(1/3) = \frac{25}{3},$$
$$u_L(b_c) = 0.358463863204305\ldots, \qquad 585u_L(b_c)^4 + 180u_L(b_c)^3 - 114u_L(b_c)^2 - 12u_L(b_c) + 1 = 0. \tag{443}$$

The upper origin $u_U(0)$ is the unique root in its stated rational box of $225u^4-900u^3-90u^2+60u+1=0$. At $f=b_c$ the lower state reduces to the C_4 state

$$|\chi(b_c,u_L(b_c))\rangle=\frac{\sqrt{1+b_c}D_0+\sqrt{3-b_c}D_4}{2}. \tag{444}$$

For $f>1/3$, the upper branch of this theorem is only a *restricted* upper optimum: the already attained paired rim $S_2=7+2f^2$ lies above it. Its construction in Theorem 8 is retained and is not relabeled as a C_2 family.

29.2 Why phases and the other character add no missing points

Proof of the reduction and coverage. Choose a symmetry-axis frame. An even-character state is $AD_0+BD_2+CD_4+DD_6$. Its nullness condition is $AD+BC=0$; therefore the matrix $\left(\begin{smallmatrix}A & B\\ C & -D\end{smallmatrix}\right)$ has rank one. Normalization gives

$$A=ac,\ B=as,\ C=bc,\ D=-bs,\qquad |a|^2+|b|^2=|c|^2+|s|^2=1.$$

Temporarily allow complex a,b,c,s. An axial collective rotation and a global phase leave one phase ϕ, so we can take

$$a=\sqrt{(1+e)/2},\quad b=e^{i\phi}\sqrt{(1-e)/2},\quad c=\sqrt{(1+\nu)/2},\quad s=\sqrt{(1-\nu)/2}.$$

The transverse dipole vanishes; a half-turn chooses $F_z=f\ge 0$. Consequently $\nu=f-2e$ and $e\in[(f-1)/2,(f+1)/2]$. Direct spin-multipole matrix elements give

$$\begin{aligned} Q_{zz}&=1+4e\nu,\qquad Q_{xz}=Q_{yz}=0,\\ \langle J_+^2\rangle&=\sqrt{1-\nu^2}\big(\sqrt{15}e+3\sqrt{1-e^2}e^{i\phi}\big),\\ \tau&=\frac{3}{2}(1+4e\nu)^2+\frac{1-\nu^2}{2}\big(15e^2+9(1-e^2)+6\sqrt{15}e\sqrt{1-e^2}\cos\phi\big). \end{aligned} \tag{445}$$

Thus the objective is affine in $\cos\phi$. An extremum occurs at $\phi=0$ or π; a signed u represents both possibilities in (438). The phase-flat case $e=0$ has value $6-9f^2/2$ and is not extremal: a two-level endpoint has value at most $3/2$, whereas the test state $\nu=0,e=f/2$ has value >6 for $f>0$. At $f=0$ the choice $e=7/47,\nu=-14/47$ gives $32867511/4879681>6$. At $\nu=\pm1$ only two Dicke levels remain and the relative phase is removable by an axial rotation.

Feasibility in the real parameter is exactly

$$\frac{1-f}{15(3+f)}\le u^2\le\frac{3-f}{15(1+f)}. \tag{446}$$

Differentiation gives (440). Hence an elementary finite exact answer, even before ordering branches, is to evaluate $\mathcal{T}$ at every feasible real root of P_f and at $E_\mp=\frac{3}{2}(2f\mp1)^2$, and take the minimum and maximum. The polynomial is never identically zero: its leading coefficient is $50625(f+1)(f+3)>0$. There are at most eight stationary candidates. The exact ordering established below selects u_L,u_U.

For the odd character the null equation and the same phase freedoms give, with $t=f/2$, the representative

$$\frac{1+t}{2}D_1+i\sqrt{\frac{1-t^2}{2}}D_3+\frac{1-t}{2}D_5,\qquad \tau_{\rm odd}=6+\frac{9}{2}f^2-\frac{3}{2}f^4. \tag{447}$$

This is strictly inside the even-character interval: a two-level even state gives at most $3/2$, and the even test state above gives

$$\tau_{\rm test}=6+\frac{3}{4}f^2+\frac{3\sqrt{15}}{2}f\sqrt{1-f^2/4}>\tau_{\rm odd}\qquad(0<f\le b_c).$$

The displayed inequality follows already from $f < 3/8$, $\sqrt{15} > 3$ and $\sqrt{1 - f^2/4} > 1/2$; at $f = 0$ use (443).

Finally u_U, u_L are consecutive roots, the sixth and seventh among the eight real roots. The sign of P_f between them is positive. Equation (440) makes $\mathcal{T}$ strictly decreasing there. The whole connecting interval is feasible, since e varies monotonically and $\nu = f - 2e$ remains between its endpoint values. This proves the complete interval and the unique scalar recovery, once the following finite ordering check is supplied. □

29.3 Exact ordering and equality orbits

For transparency we specify the finite arithmetic behind the branch selection, rather than infer it from sampled curves. This part is a computer-assisted *rational-polynomial* proof, not an SOS or numerical optimization argument. The accompanying exact supplement in Section 31.3 contains all integer polynomials, rational isolating intervals and Bernstein coefficients. They are reconstructed from (439)–(440).

The discriminant of P_f with respect to u is a positive constant times a degree-12 polynomial $D_{12}(f^2)$. Its thirteen Bernstein coefficients on $0 \le f^2 \le 9/64$ are positive. Exact root isolation at $f = 1/10$ gives eight real roots; therefore no root merges or disappears on $0 \le f \le 3/8$. To compare *values*, write $\mathcal{T} = N_T/D_T$ in reduced form and take the primitive integer part of

$$\operatorname{Res}_u(P_f(u), N_T(u, f) - vD_T(u, f)) = R(v, f^2).$$

The resulting R has degree eight in v, leading coefficient 16384, and discriminant a positive constant times $w^{10}D_{12}(w)^3D_{17}(w)^2$, with $w = f^2$. All eighteen Bernstein coefficients of D_{17} on $[0, 9/64]$ are negative. Thus stationary values never cross for $f > 0$. For any polynomial $\sum_{j=0}^{n} a_j w^j$, the coefficients checked here are the explicit rational sums

$$b_k = \sum_{j=0}^{k} a_j (9/64)^j \frac{\binom{k}{j}}{\binom{n}{j}}, \qquad 0 \le k \le n. \tag{448}$$

At $f = 1/10$, disjoint rational interval evaluations give, for $u_1 < \cdots < u_8$,

$$\mathcal{T}_1 < \mathcal{T}_7 < \mathcal{T}_3 < \mathcal{T}_5 < \mathcal{T}_2 < \mathcal{T}_6 < \mathcal{T}_4 < \mathcal{T}_8. \tag{449}$$

The feasible indices there are $2, 3, 6, 7$. Eliminating u with the two boundary quadratics in (446) shows that the only feasibility changes occur at b_c and at the root $g_c = 0.3296162566349908\ldots$ of $16f^4 + 78f^3 + 92f^2 - 6f - 11 = 0$. At g_c root u_5 enters the feasible set; it is not extremal by (449). At b_c root u_7 reaches $\nu = 1$. The boundary remainders give the respective roots exactly as

$$u_7 = -\frac{(f-3)(f-1)}{7f^2 - 18f + 1}, \qquad u_5 = -\frac{(f-1)(f+1)}{7f^2 + 18f + 1}.$$

Neither two-level endpoint is a maximum: both have value below $147/32 < 6$, whereas the even test state has value at least six. Neither is a minimum before b_c: at the smaller endpoint $\nu = 1$ the inward derivative, choosing the decreasing phase, is

$$-21f^2 + 54f - 3 - 3\sqrt{15}(1 - f)\sqrt{3 + 2f - f^2} < 0.$$

Its sign follows by squaring from

$$(-7f^2 + 18f - 1)^2 - 15(1 - f)^2(3 + 2f - f^2) = 4(16f^4 - 78f^3 + 92f^2 + 6f - 11) < 0.$$

At b_c that endpoint equals u_7. These finite checks prove the selection of $u_L = u_7$ and $u_U = u_6$ over the entire range.

There is exactly one root in each fixed rational box of the theorem: the endpoint substitutions have no zero on $[0, 3/8]$, and the discriminant and anchor root counts fix their number. The discriminant of

$P_f(u)$ as a quadratic in f is $4d_u^2 Z_{12}(u)$. Exact Sturm counts give no zero of Z_{12} in either box. Hence $\partial_f P_f$ is nonzero along each branch, as is $\partial_u P_f$. At the anchor both are negative on the lower branch and both positive on the upper branch. Implicit differentiation proves $du_L/df < 0$ and $du_U/df < 0$. The factorizations at $f = 0$ and $\nu = 1$ give (443).

For $f > 0$ the symmetry axis is the nonzero dipole axis. Phase reduction and the distinct stationary values leave only the single canonical orbit per extremum. At $f = b_c$ the relative phase of the two-level C_4 state is removable. At $f = 0$ the paired feasible roots are related by $u \mapsto -1/(15u)$ and

$$|\chi(0, -1/(15u))\rangle = iR_z(\pi/2)R_y(\pi)|\chi(0, u)\rangle. \tag{450}$$

Thus the paired roots do not create extra origin equality orbits. This completes the equality-state classification within the stated complex-C_2 class.

30 The entire C_2 arcs are physically locally stable

The restricted theorem is strengthened here to a *local* statement among arbitrary pure symmetric states. This is a useful compatibility test, but it is logically different from an unrestricted global bound.

Theorem 23 (All six internal modes)**.** *At fixed $f = |F|$ and $\Theta = 0$, the lower canonical state of Theorem 22 is a strict local minimum of τ for $0 < f < b_c$, modulo collective rotations and global phase. The upper canonical state is a strict local maximum for $0 < f < 1/3$. Perturbations breaking C_2 symmetry are included.*

30.1 A native multipole Hessian, with no support restriction

Write the Dicke coefficients as $c_k = \sqrt{\binom{6}{k}}(x_k + iy_k)$ and put $X = (x_0, \ldots, x_6, y_0, \ldots, y_6)^T$. If $S = \operatorname{diag}\sqrt{\binom{6}{k}}$, a Hermitian observable A has the real symmetric representation

$$\mathcal{R}(A) = \begin{pmatrix} \Re(SAS) & -\Im(SAS) \\ \Im(SAS) & \Re(SAS) \end{pmatrix}.$$

Let $C_N = \operatorname{diag}(S^2, S^2)$, $C_i = \mathcal{R}(J_i)$, and define the singlet matrices by

$$B_{kl} = (-1)^k \binom{6}{k} \delta_{l,6-k}, \quad C_{\Theta R} = \begin{pmatrix} B & 0 \\ 0 & -B \end{pmatrix}, \quad C_{\Theta I} = \begin{pmatrix} 0 & B \\ B & 0 \end{pmatrix}.$$

Finally set $A_{ij} = \mathcal{R}(\{J_i, J_j\}/2 - 4\delta_{ij} I)$. All these matrices are rational. They give $X^T C_N X = 1$, $X^T C_i X = F_i$, $X^T C_{\Theta R} X = \Re\Theta$, $X^T C_{\Theta I} X = \Im\Theta$ and $X^T A_{ij} X = Q_{ij}$. This is pure-state spin-multipole compatibility expressed in real coordinates, not an independent-moment relaxation.

For the canonical state use the unnormalized vector

$$\begin{gathered} Z = (1, 0, k/15, 0, u, 0, -uk, 0, \ldots, 0)^T, \\ N = (1 + 15u^2)(1 + k^2/15), \quad X = Z/\sqrt{N}, \quad k = \sqrt{\frac{15(1-\nu)}{1+\nu}} > 0. \end{gathered} \tag{451}$$

The stationarity multipliers λ, μ, κ solve

$$4 \sum_{ij} Q_{ij} A_{ij} Z = 2\lambda C_N Z + 2\mu C_z Z + 2\kappa C_{\Theta R} Z. \tag{452}$$

The even real coordinates $0, 2, 4$ determine them; their coefficient determinant is $16[(15u^2 - 1)k^2 + 450u^2]$. The remaining component vanishes by $P_f(u) = 0$ after $\nu = (15 - k^2)/(15 + k^2)$. Every odd or imaginary component vanishes by parity and conjugation. Thus this is *full* stationarity, not just stationarity in a four-level ansatz.

The real Lagrangian Hessian is

$$\mathcal{H} = \frac{8}{N}\sum_{ij}(A_{ij}Z)(A_{ij}Z)^T + 4\sum_{ij}Q_{ij}A_{ij} - 2\lambda C_N - 2\mu C_z - 2\kappa C_{\Theta R}. \tag{453}$$

The sum includes all nine Cartesian pairs; off-diagonal components are counted twice, as required for $\operatorname{Tr} Q^2$. Restrict it to the linearized norm, two real nullness and three fixed-vector equations. Fixing $F = (0, 0, f)$ rather than only its length removes collective tilt directions, without changing internal stability for $f > 0$. The eight-dimensional constrained tangent splits as follows.

Subspace	Constrained dimension	Internal dimension
Even real	1	1
Even imaginary	3	1 (phase and axial rotation removed)
Odd real	2	2
Odd imaginary	2	2

The cross-block entries vanish exactly. To construct bases without numerical nullspaces, take cofactors of the three even real constraint rows. In the even imaginary sector take cofactors of the imaginary nullness row and the two weighted gauge rows $C_N Z, C_z Z$. For either odd sector, if its single constraint row is (g_0, g_1, g_2), use columns $g_0 e_i - g_i e_0$, $i = 1, 2$. The relevant pivots are nonzero on the open branches. The resulting physical Hessian consists of two scalars and two 2×2 matrices.

30.2 Exact signs and the limitation of the result

The entries are rational in u, k. Clear their denominators and eliminate k with the stationary equation obtained by substituting $f = 2e + (15 - k^2)/(15 + k^2)$ into $P_f(u) = 0$. Exact Sturm checks of all resultant factors leave only the following possible zeros in the fixed rational branch boxes of Theorem 22:

Physical block	Lower branch	Upper branch
Even real scalar	$585u^4 + 180u^3 - 114u^2 - 12u + 1$	None
Even imaginary scalar	None	None
Each odd determinant	$45u^2 - 10u - 3$	$5u - 1$ or $225u^4 - 900u^3 - 90u^2 + 60u + 1$

By the monotonicity already proved, these are *endpoint* events: the lower origin, the C_4 join, the upper origin and $f = 1/3$. No physical eigenvalue vanishes on either stated open arc. Constraint-minor and odd-pivot denominators are checked in the same way. These are computer-assisted exact polynomial checks; the complete factor lists and Sturm counts, including nonvanishing denominators, are supplied with reproducible matrix construction in Section 31.3.

One exact anchor fixes the inertia on each connected branch. For $u = 3/8$ take the positive root $k \in (.948, .949)$ of

$$-653656k^4 + 63281005k^2 - 56385000 = 0.$$

Rational interval evaluation makes both scalar blocks positive and both odd blocks positive definite. For $u = 2/9$ take $k \in (4.928, 4.929)$ satisfying

$$-23126k^4 + 298685k^2 + 6387750 = 0.$$

Both scalars are negative and both odd blocks negative definite. The latter assertions use the signs of the leading diagonal entries and positive determinants, not numerical eigenvalues. Continuity and absence of zero determinants preserve the inertia. This proves Theorem 23. An independent construction checks all six constraints, both gauge directions, the block decomposition and the anchor signs directly from the rational spin matrices.

Strict local stability does *not* rule out a distant competing state with smaller or larger τ. A first-order crossing of two local extrema need not make either Hessian singular. Consequently this theorem alone neither proves the unrestricted roof nor extends a global bound or a positive-q plotting collar.

31 A substantially wider globally complete null-roof strip

This section adds an *unrestricted* theorem: the supporting inequalities hold for arbitrary pure symmetric states, not merely for C_2 states or nearby perturbations. It replaces the second narrow null interval in (423). The necessary three-dimensional body $\mathcal{R}_{22}$ is unchanged: no finite positive-q width is asserted by this update.

Theorem 24 (Complete unrestricted fibers on a closed strip)**.** *Let*

$$I = \left[\frac{170799}{10^6}, \frac{175797}{10^6}\right]. \tag{454}$$

For every $f \in I$, the complete image of all *pure symmetric six-qubit states with $|F| = f$ and $\Theta = 0$ is exactly* (442)*. Each endpoint has the single collective equality orbit given in Theorem 22. Equivalently, the following conditional sector inequalities are sharp and all intermediate values are attainable:*

$$\begin{gathered} S_3 = 8 + 2S_1, \qquad \frac{9724099467}{500000000000} \le S_1 \le \frac{10301528403}{500000000000}, \\ 5 + \frac{4}{15}\tau_L\left(\sqrt{3S_1/2}\right) \le S_2 \le 5 + \frac{4}{15}\tau_U\left(\sqrt{3S_1/2}\right). \end{gathered} \tag{455}$$

Current result	Precise scope	Proof
Full C_2 null band	Both complex characters, $0 \le f \le b_c$; all fibers and restricted equality orbits	Thm. 22
Six-mode stability	Arbitrary perturbations; local minimum for $0 < f < b_c$, local maximum for $0 < f < 1/3$	Thm. 23
Widened complete strip	Arbitrary pure states, $q = 0$ and $f \in [.170799, .175797]$	Thm. 24

This interval has 499.8 times the width of the former $[.173293, .173303]$ interval. The other three intervals in (423) remain valid. Outside the four globally certified strips, the band of Theorem 22 is an attained *inner* band, not an exclusion of all other states. For $f > 1/3$ the previously attained paired-circle extension remains part of the image as well.

31.1 What is certified and why it is a global proof

The all-state harmonic-cubic chart used in Section 28 has nine real coordinates z and $0 \le t \le 1$; it is not a C_2 ansatz. Its homogeneous polynomial moments are denoted N, F_H, Θ_H, τ_H. For the lower branch the exact identity has the form

$$\begin{aligned} \frac{1}{100}\left\{\tau_H - \tau(u)N^2 + \mu(t;u)(|F_H|^2 - f(u)^2N^2)\right\} = \sum_b w_b(t)\, V_b(t,z;u)^T P_b(u) V_b(t,z;u) \\ + \Re\Theta_H\, L_R + \Im\Theta_H\, L_I. \end{aligned} \tag{456}$$

Here w_b is 1 or $t(1-t)$, the V_b are explicitly supplied polynomial feature vectors and all P_b are strictly positive definite. For the upper branch the sign of $\tau_H - \tau(u)N^2$ is reversed. At fixed f and $\Theta = 0$ all multiplier terms vanish; the nonnegative right-hand side proves the respective bound for every pure state in the chart. The existing chart coverage and overlap checks cover all states. Equality forces the feature kernel, whose physical states form the canonical orbit. This is an explicitly disclosed computer-assisted positive-Gram identity; it is not being presented as a certificate-free proof.

Matching coefficients gives 945 polynomial equations for 6743 Gram and multiplier entries. A constant rank-870 part and a variable rank-73 quotient leave two compatibility syzygies. They are proved identically from physical evaluation and fixed-f stationarity, rather than discarded using a numerical rank threshold. The eight Gram orders are $59, 44, 40, 30, 40, 30, 40, 30$.

31.2 Exact fourth-order transport and directed interval control

For clarity, write the coefficient identity as $A(u)x(u) = b(u)$. At an algebraic anchor u_0, all data lie in the exact field $\mathbb{Q}(k_0)$. Put $v = u - u_0$. The first four Taylor coefficients x_j are supplied as rational field elements and satisfy

$$A_0 x_n = b_n - \sum_{\ell=1}^{n} A_\ell x_{n-\ell}, \qquad 0 \le n \le 4. \tag{457}$$

The verifier independently reconstructs each physical right-hand side from the preceding coefficients before testing all 945 equations; checking only a stored right-hand side would not suffice. The implicit stationary quartic determines each k-jet recursively, without finite differences.

For the trial polynomial $X(v) = \sum_{j=0}^{4} v^j x_j$, the first five coefficients of the defect $b - AX$ vanish exactly. At an intermediate point u_*, its fifth Taylor coefficient is

$$b_5(u_*) - \sum_{h=0}^{4} A_{5-h}(u_*) X_h(u_*), \qquad X_h(u_*) = \sum_{j=h}^{4} \binom{j}{h} (u_* - u_0)^{j-h} x_j. \tag{458}$$

Directed interval evaluation encloses this coefficient over the entire cell; Taylor's theorem bounds the defect by v^5 times that enclosure. Thus no sampled Hessian or sampled matrix positivity is used as a uniform proof.

The variable quotient is normalized as $M(u) = M_0 + vM_d(u)$. For a fixed rational derivative approximation D, use the polynomial left preconditioner $S_p(v) = \sum_{j=0}^{p} (-vD)^j$. The exact identity

$$I - S_p M = (-vD)^{p+1} - S_p\big[(M_0 - I) + v(M_d - D)\big] \tag{459}$$

gives a directed row-norm enclosure $\eta < 1$. After multiplication by S_p, the inverse is enclosed through the fourth Neumann power and the tail bound $\eta^5 \|y\|_\infty / (1 - \eta)$. Constant back-substitution and the two exact syzygies restore all 945 equations. Positivity is checked after a fixed rational congruence of every Gram block. Each common scalar factor v^j is retained outside its point-jet congruence; only the residual correction is intervalized entrywise. This preserves the genuine matrix cancellations while giving a rigorous norm bound. The total congruenced perturbation is smaller than the certified positive Gershgorin margin in every block.

The two independently replayed cells used in this theorem are

	Lower	Upper
Anchor u_0	3/8	10643749/50000000
Radius $\lvert u - u_0 \rvert$	1/4000	3/10000
Root-rectangle slope bound	30	18
Taylor order	4	4
Preconditioner order p	7	5
Certified inner f half-width	1/400	33/10000

The algebraic central dipoles are respectively $0.17329807134637065\ldots$ and $0.1732980136031291\ldots$. Exact rational interval comparison, not rounded decimal subtraction, proves that both inner dipole images contain (454). The monotone f maps cover that interval. The lower and upper identities therefore give the two unrestricted inequalities simultaneously. The monotone canonical path in Theorem 22 fills every value between them, completing the proof of Theorem 24.

31.3 Finite proof inputs and portable verification

The additional supplement is `assets/N6_null_roof_exact_proofs_20261022.zip`. Its manifest records a SHA-256 digest for each input. It includes the exact C_2 branch audit and resultant, all rational Hessian matrices and sign audits, the physical coefficient builder, the two algebraic point

certificates, their four exact transport jets and the directed interval replay programs. The large finite arithmetic data are supplied as a sidecar rather than typeset as thousands of entries. No earlier chat, fitted witness or optimizer is a proof input. The verification-only recipe in its `README.txt` uses saved exact data and bounded replay; it is usable after the research deadline and does not restart a search.

The exact rational-polynomial audits reconstruct (440), the branch discriminants, the primitive critical-value resultant, all rational root boxes and Bernstein coefficients. The Hessian audits independently rebuild the spin matrices, tangent bases, resultants and anchor signs. The two finite interval replays independently rebuild each jet right-hand side, verify the coefficient identities and report every inverse and Gram positivity margin. Finally the intersection audit proves the exact rational endpoints in (454). These are distinct checks with distinct scopes: restricted globality, full physical *local* stability, and unrestricted globality on the stated strip.

In the figures the full band (442) is violet (or C_2 blue in the family-resolved comparison) because explicit states attain it. This does not change the defining inequalities of $\mathcal{R}_{22}$, assert a sharp unrestricted roof outside the certified strips, or assign a numerical positive-q collar. Gray therefore still means excluded *or not yet certified*.

32 Exclusion of the ten gray junctions and the stronger flow cap

The ten gray markers audited in this section are the actual junctions of the displayed necessary-condition body immediately before adopting the exact comparison integrals (394). They are not the older three points A, B, C of Section 21, which had already been superseded. None of these ten tuples is attained by a pure symmetric six-qubit state. Three exclusions use the sharp-plane equality line, six use a strict step in the null-seed disk lift, and the last uses the improved flow bound. No failed state fit is an exclusion argument.

The revised figures retain the same named states, labels, leaders and attained arcs. The former flow-sheet triple G_4 is cut away and is not retained as a current marker; any replacement junction is computed from the new active inequalities. The other nine listed junctions remain excluded even where they survive as corners of the closed outer region. Exclusion of these points does not classify every neighboring gray surface.

32.1 Which ten points are meant

Table 2 locates the marker set. Except for the exact fractions in its first row, decimals are approximate locators of analytical intersections, not rational specifications of physical states. The exact definitions and obstructions follow the table.

Node	S_1	S_2	S_3	Obstruction
G_1	7/16	185/16	45/8	Sharp plane
G_2	.421428616	12.349997816	4.757145264	Sharp plane
G_3	1.729454204	10.665328164	11.000717964	Sharp plane
G_4	.000003960276	6.863220539	7.999065782	Flow lift
G_5	.000286017674	5.000527920	7.915797506	Disk lift
G_6	.002499877202	5.004735053	7.945765669	Disk lift
G_7	.019982392626	5.028072008	8.024346202	Disk lift
G_8	.028920917175	5.035165762	8.049131341	Disk lift
G_9	.061127933341	5.041531578	8.121563439	Disk lift
G_{10}	.089489117992	5.028292839	8.178977553	Disk lift

Table 2: The ten excluded gray junctions of the preceding rendering. The revised stronger flow cap removes G_4. A marker denotes an analytical surface or curve junction, not necessarily a polyhedral vertex. No marker is inferred from a projected crossing or a mere change of attainment color.

Throughout, the manuscript spin-multipole notation is

$$F_i = \langle J_i \rangle, \quad f = |F|, \quad Q_{ij} = \left\langle \frac{J_i J_j + J_j J_i}{2} \right\rangle - 4\delta_{ij}, \quad \tau = \operatorname{Tr} Q^2, \quad q = |\Theta|^2.$$

Thus $x = 2f^2/3$, $y = 5 + 4\tau/15$ and $z = 8 + 2x - 8q$.

32.2 The three sharp-plane junctions

Proposition 9. *The junctions G_1, G_2, G_3 are unattainable.*

Proof. Each lies on $65y + 63z - 217x = 1011$. Every pure equality state must belong to the fully attained segment (293); in particular

$$539x + 395y = 4719, \qquad 0 \le q \le \frac{169}{484}, \qquad y \le \frac{618}{55}. \tag{460}$$

At G_1, $q = 13/32 > 169/484$; equivalently the left side of the line equation exceeds 4719 by 84.

For G_2 put $t_0 = (196 + \sqrt{819})/262$. Its exact coordinates are

$$x = \frac{28 - 10t_0^2}{49}, \qquad y = 5 + 10t_0^2, \qquad z = 15 + 5x - y.$$

Since $t_0 > 17/20$, one has $y > 12 > 618/55$, contradicting (460).

The retained spectral–orange–plane junction G_3 has the exact parametrization

$$x = \frac{-406 - 750t^2 + 1120t}{7}, \qquad y = 5 + 10t^2, \qquad z = \frac{1011 + 217x - 65y}{63}, \quad \frac{71}{100} < t < \frac{4}{5}.$$

Here the unique exposed root is selected by equality in (383). Its precise radical expression is unnecessary: throughout the stated isolating interval,

$$539x + 395y - 4719 = -2(10t - 7)(2690t - 2429) > 0.$$

The two factors have opposite signs there. Thus G_3 cannot satisfy the necessary equality line either. □

32.3 Why the six lower disk-lift tips cannot be attained

The input is six certified equality orbits, not a whole-arc conjecture. Let f_i, τ_i be the exact five lower contacts in Table 1, followed by $f_6 = b_c$ and $\tau_6 = 3(1 - 2b_c)^2/2$. Lemmas 8 and 9 prove that each is an unrestricted null minimum with one equality orbit. These finite equality classifications are the only unrestricted sharpness inputs used below. No statement about gaps between the certified contacts is needed.

For clarity, the exact definition of G_{i+4}, $1 \le i \le 6$, is as follows. Put $U = f_i^2$, $c_U = 30 - 15U/2$, and

$$\tau_i(q) = \left[\sqrt{\tau_i} - \sqrt{q(\tau_i + c_U)}\right]_+^2,$$
$$\mathcal{C}(U, q) = \max\left\{0, \sup_{0<v<1} \frac{3}{8}(1 - v)\left[1 - \frac{81q}{v} - \frac{16U(1 - q)}{3 + v}\right]\right\}.$$

The second expression is simply the full supporting family (6i) solved for τ at $x = 2U(1 - q)/3$. The tip overlap is the selected positive contact q_i of $\tau_i(q) = \mathcal{C}(U, q)$, on the branch where $\tau_i(q) > 0$. Its coordinates are

$$x_i = \frac{2}{3} f_i^2 (1 - q_i), \quad y_i = 5 + \frac{4}{15}\tau_i(q_i), \quad z_i = 8 + 2x_i - 8q_i. \tag{461}$$

These definitions use the exact algebraic contacts, rather than the floating-point locators in Table 2.

Lemma 10 (Strictness test for a lower bias-disk lift)**.** *Suppose $0 \le U \le 2/3$ and the null minimum at squared dipole U is $L > 0$, with a single equality orbit. In particular $c_U = 30 - 15U/2 \ge 0$. Let*

$$V_{ij} = \langle \mathsf{T}\chi | \widehat{Q}_{ij} | \chi \rangle, \qquad \widehat{Q}_{ij} = \frac{J_i J_j + J_j J_i}{2} - 4\delta_{ij} I$$

for a representative χ of this orbit. If $\Re V$ and $\Im V$ are linearly independent, then the positive lower bound (389) is strict at every $0 < q < 1$ with this U.

Proof. The actual pure-state disk in Section 26.3 is

$$Q(w) = Q_0 + w_1 D + w_2 E, \qquad |w|^2 = q, \quad A = \|Q_0\|_F^2 \ge L, \quad C_{ab} = D_a : D_b, \quad \operatorname{Tr} C = A + c_U.$$

One can take $(D, E) = (\Re V, \Im V)$, up to a rotation or sign change of the two disk coordinates. The proof of the plotted lift used

$$\sqrt{\tau} \ge \sqrt{A} - \sqrt{q \operatorname{Tr} C} \ge \sqrt{L} - \sqrt{q(L + c_U)}. \tag{462}$$

For $c_U \ge 0$ and $0 < q < 1$, the function $\sqrt{A} - \sqrt{q(A + c_U)}$ is strictly increasing in $A > 0$. Thus equality in a positive bound requires $A = L$, putting the seed in the stipulated equality orbit. But independence of D, E gives $C \succ 0$ and

$$\|w_1 D + w_2 E\|_F^2 \le q \lambda_{\max}(C) < q \operatorname{Tr} C.$$

The norm estimate in the first step of (462) is then strict, a contradiction. □

Proposition 10. *All six points $G_5, \dots, G_{10}$ are unattainable.*

Proof. Write their null equality seed in the normalized form

$$\chi = \frac{\sqrt{15} D_0 + k D_2 + 15u D_4 - \sqrt{15} u k D_6}{\sqrt{N}}, \qquad N = (15 + k^2)(1 + 15u^2). \tag{463}$$

At the first five anchors $7/20 < u < 2/5$ and $k > 0$. Direct ladder action gives

$$\Re V_{zz} = -\frac{240ku}{N}, \qquad \Im V_{xy} = -\frac{3[225u^2 - 75u - (5u + 1)k^2]}{N}. \tag{464}$$

All diagonal entries of V are real; its only off-diagonal entries are the purely imaginary $V_{xy} = V_{yx}$. Thus $\Re V : \Im V = 0$, and the first displayed entry is nonzero.

To check that the second cannot vanish, put $d_u = 1 + 15u^2$, $e = (1 - 15u^2)/d_u$, $\nu = f - 2e$ and $k^2 = 15(1 - \nu)/(1 + \nu)$. Its vanishing would force

$$f = f_E(u) := \frac{-45u^2 + 10u + 3}{15u^2 + 1}.$$

For the explicit stationary polynomial (440), direct substitution gives the short identity

$$P_{f_E(u)}(u) = 160u^2(15u^2 - 6u - 1)(45u^2 - 10u - 3). \tag{465}$$

The first quadratic is negative throughout $[7/20, 2/5]$. The only possible zero in that box is therefore $u = (5 + 4\sqrt{10})/45$, for which $f_E = 0$. Every anchor has $f_i > 0$, so $\Im V_{xy} \ne 0$.

At the critical endpoint $f = b_c$, $k = 0$, and the first entry in (464) vanishes. Instead use

$$\Re V_{xx} = \frac{225u(3u + 1)}{N} > 0, \qquad \Im V_{xy} = -\frac{225u(3u - 1)}{N} \ne 0, \quad \frac{7}{20} < u < \frac{2}{5}.$$

Thus in all six cases $\Re V$ and $\Im V$ are nonzero orthogonal tensors. This property is unchanged by collective rotations and global phase. Lemma 10 makes lift equality impossible at the positive-q tips (461). □

The spin-matrix checks in this proof need only $J_z D_j = (3 - j) D_j$ and $J_+ D_j = \sqrt{j(7 - j)} D_{j-1}$. They are finite identities, not amplitude-level polynomial elimination. The algebraic sharpness of the six null inputs is the separate, explicitly identified computer-assisted input described in Section 28.

32.4 The tiny upper triple and its replacement

Proposition 11. *The former microscopic junction G_4 is unattainable and is removed by the current bound* (395)*. Moreover equality in that stronger bound is impossible whenever $f > 0$, $r > 0$ and its guard conditions hold.*

Proof. The former triple is the analytical intersection of the old flow sheet $\sqrt{\tau} = H_{\mathrm{old}}$, the unit-dipole cap (377), and the first upper null-seed lift (390). It has $f > 0$, $r > 0$ and satisfies both guards in (393). The positive improvement ϵ in (397) therefore rules out its old flow equality. In particular it is not a state hidden by a difficult numerical reconstruction.

For the last claim, consider the actual flow used to prove (395), starting at $f_0 > 0$, $r_0 > 0$. The definition of R gives $R > r_0$: indeed $\sqrt{(1-r_0)^2 - 2f_0^2} < 1 - r_0$. At the initial time the covariance bound is consequently stronger than the scalar comparison by the positive matrix $(R-r_0)I/2$. Hence

$$\dot f(0) < -\big[a - 2f_0^2\big] f_0 = \dot g(0).$$

The comparison in Section 26.4 gives $f(t) \le g(t)$ for all $t \ge 0$, with strict inequality on a nonempty time interval. Therefore $\int f(t)\,dt < \mathcal{I}_1(f_0, a)$. Its coefficient $6\sqrt{6}\sqrt{1-q_0}$ is positive in the guard domain. The integrated quadrupole bound is strictly smaller than the right side of (395), proving strictness. Thus a new positive-f, positive-r triple on this stronger sheet is also excluded; moving the necessary boundary does not turn it into an attained family. □

The updated analytical triple is approximately

$$(x, y, z) = \big(3.960336389 \times 10^{-6},\ 6.863220505413,\ 7.999065784781\big), \qquad q \simeq 0.000117766986452.$$

It replaces, rather than overlays, the old G_4 of Table 2. Both have $f, r > 0$, so the proposition excludes the replacement as well. The approximation only locates the new root; its definition is the exposed intersection with the exact comparison bound (395).

For a scale check, in the rational locator box

$$3.96027 \times 10^{-6} \le x \le 3.96028 \times 10^{-6}, \qquad 7.99906578 \le z \le 7.99906579$$

one has $q < 0.000118$, $f > 2437/10^6$, $a > 0.988$ and $c > 0.987988$. Rational squaring gives $\sqrt{6} > 2449/1000$ and $\sqrt{1-q} > 9999/10000$. Since $ac \le 1$, dropping the positive f^4 term proves

$$\epsilon > 8\frac{2449}{1000}\frac{9999}{10000}\left(\frac{2437}{10^6}\right)^3 > 2.8 \times 10^{-7}.$$

This is a strict analytic gap, not a nearly zero floating residual. At the former triple its value is approximately 2.901×10^{-7} in $\sqrt{\tau}$, corresponding to an S_2 sheet improvement of approximately 4.09×10^{-7} at fixed (x, q). Such small distances explain why the old and new sheets look almost coincident in a full-body view.

What the exclusions do and do not imply. The pure-state domain is compact and its sector map is continuous. Every excluded tuple therefore has an open neighborhood disjoint from the exact sector image. No uniform numerical neighborhood radius is asserted for the disk tips. The stronger flow inequality does provide an explicit necessary neighborhood cut, but its actual attainment boundary is not classified. The figures continue to show a closed proved outer region with constructive attained patches, not the exact pure-state sector body.

33 Base geometry and the current finite rendering

Current geometry. The current figures sample $\mathcal{R}_4^{\mathrm{ref}}$ from (10): the strengthened 22 October system together with the refined four-copy lower condition of Section 2. At each vertical section all applicable guarded inequalities are imposed, retaining only their active minimum upper and

maximum lower boundaries. The retained paired remainders are not replaced by the older closed roof (466), and the null and side lifts can remove further parts of that base roof. The exact zero-dipole side is described in Section 27.2, including the rigorous tiny-tail bracket. Finite meshes approximate these analytical intersections; no convex hull is used. The following geometric derivations describe the *base* components and explain their historical parametrizations. They are used only where they survive the new constraints. In particular a base projection-domain proof below is not a claim that the new cuts leave that projection or all its old curves unchanged.

Geometry. The linear rows are half-spaces, and the orange condition is the convex epigraph of the support envelope (5). The full teal condition is an intersection of affine half-spaces, so its upper roof (257) is concave. The new paired-spectral roof is concave as shown below, and does not justify convexifying the lower boundary. The stronger implicit cap needs more care: its roof alone is not globally concave. The relevant object is its roof *after clipping by paired purity*, which is concave as shown next. This justifies supporting-plane rendering of the clipped cap, not a convex-hull replacement of the inequality. However, the newly raised lower sheet $L(x,y) = x[2 + 12/(4 + \sqrt{5(y-5)/2})]$ is not jointly convex: at $x > 0, y > 5$ its Hessian has $L_{xx} = 0$ and $L_{xy} \neq 0$, hence negative determinant. It is exposed on an open patch, for example near $(x, y, z) = (1, 33/5, 4)$; all other defining conditions are strict there. The updated outer region is consequently nonconvex. It must not be replaced by its convex hull, nor may local tangents to this lower sheet be treated as globally valid half-spaces. This statement concerns the necessary-condition region, not a classification of the achievable pure-state set. Curved boundaries also mean that the region is not literally a polytope.

Direct vertical sections, without convexification. The base region's exact projection onto the first two coordinates is

$$\mathcal{D} = \{(x,y) : 5 \le y \le 15,\ 0 \le x \le X(y) := (y+3)/3\}.$$

On this domain the base region is the interval between

$$Z_-(x,y) = \max\left\{5x + 5 - y, 2x + \frac{12x}{4 + \sqrt{5(y-5)/2}}\right\},$$
$$Z_+(x,y) = \min\left\{Z_{\mathrm{ps}}(x,y), \frac{45 + 10x - 2y - \Psi(y)}{3},\ \frac{1011 + 217x - 65y}{63}, Z_{\mathrm{teal}}(x,y),\ R(x,y)\right\},$$

where R is the base red first-root roof in Lemma 11. In particular the flat $q = 0$ branch is already included in Z_{teal}. The current figure further clips these graphs by the new conditions before closing the surviving sections with the two nondegenerate side faces $x = 0$ and $y = 5$. The four-copy condition raises only the lower graph, as in (28); the upper envelope is unchanged.

The original paired-spectral roof, including its paired-purity branch, is

$$\begin{aligned} B(x,y) &:= 5 + 15x - y, \\ d_{\min}(B) &:= \frac{\left(\sqrt{25 + 15[B]_+} - 5\right)^2}{90}, \\ Z_{\mathrm{ps}}(x,y) &:= 15 + 5x - y - d_{\min}(B(x,y)). \end{aligned} \tag{466}$$

Indeed $z = 15 + 5x - y - \Delta$ turns (6f) into $B - 6\Delta \le \sqrt{40\Delta}$. For $B \le 0$, the minimum allowed Δ is zero. For $B > 0$, solve the monotone equation $6\Delta + \sqrt{40\Delta} = B$ to obtain $d_{\min}$. This derivation retains all points with negative left side; it never replaces the unsquared inequality by an invalid squared condition.

At $B = 0$, $d_{\min}$ and its first derivative vanish. For $B > 0$,

$$d'_{\min}(B) = \frac{1}{6} - \frac{5}{6\sqrt{25 + 15B}}, \qquad d''_{\min}(B) = \frac{25}{4(25 + 15B)^{3/2}} > 0.$$

Thus $d_{\min}$ is convex and Z_{ps} is concave. The curved branch joins the paired plane with matching gradient, so this join is not drawn as a physical ridge or populated with nodes. Its endpoints on other physical creases are marked as curve junctions, even when the surfaces meet tangentially there.

For completeness, $q \ge 0$ and $y + z \ge 5x + 5$ imply $x \le X(y)$. Conversely the section $z = x[2 + 24/(y + 3)]$ lies in every required interval on $\mathcal{D}$. For the new lower bound this follows from $\sqrt{5(y-5)/2} \le 5$; all affine upper supports hold by their endpoint values. For the red bound, putting $r = \sqrt{1 - x/X(y)}$ gives

$$\mathcal{F}(x, r) = (b + \gamma\sqrt{X(y)})(1 - r) + \sqrt{10}\, r \ge \sqrt{10} \ge \sqrt{y - 5}.$$

Thus no disconnected interval in the projection is silently filled. The same witness also satisfies the new cap: for each fixed y, it is the straight segment between $(0, y, 0)$ and $((y + 3)/3, y, 10 + 2y/3)$. At its endpoints the new inequality reads, respectively, $5y - 85 \le \sqrt{40(15 - y)}$ and $4y - 40 \le 20$, both true for $5 \le y \le 15$. Concavity of Z_{ps} retains the entire segment. Hence the projection domain remains unchanged. The added sharp plane also retains this section: its left side at the same two endpoints is $65y$ and $413 + 104y/3$, respectively, both below 1011 for $5 \le y \le 15$. No extra projection restriction has therefore been introduced by this new facet. Before the four-copy cut, the crease separating the two lower sheets is

$$(x, y, z) = \left(\frac{2v(v + 4)}{15},\ 5 + \frac{2v^2}{5},\ \frac{4v(v + 10)}{15}\right), \qquad 0 \le v \le 5.$$

Only its segment $0 \le v \le \widehat{v}_4$ remains exposed in the current body; the two new lower creases and their excluded gray junction are described in Section 2. At $y = 5$ the complete supporting family has the exact limit $Q_*(x, 5) = [1 - 6x]_+/81$. Its new outer corner $(1/6, 5, 25/3)$ is attained by Ω_4 and is marked violet, as proved in Section 3.5. The adjoining nonzero-dipole C_4 arc is violet as well. Gray marks the remaining outer corners. The ten audited gray junctions, including the current microscopic flow contact, are excluded in Section 32; the still older superseded corners are treated in Section 21. The additional gray lower junction J_4 is excluded by the Gram-equality argument in Section 2; the entire refined lower patch and open crease arcs are excluded by Theorem 2. Other gray patches or curves can still be unresolved.

Lemma 11 (The correct root and the exposed red patch)**.** *Write $u = \sqrt{x}$, $t = \sqrt{y - 5}$ and $T(u, r) = b + dr + \gamma u\sqrt{(1 - r)/(1 + r)}$. For $5 \le y \le 15$, let $r_*(x, y) = 0$ if $t \le b + \gamma u$; otherwise let it be the first root $T(u, r_*) = t$ in $[0, 1]$. On the established domain $0 \le q \le 1$, the strong cap is equivalent to the roof $z \le R(x, y) := 8 + 2x - 8r_*^2$. The clipped roof*

$$H(x, y) := \min\{R(x, y), 15 + 5x - y\}$$

is concave. The interior of its exposed nonlinear red patch is confined to

$$0 < u < u_0 := \frac{2 - b^2}{\sqrt{3b^2 + 2\gamma^2 - 6} + b\gamma} < \frac{1}{32}, \qquad 0 < r < r_0 := \frac{2 - b^2}{8 - d^2} < \frac{1}{20}. \tag{467}$$

Every tangent plane at this patch therefore supports the entire clipped roof H; the displayed rims belong to its closure.

Proof. Root selection. If $b + \gamma u \ge \sqrt{10}$, then $T - \sqrt{10} = \sqrt{(1 - r)/(1 + r)}[\gamma u - d\sqrt{1 - r^2}] \ge 0$, so the cap is automatic. Otherwise T_r is positive at zero and can change sign only once, from positive to negative (it increases up to $r = 1/2$ and then decreases). Thus $T \ge \min\{T(u, 0), T(u, 1)\}$, which also justifies $r_* = 0$ when $t \le b + \gamma u$. Since $T(u, 1) = \sqrt{10} \ge t$, a level $b + \gamma u < t \le \sqrt{10}$ has a unique first, upward crossing; the admissible overlaps are exactly $r \ge r_*$. This proves the roof assertion, including the automatic case.

Confinement by paired purity. On a red equality sheet, paired purity requires $g := T^2 - 2 - 3u^2 - 8r^2 \le 0$. Direct expansion gives

$$g = (1 - r)[(8 - d^2)r + b^2 - 2] + 2\gamma u\sqrt{\frac{1 - r}{1 + r}}(b + dr) + \left[\gamma^2\frac{1 - r}{1 + r} - 3\right]u^2.$$

Put $r_c = (\gamma^2 - 3)/(\gamma^2 + 3) = 121/271$. For $r_0 \le r \le r_c$ every term is nonnegative. If $r_c < r < 1$ and $T \le \sqrt{10}$, then $0 \le u \le d\sqrt{1 - r^2}/\gamma$. The quadratic g is concave in u; its endpoint values are $g(0, r) \ge 0$ and $(8 - 3d^2/\gamma^2)(1 - r^2) > 0$. Thus this range is also clipped away; $r = 1$ is a domain-boundary case handled by continuity. For $r \le r_0$, g increases with u. At $u = u_0$, $g(u_0, 0) = 0$, and $g_r = 2TT_r - 16r > 0$: indeed $T \ge 4/3$ and $T_r \ge 9/5 - (3/32)/(399/400)$ throughout the stated rectangle. The bound $u_0 < 1/32$ follows already from $g(1/32, 0) > (4/3 + 7/80)^2 - 2 - 3/1024 > 0$. This excludes $u \ge u_0$ as well, proving (467).

Concavity on the exposed patch. Put $a = \gamma\sqrt{(1 - r)/(1 + r)}$, $h = T_r = d - ua/(1 - r^2) > 0$. Parametrize the graph by $X = (u^2, 5 + T^2, 8 + 2u^2 - 8r^2)$ and use its upward graph normal $N = (-R_x, -R_y, 1)$. For the curvature matrix $M_{ij} = -N \cdot X_{ij}/16$, differentiation gives

$$M_{uu} = \frac{ra(b + dr)}{uTh} > 0, \qquad \det M = \frac{ar\,\mathcal{P}}{uT[ua - d(1 - r^2)]^2},$$

where

$$\mathcal{P} = bd(1 - r^2)^2 + ua\{b(3r^2 - 2r - 1) + d(-r^3 - 2r^2 + 3r)\} - 4a^2ru^2.$$

The elementary bounds $4/3 < b < 27/20$, $9/5 < d < 37/20$ and $a \le 3$ give, throughout $0 < u < 1/32$, $0 < r < 1/20$,

$$\mathcal{P} > \frac{12}{5}\left(\frac{399}{400}\right)^2 - \frac{3}{32}\frac{27}{20}\frac{11}{10} - \frac{9}{5120} > 2.24.$$

Hence M is positive definite and the red graph is concave there. At $r = 0$ it joins the affine roof $8 + 2x$ with matching gradient. At a paired-purity seam, the local roof is the minimum of this concave graph and an affine plane, hence is concave. Elsewhere a plane dominates. Thus H is locally concave everywhere on the convex open domain $x > 0$, $5 < y < 15$, and therefore globally concave; continuity includes its boundary. Its supporting-plane assertion follows. □

Exact boundaries, nonredundant pieces only. The orange constraint is equivalently the tangent family

$$-30tx + (20t - 8)y + 9tz \le 140t + 80t^2 - 40, \qquad \frac{7}{10} \le t \le 1. \tag{468}$$

Only boundary pieces surviving all the other inequalities are drawn. In particular, the new paired-spectral roof and the other inequalities clip the orange and red surfaces. The former flat spectral sheet is not drawn. No surface continues into a region excluded by another bound.

For reproducibility, every red equality point is parametrized by

$$\begin{aligned} x &= p^2, \\ y &= 5 + \left(b + dr + \gamma p\sqrt{\frac{1 - r}{1 + r}}\right)^2, \\ z &= 8 + 2p^2 - 8r^2, \qquad p \ge 0, \quad 0 \le r \le 1. \end{aligned} \tag{469}$$

Only parameters satisfying *all* the other defining inequalities belong to the displayed sheet. Its seam with paired purity is determined by

$$\left(b + dr + \gamma p\sqrt{\frac{1 - r}{1 + r}}\right)^2 = 2 + 3p^2 + 8r^2. \tag{470}$$

This replaces the explicit quadratic seam of the simplified red roof. At $p = 0$, (469) reduces to the unchanged red branch in (276). Thus the magnified $S_1 = 0$ cross-section is identical, although the three-dimensional red sheet is stricter.

Rendering accuracy versus mathematical validity. The exact region is defined by the continuous inequalities, not by the finite mesh used to display them. The highlighted curved sheets and their seams are sampled from their analytical equations and clipped against the remaining conditions. The red sheet is tested with (6j) itself. Its supporting upper planes are justified by Lemma 11; the full teal upper planes are already the affine inequalities (6i). Finite selections of upper supports give a slightly looser outer roof. The current figures instead evaluate the full envelope (257) before sampling the upper graph; finite support selections are not used to define the displayed teal surface. The nonconvex lower sheet is evaluated directly from (6a), not represented by invalid global tangent half-spaces or a convex-hull replacement. Sampling and clipping errors are display-accuracy issues, not ingredients of any physical proof. The old spectral plane, old root cap, old lower plane, and former simplified red-roof formula are redundant corollaries, not additional boundaries in any of the three perspectives or in the current magnifications. The spectral/orange crease has two branches at the same y near its turn. Connecting roots row by row can therefore create a spurious zigzag. On the low orange branch the exact curve is instead parametrized by

$$u = \sqrt{\Delta}, \qquad y = \frac{62 + 21u^2 - 7\sqrt{40}\,u}{4}, \qquad x = \frac{21u^2 + y + 9}{35}, \qquad z = 15 + 5x - y - u^2.$$

It turns at $(x, y, z) = (6/5, 29/3, 92/9)$, a point on one smooth curve, not a crossing. The curve meets the $q = 0$ rim at $(3/2, 99/10, 11)$, where the attained $\chi_{7/10}$ junction is shown in violet by Section 3.6. Before the sharp plane was added, the distinct crease continued along the high orange branch to $(25/54, 215/18, 145/27)$ and then along the paired/orange crease to the GHZ point. The new plane clips these curves and removes that old junction. Its new exposed junctions are the two points analyzed in Section 23. Only surviving curve fragments and the new facet's exposed creases are drawn, using their exact equations. Other three-surface curve junctions are marked with the same node size; apparent crossings at different three-dimensional positions are not marked as intersections.

Visibility rays start on the exact analytical boundary. In particular, the midpoint of a drawing chord must first be returned to its upper or lower graph: a chord of a concave roof lies below the surface and would otherwise spuriously classify a visible curved edge as hidden. This correction changes no inequality or attainable-state claim. The reproducible plotting sources are `Julia/polytope_symmetric_N6_uniform_attainment.jl` and `Julia/N6_paired_sharp_cap.jl`, with the full upper and lower graphs in `Julia/N6_multipole_compatibility.jl`. Magnifications are generated by `Julia/N6_violet_magnified.jl` and `Julia/N6_violet_comparison_zooms.jl`. The nearby-state detail is generated by `Julia/N6_Estar_omega_magnified.jl`; both its family-colored and single-color modes omit the non-edge equality trace. The camera, surface geometry, node coordinates and leaders are identical. Both combined perspectives are assembled by `Julia/combine_N6_violet_panels.py`; the family-colored comparison is assembled by `Julia/combine_N6_family_color_views.py`. The separate native-scale family views and their compact legends are exported by `Julia/export_N6_family_views_with_legends.py`. The state-label leaders are clipped to the outer borders of the flat markers, checked against neighboring nodes, and drawn below the markers and label text. The fold-outs import the same verified panels, including the complete magnified neighborhood of $|E_*\rangle$ and $|\omega\rangle$, without redrawing, clipping their labels or changing their coordinates.

References

[1] E. Serrano-Ensástiga, O. Giraud and J. Martin, *Monogamy relations for symmetric multiqubit systems*, working manuscript, local version `Purity_inequalities_symmetric-011026.pdf`; Eqs. (42)–(43), and the normalized multipole and two-copy definitions.

[2] E. Serrano-Ensástiga, O. Giraud and J. Martin, *Multiqubit monogamy relations beyond shadow inequalities*, Phys. Rev. A **113**, 012415 (2026). doi:10.1103/9fkf-hm8l.

[3] K. M. R. Audenaert, *Subadditivity of q-entropies for $q > 1$*, J. Math. Phys. **48**, 083507 (2007). arXiv:0705.1276. This is background for the norm approach; the new compatibility inequalities are proved independently above.

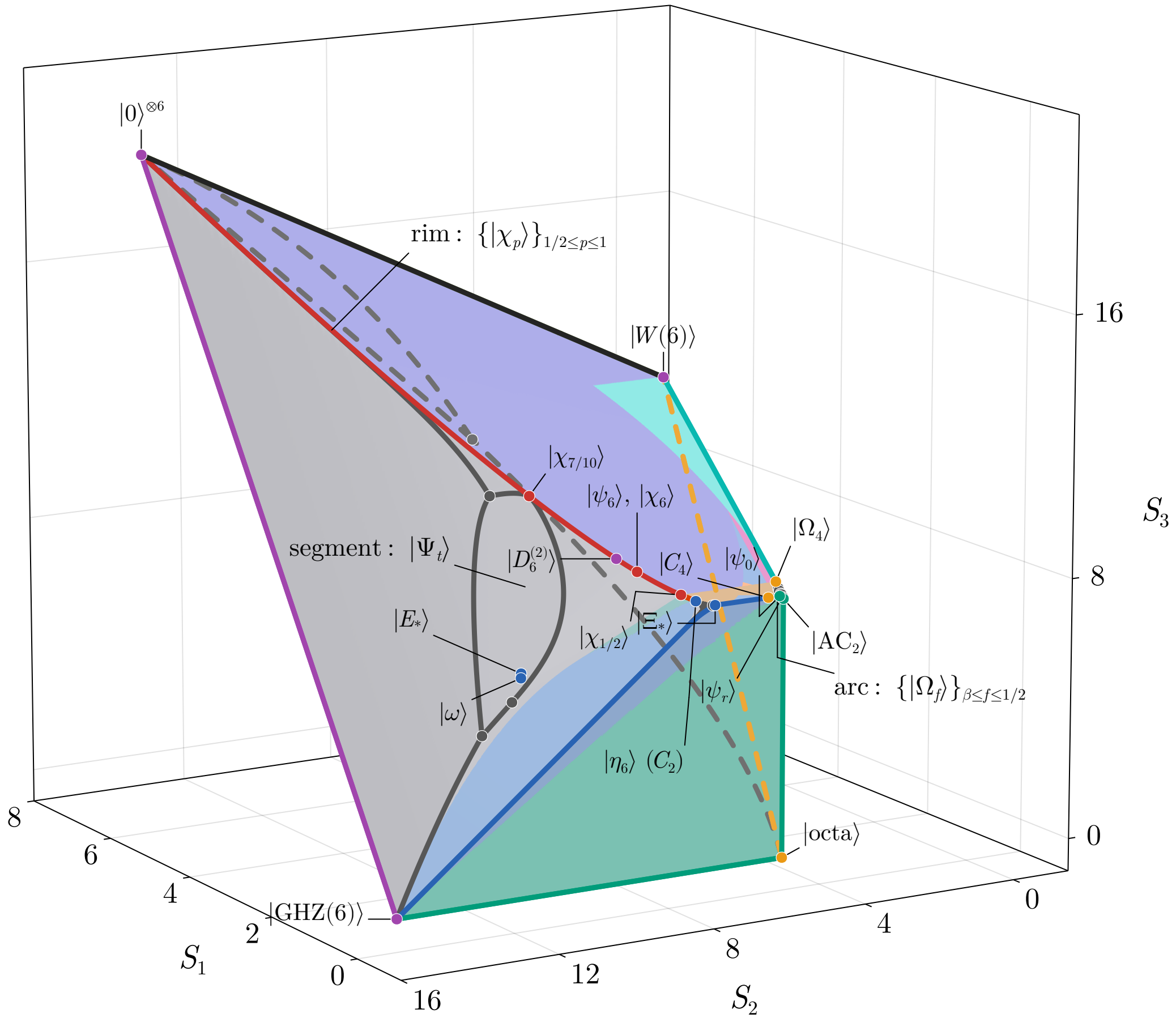
[4] Research note *Log-convexity of reduced purities*, 13 October 2026, Theorem "An even-correlation inequality", quantitative remainder `eq:even-four-quantitative`, and the proved $N = 6$ quantitative cuts. Source: `Log_convexity_reduced_purities_20261013.tex`.

[5] Research note *A spectral proof of the sharp bound $S_4 \geq 45/7$ for pure symmetric six-qubit states*, updated October 2026, `N6_S4_proof_20261014/proof.tex`.

[6] Research note *An analytical spectral lemma for pure symmetric six-qubit states*, 16 October 2026, `N6_spectral_lemma_analytical_20261016.tex`. The singular-Gram-matrix argument is reproduced in Section 18.

[7] Research note *A sharp six-qubit sector-length bound at $S_1 = 0$ and $S_3 = 8$*, 16 October 2026, `N6_S2_sharp_F0_Theta0_20261016.tex`. The scalar compatibility proof is reproduced in Lemma 1.

Attained boundary constructions (N = 6): (a) main perspective



Existing construction colors retained; the full constructive C2 null-roof band uses C2 blue.

Only true body edges are drawn; interior construction traces are omitted. All state nodes and leaders remain.

Rear edges dashed. Four-copy cap excluded except product; other gray is excluded or uncertified.

Color key Light fills: surfaces. Matching darker colors: edges and nodes.

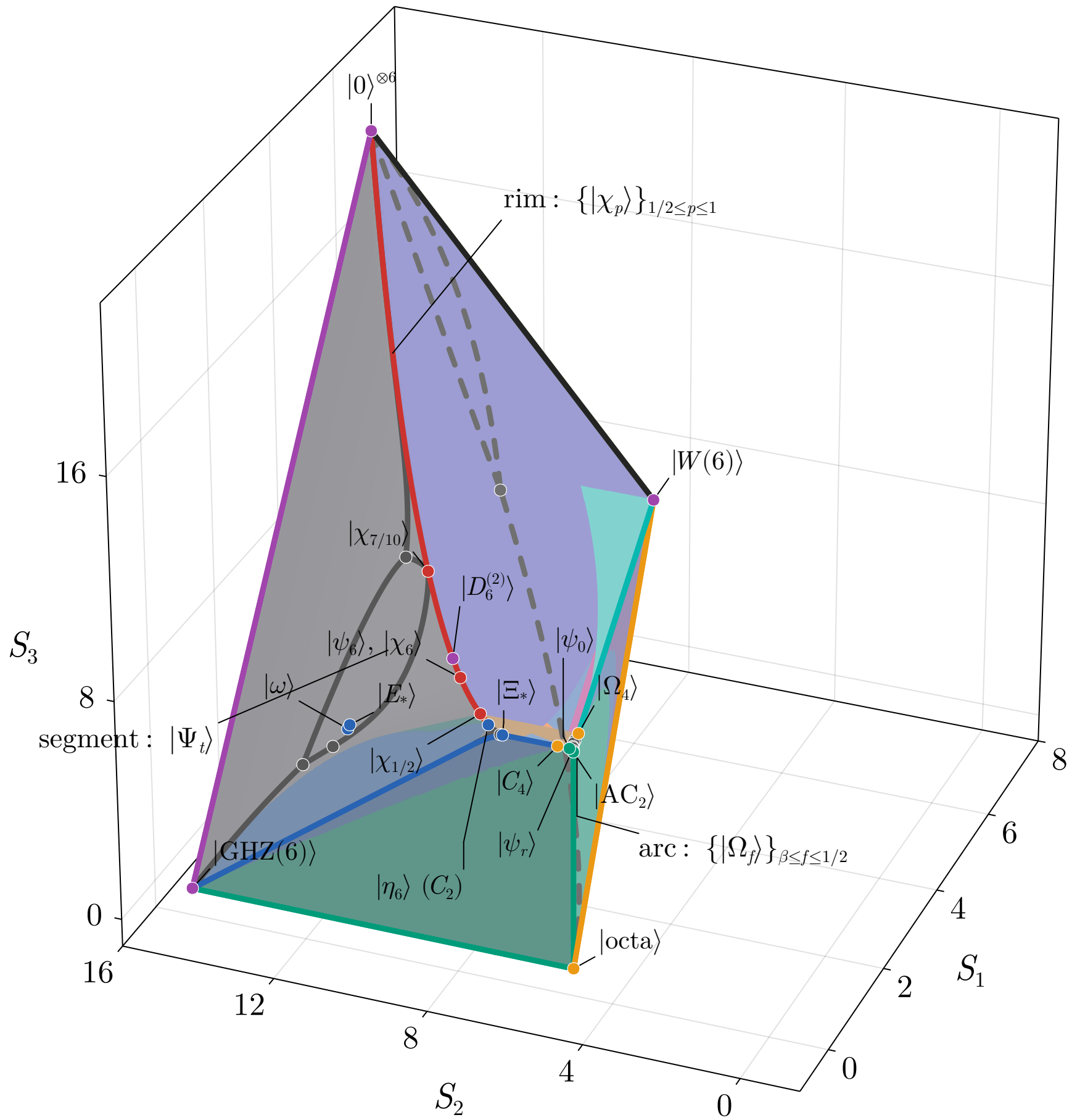
- Paired-purity face
- Quadrupole-free (Q=0)
- Null roof (q=0)
- Small-dipole four-level
- Seven-level
- Rank-one roof
- Filtered octa
- Five-level (witnessed)
- Octa-product
- GHZ-product
- Rotated-W superposition
- C2 attained patches
- C3 attained patches
- Other attained patches
- Excluded / uncertified

Additional edge / node colors

- C4 edges / nodes
- C5 rim / nodes
- C6 edge / nodes
- Global paired rim
- Other attained edges

Colors name constructions, not exclusive symmetries. Gray: excluded/uncertified faces, edges and nodes.

Attained boundary constructions (N = 6): (b) alternative perspective



Existing construction colors retained; the full constructive C2 null-roof band uses C2 blue.

Only true body edges are drawn; interior construction traces are omitted. All state nodes and leaders remain.

Rear edges dashed. Four-copy cap excluded except product; other gray is excluded or uncertified.

Color key Light fills: surfaces. Matching darker colors: edges and nodes.

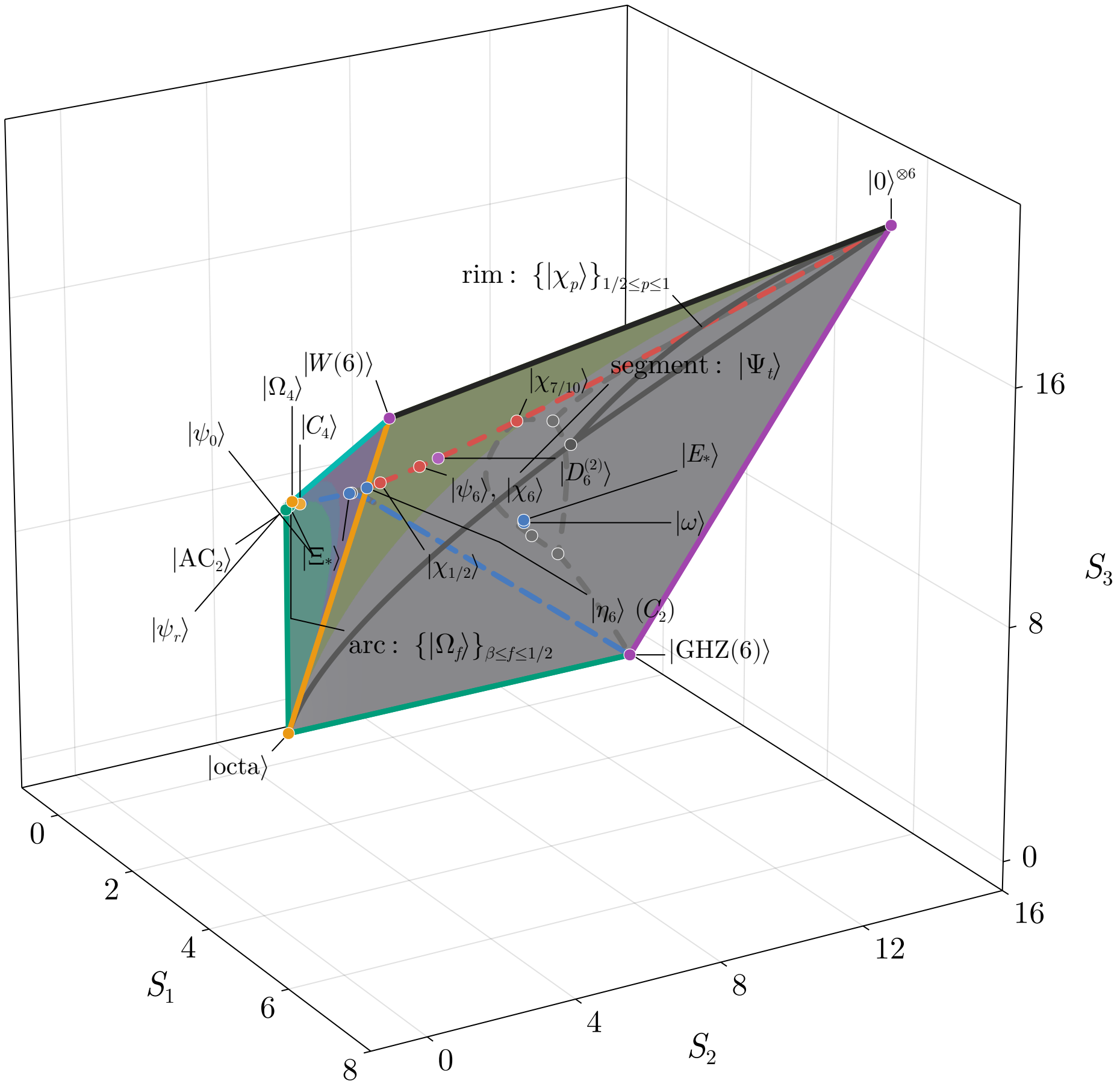
- Paired-purity face
- Quadrupole-free (Q=0)
- Null roof (q=0)
- Small-dipole four-level
- Seven-level
- Rank-one roof
- Filtered octa
- Five-level (witnessed)
- Octa-product
- GHZ-product
- Rotated-W superposition
- C2 attained patches
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- Other attained patches
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Additional edge / node colors

- C4 edges / nodes
- C5 rim / nodes
- C6 edge / nodes
- Global paired rim
- Other attained edges

Colors name constructions, not exclusive symmetries. Gray: excluded/uncertified faces, edges and nodes.

Attained boundary constructions (N = 6): (c) rear perspective



Existing construction colors retained; the full constructive C2 null-roof band uses C2 blue.

Only true body edges are drawn; interior construction traces are omitted. All state nodes and leaders remain.

Rear edges dashed. Four-copy cap excluded except product; other gray is excluded or uncertified.

Color key Light fills: surfaces. Matching darker colors: edges and nodes.

- Paired-purity face
- Quadrupole-free (Q=0)
- Null roof (q=0)
- Small-dipole four-level
- Seven-level
- Rank-one roof
- Filtered octa
- Five-level (witnessed)
- Octa-product
- GHZ-product
- Rotated-W superposition
- C2 attained patches
- C3 attained patches
- Other attained patches
- Excluded / uncertified

Additional edge / node colors

- C4 edges / nodes
- C5 rim / nodes
- C6 edge / nodes
- Global paired rim
- Other attained edges

Colors name constructions, not exclusive symmetries. Gray: excluded/uncertified faces, edges and nodes.

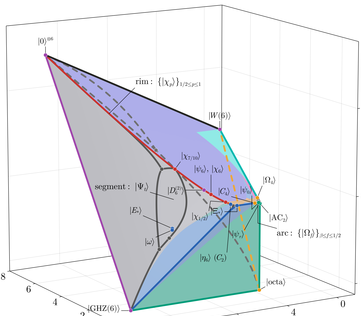
Boundary attainment: cyclic families and coherent superpositions (N = 6)

Same proved outer region: three perspectives, state-node detail and attained C2 roof band.

complete paired face: patch
small-dipole four-level: patch
filtered octa: patch
GHZ-product: patch
C3: patches
complete Q=0 side: patch
seven-level: patch
five-level: witnessed patch
rotated W: patch
complete null roof: patch
rank-one roof: patch
octa-product: patch
C2: patches
E_* and omega: C2 states
Sharp C2 endpoint: S1 = 2/27
Seven-level attained edge
Unique collective-rotation/phase orbit
Full paired rim: explicit family
C4: edges/points
C5: rim/points
C6: edge/points
Gray: excluded or not certified
Black: other attained body edges

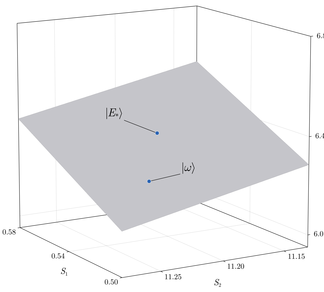
Colors give explicit attaining constructions, not exclusive symmetry classes.
Earlier constructive family colors are retained wherever they already supply a witness.
Blue adds the full constructive C2 null band; additional global sharpness is asserted only on certified strips.
Only genuine body edges are stroked; interior construction traces are omitted.
Refined four-copy cap: excluded except product; all labels and leaders are retained.

Attained boundary constructions (N = 6): (a) main perspective



Existing construction colors retained; the full constructive C2 null-roof band uses C2 blue.
Only true body edges are drawn; interior construction traces are omitted. All state nodes and leaders remain.
Rear edges dashed. Four-copy cap excluded except product; other gray is excluded or uncertified.

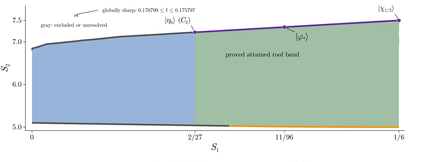
Magnified equality segment : $|E_7\rangle$ and $|\omega\rangle$



$$65S_2 + 63S_3 - 217S_1 = 1011 \qquad \text{(supporting face)}$$

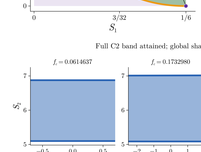
Blue: attained C2 states. Gray: supporting face. The interior equality trace is omitted.
Exact node positions; original uniform marker size. Window margins are not body edges.

$q = 0,\ S_2 = 8 + 2S_1$: current bounds and proved attained patches

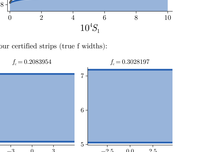


Sharp paired rim : $2/27 \le S_1 \le 1/6$; roof complete for every $S_1 \ge 2\beta^2/3$.

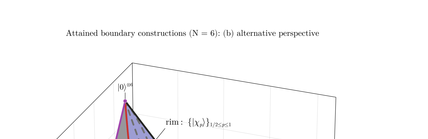
Lower rim: current cuts and the sharp C4 threshold



Upper rim: current nonredundant envelope



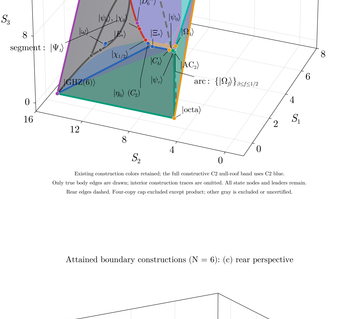
Full C2 band attained; global sharpness on four certified strips (true f widths):



Blue: C2. Green: other proved patches. Orange: C4 rim. Additional global sharpness only on certified strips; no positive-q collar.

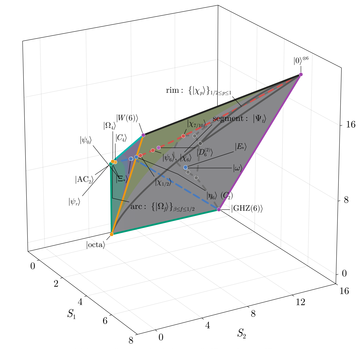
$\beta = 0.366379159399607\ldots$: $16\beta^3 - 78\beta^2 + 92\beta + 6\beta - 11 = 0$

Attained boundary constructions (N = 6): (b) alternative perspective



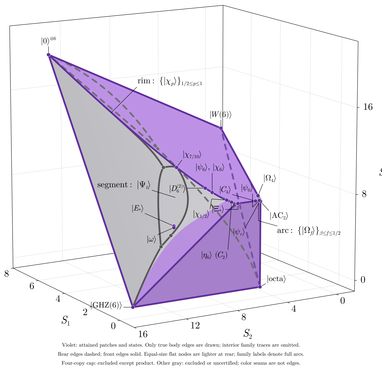
Existing construction colors retained; the full constructive C2 null-roof band uses C2 blue.
Only true body edges are drawn; interior construction traces are omitted. All state nodes and leaders remain.
Rear edges dashed. Four-copy cap excluded except product; other gray is excluded or uncertified.

Attained boundary constructions (N = 6): (c) rear perspective



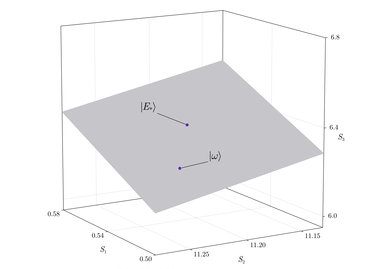
Existing construction colors retained; the full constructive C2 null-roof band uses C2 blue.
Only true body edges are drawn; interior construction traces are omitted. All state nodes and leaders remain.
Rear edges dashed. Four-copy cap excluded except product; other gray is excluded or uncertified.

Proved outer region and attained boundary patches (N = 6)



Violet: attained patches and states. Only true body edges are drawn; interior family traces are omitted.
Rose edges dashed, front edges solid. Equal-size flat nodes are lighter at rear; family labels denote full arcs.
Four-copy cap: excluded except product. Other gray: excluded or uncertified; color seams are not edges.

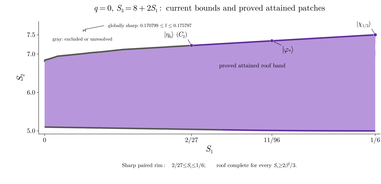
Magnified equality segment : $|E_7\rangle$ and $|\omega\rangle$



$$65S_1 + 63S_2 - 217S_3 = 1011 \qquad \text{(supporting face)}$$

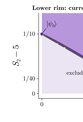
Violet: attained states. Gray: supporting face. The interior equality trace is omitted.
Exact node positions; original uniform marker size. Window margins are not body edges.

$q = 0$, $S_6 = 8 + 2S_1$: current bounds and proved attained patches

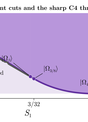


Sharp paired rim : $2/27 \le S_1 \le 1/6$; roof complete for every $S_1 \ge 2^6/3$.

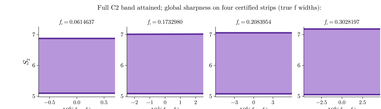
Lower rim: current cuts and the sharp C4 threshold



Upper rim: current nonredundant envelope



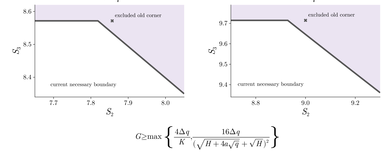
Full C2 band attained; global sharpness on four certified stripes (true f widths):



Violet: explicit states throughout the C2 band. Additional global sharpness is asserted only on certified strips; no positive-q collar.

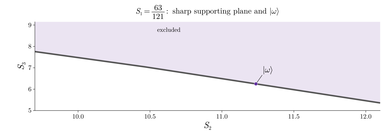
$\beta = 0.366379159399607\ldots$: $\quad 16\beta^2 - 78\beta^2 + 92\beta^2 + 63 - 11 = 0$

Current upper boundary: retained Q1/Q2 cap and supporting plane



$$G \ge \max\left\{\frac{4\Delta q}{K},\ \frac{16\Delta q}{\left(\sqrt{H + 4a\sqrt{q}} + \sqrt{H}\right)^2}\right\}$$

$S_1 = \frac{63}{121}$: sharp supporting plane and $|\omega\rangle$

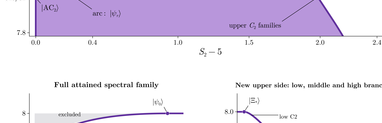


$$65S_1 + 63S_2 - 217S_3 \le 1011$$

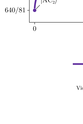
Strong Q1/Q2 bounds use the definitions in the document. Gray: not certified attained. Violet omega line on the attained equality segment.

$N = 6$, $S_1 = 0$: proved side image and attained families

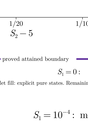
Attained side region: the old implicit cap is redundant



Full attained spectral family



New upper side: low, middle and high branches

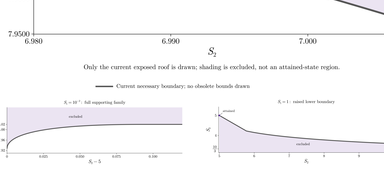


proved attained boundary — attained null segment — necessary bound (tiny residual tails)

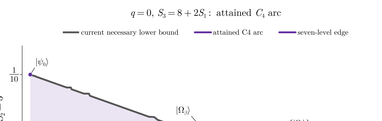
$S_1 = 0$: $\quad S_2 = 8 - 8r^2, \qquad L(r) \le S_2 \le U(r)$

Violet fill: explicit pure states. Remaining endpoint uncertainty in S2 is at most 3.53 x 10^{-8}; not extrapolated.

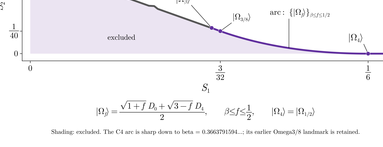
$S_1 = 10^{-4}$: magnified current necessary cap



Only the current exposed roof is drawn; shading is excluded, not an attained-state region.



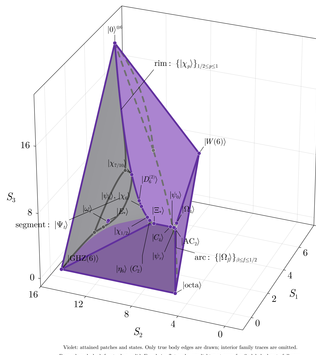
$q = 0$, $S_6 = 8 + 2S_1$: attained C_4 arc



$$|\Omega_f\rangle = \frac{\sqrt{1+f}\ D_6 + \sqrt{3-f}\ D_5}{2}, \qquad \beta \le f \le \frac{1}{2}, \qquad |\Omega_1\rangle = |\Omega_{1/2}\rangle$$

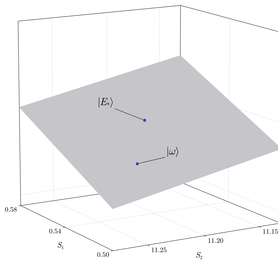
Shading: excluded. The C4 arc is sharp down to beta = 0.3663791594...; its earlier Omega3/8 landmark is retained.

Proved outer region and attained boundary patches (N = 6): alternative perspective



Violet: attained patches and states. Only true body edges are drawn; interior family traces are omitted. Bear edges dashed; front edges solid. Equal-size flat nodes are lighter at rear; family labels denote full arcs. Four-copy cap: excluded except product. Other gray: excluded or uncertified; color seams are not edges.

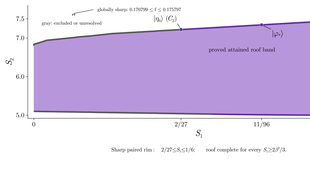
Magnified equality segment : $|E_7\rangle$ and $|\omega\rangle$



$$65S_1+63S_2-217S_3=1011 \qquad \text{(supporting face)}$$

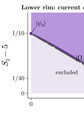
Violet: attained states. Gray: supporting face. The interior equality trace is omitted. Exact node positions; original uniform marker size. Window margins are not body edges.

$q=0,\ S_1=8+2S_2$: current bounds and proved attained patches

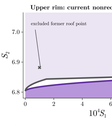


Sharp-paired rim : $2/27\le S_1\le 1/6$; roof complete for every $S_1\ge 2\beta^2/3$.

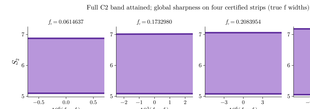
Lower rim: current cuts and the sharp C4 threshold



Upper rim: current nonredundant envelope



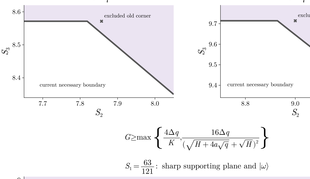
Full C2 band attained; global sharpness on four certified strips (true f widths):



Violet: explicit states throughout the C2 band. Additional global sharpness is asserted only on certified strips; no positive-q collar.

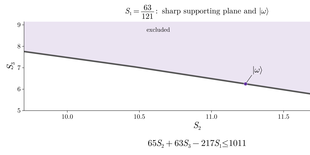
$\beta=0.366379159399607\ldots$: $16\beta^3-78\beta^2+92\beta+6\beta-11=0$

Current upper boundary: retained Q1/Q2 cap and supporting plane



$$G\ge\max\left\{\frac{4\Delta q}{K},\frac{16\Delta q}{(\sqrt{H}+4a\sqrt{q}+\sqrt{H})^2}\right\}$$

$S_1=\frac{63}{121}$: sharp supporting plane and $|\omega\rangle$

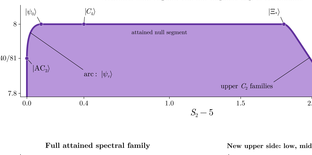


$$65S_1+63S_2-217S_3\le 1011$$

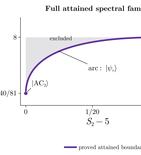
Strong Q1/Q2 bounds use the definitions in the document. Gray: not certified attained. Violet omega lies on the attained equality segment.

$N=6,\ S_1=0$: proved side image and attained families

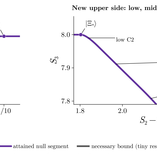
Attained side region: the odd implicit cap is redundant



Full attained spectral family



New upper side: low, middle and high branches

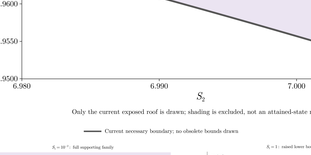


proved attained boundary — attained null segment — necessary bound (tiny residual tail)

$S_1=0$: $\quad S_3=8-8r^2, \qquad L(r)\le S_2\le U(r)$

Violet fill: explicit pure states. Remaining endpoint uncertainty in S2 is at most 3.53 x 10^{-8}; not extrapolated.

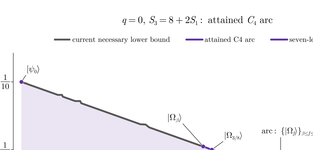
$S_1=10^{-4}$: magnified current necessary cap



Only the current exposed roof is drawn; shading is excluded, not an attained-state region.



$q=0,\ S_1=8+2S_2$: attained C_4 arc



$$|\Omega_\beta\rangle=\frac{\sqrt{1+f}\,D_a+\sqrt{3-f}\,D_b}{2}, \qquad \beta\le f\le\frac{1}{2}, \qquad |\Omega_\gamma\rangle=|\Omega_{1/2}\rangle$$

Shading: excluded. The C4 arc is sharp down to beta = 0.366379594...; its earlier Omega3/8 landmark is retained.